

Solutions to Calculus Book I

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Chapter 1 Schröder-Bernstein Theorem

1.1 PSA

1.1.1 A1) Suppose a non-empty set $X \subset \mathbb{R}$ has an upper bound, and M is an upper bound of X . The following two propositions are equivalent:

- $M = \sup X$.
- For any $\varepsilon > 0$, there exists an $x \in X$ such that $x > M - \varepsilon$.

Proof:

$$M = \sup X \iff \forall M' < M, \exists x \in X, x > M' \iff \forall \varepsilon = M - M' > 0, \exists x \in X, x > M - \varepsilon.$$

1.1.2 A2) Prove that every non-empty open interval contains infinitely many rational numbers.

Proof: We only need to find one rational number q in the interval (a, b) , then we can apply the process to (a, q) and so on. By the Archimedean rule, there is a positive integer N such that $N(b - a) > 2$, hence there exists an integer p such that $p = \lfloor bN \rfloor \in (aN, bN)$, and $q = \frac{p}{N} \in (a, b) \cap \mathbb{Q}$.

1.1.3 A3) Let (X, d) be a metric space, $Y \subset X$. We define the distance function on Y :

$$d_Y : Y \times Y \rightarrow \mathbb{R}, (y_1, y_2) \mapsto d_Y(y_1, y_2) = d(y_1, y_2).$$

Prove that d_Y is a distance function, and (Y, d_Y) is a metric space. We call d_Y the induced metric on Y , and (Y, d_Y) is called a subspace. Proof: Trivial, since $d_Y(y_1, y_2) = d(y_1, y_2)$.

1.1.4 A4) Let $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R} = \{(x_1, \dots, x_n) \mid x_i \in \mathbb{R}, \dots, x_n \in \mathbb{R}\}$, for any $x, y \in \mathbb{R}^n$, we define

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}, \quad x = (x_1, \dots, x_n), y = (y_1, \dots, y_n).$$

Prove that $(\mathbb{R}^n, d)$ is a metric space. Proof:

1. $d(x, y) = 0 \iff x_i = y_i, \forall 1 \leq i \leq n \iff x = y$.
2. $d(x, y) = d(y, x)$ is trivial.
3. $d(x, y) + d(y, z) \geq d(x, z)$ is the Minkowski inequality:

$$\left(\sqrt{\sum_{i=1}^n a_i^2} + \sqrt{\sum_{i=1}^n b_i^2} \right)^2 = \sum_{i=1}^n a_i^2 + b_i^2 + 2\sqrt{\sum_{i=1}^n a_i^2 \sum_{i=1}^n b_i^2} \geq \sum_{i=1}^n a_i^2 + b_i^2 + 2a_i b_i = \sum_{i=1}^n (a_i + b_i)^2.$$

1.1.5 A5) Given a metric space (X, d) , and $Y \subset X$. If for any $x \in X$ and $\varepsilon > 0$, there exists $y \in Y$ such that $d(y, x) < \varepsilon$, then we say Y is dense in X . Prove that the set of rational numbers is dense in $\mathbb{R}$.

Proof: For any $x \in \mathbb{R}$, let $N = \lfloor x \rfloor$, then for any $\varepsilon > 0$, let $q > 1/\varepsilon$. Then for $p \in [Nq, (N+1)q] \cap \mathbb{Z}$, choose p such that $|x - p/q|$ is minimal. Suppose $p/q < x$, then

$$2 \left| x - \frac{p}{q} \right| < \left| x - \frac{p}{q} \right| + \left| x - \frac{p+1}{q} \right| = \frac{1}{q} < \varepsilon.$$

Hence $d(x, p/q) < \varepsilon$.

1.1.6 A6) For $(x, y) \in \mathbb{R}^2$, if its coordinates x and y are rational numbers, then we call this point a rational point. Prove that $(\mathbb{R}^2, d)$ (refer to question A4) the set of rational points in $\mathbb{R}^2$ is dense.

Proof: By A5), $\overline{\mathbb{Q}} = \mathbb{R}$. Hence for any $(x, y) \in \mathbb{R}^2$ and $\varepsilon > 0$, there exists $(a, b) \in \mathbb{Q}^2$ such that $|a - x|, |b - y| < \varepsilon/2$. Then

$$d((x, y), (a, b)) = \sqrt{(a - x)^2 + (b - y)^2} < \varepsilon.$$

Hence $\mathbb{Q}^2$ is dense in $\mathbb{R}^2$

1.1.7 A7) Prove that the axiom (F) and (O), and the boundedness principle imply the Archimedean axiom (A).

Proof: Otherwise assume that $\mathbb{N}$ has an upper bound. Then $M = \sup \mathbb{N}$ exists. Let $\varepsilon = 1/2$ then there is an $n \in \mathbb{N}$ such that $n > M - \varepsilon$. Hence $n + 1 > M$, leading to contradiction.

1.1.8 A8) (Existence of irrational numbers) Let $X = \{x \in \mathbb{Q} \mid x^2 < 2\}$ be a bounded set, and $\sqrt{2} = \sup X$. Prove that $\sqrt{2}$ is an irrational number.

Proof: If $\sqrt{2} = s = p/q$ is rational, then $p^2 \geq 2q^2$, otherwise let $x = s(2 - s^2)/4 + s$, then $s < x$ and $x^2 < 2$, a contradiction. If $s^2 > 2$, then $x = s(2 - s^2)/4 < s$ and $x^2 > 2$, hence x is an upper bound of X , leading to contradiction. Therefore $s^2 = 2$ which is impossible.

1.1.9 A9) Prove that every open interval contains infinitely many irrational numbers.

Proof: Otherwise the open interval will be a countable set.

1.2 PSB: Countable and Uncountable Sets

Let $\mathbb{N}$ denote the set of natural numbers (including 0). X is a set, if there is an injective map $f : X \rightarrow \mathbb{N}$, then we say X is countable; if X is not countable, then we say X is uncountable.

1.2.1 B1) Prove that finite sets are countable.

Proof: For any finite set $X = \{a_1, \dots, a_n\}$, the map $f : a_k \mapsto k$ is an injective, hence X is countable.

1.2.2 B2) Prove that subsets of countable sets are countable.

Proof: If X is countable and $Y \subset X$, then there is an injective map $f : X \rightarrow \mathbb{N}$, so $f|_Y : Y \rightarrow \mathbb{N}$ is an injective map, hence Y is countable.

1.2.3 B3) Prove that if X is a countable set, then we can always write $X = \{x_1, x_2, x_3, \dots\}$ (that is, the elements of X can be indexed by natural numbers).

Proof: Let $I = \{n \in \mathbb{N} : f^{-1}(n) \neq \emptyset\}$, $x_k = f^{-1}(\min I \setminus \{f(x_1), \dots, (x_{k-1})\})$. Then $x_k \in X$, and for any $x \in X$, $f(x) \in I$ hence $x \in \{x_1, \dots, x_{f(x)}\}$. Therefore $X = \{x_1, \dots, x_n, \dots\}$.

1.2.4 B4) Prove that the set of rational numbers $\mathbb{Q}$ is countable.

Proof: List every positive rational number as below:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{1}{3}, \frac{2}{2}, \frac{3}{1}, \frac{1}{4}, \frac{2}{3}, \dots$$

such that p/q is before m/n if $p + q < m + n$ or $p + q = m + n$ and $p < m$, then every number in $\mathbb{Q}_{>0}$ is listed at least once. Hence $\mathbb{Q}_{>0}$ is countable and so is $\mathbb{Q}$.

1.2.5 B5) Prove that the countable union of countable sets is countable, that is, if $X_1, X_2, \dots, X_n, \dots$ are all countable sets, then their union $\bigcup_{n=1}^{\infty} X_n$ is also a countable set.

Proof: Assume X_n are disjoint. Since X_n are countable, we can write

$$X_n = \{a_1^{(n)}, a_2^{(n)}, \dots, a_m^{(n)}, \dots\}.$$

Then

$$\bigcup_{n=1}^{\infty} X_n = \{a_1^{(1)}, a_2^{(1)}, a_1^{(2)}, a_3^{(1)}, \dots\}$$

where the order is the same as in B4). Hence $\bigcup_{n \geq 1} X_n$ is countable.

1.2.6 B6) If X is countable, and the map $f : X \rightarrow Y$ is surjective, then Y is countable.

Proof: Since X is countable, there is an injective map $g : X \rightarrow \mathbb{N}$. Let

$$h : Y \rightarrow \mathbb{N}, y \mapsto \min g(f^{-1}(\{y\})).$$

then g is injective, hence Y is countable.

1.2.7 B7) Prove the following using proof by contradiction: $\mathbb{R}$ is uncountable.

B7-1) Suppose $J \subset \mathbb{R}$ is a closed interval and its length $|J| > 0$. For any $x \in \mathbb{R}$, there always exists an interval $I \subset J$ such that $|I| > 0$ and $x \notin I$. Proof: Any closed interval $J = [a, b]$ can be written in the form $J = A \cup B \cup C$, where $A = [a, \frac{2a+b}{3}]$, $B = [\frac{2a+b}{3}, \frac{a+2b}{3}]$, $C = [\frac{a+2b}{3}, b]$, and x can only be in at most 2 of these sets. Hence we can choose a set I in A, B, C . B7-2) Prove that if $\{x_1, x_2, \dots\}$ is a countable subset of $\mathbb{R}$, then there exists a nested interval sequence $I_1 \supset I_2 \supset \dots$ such that for any n , $x_n \notin I_n$. Proof: Simple application of B7-1) B7-3) Prove that $\mathbb{R}$ is uncountable. Proof: If $\mathbb{R}$ is countable, write $\mathbb{R} = \{r_1, r_2, \dots\}$, then set $I_0 = [0, 1]$. By B7-2) we can obtain a sequence $I_0 \supset I_1 \supset \dots$ such that $x_n \notin I_n$ for any n . Hence

$$\bigcap_{n=0}^{\infty} I_n = \emptyset,$$

leading to contradiction.

1.2.8 B8) Prove that if X is an uncountable set, and A is a countable subset of X , then $X - A$ is uncountable.

Proof: Otherwise suppose that both A and $X - A$ is countable, then there exist injective mappings $f : A \rightarrow \mathbb{N}$ and $g : X - A \rightarrow \mathbb{N}$. Define

$$h : X \rightarrow \mathbb{N}, x \mapsto \begin{cases} 2f(x), & x \in A, \\ 2g(x) + 1, & x \notin A. \end{cases}$$

Then h is injective, hence X is countable.

1.2.9 B9) Prove that any interval of non-zero length (open or closed) is uncountable.

Proof: Same as B7). Or use the fact that $\mathbb{R}$ is the countable union of intervals of the same length, and the countable union of countable sets is still countable.

1.2.10 B10) Prove that the set of complex numbers $\mathbb{C}$ is uncountable.

Proof: $\mathbb{C}$ has an uncountable subset $\mathbb{R}$.

1.2.11 B11) Suppose $\mathcal{I}$ is a collection of non-overlapping closed intervals, satisfying the following property: for any $I, J \in \mathcal{I}$, if $I \neq J$, then their intersection is empty, i.e., $I \cap J = \emptyset$. Prove that $\mathcal{I}$ is countable.

Proof: For any $I \in \mathcal{I}$, there exists a rational number $r_I \in I$. Consider $f : \mathcal{I} \rightarrow \mathbb{Q}$, $I \mapsto r_I$, then f is injective. Since $\mathbb{Q}$ is countable, so is $\mathcal{I}$.

1.3 PSC: Schröder-Bernstein Theorem

Suppose X and Y are two sets, and mappings $f : X \rightarrow Y$ and $g : Y \rightarrow X$ are both injective. Let $X' = X - g(Y)$.

1.3.1 C1) If X is a finite set, prove that there exists a bijection $\varphi : X \rightarrow Y$.

Proof: $g : Y \rightarrow X$ is injective and X is finite, $\implies Y$ is finite. Hence $|X| \leq |Y|$, and $|X| \geq |Y|$, so $|X| = |Y|$. Therefore we can write $X = \{x_1, x_2, \dots, x_n\}$ and $Y = \{y_1, y_2, \dots, y_n\}$, and obtain

$$\varphi : X \rightarrow Y, x_k \mapsto y_k.$$

1.3.2 C2) If X is countable, prove that there exists a bijection $\varphi : X \rightarrow Y$.

Proof: Assume X is infinite, then Y is countable (by g) and infinite (by f). Hence we can list $X = \{x_1, x_2, \dots\}$ and $Y = \{y_1, y_2, \dots\}$ and define

$$\varphi : X \rightarrow Y, x_k \mapsto y_k.$$

From now on, we impose no restrictions on X . Let $h : X \rightarrow X$ be the composite map $h = g \circ f$.

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow h & & \downarrow g \\ X & \leftarrow & \end{array}$$

1.3.3 C3) Consider the set family $\mathcal{F} = \{A \subset X \mid X' \cup h(A) \subset A\}$. Prove that $\mathcal{F}$ is non-empty.

Proof: $X \in \mathcal{F}$.

1.3.4 C4) Prove that if $A \in \mathcal{F}$, then $X' \cup h(A) \in \mathcal{F}$.

Proof: If $A \in \mathcal{F}$ then $X' \cup h(A) \subset A$, hence (let B denote $X' \cup h(A)$)

$$X' \cup h(B) \subset X' \cup h(A) = B.$$

1.3.5 C5) We define

$$A_0 = \bigcap_{A \in \mathcal{F}} A = \{x \in X \mid \text{for any } A \in \mathcal{F}, \text{ we have } x \in A\}.$$

Prove that $A_0 \in \mathcal{F}$. Proof:

$$X' \cup h(A_0) \subset X' \cup \left(\bigcap_{A \in \mathcal{F}} h(A) \right) = \bigcap_{A \in \mathcal{F}} X' \cup h(A) \subset \bigcap_{A \in \mathcal{F}} A = A_0.$$

Hence $A_0 \in \mathcal{F}$.

1.3.6 C6) Prove that $X' \cup h(A_0) = A_0$.

Proof:

$$A_0 \in \mathcal{F} \implies X' \cup h(A_0) \in \mathcal{F} \implies A_0 \subset X' \cup h(A_0).$$

The other side is proved in C5).

1.3.7 C7) Let $B_0 = X - A_0$. Prove that $f(A_0) \cap g^{-1}(B_0) = \emptyset$ and $f(A_0) \cup g^{-1}(B_0) = Y$.

Proof: If $f(A_0) \cap g^{-1}(B_0) \neq \emptyset$, then there exist $a \in A_0, b \in B_0$ such that $f(a) = g^{-1}(b)$, i.e. $b = h(a)$. Since $a \in A_0$, for any $A \in \mathcal{F}$, $a \in A$, hence $b = h(a) \in X' \cup h(A) \subset A$. Therefore $b \in A_0$, a contradiction. Otherwise if there exists $y \in Y$ such that $y \notin f(A_0) \cup g^{-1}(B_0)$, then $g(y) \notin B_0 \implies g(y) \in A_0$. Let $z = g(y) \in A_0 \cap g(Y)$, then $z \notin X'$ so $z \in h(A_0)$ by C6). Let $z = h(t)$ then $y = f(t) \in f(A_0)$ since g is injective, leading to contradiction.

1.3.8 C8) We define the map $\varphi : X \rightarrow Y$. For $x \in X$, we require

$$\varphi(x) = \begin{cases} f(x), & \text{if } x \in A_0; \\ g^{-1}(x), & \text{if } x \in B_0. \end{cases}$$

Prove that this is a bijection. Proof:

1. φ is injective: for any $x, y \in A_0, x \neq y$, $\varphi(x) \neq \varphi(y)$ since f is injective. For any $x, y \in B_0, x \neq y$, $\varphi(x) \neq \varphi(y)$ since g is a mapping. For any $x \in A_0, y \in B_0$, $\varphi(x) \neq \varphi(y)$ since $f(A_0) \cap g^{-1}(B_0) = \emptyset$.
2. φ is surjective: $\varphi(X) = \varphi(A_0 \cup B_0) = f(A_0) \cup g^{-1}(B_0) = Y$.

Based on the above, we have proved: **Theorem (Schroeder-Bernstein)**. If there exist injective maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$, then there exists a bijection $\varphi : X \rightarrow Y$ between the two sets.

1.4 PSD: Details of Dedekind Cut

The goal of this part of the exercise is to complete the part of the Dedekind cut construction method taught in class, thereby providing a complete proof for the construction of real numbers.

1.4.1 D1) Prove that if X and Y are both Dedekind cuts, then the product $X \cdot Y$ as defined in the lecture is also a Dedekind cut, i.e.,

$$\times : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}, (X, Y) \mapsto X \cdot Y,$$

is well-defined. (Hint: You only need to prove the case where $X > 0, Y > 0$.) Proof: The set $X \cdot Y$ is define as $Z = \bar{0} \cup \{x \cdot y : x, y \geq 0, x \in X, y \in Y\}$. Let $Z' = \mathbb{Q} - Z$, then

1. $Z \neq \emptyset, Z' \neq \emptyset$, since for any $x \in X', y \in Y', x \cdot y \notin Z$.
2. For any $z \in Z, z' \in Z'$, if $z' < z$ then $z > 0$. So assume $z = x \cdot y, x \in X, y \in Y, x, y \geq 0$, then $z' = x \cdot (yz'/z) \in Z$, a contradiction.
3. If Z has a maximal element $z = x \cdot y, x, y \geq 0, x \in X, y \in Y$, then since x, y are both not maximal, there exists $x' \in X, y' \in Y$, such that $x < x', y < y'$ so $z < z' = x' \cdot y' \in Z$, a contradiction.

1.4.2 D2) Prove that $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$. ($\implies$ (F5))

Proof: We only need to verify the case where $X, Y, Z > 0$. Then both $(X \cdot Y) \cdot Z$ and $X \cdot (Y \cdot Z)$ are the set

$$\bar{0} \cup \{x \cdot y \cdot z : x, y, z \geq 0, x \in X, y \in Y, z \in Z\}.$$

1.4.3 D3) Prove that $X \cdot Y = Y \cdot X$. ($\implies$ (F6))

Proof: Same as D2).

1.4.4 D4) Prove that $X \cdot (Y + Z) = X \cdot Y + X \cdot Z$. ($\implies$ (F9))

Proof: We can assume that $X, Y, Z > 0$, then

$$X \cdot (Y + Z) = \{xy + xz : x \in X, y \in Y, z \in Z\}$$

while

$$X \cdot Y + X \cdot Z = \{xy + x'z : x, x' \in X, y \in Y, z \in Z\}.$$

Hence $X \cdot (Y + Z) \subset X \cdot Y + X \cdot Z$. For any $xy + x'z \in X \cdot Y + X \cdot Z$, suppose $x \geq x'$, then $xy + xz \in X \cdot (Y + Z)$ and $xy + x'z \leq xy + xz$, so $xy + x'z \in X \cdot Y + X \cdot Z$, therefore $X \cdot Y + X \cdot Z = X \cdot (Y + Z)$.

1.4.5 D5) Prove that $\bar{1} \cdot X = X$ and $\bar{1} \neq \bar{0}$. ($\implies$ (F7))

Proof: Assume that $X > 0$, then $\bar{1} \cdot X = \{u \cdot v : u < 1, v \in X\}$. Foy any $u < 1, v \in X, u \cdot v < v$ hence $u \cdot v \in X$. For any $x \in X$, there exists $x' \in X, x' > x$, then $x = x' \cdot (x/x') \in \bar{1} \cdot X$. Therefore $\bar{1} \cdot X = X$ and $1/2 \in \bar{1} \setminus \bar{0}$, so $\bar{1} \neq \bar{0}$.

1.4.6 D6) Prove that if $X \cdot Y = \bar{0}$, then $X = \bar{0}$ or $Y = \bar{0}$; conversely, if $X \geq \bar{0}, Y \geq \bar{0}$, then $X \cdot Y \geq \bar{0}$. ($\implies$ (O5))

Proof: Otherwise there exists $x, x' \in X, y, y' \in Y$, such that $x, y > 0, x', y' < 0$. Hence $xy, x'y \in X \cdot Y$, where $xy > 0 > x'y$, so $X \cdot Y \neq \bar{0}$. Suppose $X, Y > 0$, then there exists $x \in X, y \in Y$ such that $x, y > 0$, hence $0 < xy \in X \cdot Y$, so $X \cdot Y > \bar{0}$.

1.4.7 D7) X is a positive Dedekind cut. Prove that for any integer n , there exist $x \in X, x' \in X'$ such that

$$1 < \frac{x'}{x} < 1 + \frac{1}{n}.$$

Proof: Let $l_0 = x \in X, r_0 = x' \in X'$. Define l_n, r_n as follows: If $(l_{n-1} + r_{n-1})/2 \in X$, then $l_n = (l_{n-1} + r_{n-1})/2, r_n = r_{n-1}$, otherwise $l_n = l_{n-1}, r_n = (l_{n-1} + r_{n-1})/2$. Then

$$0 < \frac{r_n - l_n}{l_n} \leq \frac{1}{2} \frac{r_{n-1} - l_{n-1}}{l_{n-1}}.$$

Hence there exist such x, x' .

1.4.8 D8) Prove that for any Dedekind cuts X and Y , if $Y \neq \bar{0}$, there exists a unique Dedekind cut Z such that

$$Y \cdot Z = X.$$

We denote Z as $\frac{X}{Y}$. When $X = \bar{1}$, we also denote it as Y^{-1} . ($\implies$ (F8)) Proof: By D6), Z is unique. By D2) we can assume that $X = \bar{1}$, and $Y > 0$. Let

$$Z = \left\{ \frac{1}{y} : y \in Y' \right\} \cup \bar{0} \cup \{0\}.$$

Then by D7), $Y \cdot Z = \bar{1}$.

Chapter 2 Cesàro sum

2.1 PSA

2.1.1 A1) $\{x_n\}_{n \geq 1}$ is a bounded real sequence. Prove that there is a subsequence $\{x_{n_i}\}_{i \geq 1}$ such that $\lim_{i \rightarrow \infty} x_{n_i}$ exists and

$$\lim_{i \rightarrow \infty} x_{n_i} = \limsup_{n \rightarrow \infty} x_n.$$

Proof: Let $M = \limsup_{n \rightarrow \infty} x_n < \infty$, then for any $\varepsilon = 1/i > 0$ there exists $N \geq n_{i-1}$ such that $M \leq \sup_{k \geq N} x_k < M + \varepsilon$. Hence there exists $n_i \geq N$ such that $x_{n_i} \in (M - \varepsilon, M + \varepsilon)$. Take the sequence $\{x_{n_i}\}_{i \geq 1}$ then $\lim_{i \rightarrow \infty} x_{n_i} = \limsup_{n \rightarrow \infty} x_n$.

2.1.2 A2) $\{x_n\}_{n \geq 1}$ is a real sequence. Prove that $\{x_n\}_{n \geq 1}$ converges iff
$$\limsup_{n \rightarrow \infty} x_n = \liminf_{n \rightarrow \infty} x_n.$$

Proof: Since a sub-sequence of a Cauchy sequence converge to the same value as the original sequence, $\Rightarrow$ is trivial by A1). $\Leftarrow$ $\lim_{n \rightarrow \infty} \sup_{k \geq n} x_k - \inf_{k \geq n} x_k = 0$ implies x_n is Cauchy, hence convergent.

2.1.3 A3) $\{x^{(k)}\}_{k \geq 1} \subset \mathbb{R}^n$, where $x^{(k)} = (x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)})$. Then $\{x^{(k)}\}_{k \geq 1}$ converges in $\mathbb{R}^n$ iff for any $i = 1, 2, \dots, n$, $\{x_i^{(k)}\}_{k \geq 1}$ converges.

Proof: Use Cauchy sequences and the fact that for $x = (x_1, x_2, \dots, x_n)$,

$$\max\{|x_k| : 1 \leq k \leq n\} \leq \|x\| \leq \sum_{k=1}^n |x_k|.$$

2.1.4 A4) Suppose $\{z_n\}_{n \geq 1}, \{w_n\}_{n \geq 1}$ are two convergent complex sequences. Prove that if $\lim_{n \rightarrow \infty} w_n \neq 0$, then the sequence $\{z_n/w_n\}_{n \geq 1}$ converges.

Proof: Suppose $z = \lim_{n \rightarrow \infty} z_n$ and $w = \lim_{n \rightarrow \infty} w_n$, then

$$\left| \frac{z_n}{w_n} - \frac{z}{w} \right| \leq \frac{|w| \cdot |z_n - z|}{|w \cdot w_n|} + \frac{|z| \cdot |w_n - w|}{|w \cdot w_n|}.$$

Hence $\left| \frac{z_n}{w_n} - \frac{z}{w} \right| \rightarrow 0$, so $\lim_{n \rightarrow \infty} z_n/w_n = z/w$.

2.1.5 A5) Suppose $\{a_n\}_{n \geq 1}$ is a monotonically decreasing sequence of positive reals, and $\lim_{n \rightarrow \infty} a_n = 0$. Prove that the series

$$a_1 - a_2 + a_3 - a_4 + \dots + (-1)^{n-1} a_n + \dots$$

converges. Proof: Suppose $a_n = a_1 - \sum_{k=1}^n b_k$, then $b_k \geq 0$ and $\sum_{k=1}^{\infty} b_k = a_1$. The series

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n = \sum_{n=1}^{\infty} b_{2n} < a_1$$

clearly converges.

2.1.6 A6) $\{a_n\}_{n \geq 1} \subset \mathbb{C}$. Prove that if $\sum_{k=1}^{\infty} |a_k|$ converges, then $\sum_{k=1}^{\infty} a_k$ converges.

Proof: $\sum_{k=1}^{\infty} |a_k|$ converges implies for any $\varepsilon > 0$, there exists N such that for any $n \geq N, p \geq 0, \sum_{k=n}^{n+p} |a_k| < \varepsilon$. Note that $\left| \sum_{k=n}^{n+p} a_k \right| \leq \sum_{k=n}^{n+p} |a_k|$, so $\sum_{k=1}^{\infty} a_k$ converges.

2.1.7 A7) Prove that we can define the exponential function on $\mathbb{C}$:

$$\exp : \mathbb{C} \rightarrow \mathbb{C}, z \mapsto \exp(z) = e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

Proof: Use A6).

2.1.8 A8) $\{a_n\} \subset \mathbb{C}$. Suppose for any $n \in \mathbb{N}, a_n \neq 0$. Let $P_n = a_1 \cdot a_2 \cdots a_n$. If $\lim_{n \rightarrow \infty} P_n$ exists and is not 0, we call $\prod_{n=1}^{\infty} a_n$ convergent and let $\prod_{n=1}^{\infty} a_n = \lim_{n \rightarrow \infty} P_n$. Prove that $\prod_{n=1}^{\infty} a_n$ converges iff for any $\varepsilon > 0$, there exists N such that for any $n \geq N, p \geq 0$,

$$|a_n \cdot a_{n+1} \cdots a_{n+p} - 1| < \varepsilon.$$

Proof: If $\lim_{n \rightarrow \infty} P_n = P$ exists and is non-zero, then for any $\varepsilon > 0$, there exists N such that for any $n \geq N, |P_n - P| < \varepsilon P/4$ and $|P_n| > P/2$. Then for any $n \geq N, p \geq 0, |P_{n+p}/P_n - 1| < \varepsilon$. If for any $\varepsilon > 0$, there exists N such that for any $n \geq N, p \geq 0, |P_{n+p} - P_n| < \varepsilon |P_n|$, then let $\varepsilon = 1$ we infer that P_n is bounded by some constant M . Hence the sequence $\{P_n\}$ is Cauchy, and $P = \lim_{n \rightarrow \infty} P_n$ cannot be zero, otherwise there is no such N for $\varepsilon = 1/2$.

2.1.9 A9) Prove that $\exp(x)$ is monotonically increasing on $\mathbb{R}$.

Proof: For $x, y \in \mathbb{R}$,

$$\exp(x) \cdot \exp(y) = \sum_{n=0}^{\infty} \frac{x^n}{n!} \cdot \sum_{m=0}^{\infty} \frac{y^m}{m!} = \sum_{k=0}^{\infty} \sum_{n+m=k} \frac{x^n y^m}{k!} = \sum_{k=0}^{\infty} \frac{(x+y)^k}{k!} = \exp(x+y).$$

$\exp(x) \cdot \exp(-x) = \exp(0) = 1$ implies $\exp(x) > 0$ for all $x \in \mathbb{R}$, so if $x > y$, $\exp(x)/\exp(y) = \exp(x-y) > 1 \implies \exp(x) > \exp(y)$.

2.1.10 A10) Suppose $P(x)$ and $Q(x)$ are polynomials of degree n, m , where $m > n$. Prove that

$$\lim_{n \rightarrow \infty} \frac{Q(n)}{P(n)} = 0, \quad \lim_{n \rightarrow \infty} \frac{Q(n)}{e^n} = 0.$$

Proof: Suppose $P(x) = \sum_{k=0}^n a_k x^k$ and $Q(x) = \sum_{k=0}^m b_k x^k$, then there exists N such that for any $x \geq N, |P(x)| > |a_n| x^n/2, |Q(x)| \leq \sum_{k=0}^m |b_k| \cdot x^m$, and $e^x \geq x^{m+1}/(m+1)!$, hence

$$\left| \frac{Q(x)}{P(x)} \right| \leq \frac{2 \sum_{k=0}^m |b_k|}{|a_n|} \cdot x^{m-n} \rightarrow 0, \quad \left| \frac{Q(x)}{e^x} \right| \leq (m+1)! \sum_{k=0}^m |b_k| \cdot x^{-1} \rightarrow 0.$$

2.2 PSB: Calculation of Limits**2.2.1 B1)**

$$\lim_{n \rightarrow \infty} \frac{n+10}{2n-1} = \frac{1}{2}.$$

2.2.2 B2)

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n} + 10}{2\sqrt{n} - 1} = \frac{1}{2}.$$

2.2.3 B3)

$$\lim_{n \rightarrow \infty} 0.\underbrace{99 \cdots 9}_{n \text{ times}} = 1.$$

2.2.4 B4)

$$\lim_{n \rightarrow \infty} \frac{1}{n(n+3)} = 0.$$

2.2.5 B5)

$$\lim_{n \rightarrow \infty} \frac{\cos n}{n} = 0.$$

2.2.6 B6)

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0.$$

2.2.7 B7)

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0.$$

2.2.8 B8)

$$\lim_{n \rightarrow \infty} \sqrt{n+10} - \sqrt{n+1} = 0.$$

2.2.9 B9)

$$\lim_{n \rightarrow \infty} \frac{1+2+\cdots+n}{n^2} = \frac{1}{2}.$$

2.2.10 B10)

$$\lim_{n \rightarrow \infty} \frac{1^2+2^2+\cdots+n^2}{n^3} = \frac{1}{3}.$$

2.2.11 B11) $a > 0$

$$\lim_{n \rightarrow \infty} a^{1/n} = 1.$$

2.2.12 B12) $a > 1$

$$\lim_{n \rightarrow \infty} \frac{n^{10000}}{a^n} = 0.$$

2.2.13 B13)

$$\lim_{n \rightarrow \infty} \frac{2^n + n}{3^n + n^2} = 0.$$

2.2.14 B14)

$$\lim_{n \rightarrow \infty} \frac{3^n + 2^n}{3^n + n^2} = 1.$$

2.2.15 B15)

$$\lim_{n \rightarrow \infty} \sqrt{n}(\sqrt{n+1} - \sqrt{n}) = \frac{1}{2}.$$

2.2.16 B16) same as B12)**2.2.17 B17)**

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = e^{-1}.$$

2.2.18 B18)

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{5n}\right)^{n+2019} = e^{-1/5}.$$

2.2.19 B19)

$$\lim_{n \rightarrow \infty} (n^3 + n^2 + 9n + 1)^{1/n} = 1.$$

2.2.20 B20)

$$\lim_{n \rightarrow \infty} (2018^n + 2019^n)^{1/n} = 2019.$$

2.3 PSC: Riemann Rearrangement Theorem

Suppose $\sum_{n=1}^{\infty} a_n$ is conditionally convergent, we will prove that for and $\alpha \in \mathbb{R} \cup \{-\infty, \infty\}$, we can rearrange the sequence such that the new series sums to α . Suppose $\varphi : \mathbb{Z}_{\geq 1} \rightarrow \mathbb{Z}_{\geq 1}$ is a bijection, let $b_k = a_{\varphi(k)}$, then the sequence $\{b_k\}_{k \geq 1}$ is called a rearrangement of $\{a_n\}_{n \geq 1}$. Let all non-negative terms of $\{a_n\}_{n \geq 1}$, listed in the same order as in $\{a_n\}$ be $c_1, c_2, \dots$, and the negative terms be $d_1, d_2, \dots$.

2.3.1 C1) Prove that $\lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} d_n = 0$.

Proof: Since $\sum_{n=1}^{\infty} a_n$ is conditionally convergent, c_n, d_n both have infinite terms and $\lim_{n \rightarrow \infty} a_n = 0$. Therefore $\lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} d_n = 0$.

2.3.2 C2) Prove that $\sum_{n=1}^{\infty} c_n = \sum_{n=1}^{\infty} b_n = \infty$.

Proof: Since $\sum_{n=1}^{\infty} a_n$ is not absolutely convergent, the two series can not be both convergent. If one converges and the other doesn't, then $\sum_{n=1}^{\infty} a_n$ will diverge. Hence they both diverge.

2.3.3 C3) Prove that for any $\alpha \in \mathbb{R}$, there exists a rearrangement $\{b_n\}$ of $\{a_n\}$ such that

$$\sum_{k=1}^{\infty} b_k = \alpha.$$

Proof: Suppose $\alpha \geq 0$. Inductively define the indices u_i and v_i as follows ($u_0 = v_0 = 0$): For $i \geq 1$, let u_i be the least index such that $u_i > u_{i-1}$ and

$$\sum_{j=1}^{u_i} c_j - \sum_{j=1}^{v_{i-1}} d_j \geq \alpha,$$

and v_i be the least index such that $v_i > v_{i-1}$ and

$$\sum_{j=1}^{u_i} c_j - \sum_{j=1}^{v_i} d_j \leq \alpha.$$

Let φ be the permutation such that

$$b_1 = c_1, b_2 = c_2, \dots, b_{u_1} = c_{u_1}, b_{u_1+1} = -d_1, \dots, b_{u_1+v_1} = -d_{v_1}, \dots$$

Since $\sum_{n=1}^{\infty} c_n = \sum_{n=1}^{\infty} d_n = \infty$, u_i and v_i all exists, so φ is indeed a bijection. By definition we know that

$$\left| \sum_{j=1}^{u_i} c_j - \sum_{j=1}^{v_{i-1}} d_j - \alpha \right| \leq c_{u_i-1},$$

and

$$\left| \sum_{j=1}^{u_i} c_j - \sum_{j=1}^{v_i} d_j - \alpha \right| \leq d_{v_i-1}.$$

Since $\lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} d_n = 0$, the two values above both tend to 0. Note that the series $\sum_{n=1}^{\infty} b_n$ is monotonic between these indices, hence $\sum_{n=1}^{\infty} b_n = \alpha$.

2.3.4 C4) Prove that there exists a rearrangement $\{x_k\}$ of $\{a_n\}$ such that $\sum_{k=1}^{\infty} x_k = \infty$.

Proof: Define u_i and v_i as in C3), such that

$$\sum_{j=1}^{u_i} c_j - \sum_{j=1}^{v_{i-1}} d_j \geq i \geq \sum_{j=1}^{u_i} c_j - \sum_{j=1}^{v_i} d_j.$$

Same as C3) define the sequence x_k and clearly $\sum_{n=1}^{\infty} x_k = \infty$.

2.4 PSD: Cesàro Sum

For a real sequence $\{a_n\}_{n \geq 1}$, let $\sigma_n = \frac{a_1 + a_2 + \dots + a_n}{n}$.

2.4.1 D1) Suppose $\lim_{n \rightarrow \infty} a_n = a$, prove that $\lim_{n \rightarrow \infty} \sigma_n = a$.

Proof: For any $n > 0$,

$$|\sigma_n - a| \leq \sum_{i=1}^N \frac{|a_i - a|}{n} + \sum_{i=N+1}^n \frac{|a_i - a|}{n} \leq \frac{MN}{n} + \varepsilon(N),$$

where $M = |a| + \sup_{i \leq N} |a_i|$, and $\varepsilon(N) = \sup_{i > N} |a_i - a|$. By $\lim_{n \rightarrow \infty} a_n = a$ we know $\varepsilon(N) \rightarrow 0$, hence $\lim_{n \rightarrow \infty} \sigma_n = a$.

2.4.2 D2) Construct a divergent sequence $\{a_n\}$ such that $\lim_{n \rightarrow \infty} \sigma_n = 0$.

Solution: $a_n = (-1)^{n-1}$, $\sigma_n \in [0, 1/n]$.

2.4.3 D3) Determine whether there exists $\{a_n\}_{n \geq 1}$ such that for any $n \geq 1$, $a_n > 0$ and $\limsup_{n \rightarrow \infty} a_n = \infty$ but $\lim_{n \rightarrow \infty} \sigma_n = 0$.

Solution: Let

$$a_n = \begin{cases} 2^{-n}, & n \neq 2^k, \\ k, & n = 2^k. \end{cases}$$

Then $\limsup_{n \rightarrow \infty} a_n = \infty$ and $a_n > 0$, but for any n , suppose $n \in [2^{k-1}, 2^k]$, then

$$\sigma_n \leq \frac{1}{n} \cdot \left(1 + \frac{k(k+1)}{2}\right) \leq \frac{k(k+1)}{2^{k-1}}.$$

Hence $\lim_{n \rightarrow \infty} \sigma_n = 0$.

2.4.4 D4) For $k \geq 1$, denote $b_k = a_{k+1} - a_k$. Prove that for any $n \geq 2$, $a_n - \sigma_n = \sum_{k=1}^{n-1} kb_k/n$.

Proof:

$$\sum_{k=1}^{n-1} kb_k = \sum_{k=1}^{n-1} k(a_{k+1} - a_k) = (n-1)a_n - \sum_{k=1}^{n-1} a_k = n(a_n - \sigma_n).$$

2.4.5 D5) Suppose $\lim_{k \rightarrow \infty} kb_k = 0$ and $\{\sigma_n\}_{n \geq 1}$ converges. Prove that $\{a_n\}_{n \geq 1}$ also converges.

Proof: By D1), $\lim_{k \rightarrow \infty} kb_k = 0$ implies

$$\lim_{n \rightarrow \infty} a_n - \sigma_n = \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^{n-1} kb_k}{n} = \lim_{k \rightarrow \infty} kb_k = 0.$$

Therefore $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sigma_n$ exists.

2.4.6 D6) Suppose $\{kb_k\}_{k \geq 1}$ is bounded, i.e. $b_k = O(k^{-1})$, and $\lim_{n \rightarrow \infty} \sigma_n = \sigma$. Prove that $\lim_{n \rightarrow \infty} a_n = \sigma$.

Proof: Note that for $m < n$,

$$a_n - \sigma_n = \frac{m}{n-m}(\sigma_n - \sigma_m) + \frac{1}{n-m} \sum_{k=m+1}^n a_n - a_k.$$

Therefore since σ_n is a Cauchy sequence, and $|a_n - a_k| \leq M(n-k)/k$, we can choose n, m to show that $\lim_{n \rightarrow \infty} a_n - \sigma_n = 0$.

2.5 PSE: Definition of $\sqrt[n]{x}$ and b^x

2.5.1 E1) Given $n \in \mathbb{N}$ and $x > 0$, prove that if $y_1, y_2 > 0$ satisfy $y_1^n = x = y_2^n$, then $y_1 = y_2$.

Proof: Note that $y_1^{n-1} + y_1^{n-2}y_2 + \cdots + y_2^{n-1} > 0$, and

$$0 = y_1^n - y_2^n = (y_1 - y_2) \cdot (y_1^{n-1} + y_1^{n-2}y_2 + \cdots + y_2^{n-1}).$$

Hence $y_1 = y_2$.

2.5.2 E2) Prove that if $x > 0$, then the set $E(x) = \{t \in \mathbb{R} : t^n < x\}$ is non-empty and has an upper-bound.

Proof: Note that $0 \in E(x)$ and $E(x)$ has the upper-bound $\max\{1, x\}$.

2.5.3 E3) Prove that $y = \sup E(x)$ satisfy $y^n = x$ and $y > 0$.

Proof: $y = \sup E(x) \implies y^n = x$ since t^n is continuous on $\mathbb{R}$, and $y^n = x$ and $0 \in E(x)$ implies $y > 0$.

2.5.4 E4) Prove that the mapping $\sqrt[n]{\cdot} : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}, x \mapsto \sqrt[n]{x} = y$ is well-defined. Denote $\sqrt[n]{x}$ as $x^{1/n}$.

Proof: Use E3).

2.5.5 E5) Prove the $\sqrt[n]{\cdot} : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ is a bijection.

Proof: By E1) it is injective, and $\sqrt[n]{y^n} = y$ implies it is surjective. Hence it is a bijection.

2.5.6 E6) $a, b > 0, n \in \mathbb{N}$, prove that $(ab)^{1/n} = a^{1/n}b^{1/n}$.

Proof: Use E5) and $(xy)^n = x^n y^n$.

2.5.7 E7) Suppose $b > 1, m, n, p, q \in \mathbb{Z}$ where $n, q > 0$. Let $r = \frac{m}{n} = \frac{p}{q}$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Proof: Use $(b^m)^q = (b^p)^n$ and E5).

2.5.8 E8) Prove that for any $r \in \mathbb{Q}, r \mapsto b^r$ is well-defined.

Proof: For $r = p/q$, where $q > 0, \gcd(p, q) = 1$, let $b^r = (b^p)^{1/q}$, then for any $r = m/n, b^r = (b^m)^{1/n}$.

2.5.9 E9) Prove that for $r, s \in \mathbb{Q}, b^{r+s} = b^r b^s$.

Proof: Suppose $r = p/q, s = m/n$, where $n, q > 0$, then

$$b^{r+s} = b^{(mq+np)/nq} = (b^{mq} \cdot b^{np})^{1/nq} = (b^m)^{1/n} \cdot (b^p)^{1/q} = b^r b^s.$$

2.5.10 E10) For $x \in \mathbb{R}$, let $B(x) = \{b^t : t \in \mathbb{Q}, t \leq x\}$. Prove that $B(x)$ is non-empty and has an upper-bound. Define $b^x = \sup B(x)$.

Proof: $B(x)$ is clearly non-empty and bounded by $b^{\lfloor x \rfloor + 1}$.

2.5.11 E11) Prove that if $r \in \mathbb{Q}$, then

$$b^r = \sup B(r), \forall r \in \mathbb{Q}.$$

Proof: $b^r \in B(r)$ and since b^t is monotonically increasing, $b^r \geq \sup B(r)$, hence $b^r = \sup B(r)$.

2.5.12 E12) Prove that for any $x, y \in \mathbb{R}$, $b^{x+y} = b^x b^y$.

Proof: For any $b^t \in B(x)$, $b^s \in B(y)$, $t \leq x$ and $s \leq y$, so $t + s \leq x + y$ and $b^{t+s} \in B(x + y)$, hence $b^{x+y} \geq b^x b^y$. For any $b^t \in B(x + y)$, t can be written in the form $t = u + v$ where $b^u \in B(x)$, $b^v \in B(y)$, so $b^{x+y} \leq b^x b^y$.

2.5.13 E13*) Prove that when $b = e$, the function derived from E10) (denoted as e^x) is the same as $\exp(x) = \sum_{n=0}^{\infty} x^n/n!$.

Proof: From $\exp(1) = e$, $\exp(0) = 1$ and $\exp(x + y) = \exp(x) \cdot \exp(y)$ we know that for $n \in \mathbb{Z}$, $\exp(n) = e^n$. For $r = p/q \in \mathbb{Q}$,

$$(e^r)^q = e^p = \exp(p) = \exp(r)^q,$$

so by E5) $e^r = \exp(r)$. Since $\exp$ is continuous, for any $x \in \mathbb{R}$, $e^x = \exp(x)$.

2.6 PSF

Given $\alpha > 0$ and $x_1 > \sqrt{\alpha}$, we define inductively $\{x_n\}_{n \geq 1}$:

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right), n \geq 1.$$

2.6.1 F1) Prove that $\{x_n\}$ is monotonically decreasing and $\lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}$ (which is defined in E).

Proof: Note that

$$x_{n+1} - x_n = \frac{\alpha - x_n^2}{2x_n}.$$

Hence we can prove by induction that $x_n > \sqrt{\alpha}$ and $x_n > x_{n+1}$. x_n is decreasing and bounded, so $\lim_{n \rightarrow \infty} x_n = A$ exists, and $A = (A + \alpha/A)/2$. Therefore $\lim_{n \rightarrow \infty} x_n = A = \sqrt{\alpha}$.

2.6.2 F2) Let $\varepsilon_n = x_n - \sqrt{\alpha}$. Prove that $\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$.

Proof:

$$\frac{\varepsilon_n^2}{2x_n} = \frac{x_n^2 + \alpha - 2x_n\sqrt{\alpha}}{2x_n} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) - \sqrt{\alpha} = x_{n+1} - \sqrt{\alpha} = \varepsilon_{n+1}.$$

2.6.3 F3) Prove that if $\beta = 2\sqrt{\alpha}$, then $\varepsilon_{n+1} < \beta(\varepsilon_1/\beta)^{2^n}$.

Proof: $\varepsilon_{n+1}/\beta < (\varepsilon_n/\beta)^2$, hence $\varepsilon_{n+1} < \beta(\varepsilon_1/\beta)^{2^n}$.

2.6.4 F4) Let $\alpha = 3$, $x_1 = 2$. Verify that $\varepsilon_1/\beta < 0.1$, $\varepsilon_5 < 4 \cdot 10^{-16}$, $\varepsilon_6 < 4 \cdot 10^{-32}$.

Now we consider $\alpha > 1$ and $y_1 > \sqrt{\alpha}$, and define

$$y_{n+1} = \frac{\alpha + y_n}{1 + y_n} = y_n + \frac{\alpha - y_n^2}{1 + y_n}, n \geq 1$$

2.6.5 F6) Prove that $\{y_{2k-1}\}$ is monotonically decreasing.

Proof: Note that

$$y_{n+2} = \frac{\alpha + y_{n+1}}{1 + y_{n+1}} = \frac{\alpha + \frac{\alpha + y_n}{1 + y_n}}{1 + \frac{\alpha + y_n}{1 + y_n}} = \frac{2\alpha + (\alpha + 1)y_n}{(\alpha + 1) + 2y_n},$$

hence

$$y_{n+2} - y_n = \frac{2(\alpha - y_n^2)}{(\alpha + 1) + 2y_n}, \quad y_{n+2} - \sqrt{\alpha} = \frac{(\sqrt{\alpha} - 1)^2}{(\alpha + 1) + 2y_n} (y_n - \sqrt{\alpha}).$$

Therefore $y_1 > \sqrt{\alpha}$ implies $\sqrt{\alpha} < y_{2n+1} < y_{2n-1}$.

2.6.6 F7) Prove that $\{y_{2k}\}$ is monotonically increasing.

Proof: $y_2 = (\alpha + y_1)/(1 + y_1) < \sqrt{\alpha}$, so same as F6), $y_{2k} > y_{2k-2}$ and $y_{2k} < \sqrt{\alpha}$.

2.6.7 F8) Prove that $\lim_{n \rightarrow \infty} y_n = \sqrt{\alpha}$.

Proof: $\{y_{2n-1}\}$ is decreasing and bounded by $\sqrt{\alpha}$, so $\lim_{n \rightarrow \infty} y_{2n-1} = A$ exists and $A = (2\alpha + (\alpha + 1)A)/((\alpha + 1) + 2A)$, so $A = \sqrt{\alpha}$. Likewise $\lim_{n \rightarrow \infty} y_{2n} = \sqrt{\alpha}$, hence $\lim_{n \rightarrow \infty} y_n = \sqrt{\alpha}$.

2.6.8 F9) Compare the rates of convergence between x_n and y_n .

Solution: Let $\delta_n = |y_n - \sqrt{\alpha}|$, then $\delta_n \sim c^n \delta_1$, hence x_n converges faster than y_n .

2.7 PSG: Banach-Mazur Game

Alice and Bob are playing a game: Alice selects a closed interval W_1 first, then Bob choose a subinterval L_1 of W_1 , such that the length of L_1 is less than half of the length of W_1 ; they take turns choosing intervals W_n and L_n , such that $L_n \subset W_n \subset L_{n-1}$ and $|L_n| < |W_n|/2 < |L_{n-1}|/4$, obtaining

$$W_1 \supset L_1 \supset W_2 \supset L_2 \supset \cdots \supset W_n \supset L_n \supset \cdots$$

Alice and Bob find that

$$\bigcap_{n \geq 1} W_n = \bigcap_{n \geq 1} L_n = \{x\}$$

is a real number. If $x \in \mathbb{Q}$ then Alice wins, otherwise Bob wins. Who has a winning strategy? **Solution:** Bob will win. We show that if $\mathbb{Q}$ is replaced with any set M that is of first category, Bob can still win. M can be written as the union of a countable number of nowhere dense sets. Then in every move of Bob, he can choose L_n such that it does not intersect the n th such nowhere dense set. Hence the final number x is not in M .

2.8 Problem H

Consider the set $\mathcal{P} = \{\{p_n\}_{n \geq 1} : p_n \in \mathbb{Z}, p_1 \geq 2, p_{n+1} \geq p_n^2\}$.

2.8.1 H1) For any $p = \{p_n\}_{n \geq 1} \in \mathcal{P}$, define the sequence

$$a_n = \prod_{k=1}^n \left(1 + \frac{1}{p_k}\right).$$

Prove that $f(p) = \lim_{n \rightarrow \infty} a_n$ exists and $f(p) \in (1, 2]$. Proof: Note that $p_n \geq p_1^{2^{n-1}}$, then

$$a_n \leq \prod_{k=1}^n \left(1 + \frac{1}{p_1^{2^{k-1}}}\right) = \frac{1 - p_1^{-2^n}}{1 - p_1^{-1}} < \frac{1}{1 - p_1^{-1}}.$$

So the sequence $\{a_n\}$ is monotonic and bounded, hence $f(p) = \lim_{n \rightarrow \infty} a_n$ exists. Since $a_n \in (1 + 1/p_1, \frac{1}{1-p_1^{-1}})$, we obtain $f(p) \in [1 + 1/p_1, \frac{1}{1-p_1^{-1}}] \subset (1, 2]$.

2.8.2 H2) Prove that $f : \mathcal{P} \rightarrow (1, 2]$ is a bijection.

Proof: For any $p = \{p_n\}, q = \{q_n\} \in \mathcal{P}$, if $p \neq q$, take the least k such that $p_k \neq q_k$ and suppose $q_k \geq p_k + 1$, then for any $n > k$,

$$a_n = \prod_{t=1}^n \left(1 + \frac{1}{p_t}\right) \geq \prod_{t=1}^k \left(1 + \frac{1}{p_t}\right) \cdot \left(1 + \frac{1}{p_{k+1}}\right) \quad (2.1)$$

$$b_n = \prod_{t=1}^n \left(1 + \frac{1}{q_t}\right) \leq \prod_{t=1}^{k-1} \left(1 + \frac{1}{p_t}\right) \cdot \frac{1 - q_k^{-2^{n-k}}}{1 - q_k^{-1}} \quad (2.2)$$

Therefore

$$b_n \leq \prod_{t=1}^k \left(1 + \frac{1}{p_t}\right) \leq (1 + C)a_n$$

for all $n > k$ where $C = p_{k+1}^{-1} > 0$, hence $f(q) \leq (1 + C)f(p) < f(p)$, hence f is injective. For any $x \in (1, 2]$, inductively define $p = \{p_n\} \in \mathcal{P}$ as follows: For any $n \geq 1$, Let t be the least integer such that $a_n \leq x$ and $t \geq p_{n-1}^2$ (clearly such t exists). If $a_n = x$, then let $p_n = t - 1, p_m = p_n^{2^{m-n}}$ for all $m > n$, then $f(p) = x$. Otherwise let $p_n = t$. Note that for any n such that $p_n > p_{n-1}^2$,

$$|x - a_n| \leq 2^{-2^n},$$

therefore $f(p) = x$, and f is surjective.

2.8.3 H3) Prove that $\mathcal{P}$ is uncountable.

Proof: By H2) and the fact that $(1, 2]$ is uncountable.

2.9 Problem I: Binary Expansion

Consider the set $\mathcal{S} = \{\{s_n\}_{n \geq 0} : s_n \in \{-1, 1\}\}$.

2.9.1 I1) For any $s = \{s_n\}_{n \geq 0} \in \mathcal{S}$, define the sequence

$$c_n = \sum_{k=0}^n \frac{s_0 s_1 \cdots s_k}{2^k}.$$

Prove that $h(s) = \lim_{n \rightarrow \infty} c_n$ exists and $h(s) \in [-2, 2]$. Proof: $h(s)$ exists since c_n is clearly a Cauchy sequence, and $c_n \in [-2, 2]$ hence $h(s) \in [-2, 2]$.

2.9.2 I2) Prove that $h : \mathcal{S} \rightarrow [-2, 2]$ is surjective. Determine whether it is injective.

Proof: Consider any $x \in [-2, 2]$, we can choose s_n such that $|c_n - x| \leq 2^{-n}$. Hence there exists $s = \{s_n\} \in \mathcal{S}$ such that $h(s) = \lim_{n \rightarrow \infty} c_n = x$, so h is surjective. Consider $s = \{1, -1, 1, 1, 1, \dots\} \in \mathcal{S}$ and $s' = \{-1, -1, 1, 1, \dots\}$, then $h(s) = h(s') = 0$, hence h is not injective.

2.9.3 I3) For $s = \{s_n\}_{n \geq 0} \in \mathcal{S}$, prove that

$$2 \sin\left(\frac{\pi}{4} c_n\right) = s_0 \sqrt{2 + s_1 \sqrt{2 + \cdots + s_n \sqrt{2}}}.$$

Proof: We prove by induction on n . The base $n = 0$ is trivial. If the statement holds for $n-1$, then let $s' = \{s_{n+1}\}_{n \geq 0} \in \mathcal{S}$, we have

$$2 \sin\left(\frac{\pi}{4}c_n\right) = 2 \sin s_0 \left(\frac{\pi}{4} + \frac{1}{2} \cdot \frac{\pi}{4}c'_{n-1}\right) = s_0 \sqrt{2 + \sin\left(\frac{\pi}{4}c'_{n-1}\right)}.$$

By the induction hypothesis, the statement also holds for n .

2.9.4 I4) Calculate the limit

$$\lim_{n \rightarrow \infty} \sqrt{2 + \sqrt{2 + \cdots + \sqrt{2}}}$$

Solution: Consider $s = \{s_n = 1\}_{n \geq 0} \in \mathcal{S}$, then $c_n = 2 - 2^n$ hence $\lim_{n \rightarrow \infty} 2 \sin(\pi c_n/4) = 2$.

2.10 Problem J

2.10.1 Problem: $k \geq 2$ is a given integer. Define the sequence $\{a_n\}$ as follows:

$$a_0 > 0 \text{ already given, } a_{n+1} = a_n + a_n^{-1/k}, n \geq 0.$$

Calculate $\lim_{n \rightarrow \infty} a_n^{k+1}/n^k$. Solution: It is easy to see that $a_n \rightarrow \infty$, hence

$$\lim_{n \rightarrow \infty} \frac{a_n^{\frac{k+1}{k}}}{n} = \lim_{n \rightarrow \infty} a_{n+1}^{\frac{k+1}{k}} - a_n^{\frac{k+1}{k}} = \lim_{n \rightarrow \infty} a_n^{\frac{k+1}{k}} \left(\left(1 + \frac{a_{n+1} - a_n}{a_n}\right)^{\frac{k+1}{k}} - 1 \right) \quad (2.3)$$

$$= \lim_{n \rightarrow \infty} a_n^{\frac{k+1}{k}} \left(\left(1 + a_n^{-\frac{k+1}{k}}\right)^{\frac{k+1}{k}} - 1 \right) = \frac{k+1}{k}. \quad (2.4)$$

Therefore

$$\lim_{n \rightarrow \infty} \frac{a_n^{k+1}}{n^k} = \left(1 + \frac{1}{k}\right)^k.$$

Chapter 3 Basel Problem

3.1 PSA

3.1.1 A1) Given $f : (a, x_0) \cup (x_0, b) \rightarrow \mathbb{R}$, then $\lim_{x \rightarrow x_0} f(x)$ exists iff for any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x_1, x_2 \in (x_0 - \delta, x_0 + \delta)$, $|f(x_1) - f(x_2)| < \varepsilon$.

Proof: $\Leftarrow$ Let $x_n = x_0 + 1/n$, then $\{f(x_n)\}$ form a Cauchy sequence, hence $f(x_0) = \lim_{n \rightarrow \infty} f(x_n)$ exists. For any $\varepsilon > 0$, there exists $N, \delta > 0$ such that for any $x, y \in (x_0 - \delta, x_0 + \delta)$, $|f(x) - f(y)| < \varepsilon$ and for any $n > N$, $|f(x_n) - f(x_0)| < \varepsilon$, hence let $\delta' = \min\{\delta, 1/N\}$, then for any $x \in (x_0 - \delta', x_0 + \delta')$,

$$|f(x) - f(x_0)| \leq |f(x) - f(x_N)| + |f(x_N) - f(x_0)| < 2\varepsilon.$$

Hence $\lim_{x \rightarrow x_0} f(x) = f(x_0)$ exists. $\Rightarrow$ For any $\varepsilon > 0$ there exists $\delta > 0$ such that for any $x \in (x_0 - \delta, x_0 + \delta)$, $|f(x) - f(x_0)| < \varepsilon$, hence for any $x, y \in (x_0 - \delta, x_0 + \delta)$, $|f(x) - f(y)| < 2\varepsilon$.

3.1.2 A2) Suppose I is an interval (not a point), prove that the linear space $C(I)$ on $\mathbb{R}$ is of infinite dimension.

Proof: $C(I)$ contains the subspace of all polynomials, hence is of infinite dimension.

3.1.3 A3) Suppose $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are both continuous, prove that $g \circ f : X \rightarrow Z$ is also continuous.

Proof: For any open set $U \in Z$, $g^{-1}(U) \subset Y$ is an open set, and $f^{-1}(g^{-1}(U)) \subset X$ is an open set, hence $(g \circ f)^{-1}(U)$ is an open set in X and therefore $g \circ f$ is continuous on X .

3.1.4 A4) Suppose (X, d_X) and (Y, d_Y) are metric spaces, $f : X \rightarrow Y$ is continuous. If d'_X and d_X are equivalent metrics, and so are d'_Y and d_Y , then in the spaces (X, d'_X) and (Y, d'_Y) , f is also continuous.

Proof: The topology generated by equivalent metrics are the same.

3.1.5 A5) The mapping $f : X \rightarrow \mathbb{R}^n$ can be written in the form

$$f : X \rightarrow \mathbb{R}^n, x \mapsto f(x) = (f_1(x), f_2(x), \dots, f_n(x)).$$

Prove that f is continuous iff f_i is continuous for every $i = 1, 2, \dots, n$. Proof: Since f is continuous iff $\forall x_n \rightarrow x, f(x_n) \rightarrow f(x)$, and $\{x_k = (x_k^{(1)}, \dots, x_k^{(n)})\}_{k \geq 1}$ converges iff every $\{x_k^{(i)}\}_{k \geq 1}$ converges, f is continuous iff every f_i is continuous.

3.1.6 A6) Suppose (X, d_X) is a metric space, $(V, \|\cdot\|)$ is a normed linear space. $f : X \rightarrow V$ and $g : X \rightarrow V$ are continuous mappings. Prove that $f \pm g : X \rightarrow V$ is continuous. If $V = \mathbb{C}$ then $f \cdot g : X \rightarrow \mathbb{C}$ is continuous. If $V = \mathbb{C}$ and for any $x \in X$, $g(x) \neq 0$, then $f/g : X \rightarrow \mathbb{C}$ is continuous.

(Choose one statement to prove.) Proof: Since for $\{x_n\}, \{y_n\} \subset \mathbb{C}$, $\lim_{n \rightarrow \infty} x_n y_n = \lim_{n \rightarrow \infty} x_n \cdot \lim_{n \rightarrow \infty} y_n$ and if $y_n \neq 0$, then

$$\lim_{n \rightarrow \infty} x_n / y_n = \lim_{n \rightarrow \infty} x_n / \lim_{n \rightarrow \infty} y_n.$$

Hence $f \cdot g, f/g$ are both continuous. For $\{x_n\}, \{y_n\} \subset V$, if $A = \lim_{n \rightarrow \infty} x_n$ and $B = \lim_{n \rightarrow \infty} y_n$ then

$$\|x_n + y_n - A - B\| \leq \|x_n - A\| + \|y_n - B\| \rightarrow 0.$$

Hence $f \pm g$ is continuous.

3.1.7 A7) Find all discontinuities of the function

$$f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \begin{cases} 1/q, & \text{if } x = p/q \in \mathbb{Q}, \text{ where } q \geq 1, (p, q) = 1. \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Solution: For any $x \in \mathbb{Q}$, $f(x) \neq 0$ but for any $\delta > 0$ there exists $y \in (x - \delta, x + \delta)$ such that $y \notin \mathbb{Q}$. Hence $|f(x) - f(y)| = f(x)$, so f is not continuous at x . For any $x \notin \mathbb{Q}$, and any $\varepsilon > 0$, let $N = \lfloor 1/\varepsilon \rfloor + 1$ and $\delta = \inf_{n \leq N} \|xn\|/n$, then for any $y \in (x - \delta, y + \delta)$, if $y \notin \mathbb{Q}$ then $f(x) = f(y) = 0$, if $y = p/q \in \mathbb{Q}$ then $q > N > 1/\varepsilon$, hence $|f(x) - f(y)| = f(y) = 1/q < \varepsilon$. Therefore f is continuous at x iff $x \notin \mathbb{Q}$.

3.1.8 A8) Calculate

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \sum_{n=1}^{\infty} \frac{x^{n-1}}{n!} = 1.$$

3.1.9 A9) Calculate

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

Since $\lim_{n \rightarrow \infty} (1 + 1/n)^n = e$ and $(1 + 1/x)^x$ is monotonic on $[100, \infty)$.

3.1.10 A10) Calculate

$$\lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x}\right)^x = e.$$

Since $\lim_{x \rightarrow \infty} (1 - 1/x)^x = \lim_{x \rightarrow \infty} (1 - 1/x)^{x-1} = e$.

3.2 PSB

3.2.1 B1) Calculate the following series:

1.

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n} - \frac{1}{n+1} = 1.$$

2.

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{2n-1} - \frac{1}{2n+1} = \frac{1}{2}.$$

3.

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} = \frac{1}{4}.$$

4.

$$\sum_{n=1}^{\infty} \arctan \frac{1}{n^2 + n + 1} = \sum_{n=1}^{\infty} \arctan \frac{1}{n} - \arctan \frac{1}{n+1} = \frac{\pi}{4}.$$

5.

$$\sum_{n=0}^{\infty} \frac{(-1)^n + 2}{3^n} = \frac{1}{1+1/3} + \frac{2}{1-1/3} = \frac{3}{4} + 3 = \frac{15}{4}.$$

6.

$$\sum_{n=1}^{\infty} \frac{n}{3^n} = \frac{3}{4}.$$

7.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2^{n-1}} = \frac{1}{1+1/2} = \frac{2}{3}.$$

8.

$$\sum_{n=1}^{\infty} \frac{2n-1}{2^n} = 3.$$

9.

$$\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} - \frac{1}{(n+1)^2} = 1.$$

10.

$$\sum_{n=1}^{\infty} \sqrt{n+2} - 2\sqrt{n+1} + \sqrt{n} = 1 - \sqrt{2}.$$

11.

$$\sum_{n=1}^{\infty} \log \left(\frac{n(2n+1)}{(n+1)(2n-1)} \right) = \lim_{n \rightarrow \infty} \log \left(\frac{2n+1}{n+1} \right) = \log 2.$$

12.

$$\sum_{n=1}^{\infty} \frac{1}{n(n+m)} = \frac{1}{m} \sum_{n=1}^{\infty} \frac{1}{n} - \frac{1}{n+m} = \frac{1}{m} \sum_{n=1}^m \frac{1}{n}.$$

3.2.2 B2) Determine whether the following series converge:

1.

$$\sum_{n=1}^{\infty} \sqrt{n+1} - \sqrt{n} = \lim_{n \rightarrow \infty} \sqrt{n+1} - 1 = \infty.$$

2.

$$\sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{n} \leq \sum_{n=1}^{\infty} \frac{1}{2n\sqrt{n}}$$

converges. 3.

$$\sum_{n=2}^{\infty} (\sqrt[n]{n} - 1)^n$$

converges, since $\limsup_{n \rightarrow \infty} \sqrt[n]{(\sqrt[n]{n} - 1)^n} = 0 < 1$. 4.

$$\sum_{n=1}^{\infty} \frac{1}{1+x^n}$$

converges if $|x| > 1$ and diverges if $|x| \leq 1$. 5.

$$\sum_{n=1}^{\infty} \frac{1}{n2^n} \leq \sum_{n=1}^{\infty} \frac{1}{2^n} = 1.$$

converges 6.

$$\sum_{n=1}^{\infty} \left(\frac{n^2}{3n^2 + 1} \right)^n \leq \sum_{n=1}^{\infty} \frac{1}{3^n} < 1.$$

converges. 7.

$$\sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}} \geq \sum_{n=1}^{\infty} \frac{1}{2n}$$

diverges. 8.

$$\sum_{n=2}^{\infty} \frac{1}{(\log n)^{\log n}} = \sum_{n=2}^{\infty} \frac{1}{n^{\log \log n}} \leq C + \sum_{n=100}^{\infty} \frac{1}{n^2}$$

converges. 9.

$$\sum_{n=1}^{\infty} \frac{n^{n+1/n}}{\left(n + \frac{1}{n}\right)^n}$$

diverges, since

$$\lim_{n \rightarrow \infty} \frac{n^{n+1/n}}{\left(n + \frac{1}{n}\right)^n} = \exp \lim_{n \rightarrow \infty} \left(\frac{\log n}{n} - n \log \left(1 + \frac{1}{n^2} \right) \right) = 1.$$

10.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \sqrt{n}}{n+1}$$

converges (conditionally), since the partial sum of $(-1)^{n-1}$ is bounded and $\frac{\sqrt{n}}{n+1}$ monotonically tends to 0. 11.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt[n]{n}}$$

diverges since $(-1)^{n-1} n^{-1/n}$ does not tend to 0. 12. Let $H_n = 1 + 1/2 + \cdots + 1/n$.

$$\sum_{n=1}^{\infty} \frac{H_n \sin nx}{n}$$

converges since the partial sum of $\sin nx$ is bounded and $\frac{H_n}{n}$ monotonically tends to 0.

3.2.3 B3) Determine whether the following series converge (absolutely):

1.

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{n \log n}$$

converges since the partial sum of $(-1)^n$ is bounded and $\frac{1}{n \log n}$ monotonically tends to 0, but only conditionally by C3).

2.

$$\sum_{n=2}^{\infty} \frac{\sin(n\pi/4)}{\log n}$$

converges since the partial sum of $\sin(n\pi/4)$ is bounded and $\frac{1}{\log n}$ monotonically tends to 0, but only conditionally since $\sum_{n=2}^{\infty} \frac{1}{\log(4n+2)}$ tends to infinity. 3.

$$\sum_{n=1}^{\infty} (-1)^n \frac{n-1}{n+1} \frac{1}{\sqrt[n]{n}}$$

converges since $\frac{n-1}{(n+1)\sqrt[n]{n}}$ monotonically tends to 0, but only conditionally since $\sum_{n=1}^{\infty} n^{-1/3}$ diverges. 4. $a > 1$.

$$\sum_{n=1}^{\infty} (-1)^{n(n-1)/2} \frac{n^{10}}{a^n}$$

converges absolutely since there exists $C > 0$ such that for $n > C$, $n^{10} a^{-n} \leq a^{-n/2}$, and $\sum_{n=1}^{\infty} a^{-n/2}$ converges.

3.3 PSC

Suppose the integer $b \geq 2$, $f : [1, \infty) \rightarrow \mathbb{R}_{>0}$ is monotonically decreasing.

3.3.1 C1) Prove that

$$(b-1)b^{k-1}f(b^k) \leq \sum_{j=b^{k-1}}^{b^k-1} f(j) \leq (b-1)b^{k-1}f(b^{k-1}).$$

Proof: There are $(b-1)b^{k-1}$ integers in $[b^{k-1}, b^k - 1]$, and since f is monotonically decreasing, for any $j \in [b^{k-1}, b^k - 1]$, $f(j) \in [f(b^k), f(b^{k-1})]$.

3.3.2 C2) Prove that the series

$$\sum_{n=1}^{\infty} f(n) \text{ and } \sum_{n=1}^{\infty} b^n f(b^n)$$

converge or diverge simultaneously. Proof: From C1),

$$\sum_{k=1}^{\infty} (b-1)b^{k-1}f(b^k) \leq \sum_{n=1}^{\infty} f(n) = \sum_{k=1}^{\infty} \sum_{j=b^{k-1}}^{b^k-1} f(j) \leq \sum_{k=1}^{\infty} (b-1)b^{k-1}f(b^{k-1}).$$

Therefore the two series converge or diverge simultaneously.

3.3.3 C3) Prove that $\sum_{n=2}^{\infty} \frac{1}{n \log n}$ diverges.

Proof: Consider $f(x) = \frac{1}{x \log x}$ which is monotonically decreasing. Note that

$$\sum_{n=2}^{\infty} 2^n f(2^n) = \sum_{n=2}^{\infty} \frac{1}{n \log 2} = \infty.$$

From C2) we know that $\sum_{n=2}^{\infty} f(n)$ diverges.

3.3.4 C4) Prove that $\sum_{n=100}^{\infty} \frac{1}{n \log n \log \log n}$ diverges.

Proof: Consider $f(x) = \frac{1}{x \log x \log \log x}$ which is monotonically decreasing. From C3),

$$\sum_{n=100}^{\infty} 2^n f(2^n) = \sum_{n=100}^{\infty} \frac{1}{n \log 2 \cdot \log(n \log 2)}$$

diverges. Hence from C2) we know that $\sum_{n=100}^{\infty} f(n)$ diverges.

3.3.5 C5) Prove that $\sum_{n=1}^{\infty} n^{-s}$ converges iff $s > 1$.

Proof: Consider $f(x) = x^{-s}$ which is monotonically decreasing. Note that

$$\sum_{n=1}^{\infty} 2^n f(2^n) = \sum_{n=1}^{\infty} 2^{-n(s-1)} = \frac{2^{1-s}}{1 - 2^{1-s}}.$$

3.3.6 C6) Suppose $s > 1$, prove that $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^s}$ and $\sum_{n=10}^{\infty} \frac{1}{n \log n (\log \log n)^s}$ converges.

Proof: Same as C3) and C4).

3.4 PSD

For $\{a_n\}_{n \geq 1} \subset \mathbb{R}$,

- $\alpha \in \mathbb{R}$, if for any $\varepsilon > 0$, there are infinitely many n such that $a_n \in (\alpha - \varepsilon, \alpha + \varepsilon)$, then we call α a limit point of $\{a_n\}_{n \geq 1}$.
- Likewise define limit points for $\alpha = \pm\infty$.

3.4.1 D1) Prove that $\alpha \in \mathbb{R}$ is a limit point of $\{a_n\}_{n \geq 1}$ iff there is a sub-sequence $\{a_{n_k}\}_{k \geq 1}$ which converges to α .

Proof: $\Leftarrow$ is trivial. $\Rightarrow$ Let $\varepsilon = 1/k$ then there exists a_{n_k} such that $|a_{n_k} - \alpha| < \varepsilon$. Hence $\lim_{k \rightarrow \infty} a_{n_k} = \alpha$.

3.4.2 D2) Prove that $+\infty$ is a limit point of $\{a_n\}_{n \geq 1}$ iff there is a sub-sequence $\{a_{n_k}\}_{k \geq 1}$ such that $\lim_{k \rightarrow \infty} a_{n_k} = \infty$.

Proof: Same as D1).

3.4.3 D3) Let $E = \{\alpha \in \mathbb{R} \cup \{\pm\infty\} : \alpha \text{ is a limit point of } \{a_n\}\}$. Prove that $E \neq \emptyset$.

Proof: If $\{a_n\}$ is unbounded, then by D2) $E \cap \{\pm\infty\} \neq \emptyset$. If $\{a_n\}$ is bounded, then by Bolzano-Weierstrass theorem, $E \neq \emptyset$.

3.4.4 D4) Prove that $E \subset \mathbb{R}$ iff $\{a_n\}$ is bounded.

Proof: Use D2)

3.4.5 D5) Suppose $\{a_n\}_{n \geq 1}$ is bounded. Prove that $\sup E = \limsup_{n \rightarrow \infty} a_n$, $\inf E = \liminf_{n \rightarrow \infty} a_n$.

Proof: Let $M = \limsup_{n \rightarrow \infty} a_n$, then for any $\varepsilon > 0$, there exists n such that $M \leq \sup_{k \geq n} a_k < M + \varepsilon$, hence there exists $k \geq n$ such that $|a_k - M| < \varepsilon$, so $M \in E$. For any $\alpha \in E$, there is a sub-sequence $\{a_{n_k}\} \rightarrow \alpha$, hence

$$\alpha = \lim_{k \rightarrow \infty} a_{n_k} \leq \lim_{k \rightarrow \infty} \sup_{m \geq n_k} a_m = \limsup_{n \rightarrow \infty} a_n = M.$$

Therefore $M = \sup E$. Substitute $a_n \rightarrow -a_n$ and we obtain $\inf E = \liminf_{n \rightarrow \infty} a_n$.

3.4.6 D6) Suppose $\{a_n\}_{n \geq 1}$ is bounded. Let $a^* = \limsup_{n \rightarrow \infty} a_n$. Prove that

i) $a^* \in E$, i.e. $\sup E \in E$. Proof: See the proof of D5). ii) For any $x > a^*$, there exists $N \in \mathbb{Z}_{\geq 1}$ such that for any $n > N$, $a_n < x$. Proof: If there is an infinite sub-sequence $\{a_{n_k}\}_{k \geq 1}$ such that $a_{n_k} \geq x$, then $\{a_{n_k}\}$ has a limit point $a' > x > a^*$, contradicting $a^* = \sup E$.

3.4.7 D7) Construct an example of $\{a_n\}_{n \geq 1}$ such that $E \cap \mathbb{R} \neq \emptyset$ and $E \not\subset \mathbb{R}$.

Solution: Since $\mathbb{Q}$ is countable, let $\{a_n\}_{n \geq 1}$ iterate every element of $\mathbb{Q}$, then $E = \mathbb{R} \cup \{\pm\infty\}$ is an infinite set.

3.4.8 D8) Construct $\{a_n\}_{n \geq 1}$ such that E is an infinite set.

Solution: Same as D7).

3.5 PSE: Reciprocal Sum of Primes

Define the ζ -function:

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}.$$

We have proved the formula:

$$\zeta(s) = \prod_{p \in \mathcal{P}} \frac{1}{1 - p^{-s}}.$$

Prove that the series

$$\sum_{p \in \mathcal{P}} p^{-s}$$

converges when $s > 1$, and diverges when $0 < s \leq 1$. **Proof:** We know that for $|a_n| < 1$, $\prod_{n=1}^{\infty} (1 - a_n)$ converges iff $\sum_{n=1}^{\infty} a_n$ converges. Hence by $\zeta(s)^{-1} = \prod_{p \in \mathcal{P}} (1 - p^{-s})$, we obtain $\sum_{p \in \mathcal{P}} p^{-s}$ converges iff $s > 1$.

3.6 PSF: Euler's "Proof" of the Basel Problem

For any $\theta \in \mathbb{R}, n \in \mathbb{Z}$, prove the identity

$$\frac{\sin((2n+1)\theta)}{(2n+1)\sin\theta} = \prod_{k=1}^n \left(1 - \frac{\sin^2\theta}{\sin^2(k\pi/(2n+1))}\right).$$

Further prove that for any $x \in \mathbb{R}$,

$$\frac{\sin(\pi x)}{\pi x} = \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right).$$

Proof: (1) By induction there is a polynomial $P_n(x)$ such that $P_n(\sin\theta) = \sin((2n+1)\theta)$ for any $\theta \in \mathbb{R}$ and $\deg P_n = 2n+1$. For any $k = 1, 2, \dots, n$, and $\theta = \pm k\pi/(2n+1)$, $\sin((2n+1)\theta) = 0$, hence P_n has roots 0 and $\pm \sin(k\pi/(2n+1))$ for $k = 1, 2, \dots, n$. Since $\deg P_n = 2n+1$,

$$P_n(x) = Cx \prod_{k=1}^n \left(1 - \frac{x^2}{\sin^2(k\pi/(2n+1))}\right)$$

for some $C \in \mathbb{R}$. Let $x = \sin\theta$ and consider the derivatives on both sides when $\theta = 0$, then we obtain $C = 2n+1$, therefore

$$\frac{\sin((2n+1)\theta)}{(2n+1)\sin\theta} = \prod_{k=1}^n \left(1 - \frac{\sin^2\theta}{\sin^2(k\pi/(2n+1))}\right).$$

(2) Let $m = 2n+1$. From (1) we know that for any $x \in \mathbb{C}$ and $k < n$, $\sin x = U_k^{(n)} \cdot V_k^{(n)}$, where

$$U_k^{(n)} = m \sin \frac{x}{m} \prod_{j=1}^k \left(1 - \frac{\sin^2(x/m)}{\sin^2(j\pi/m)}\right), \quad (3.1)$$

$$V_k^{(n)} = \prod_{j=k+1}^n \left(1 - \frac{\sin^2(x/m)}{\sin^2(j\pi/m)}\right). \quad (3.2)$$

Clearly, for any $k \in \mathbb{N}$,

$$\lim_{n \rightarrow \infty} U_k^{(n)} = U_k = x \prod_{j=1}^k \left(1 - \frac{x^2}{j^2\pi^2}\right).$$

and for any $x \in \mathbb{C}$ and $j \in \mathbb{N}$,

$$\left| \frac{\sin^2(x/m)}{\sin^2(j\pi/m)} \right| \leq \frac{x^2}{4j^2} \cdot K(|x|/m)^2,$$

where $K(x) = \sum_{n=0}^{\infty} |x|^n / (2n+1)!$ is monotonic on $[0, \infty)$ and $K(0) = 1$. Note that for $\alpha_i \in \mathbb{C}$,

$$\left| 1 - \prod_{j=1}^n (1 - \alpha_n) \right| \leq \sum_{j=1}^n \left(\sum_{k=1}^n |\alpha_k| \right)^j.$$

Hence for any $x \in \mathbb{C}$ and $\varepsilon > 0$, there exists N such that for any $k \geq N$, and any $n > k$, $|V_k^{(n)} - 1| < \varepsilon$, since

$$|V_k^{(n)} - 1| \leq \sum_{j=1}^{\infty} \left(\sum_{l=k+1}^{\infty} \frac{x^2}{4l^2} K(|x|/m)^2 \right)^j \leq \sum_{j=1}^{\infty} \left(K(|x|/(2k+1))^2 \cdot \frac{x^2}{k} \right)^j \rightarrow 0.$$

i.e. for any $x \in \mathbb{C}$

$$\lim_{k \rightarrow \infty} \sup_{n > k} |V_k^{(n)} - 1| = 0.$$

And likewise we know that there is a constant M such that for any $n > k$, $|x| < k$, $|U_k^{(n)}| \leq M$. Therefore for any $x \in \mathbb{C}$,

$$\sin x = x \lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 - \frac{x^2}{k^2 \pi^2} \right) = x \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2 \pi^2} \right).$$

Note: From the formula above, we can formally deduce that

$$\sin(\pi x) = \pi x (1 - \zeta(2)x^2 + \zeta(4)x^4 + \dots).$$

Compare it to $\sin z = x - x^3/6 + \dots$, and we get $\zeta(2) = \pi^2/6$.

Chapter 4 Topology

4.1 PSA: Topology on Metric Spaces

4.1.1 A1) Suppose (X, d_x) and (Y, d_Y) are metric spaces, $f : X \rightarrow Y$ is a mapping. Prove that the two following definitions of continuity is equivalent:

- Suppose $x_0 \in X$, if for any $\{x_n\}_{n \geq 1} \subset X$ such that $\lim_{n \rightarrow \infty} x_n = x_0$, we have $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$, then we say f is continuous at x_0 . If f is continuous at every point $x \in X$, then f is a continuous mapping.
- Suppose $x_0 \in X, y_0 = f(x_0) \in Y$. If for any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $d_X(x, x_0) < \delta, x \in X$, we have $d_Y(f(x), f(x_0)) < \varepsilon$, we call f continuous at x_0 . If f is continuous at every point $x \in X$, then f is a continuous mapping.

Proof: $1 \Rightarrow 2$: If there exists $\varepsilon > 0$ such that for any $n \geq 1$, there exists x_n such that $d_X(x_0, x_n) < 1/n$ but $d_Y(f(x_n), f(x_0)) > \varepsilon$, then $\lim_{n \rightarrow \infty} x_n = x_0$ but $\lim_{n \rightarrow \infty} f(x_n) \neq f(x_0)$, a contradiction. $2 \Rightarrow 1$: For any $\{x_n\}_{n \geq 1} \subset X$ such that $\lim_{n \rightarrow \infty} x_n = x_0$, and any $\varepsilon > 0$, take the corresponding δ and N such that $n > N \implies d(x_n, x_0) < \delta$. Then for any $n > N, d(x_n, x_0) < \delta$ so $d(f(x_n), f(x_0)) < \varepsilon$, hence $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$.

4.1.2 A2) (X, d) is a metric space. For any $x \in X, r > 0$, let $B(x, r) = \{y \in X : d(x, y) < r\}$. Proved that for any $x \in X, r > 0$, if $x' \in B(x, r)$, then there exists $r' > 0$ such that $B(x', r') \subset B(x, r)$.

If $U = \bigcup_{\alpha \in \mathcal{A}} B(x_\alpha, r_\alpha)$, then we call U an open set. Prove that $U \subset X$ is open iff for any $x \in U$, there exists $\delta_x > 0$ such that $B(x, \delta_x) \subset U$. Proof: If $x' \in B(x, r)$, let $r' = r - d(x, x')$, then for any $y \in B(x', r')$, $d(x, y) \leq d(x, x') + d(x', y) < d(x, x') + r' = r$, hence $y \in B(x, r)$ so $B(x', r') \subset B(x, r)$. If for any $x \in U$, there exists $\delta_x > 0$ such that $B(x, \delta_x) \subset U$, then $U = \bigcup_{x \in U} B(x, \delta_x)$ is open. If U is open then for any $x \in U$, suppose $x \in B(x_\alpha, r_\alpha)$ for some $\alpha \in \mathcal{A}$, then there exists $\delta_x > 0$ such that $B(x, \delta_x) \subset B(x_\alpha, r_\alpha) \subset U$.

4.1.3 A3) Let $\mathcal{T}$ denote all open sets on (X, d) . Prove that $\mathcal{T}$ is a topology.

Proof: 1. $\emptyset \in \mathcal{T}, X = \bigcup_{x \in X} B(x, 1) \in \mathcal{T}$. 2. If $\{U_\alpha : \alpha \in J\} \subset \mathcal{T}$, where $U_\alpha = \bigcup_{x \in \mathcal{A}_\alpha} B(x, r_{\alpha, x})$ then let $\mathcal{A} = \bigcup_{\alpha \in J} \mathcal{A}_\alpha$,

$$\bigcup_{\alpha \in J} U_\alpha = \bigcup_{x \in \mathcal{A}} B(x, \sup_{\alpha \in J} r_{\alpha, x}) \in \mathcal{T}.$$

1. If $U_1, \dots, U_n \in \mathcal{T}$, where $U_k = \bigcup_{x \in \mathcal{A}_k} B(x, r_{k, x})$, then let $\mathcal{A} = \bigcup_{k=1}^n \mathcal{A}_k$

$$\bigcap_{k=1}^n U_k = \bigcup_{x \in \mathcal{A}} B(x, \min_{k=1, \dots, n} r_{k, x}) \in \mathcal{T}.$$

Therefore $\mathcal{T}$ is a topology on X .

4.1.4 A4) (X, d) is a metric space. If $F \subset X$ and F^C is open, then we call F a closed set. Prove that F is closed iff for any sequence $\{x_n\}_{n \geq 1} \in F$, if $\lim_{n \rightarrow \infty} x_n = x$ then $x \in F$.

Proof: Suppose F is closed, if a sequence $\{x_n\}_{n \geq 1}$ satisfy $\lim_{n \rightarrow \infty} x_n = x$ and $x \in F^C$, then there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset F^C$. However $B(x, \varepsilon) \cap \{x_n\} \neq \emptyset$, leading to contradiction. If for any sequence $\{x_n\}_{n \geq 1}$ such that $\lim_{n \rightarrow \infty} x_n = x$, there is $x \in F$, then for any $x \in F^C$, if for any $\varepsilon > 0, B(x, \varepsilon) \not\subset F^C$, then for any $n \geq 1$, take $x_n \in B(x, \varepsilon) \cap F$. The sequence $\{x_n\}$ has the limit $\lim_{n \rightarrow \infty} x_n = x$ but $x \in F^C$, a contradiction. Hence F is closed.

4.1.5 A5) Prove that

1) $\emptyset$ and X are closed sets. 2) Any intersection of closed sets are still closed. 3) Finite unions of closed sets are still closed. Proof: Use A3) and de Morgan's theorem.

4.1.6 A6) Suppose (X, d_X) and (Y, d_Y) are metric spaces and $f : X \rightarrow Y$, then the following statements are equivalent:

1) f is continuous. 2) For any $U \subset Y$ open, $f^{-1}(U)$ is an open set in X . 3) For any $F \subset Y$ closed, $f^{-1}(F)$ is a closed set in X . Proof: $1 \Rightarrow 2$: If f is continuous, then for any $U \subset Y$ open, consider any point $x \in f^{-1}(U)$. Let $y = f(x) \in U$, then there exists $\varepsilon > 0$ such that $B(y, \varepsilon) \subset U$. Since f is continuous, there exists $\delta > 0$ such that for any $x' \in B(x, \delta)$, $f(x') \in B(y, \varepsilon) \subset U$, hence $B(x, \delta) \subset f^{-1}(U)$. Therefore $f^{-1}(U)$ is an open set in X . $2 \Rightarrow 1$: For any $x \in X$ and $\varepsilon > 0$, consider the open set $U = B(y, \varepsilon)$, where $y = f(x)$. Since $x \in f^{-1}(U)$ and $f^{-1}(U)$ is an open set, there exists $\delta > 0$ such that $B(x, \delta) \subset f^{-1}(U)$, therefore f is continuous. $2 \Leftrightarrow 3$: Note that $f^{-1}(Y \setminus A) = X \setminus f^{-1}(A)$.

4.1.7 A7) Let A' be the set of limit points of A . Prove that $\bar{A} = A' \cup A$.

Proof: For any closed set $F \supset A$, by A4) we know $A' \subset F$, hence $A' \cup A \subset \bar{A}$. Consider a sequence $\{x_n\}_{n \geq 1} \subset A' \cup A$ such that $\lim_{n \rightarrow \infty} x_n = x$ exists, for any $n \geq 1$ we can find a $y_n \in A$ such that $d(x_n, y_n) \leq 2^{-n}$, hence $\lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} x_n = x$ so $x \in A' \cup A$. Therefore $A' \cup A$ is closed, and hence $\bar{A} = A' \cup A$.

4.1.8 A8) Suppose (Y, d_Y) and (Z, d_Z) are metric spaces, define the metric on $Y \times Z$:

$$d_{Y \times Z} : (Y \times Z)^2 \rightarrow \mathbb{R}_{\geq 0}, ((y_1, z_1), (y_2, z_2)) \rightarrow \sqrt{d_Y(y_1, y_2)^2 + d_Z(z_1, z_2)^2}.$$

Prove that this defines a metric and the projection mappings are continuous:

$$\pi_Y : Y \times Z \rightarrow Y, (y, z) \mapsto y; \pi_Z : Y \times Z \rightarrow Z, (y, z) \mapsto z.$$

Given a mapping $F : X \rightarrow Y \times Z$, then F is continuous iff $\pi_Y \circ F$ and $\pi_Z \circ F$ are both continuous. Proof: $d((y_1, z_1), (y_2, z_2)) = 0 \iff (y_1, z_1) = (y_2, z_2)$, $d((y_1, z_1), (y_2, z_2)) = d((y_2, z_2), (y_1, z_1))$, and $d((y_1, z_1), (y_2, z_2)) \leq d((y_1, z_1), (y_3, z_3)) + d((y_3, z_3), (y_2, z_2))$ (since $\sqrt{(x+y)^2 + (u+v)^2} \leq \sqrt{x^2 + u^2} + \sqrt{y^2 + v^2}$), hence $d_{Y \times Z}$ is a metric. Note that $d((y_1, z_1), (y_2, z_2)) \geq d(y_1, y_2)$, hence π_Y and π_Z are continuous. $d((y_1, z_1), (y_2, z_2)) \leq d(y_1, y_2) + d(z_1, z_2)$, hence F is continuous iff $\pi_Y \circ F$ and $\pi_Z \circ F$ are both continuous.

4.1.9 A9) Prove that the operators $+$ and $\cdot$ on real numbers are continuous.

Proof: For any $(x, y), (u, v) \in \mathbb{R}^2$,

$$|(x + y) - (u + v)| \leq |x - u| + |y - v| \leq 2|(x, y) - (u, v)|.$$

Hence $+$ is uniformly continuous.

$$|x \cdot y - u \cdot v| \leq |x| \cdot |y - v| + |v| \cdot |x - u|.$$

Therefore $\cdot$ is continuous.

4.1.10 A10) Prove that the operators $+$ and $\cdot$ on $M_n(\mathbb{R})$ are continuous.

Proof: The proof of A9) only uses the properties of norms, and the fact that $\|A \cdot B\| \leq \|A\| \cdot \|B\|$. This also holds for the norm $\|A\| = \sup_{|x|=1} |Ax|$ on $M_n(\mathbb{R})$, therefore $+$ and $\cdot$ are continuous on $M_n(\mathbb{R})$.

4.1.11 A11) Prove that $\mathbf{GL}_n(\mathbb{R})$ is an open set on $\mathbf{M}_n(\mathbb{R})$.

Proof: The mapping $\det : \mathbf{M}_n(\mathbb{R}) \rightarrow \mathbb{R}$ is continuous, since view as $\det : \mathbb{R}^{n^2} \rightarrow \mathbb{R}$ it is a multi-linear mapping. The set $\mathbf{GL}_n(\mathbb{R}) = \det^{-1}(\{x \in \mathbb{R} : x \neq 0\})$, where $\{x \in \mathbb{R} : x \neq 0\}$ is an open set on $\mathbb{R}$, therefore $\mathbf{GL}_n(\mathbb{R})$ is an open set on $\mathbf{M}_n(\mathbb{R})$.

4.1.12 A12) Prove that $\text{Inv} : \mathbf{GL}_n(\mathbb{R}) \rightarrow \mathbf{GL}_n(\mathbb{R}), A \mapsto A^{-1}$ is continuous.

Proof: Note that for any $A, B \in \mathbf{GL}_n(\mathbb{R})$,

$$\|A^{-1} - B^{-1}\| \leq \frac{\|A - B\|}{\|A\| \cdot \|B\|}.$$

Hence Inv is continuous.

4.2 PSB

Prove the following equalities:

4.2.1 B1) $\lambda > 0, \lim_{x \rightarrow \infty} \frac{x^n}{e^{\lambda x}} = 0$.

Proof: By definition, for $x > 0, e^{\lambda x} \geq (\lambda x)^{n+1}/(n+1)!$. Hence for any $\varepsilon > 0$, let $M = \frac{(n+1)!}{\lambda^{n+1}\varepsilon}$, then for any $x > M$,

$$\left| \frac{x^n}{e^{\lambda x}} \right| \leq \frac{(n+1)!}{\lambda^{n+1}x} < \varepsilon.$$

Therefore

$$\lim_{x \rightarrow \infty} \frac{x^n}{e^{\lambda x}} = 0.$$

4.2.2 B2) $\alpha > 0$, then

$$\lim_{x \rightarrow \infty} x^\alpha \log \left(1 + \frac{1}{x} \right) = \begin{cases} \infty, & \alpha > 1; \\ 1, & \alpha = 1; \\ 0, & 0 < \alpha < 1. \end{cases}$$

Proof: If $0 < \alpha < 1$, then for any $\varepsilon > 0$, there exists $\delta = \varepsilon^{1/(\alpha-1)}$ such that for any $x > \delta$,

$$\left| x^\alpha \log \left(1 + \frac{1}{x} \right) \right| \leq x^{\alpha-1} < \varepsilon.$$

If $\alpha > 1$, then for any $\varepsilon > 0$, there exists $\delta = (2\varepsilon)^{1/\alpha-1}$ such that for any $x > \delta$,

$$\left| x^\alpha \log \left(1 + \frac{1}{x} \right) \right| \geq \frac{x^\alpha}{x+1} \geq \frac{1}{2}x^{\alpha-1} > \varepsilon.$$

If $\alpha = 1$, then for any $\varepsilon > 0$, there exists $\delta = 1/\varepsilon$ such that for any $x > \delta$,

$$1 - \varepsilon \leq \frac{x}{x+1} \leq x \log \left(1 + \frac{1}{x} \right) \leq 1.$$

Therefore

$$\lim_{x \rightarrow \infty} x^\alpha \log \left(1 + \frac{1}{x} \right) = \begin{cases} \infty, & \alpha > 1; \\ 1, & \alpha = 1; \\ 0, & 0 < \alpha < 1. \end{cases}$$

4.2.3 B3) $\lim_{x \rightarrow 0^+} x^{-n} e^{-1/x^2} = 0$.

Proof: If $x < 1$, then $e^{-1/x^2} \leq e^{-1/x} \leq (n+1)!x^{n+1}$, hence for any $\varepsilon > 0$, let $\delta = \varepsilon/(n+1)!$, then for any $x \in (0, \delta)$, $x^{-n} e^{-1/x^2} \leq (n+1)!x \leq \varepsilon$. Therefore

$$\lim_{x \rightarrow 0^+} x^{-n} e^{-1/x^2} = 0.$$

4.2.4 B4) We know that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. Calculate

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x}, \text{ and } \lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2/2}.$$

Solution: For any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $|x| < \delta$, $\sin x \in ((1-\varepsilon)x, (1+\varepsilon)x)$. Hence

$$\left| \frac{\cos x - 1}{x} \right| \leq \left| \frac{\sqrt{1 - \sin^2 x} - 1}{x} \right| \leq \left| \frac{\sin^2 x}{x(\sqrt{1 - \sin^2 x} + 1)} \right| \leq (1+\varepsilon)^2 x \leq \delta(1+\varepsilon)^2.$$

Therefore

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0.$$

Likewise

$$\left| \frac{\cos x - 1}{x^2/2} + 1 \right| \leq \left| \frac{\sin^2 x - x^2(1 + \sqrt{1 - \sin^2 x})/2}{x^2/2 \cdot (\sqrt{1 - \sin^2 x} + 1)} \right| \leq (2\varepsilon + \sqrt{1 - \sin^2 x} - 1).$$

Therefore

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2/2} = -1.$$

4.3 PSC: Root of Function:**4.3.1 C1) Prove that $x^3 + 2x - 1 = 0$ has exactly one root which lies in $(0, 1)$.**

Proof: Let $f(x) = x^3 + 2x - 1$, then $f(0) = -1$ and $f(1) = 2$, so $f(0) < 0 < f(1)$. Since f is continuous and monotonically increasing on $(0, 1)$, there is exactly one root in $(0, 1)$.

4.3.2 C2) Suppose $0 \leq \lambda < 1$, $b > 0$, determine whether the equation $x - \lambda \sin x = b$ has a solution.

Solution:

4.3.3 C3) Prove that $\sin x = 1/x$ has infinitely many roots.

Proof: For any $n \in \mathbb{N}$, let $x_n = (2n+1/2)\pi$, $y_n = (2n+3/2)\pi$, and $f(x) = \sin x - 1/x$, then $f(x_n) = 1 - 1/x_n > 0$, $f(y_n) = -1 - 1/y_n < 0$, therefore f has a root in (x_n, y_n) , and hence f has infinitely many roots.

4.3.4 C4) Assume $f \in C([0, 2])$ and $f(0) = f(2)$. Prove that $f(x) - f(x+1) = 0$ has a root in $[0, 1]$.

Proof: Let $g(x) = f(x) - f(x+1)$, then $g(0) = f(0) - f(1) = -g(1)$ and $g \in C([0, 1])$. Therefore g has a root in $[0, 1]$.

4.3.5 C5) Prove that $x^3 + 3 = e^x$ has a solution in $\mathbb{R}$.

Proof: Let $f(x) = e^x - x^3 - 3$, then $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow -\infty} f(x) = -\infty$, therefore f has a root in $\mathbb{R}$.

4.3.6 C6) Suppose $f : [0, 2] \rightarrow \mathbb{R}$ is continuous and $f(0) = f(2)$ then there exists $x \in [1, 2]$ such that $f(x) = f(x - 1)$.

Exactly the same as C4)?

4.3.7 C7) $f : \mathbb{R} \rightarrow \mathbb{R}$, Prove that if for any $c \in \mathbb{R}$, $|f^{-1}(c)| = 2$, then f is not continuous.

Proof: If f is continuous on $\mathbb{R}$, suppose $f^{-1}(0) = \{a < b\}$, then $f|_{[a,b]}$ is bounded. Suppose $f\left(\frac{a+b}{2}\right) > 0$, then for any $t \in (a, b)$, $f(t) > 0$ (otherwise $|f^{-1}(0)| > 2$). Consider an arbitrary $M > y = \sup_{x \in [a,b]} f(x)$, and take $t \in f^{-1}(M)$. Assume $t < a$, then $f(t) = M > y/2 > f(a) = 0$, hence there exists $s \in (t, a)$ such that $f(s) = y/2$. However there are at least two elements of $f^{-1}(y/2)$ in (a, b) , leading to contradiction.

4.3.8 C8) Suppose the continuous function $f : [a, b] \rightarrow \mathbb{R}$ is injective. If $f(a) < f(b)$, prove that f is monotonically increasing.

Proof: Otherwise suppose $f(u) > f(v)$ for some $u < v$. Note that for any $c \in (a, b)$, $f(a) < f(c) < f(b)$, otherwise $f(c) < f(a) \implies \exists d \in (c, b), f(d) = f(a)$, or $f(c) > f(b) \implies \exists d \in (a, c), f(d) = f(b)$. Hence $a < u < v < b$. Likewise consider $u < v < b$ we get $f(u) > f(v) > f(b)$, and by $a < u < v$ we get $f(a) > f(u) > f(v)$, therefore $f(a) > f(b)$, a contradiction.

4.4 PSD: Calculation of Limits

n, m are positive integers.

4.4.1 (1)

$$\lim_{x \rightarrow \infty} \frac{a_0 x^n + a_1 x^{n-1} + \cdots + a_n}{b_0 x^m + b_1 x^{m-1} + \cdots + b_m} = \begin{cases} 0, & m > n, \\ \infty, & m < n, a_0 > 0, \\ -\infty, & m < n, a_0 < 0, \\ \frac{a_0}{b_0}, & m = n. \end{cases}$$

4.4.2 (2) $a > 1, b > 0$

$$\lim_{x \rightarrow \infty} \frac{x^b}{a^x} = 0.$$

4.4.3 (3) $a > 0$

$$\lim_{x \rightarrow \infty} \frac{\log x}{x^a} = 0.$$

4.4.4 (4) $a > 0$

$$\lim_{x \rightarrow 0^+} x^a \log x = 0.$$

4.4.5 (5)

$$\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x^2 - 2} \right)^{x^2} = \lim_{x \rightarrow \infty} \left(\frac{x + 1}{x - 2} \right)^x = e^3.$$

4.4.6 (6)

$$\lim_{x \rightarrow \infty} (x - \sqrt{x^2 - a}) = \lim_{x \rightarrow \infty} \frac{a}{x + \sqrt{x^2 - a}} = 0.$$

4.4.7 (7)

$$\lim_{x \rightarrow \infty} \sqrt{x + 1} - \sqrt{x - 1} = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{x + 1} + \sqrt{x - 1}} = 0.$$

4.4.8 (8)

$$\lim_{x \rightarrow 0} \frac{(1 + x)(1 + 2x)(1 + 3x) - 1}{x} = 1 + 2 + 3 = 6.$$

4.4.9 (9)

$$\lim_{x \rightarrow 1} \frac{x + x^2 + \cdots + x^n - n}{x - 1} = \frac{n(n + 1)}{2}.$$

4.4.10 (10)

$$\lim_{x \rightarrow 1} \frac{x^{100} - 2x + 1}{x^{50} - 2x + 1} = \frac{49}{24}.$$

4.4.11 (11)

$$\lim_{x \rightarrow 1} \left(\frac{m}{1 - x^m} - \frac{n}{1 - x^n} \right) = \frac{m - n}{2}.$$

Proof: Note that

$$\lim_{x \rightarrow 1} \left(\frac{m}{1 - x^m} - \frac{n}{1 - x^n} \right) = \lim_{x \rightarrow 1} \frac{m(1 + x + \cdots + x^{n-1}) - n(1 + x + \cdots + x^{m-1})}{(1 + x + \cdots + x^{m-1})(1 + x + \cdots + x^{n-1})(1 - x)} \quad (4.1)$$

$$= \frac{1}{mn} \cdot \lim_{x \rightarrow 1} \frac{m(x - 1 + \cdots + x^{n-1} - 1) - n(x - 1 + \cdots + x^{m-1} - 1)}{1 - x} \quad (4.2)$$

$$= \frac{1}{mn} \cdot (-m(1 + 2 + \cdots + (n - 1)) + n(1 + 2 + \cdots + (m - 1))) \quad (4.3)$$

$$= \frac{m - n}{2}. \quad (4.4)$$

4.4.12 (12)

$$\lim_{x \rightarrow 0} \frac{(1 + x)^a - 1}{x} = a.$$

(diverges if $a = 0$).

4.4.13 (13)

$$\lim_{x \rightarrow 1} \frac{x^a - 1}{x^b - 1} = \frac{a}{b}.$$

4.4.14 (14)

$$\lim_{x \rightarrow \infty} (\log x)^{1/x} = \lim_{x \rightarrow \infty} e^{(\log \log x)/x} = 1.$$

4.4.15 (15) $a, b > 0$

$$\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2} \right)^{1/x} = \sqrt{ab}.$$

4.4.16 (16)

$$\lim_{x \rightarrow \infty} \sqrt[k]{(x + a_1)(x + a_2) \cdots (x + a_k)} - x$$

Proof: Let $y = (x + a_1)(x + a_2) \cdots (x + a_k)$ and $s = a_1 + \cdots + a_k$, then

$$\frac{sx^{k-1}}{ky^{(k-1)/k}} \leq \sqrt[k]{y} - x = \frac{y - x^k}{y^{(k-1)/k} + \cdots + x^{k-1}} \leq \frac{sx^{k-1} + \prod_{i=1}^k (1 + a_i)x^{k-2}}{kx^{k-1}}.$$

Therefore

$$\lim_{x \rightarrow \infty} \sqrt[k]{y} - x = s = \sum_{i=1}^k a_i.$$

4.4.17 (17)

$$\lim_{x \rightarrow 0} \frac{(\sqrt{1+x^2} + x)^n - (\sqrt{1+x^2} - x)^n}{x} = 2n.$$

4.4.18 (18)

$$\lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x} = 1.$$

4.4.19 (19)

$$\lim_{x \rightarrow \infty} \left(\sin \frac{1}{x} + \cos \frac{1}{x} \right)^x = e.$$

4.4.20 (20) $\alpha > 0$,

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x + \sqrt{x + \sqrt{x}}}}{x^\alpha} = \begin{cases} 0, & \alpha > \frac{1}{2}, \\ 1, & \alpha = \frac{1}{2}, \\ \infty, & \alpha < \frac{1}{2}. \end{cases}$$

4.4.21 (21) $\alpha > 0$,

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x + \sqrt{x + \sqrt{x}}}}{x^\alpha} = \begin{cases} 0, & \alpha < \frac{1}{8}, \\ 1, & \alpha = \frac{1}{8}, \\ \infty, & \alpha > \frac{1}{8}. \end{cases}$$

Proof: Note that for $x \in (0, 1)$,

$$x^{1/8-\alpha} \leq \frac{\sqrt{x + \sqrt{x + \sqrt{x}}}}{x^\alpha} \leq 2x^{1/8-\alpha}.$$

And for any $\varepsilon > 0$ there exists $\delta = (1 + \varepsilon)\varepsilon$ such that for any $x < \delta$, $\sqrt{x + \sqrt{x + \sqrt{x}}} < \varepsilon x^{1/8}$. Therefore

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x + \sqrt{x + \sqrt{x}}}}{x^\alpha} = \begin{cases} 0, & \alpha < \frac{1}{8}, \\ 1, & \alpha = \frac{1}{8}, \\ \infty, & \alpha > \frac{1}{8}. \end{cases}$$

4.5 Problem E

Prove that for any $A \subset \mathbb{R}$ that is countable, there exists a monotonic function $f : \mathbb{R} \rightarrow \mathbb{R}$, such that the set of discontinuities of f is exactly A . Proof: Let $A = \{x_1, x_2, \dots\}$ and $f(x) = \sup\{1 - 2^n : x_n < x\}$, (define $\sup \emptyset = 0$) then f is monotonically increasing and the set of discontinuities is exactly A .

4.6 Problem F

$f : [0, 1] \rightarrow [0, 1]$ is monotonic. Prove that f has a fixed point. Proof: Otherwise suppose that f has no fixed points. Let $S = \{t \in [0, 1] : f(t) > t\}$ and $x = \sup S$. Note that $0 \in S$ so S is non-empty. If $x \in S$, then $f(x) > x$ so $f(f(x)) > f(x)$ (f is monotonic) then $x < f(x) \in S$ which leads to contradiction. If $x \notin S$, then $f(x) < x$. Take $y \in (f(x), x) \cap S$, (y exists since $x = \sup S$) then $f(x) > f(y) > y > f(x)$, a contradiction.

4.7 Problem G

Consider all self-homeomorphisms of $[0, 1]$, i.e.

$$\text{Homeo}([0, 1]) = \{f : [0, 1] \rightarrow [0, 1] : f \text{ is a continuous bijective}\}$$

We know that for any $f \in \text{Homeo}([0, 1])$, $f^{-1} \in \text{Homeo}([0, 1])$. Suppose $f, g \in \text{Homeo}([0, 1])$ and the only fixed points of f, g are 0, 1. Prove that there exists $h \in \text{Homeo}([0, 1])$, such that

$$h \circ f \circ h^{-1} = g.$$

Proof: Take $x_0 = 1/2$, and let $I_n = [f^n(x_0), f^{n+1}(x_0)]$, $J_n = [g^n(x_0), g^{n+1}(x_0)]$. Note that $(0, 1) = \bigcup_{n \in \mathbb{Z}} I_n = \bigcup_{n \in \mathbb{Z}} J_n$. Define $h_0 : I_0 \rightarrow J_0$, $x \mapsto kx + b$ such that the line h_0 passes (x_0, x_0) and $(f(x_0), g(x_0))$, i.e. $x \mapsto \frac{g(x_0) - x_0}{f(x_0) - x_0}(x - x_0) + x_0$. Define $h_n : I_n \rightarrow J_n$, $x \mapsto g^n \circ f^{-n}(x)$, and $h : [0, 1] \rightarrow [0, 1]$ such that

$$h(x) = \begin{cases} x, & x \in \{0, 1\}, \\ h_n(x), & x \in I_n. \end{cases}$$

Then for any $x \in I_n$, $f(x) \in I_{n+1}$ hence $h(f(x)) = g^{n+1} \circ f^{-n}(x) = g(h(x))$. Since h maps I_n to J_n bijectively, h is a bijection on $[0, 1]$. For any $x \in I_n \cap I_{n+1}$ the value of h does not depend on which interval we choose, and h is continuous on any interval I_n , therefore h is a continuous bijection.

Chapter 5 Infinity of Primes

5.1 PSA

5.1.1 A1) Prove that e^x is uniformly continuous on $(-\infty, 0]$ but not on $\mathbb{R}$.

Proof: For $y < x \leq 0$ and $|x - y| < \varepsilon$,

$$e^x - e^y = e^y(e^{y-x} - 1) \leq e^\varepsilon - 1.$$

Hence e^x is uniformly continuous on $(-\infty, 0]$. But for any $\delta > 0$, there exists y and $x = y + \delta$ such that

$$e^x - e^y = e^y \cdot (e^\delta - 1) > 1.$$

Therefore e^x is not uniformly continuous on $\mathbb{R}$.

5.1.2 A2) Prove that the function $f : \mathbb{R}_{>0} \times \mathbb{R} \rightarrow \mathbb{R}$, $(x, \alpha) \mapsto x^\alpha$ is continuous on $\mathbb{R}_{>0} \times \mathbb{R}$.

Proof: For $(x, \alpha), (y, \beta)$,

$$|x^\alpha - y^\beta| \leq |x^\alpha - y^\alpha| + |y^\alpha - y^\beta|.$$

Since x^α and a^x are both continuous (as functions of x), so is $x^\alpha : \mathbb{R}_{>0} \times \mathbb{R} \rightarrow \mathbb{R}$.

5.1.3 A3) Prove that for any $x, y > 0$ and α, β , $(xy)^\alpha = x^\alpha y^\alpha$, $(x^\alpha)^\beta = x^{\alpha\beta}$, $a^{\log_a x} = x$. If $x > 0, y > 0$, then $a^{x+y} = a^x a^y$, $\log_a(x \cdot y) = \log_a x + \log_a y$.

Proof: See PSE of HW2.

5.1.4 A4) Consider the sequence of functions $\{f_n(x)\}_{n \geq 1}$ defined on $[0, 1]$, where $f_n(x) = x^n$. Prove that for any $a < 1$, $\{f_n(x)\}_{n \geq 1}$ converges uniformly to 0 on $[0, a]$, but $\{f_n(x)\}_{n \geq 1}$ does not converge uniformly on $[0, 1)$.

Proof: For any $a < 1$, and any $\varepsilon > 0$, let $N = \log_a x$, then for any $n > N$, $f_n(x) < \varepsilon$, hence $\{f_n(x)\}_{n \geq 1}$ converges uniformly to 0 on $[0, a]$. Let $\varepsilon = 1/2$, then for any $N \in \mathbb{N}$, there exists $1 > x > 2^{-1/N}$ such that $f_N(x) > \varepsilon$. Hence $\{f_n(x)\}_{n \geq 1}$ is not uniformly convergent on $[0, 1)$.

5.1.5 A5) Consider the sequence of functions $\{f_n(x)\}_{n \geq 1}$, where $f_n(x) = \frac{nx}{1+n^2x^2}$. Prove that $\{f_n(x)\}_{n \geq 1}$ converges point-wise to 0 on $\mathbb{R}$, but does not converge uniformly.

Proof: For any $x \in \mathbb{R}$, and any $\varepsilon > 0$, there exists $N = 1/(x\varepsilon)$ such that for any $n \geq N$,

$$\left| \frac{nx}{1+n^2x^2} \right| \leq \frac{1}{|nx|} < \varepsilon.$$

Hence $f_n(x)$ converges to 0 for any $x \in \mathbb{R}$. Let $\varepsilon = 1/2$, then for any $n \in \mathbb{N}$, there exists $x = 1/n$ such that $f_n(x) = \varepsilon$, so f is not uniformly continuous on $\mathbb{R}$.

5.1.6 A6) Consider the sequence of functions $\{f_n(x)\}_{n \geq 1}$, where

$$f_n(x) = \begin{cases} \frac{nx^2}{1+nx}, & x > 0; \\ \frac{nx^3}{1+nx^2}, & x \leq 0. \end{cases}$$

Determine the convergence of $\{f_n(x)\}_{n \geq 1}$ on $\mathbb{R}$ (both point-wise and uniformly). Proof: For any $\varepsilon > 0$, let $N = \max\{1/\varepsilon, 1/4\varepsilon^2\}$, then for any $x > 0$ and $n > N$,

$$|f_n(x) - x| = \left| \frac{x}{1 + nx} \right| < \frac{1}{n} < \varepsilon.$$

For any $x < 0$,

$$|f_n(x) - x| = \left| \frac{x}{1 + nx^2} \right| \leq \frac{1}{2\sqrt{n}} < \varepsilon.$$

Hence $\{f_n\}_{n \geq 1}$ converges uniformly to x .

5.1.7 A7) Given $\varphi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ such that $\varphi(0) = 0$, $\lim_{x \rightarrow \infty} \varphi(x) = 0$, φ is continuous and not identically zero. Prove that the sequences $\{f_n(x)\}_{n \geq 1}$ and $\{g_n(x)\}_{n \geq 1}$ converge point-wise to 0, but uniformly, where $f_n(x) = \varphi(nx)$, $g_n(x) = \varphi(x/n)$.

Proof: Point-wise convergence is trivial. Let $\varepsilon = |\varphi(1)| > 0$, then for any n there exists $x = 1/n > 0$ such that $|f_n(x)| = \varepsilon$, hence $\{f_n(x)\}_{n \geq 1}$ is not uniformly convergent. Likewise $\{g_n(x)\}_{n \geq 1}$ is not uniformly continuous.

5.1.8 A8) $f \in C([a, b])$. For $n \geq 1$, let $a_k = a + (k-1)(b-a)/n$. Define

$$S_n = \sum_{k=1}^n \frac{b-a}{n} f(a_k).$$

Prove that $\{S_n\}_{n \geq 1}$ converges, and denote this limit by $\int_a^b f$. Further show that the mapping

$$\int_a^b : C([a, b]) \rightarrow \mathbb{R}, f \mapsto \int_a^b f$$

is linear and continuous with metric d_∞ on $C([a, b])$. Proof: For any $n, m \in \mathbb{N}$, note that $|S_n - S_m| \leq |S_n - S_{nm}| + |S_{nm} - S_m|$, and

$$|S_n - S_{nm}| \leq \sum_{k=1}^n \frac{b-a}{n} \left| f(a_k^{(n)}) - \frac{1}{m} \sum_{j=1}^m f(a_{n(k-1)+j}^{(nm)}) \right| \leq (b-a) \sup_{|x-y| < 1/n} |f(x) - f(y)|.$$

Since f is uniformly continuous on $[a, b]$, the sequence $\{S_n\}_{n \geq 1}$ is Cauchy. Obviously $\int_a^b \cdot$ is linear, and for $f, g \in C([a, b])$,

$$\left| \int_a^b f - \int_a^b g \right| = \lim_{n \rightarrow \infty} |S_n(f) - S_n(g)| \leq (b-a) \|f - g\|_\infty.$$

Hence $\int_a^b \cdot$ is continuous on $C([a, b])$ with metric d_∞ .

5.1.9 A9) For any $f : [a, \infty) \rightarrow \mathbb{R}$, suppose f is bounded on any closed interval $[a, b]$, then when the limits in RHS exist,

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} f(x+1) - f(x). \quad (5.1)$$

$$\lim_{x \rightarrow \infty} f(x)^{1/x} = \lim_{x \rightarrow \infty} \frac{f(x+1)}{f(x)}, \text{ if for any } x \in [a, \infty), f(x) \geq c > 0. \quad (5.2)$$

Proof: Suppose $\lim_{x \rightarrow \infty} f(x+1) - f(x) = A$, then for any $\varepsilon > 0$ there exists M such that for any $x > M$, $|f(x+1) - f(x) - A| < \varepsilon$, so for any $n \geq 1$, $|f(x+n) - f(x) - nA| < n\varepsilon$. Hence

$$\left| \frac{f(n+x)}{n+x} - A \right| \leq \left| \frac{f(n+x) - f(x) - nA}{n+x} \right| + \left| \frac{f(x) - xA}{n+x} \right| \leq \varepsilon A + \frac{|f(x) - xA|}{n} \rightarrow 0.$$

For any $x > M$. Therefore (since f is bounded on any closed interval) there exists N such that for any $x > N$, $|f(x)/x - A| < 2\varepsilon A$, and hence

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = A = \lim_{x \rightarrow \infty} f(x+1) - f(x).$$

Substitute f by $\log f$ and we obtain the second identity.

5.2 PSB: Uniform Continuity

Determine whether the following functions f are uniformly continuous on I :

5.2.1 B1) $f(x) = x^{1/3}$, $I = (0, \infty)$

For any $\varepsilon > 0$ and $x - y \in (0, \varepsilon)$,

$$x^{1/3} - y^{1/3} = \frac{x - y}{x^{2/3} + x^{1/3}y^{1/3} + y^{2/3}} \leq \frac{\varepsilon}{\varepsilon^{2/3}} = \varepsilon^{1/3}.$$

Hence $f(x)$ is uniformly continuous on I .

5.2.2 B2) $f(x) = \log x$, $I = (0, 1)$

For any $\varepsilon > 0$ and $x - y \in (0, \varepsilon)$,

$$\log x - \log y = \log \left(1 + \frac{x - y}{y} \right).$$

When $y \rightarrow 0$ and $x - y$ is constant, $\log x - \log y \rightarrow \infty$, hence $\log x$ is not uniformly continuous on I .

5.2.3 B3) $f(x) = \cos x^{-1}$, $I = (0, 1)$

Note that for $x_n = 1/(2n\pi)$ and $y_n = 1/(2n\pi + \pi)$, $f(x_n) = 1$ and $f(y_n) = -1$. Hence for $\varepsilon = 1$ and any $\delta > 0$, there exists n such that $|x_n - y_n| < \delta$ but $|f(x_n) - f(y_n)| = 2 > \varepsilon$. Therefore f is not uniformly continuous on I .

5.2.4 B4) $f(x) = x \cos x^{-1}$, $I = (0, \infty)$

For $x > y > 1$ and $|x - y| < \varepsilon$,

$$|x \cos x^{-1} - y \cos y^{-1}| \leq |x - y| |\cos x^{-1}| + |y| \cdot |\cos x^{-1} - \cos y^{-1}| \quad (5.3)$$

$$\leq \varepsilon + 2|y| \cdot |\sin(x^{-1} + y^{-1})/2 \sin(x^{-1} - y^{-1})/2| \leq \varepsilon + \frac{y}{2} \left(\frac{1}{y^2} - \frac{1}{x^2} \right) \leq 2\varepsilon. \quad (5.4)$$

For $1 > x > y$ and $|x - y| < \varepsilon$,

$$|x \cos x^{-1} - y \cos y^{-1}| \leq |x| + |y| < 2\varepsilon.$$

Hence f is uniformly continuous on I .

5.3 PSC: Existence of Limits

5.3.1 C1) $\alpha > 0$,

$$\lim_{x \rightarrow 1} \frac{\log x}{(x - 1)^\alpha} = \lim_{t \rightarrow 0} \frac{\log(1 + t)}{t^\alpha} = \lim_{t \rightarrow 0} t^{1-\alpha}$$

exists iff $\alpha \leq 1$.

5.3.2 C2) $\alpha > 0$,

$$\lim_{x \rightarrow 1} \frac{e^x - e}{(x - 1)^\alpha} = e \lim_{t \rightarrow 0} \frac{e^t - 1}{t^\alpha} = \lim_{t \rightarrow 0} e t^{1-\alpha}.$$

exists iff $\alpha \leq 1$.

5.3.3 C3) $\alpha > 0$,

$$\lim_{x \rightarrow 1} \frac{x^x - 1}{(x - 1)^\alpha} = \lim_{x \rightarrow 1} \frac{x^x (\log x + 1)}{\alpha (x - 1)^{\alpha-1}}$$

exists iff $\alpha \leq 1$.

5.3.4 C4) $\alpha > 0$,

$$\lim_{x \rightarrow 1} \frac{\sqrt[3]{1 - \sqrt{x}}}{(x - 1)^\alpha}$$

exists iff $\alpha \leq 1/3$.

5.3.5 C5)

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + x^2} - 1}{1 - \cos x} = 1.$$

5.3.6 C6)

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + x^4} - 1}{1 - \cos^2 x} = 0.$$

5.3.7 C7) $\alpha > 0$,

$$\lim_{x \rightarrow 1} \frac{(x - 1)^\alpha}{\sin(\pi x)}$$

exists iff $\alpha \geq 1$.

5.4 PSD: Problems on Uniform Continuity**5.4.1 D1) If f is continuous, monotonic and bounded on the open interval I , then f is uniformly continuous on I .**

Proof: Otherwise if there exists $\varepsilon > 0$ such that for any $\delta > 0$ there exists $|x - y| < \delta$ such that $|f(x) - f(y)| > \varepsilon$. We define x_n, y_n inductively as follows: Let $L = \min\{x_1, \dots, x_{n-1}\}$, $R = \max\{y_1, \dots, y_{n-1}\}$. Since f is uniformly continuous on $[L, R]$, there exists $\delta > 0$ such that for any $|s - t| < \delta$, $|f(s) - f(t)| < \varepsilon$. Hence there exists $x < y$ such that $x, y \notin [L, R]$, $|x - y| < \delta$ and $|f(x) - f(y)| > \varepsilon$. Let $x_n = x, y_n = y$, then (x_n, y_n) are disjoint intervals and $|f(x_n) - f(y_n)| > \varepsilon$. Which contradicts the fact that f is monotonic and bounded. Therefore f is uniformly continuous on I .

5.4.2 D2) I is an interval with finite length. Prove that the function f on I is uniformly continuous iff for any Cauchy sequence $\{x_n\}_{n \geq 1} \subset I$, $\{f(x_n)\}_{n \geq 1}$ is also a Cauchy sequence.

(f should be continuous, otherwise after changing the value of f at one point, $\{f(x_n)\}$ remains a Cauchy sequence.)

Proof: $\Rightarrow$ If $\{x_n\}_{n \geq 1}$ is a Cauchy sequence, then for any $\varepsilon > 0$ there exists $\delta > 0$ such that for all $|x - y| < \delta$, $|f(x) - f(y)| < \varepsilon$. There exists N such that for all $n, m > N$, $|x_n - x_m| < \delta$, hence $|f(x_n) - f(x_m)| < \varepsilon$, so $\{f(x_n)\}_{n \geq 1}$ is a Cauchy sequence. $\Leftarrow$ If $I = (a, b)$ is open we can take $x_n \rightarrow a$ and define $f(a) = \lim_{n \rightarrow \infty} f(x_n)$, hence we can assume that I is closed. Therefore f is uniformly continuous.

5.4.3 D3) f is uniformly continuous on $\mathbb{R}$. Prove that there exists $a, b \in \mathbb{R}_{>0}$ such that for any $x \in \mathbb{R}$,

$$|f(x)| \leq a|x| + b.$$

Proof: For $\varepsilon = 1$, there exists $\delta > 0$ such that $|x - y| < \delta \implies |f(x) - f(y)| < 1$. Hence let $C = \sup_{x \in [0, \delta]} |f(x)|$, then $|f(x)| \leq C + |x| \cdot (\frac{1}{\delta} + 1)$.

5.4.4 D4) Suppose f is uniformly continuous on $[0, \infty)$ and for any $x \in [0, 1]$, $\lim_{n \rightarrow \infty} f(x + n) = 0$. Prove that

$$\lim_{x \rightarrow \infty} f(x) = 0.$$

If we change the condition to f is continuous, will the statement still hold? Proof: For any $\varepsilon > 0$, there exists $\delta > 0$ such that $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$. Let $N = \lfloor 1/\delta \rfloor + 1$, then for any $1 \leq n \leq N$, there exists M_n such that for all $m > M_n$, $|f(m + n/N)| < \varepsilon$. Let $M = \max\{M_1, \dots, M_N\}$, then for all $x > M$, there exists $m \in \mathbb{Z}_{>M}$ and $1 \leq n \leq N$ such that $|x - m - n/N| < \delta$. Hence

$$|f(x)| \leq \varepsilon + |f(m + n/N)| < 2\varepsilon.$$

Therefore $\lim_{x \rightarrow \infty} f(x) = 0$.

5.4.5 D5) Suppose X is an interval, $f : X \rightarrow \mathbb{R}$ is continuous. If there is a constant $L > 0$ such that for any $x, y \in X$,

$$|f(x) - f(y)| \leq L|x - y|.$$

We say f satisfy the Lipschitz condition on X .

1. Prove that f satisfy the Lipschitz condition implies f is uniformly continuous.

Proof: For any $\varepsilon > 0$, let $\delta = \varepsilon/L$, then for any $|x - y| < \delta$, $|f(x) - f(y)| \leq L|x - y| < \varepsilon$.

1. Determine whether the reversed statement holds.

Consider the function $f(x) = x^{1/2}$, then f is uniformly continuous but $\frac{f(x) - f(y)}{x - y} = \frac{1}{\sqrt{x} + \sqrt{y}}$ is unbounded, hence does not satisfy the Lipschitz condition.

1. If f satisfy the Lipschitz condition on $[a, \infty)$, where $a > 0$, prove that $f(x)/x$ is uniformly continuous on $[a, \infty)$.

Proof: Same as D3), there exists C such that $|f(x)| \leq C|x|$ for $x \in [a, \infty)$, then for $a < x < y$,

$$\left| \frac{f(x)}{x} - \frac{f(y)}{y} \right| = \frac{|xf(y) - yf(x)|}{xy} \leq \frac{x|f(y) - f(x)| + |f(x)|(y - x)}{xy} \quad (5.5)$$

$$\leq \frac{L + C}{y} \cdot |x - y|. \quad (5.6)$$

Hence $f(x)/x$ satisfy the Lipschitz condition.

5.5 PSE:

Exactly the same as PSC in HW4?

5.6 PSF: Calculate Limits

5.6.1 F1)

$$\lim_{x \rightarrow \pi} \frac{\sin mx}{\sin nx} = \frac{m(-1)^m}{n(-1)^n}.$$

5.6.2 F2)

$$\lim_{x \rightarrow 0} \frac{1 - \cos x \sqrt{\cos 2x} \sqrt[3]{\cos 3x}}{x^2} = 3.$$

5.6.3 F3)

$$\lim_{x \rightarrow \infty} \sin \sqrt{1+x} - \sin \sqrt{x} = 0.$$

Since the function $\sin x$ is uniformly continuous and $\lim_{x \rightarrow \infty} \sqrt{1+x} - \sqrt{x} = 0$.

5.6.4 F4)

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x \sin x} - 1}{e^{x^2} - 1} = \frac{1}{2}.$$

Since $\lim_{x \rightarrow 0} x^2/(e^{x^2} - 1) = 1$, $\lim_{x \rightarrow 0} x \sin x/x^2 = 1$ and $\lim_{x \rightarrow 0} 1/(1 + \sqrt{1+x \sin x}) = 1/2$.

5.6.5 F5)

$$\lim_{n \rightarrow \infty} \sin^{(n)}(x) = 0.$$

Since the sequence $\{a_n = \sin^{(n)}(x)\}_{n \geq 1}$ is decreasing and bounded by 0, and its limit A satisfy $A = \sin A$. Therefore $\lim_{n \rightarrow \infty} \sin^{(n)}(x) = 0$.

5.7 Problem G

The continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the following property: for any $\delta > 0$,

$$\lim_{n \rightarrow \infty} f(n\delta) = 0.$$

Prove that $\lim_{x \rightarrow \infty} f(x) = 0$. Proof: Consider any $\varepsilon > 0$. For any $N \in \mathbb{N}$,

$$A_N = \{\delta > 0 : \forall n \geq N, |f(n\delta)| < \varepsilon\}.$$

Then by the continuity of f , A_N is closed, and by $\lim_{n \rightarrow \infty} f(n\delta) = 0$ for any $\delta > 0$, $\bigcup_{N \geq 1} A_N = \mathbb{R}_{>0}$. Hence by Baire Category Theorem, there exists an $N > 0$ such that $(a, b) \subset A_N$ for some interval (a, b) . Let $X = \{x \in \mathbb{R}_{>0} : |f(x)| < \varepsilon\}$, then since $(a, b) \subset A_N$, for any $n \geq N$, $(na, nb) \subset X$. Note that when $n > b/(b-a)$, $nb > (n+1)a$, hence there exists $M > 0$ such that $(M, \infty) \subset X$. Therefore $\lim_{x \rightarrow \infty} f(x) = 0$.

5.8 Problem H

The continuous function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the following properties: 1) $\lim_{x \rightarrow \infty} (\varphi(x) - x) = \infty$. 2) $\{x \in \mathbb{R} : \varphi(x) = x\}$ is a non-empty finite set.

Prove that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $f \circ \varphi = f$, then f is constant. (Probably need the condition $\lim_{x \rightarrow -\infty} \varphi(x) - x = -\infty$). Proof: Suppose $\{x \in \mathbb{R} : \varphi(x) = x\} = \{a_1, \dots, a_n\}$ where $a_1 < \dots < a_n$. For any $x \in \mathbb{R}$, we will show that $f(x) \in \{f(a_1), \dots, f(a_n)\}$ hence f is constant. If $a_i < x < a_{i+1}$. Suppose $\varphi(x) > x$, then let $x_0 = x$, and inductively define x_k as a point in (a_k, x_{k-1}) such that $\varphi(a_i) = a_i < \varphi(x_k) = x_{k-1} < \varphi(x_{k-1})$. Since φ is continuous and $a_1, \dots, a_n$ are all the roots of $\varphi(x) = x$, we know that $\varphi(x_k) > x_k$ for all $k \geq 0$. The sequence $\{x_k\}_{k \geq 0}$ is decreasing and bounded by a_i , hence has a limit A . From $\varphi(x_k) = x_{k-1}$ we know that $\varphi(A) = A$, so $A = a_i$. Note that $f(x_k) = f(\varphi(x_k)) = f(x_{k-1})$, hence $f(x) = f(x_k) = \lim_{k \rightarrow \infty} f(x_k) = f(a_i)$. The case $\varphi(x) < x$ is the same, by constructing a sequence which tends to a_{i+1} . If $x > a_n$, then $\varphi(x) > x$, likewise we can construct a sequence x_k such that $x_{k-1} = \varphi(x_k)$ and $\lim_{k \rightarrow \infty} x_k = a_n$. The case $x < a_1$ is the same. Hence for all $x \in \mathbb{R}$, $f(x) \in \{f(a_1), \dots, f(a_n)\}$.

5.9 Problem I

The continuous function $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ satisfy $\lim_{x \rightarrow \infty} f(x)/x = 0$. Suppose $\{a_n\}_{n \geq 1}$ is a sequence of non-negative real numbers and the sequence $\{a_n/n\}_{n \geq 1}$ is bounded. Prove that $\lim_{n \rightarrow \infty} f(a_n)/n = 0$. Proof: Suppose $\{a_n/n\}$ is bounded by M . For any $\varepsilon > 0$, we need to find N such that $n \geq N \implies |f(a_n)| < \varepsilon n$. For $C > 0$, we can divide n into two parts: If $a_n \leq C$, then $|f(a_n)| \leq \sup_{x \in [0, C]} |f(x)|$, otherwise $a_n \geq C$, then $|f(a_n)| \leq \sup_{x \geq C} |f(x)/x| \cdot Mn$. Therefore, if we choose $C > 0$ such that $\sup_{x \geq C} |f(x)/x| < \varepsilon/M$, and choose N such that $N > \sup_{x \in [0, C]} |f(x)|/\varepsilon$, then for any $n \geq N$, $|f(a_n)| < \varepsilon n$, hence

$$\lim_{n \rightarrow \infty} \frac{f(a_n)}{n} = 0.$$

5.10 Ex: Proof of the infinity of primes using topology

Proof: Assume otherwise that the set $\mathcal{P}$ of primes is finite. Let $L_{a,b} = \{at + b : t \in \mathbb{Z}\}, \forall (a, b) \in I = \mathbb{Z}_{>0} \times \mathbb{Z}$. Then

$$\mathbb{Z} \subset \bigcup_{b \in \mathbb{Z}} L_{1,b} \subset \bigcup_{(a,b) \in I} L_{a,b} \subset \mathbb{Z} \implies \bigcup_{(a,b) \in I} L_{a,b} = \mathbb{Z}.$$

and for any $x \in \bigcap_{i=1}^n L_{a_i, b_i}$, let $a = \text{lcm}(a_1, \dots, a_n)$, then

$$x \in L_{a,x} \subset \bigcap_{i=1}^n L_{a_i, b_i}.$$

Hence $L_{a,b}$ form a base. Consider the topology $\mathcal{T}$ on $\mathbb{Z}$ generated by the base $\{L_{a,b} : (a,b) \in I\}$. Note that

$$L_{a,b} = \mathbb{Z} \setminus \bigcup_{r=1}^{a-1} L_{a, b+r}$$

so $L_{a,b}$ is also closed. Since $\mathcal{P}$ is finite, the set

$$\bigcup_{p \in \mathcal{P}} L_{p,0} = \mathbb{Z} \setminus \{-1, 1\}$$

is closed, hence $\{-1, 1\}$ is open. However, an open set G is the union of $L_{a,b}$ which is infinite, so G is infinite, leading to contradiction.

Quote: As for everything else, so for a mathematical theory: beauty can be perceived but not explained.

——A. Cayley

Chapter 6 Takagi Function

6.1 PSA: Calculating Derivatives

6.1.1 A1) Consider the function

$$f : \mathbb{R} \rightarrow \mathbb{R}^n, x \mapsto f(x) = (f_1(x), \dots, f_n(x)).$$

Prove that f is differentiable at x_0 iff every f_k is differentiable at x_0 and

$$f'(x) = (f'_1(x), \dots, f'_n(x)).$$

Proof: For any $h \in \mathbb{R}$,

$$\left\| \frac{f(x+h) - f(x)}{h} - (f'_1(x), \dots, f'_n(x)) \right\|_2 \leq n \max_{1 \leq k \leq n} \left\| \frac{f_k(x+h) - f_k(x)}{h} - f'_k(x) \right\| \rightarrow 0.$$

Therefore $f'(x) = (f'_1(x), \dots, f'_n(x))$.

6.1.2 A2) Consider the function

$$f : \mathbb{R} \rightarrow \mathbb{C}, x \mapsto e^{ix}.$$

Prove by definition, $f'(0) = i$ and $(e^{ix})' = ie^{ix}$. Proof: For any $h \in \mathbb{R}$,

$$\left| \frac{f(h) - f(0)}{h} - i \right| = \left| \frac{e^{ih} - ih - 1}{h} \right| \leq \sum_{n=2}^{\infty} \left| \frac{1}{h} \frac{(ih)^n}{n!} \right| \leq |h| \rightarrow 0.$$

Therefore $f'(0) = i$. For any $x \in \mathbb{R}$,

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} e^{ix} \frac{f(h) - f(0)}{h} = ie^{ix}.$$

Hence $(e^{ix})' = ie^{ix}$.

6.1.3 A3) Calculate the derivatives of $\sin x$ and $\cos x$.

Solution: $\sin x = (e^{ix} - e^{-ix})/2i$, so $(\sin x)' = (e^{ix} + e^{-ix})/2 = \cos x$. Likewise $(\cos x)' = -\sin x$.

6.1.4 A4) Prove Faà di Bruno's formula for $n = 3$.

Proof:

$$\frac{d}{dx}(f \circ g) = f'(g) \cdot g'. \quad (6.1)$$

$$\frac{d^2}{dx^2}(f \circ g) = f'(g) \cdot g'' + f''(g) \cdot (g')^2. \quad (6.2)$$

$$\frac{d^3}{dx^3}(f \circ g) = f'(g) \cdot g''' + f''(g) \cdot g'' \cdot g' + f'''(g) \cdot (g')^3 + f''(g) \cdot 2g'g''. \quad (6.3)$$

6.1.5 A5) Define the map

$$E : \mathbb{R} \rightarrow \mathbb{C} = \mathbb{R}^2, \theta \mapsto (\cos \theta, \sin \theta).$$

Prove that the points in $\mathbf{S}^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ can be written in the form $(\sin \theta, \cos \theta)$, i.e. $E(\mathbb{R}) = \mathbf{S}^1$. Calculate $E'(\theta)$ and show that Rolle's mean-value theorem is invalid for E . Proof: Obviously $E(\mathbb{R}) \subset \mathbf{S}^1$. Consider any $(x, y) \in \mathbf{S}^1$, then $x \in [-1, 1]$. Note that $\cos 0 = 1, \cos \pi = -1$, hence there exists $\theta \in [0, \pi]$ such that $\cos \theta = x$,

and $|\sin \theta| = |y|$. If $\sin \theta = y$ then $(x, y) = (\cos \theta, \sin \theta) \in E(\mathbb{R})$. Otherwise $(x, y) = (\cos(-\theta), \sin(-\theta)) \in E(\mathbb{R})$, therefore $E(\mathbb{R}) = \mathbb{S}^1$. By A1) and A3), $E'(\theta) = (-\sin \theta, \cos \theta)$. Since $E'(\theta) \neq 0$ for all $\theta \in \mathbb{R}$ and $E'(\theta) = E'(\theta + 2\pi)$, Rolle's mean-value theorem is invalid.

6.1.6 A6) Calculate the derivatives of the following functions:

(1) $f(x) = a^x, a > 0$.

$$f'(x) = (e^{x \log a})' = a^x \log a.$$

(2) $f(x) = \arcsin x$. Let $y = \arcsin x$, then $x = \sin y$, so $1 = \cos y \cdot y'$, hence

$$f'(x) = \frac{1}{\cos(\arcsin x)} = \frac{1}{\sqrt{1-x^2}}.$$

(3) $f(x) = \arctan x$. Let $y = \arctan x$, then $x = \tan y$, so $1 = \sec^2 y \cdot y'$, hence

$$f'(x) = \cos^2 y = \frac{1}{1+x^2}.$$

(4) $f(x) = x^{x^x}, x > 0$. Let $y = x^x, z = x^y$, then $\log y = x \log x$, so $y'/y = \log x + 1, y' = x^x(1 + \log x)$. $\log z = y \log x$, so $z'/z = y' \log x + y/x = x^x \log x(1 + \log x) + x^{x-1}$. Therefore

$$f'(x) = x^{x^x} \cdot x^x \cdot (\log x + \log^2 x + x^{-1}).$$

(5) $f(x) = \log(\log(\log x))$.

$$f'(x) = \frac{(\log \log x)'}{\log \log x} = \frac{1}{x \log x \log \log x}.$$

(6) $f(x) = \sqrt{x + \sqrt{x + \sqrt{x}}}$.

$$f'(x) = \frac{(x + \sqrt{x + \sqrt{x}})'}{2\sqrt{x + \sqrt{x + \sqrt{x}}}} = \left(1 + \frac{1 + \frac{1}{2\sqrt{x}}}{2\sqrt{x + \sqrt{x}}}\right) / 2\sqrt{x + \sqrt{x + \sqrt{x}}} \quad (6.4)$$

$$= \frac{2\sqrt{x + \sqrt{x}} + 1 + 1/2\sqrt{x}}{4\sqrt{x + \sqrt{x}}\sqrt{x + \sqrt{x + \sqrt{x}}}}. \quad (6.5)$$

(7) $f(x) = |x|$. If $x > 0, f'(x) = (x)' = 1$. If $x < 0, f'(x) = (-x)' = -1$. If $x = 0, f$ is not differentiable at x . (8) $f(x) = \log|x|$. If $x > 0, f'(x) = \frac{1}{x}$. If $x < 0, f'(x) = -\frac{1}{x}$. If $x = 0, f$ is not differentiable at x . (9)

$$f(x) = \begin{cases} x^n \sin \frac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases} \quad n = 1, 2, \dots$$

For $x \neq 0, f'(x) = nx^{n-1} \sin \frac{1}{x} - x^{n-2} \cos \frac{1}{x}$. When $x = 0$,

$$f'(0) = \lim_{h \rightarrow 0} h^{n-1} \sin \frac{1}{h} = \begin{cases} 0, & n \geq 2; \\ \text{diverges}, & n = 1. \end{cases}$$

6.1.7 A7) Calculate $f^{(3)}(x)$:

(1) $f(x) = \log(x+1)$.

$$\frac{d^3}{dx^3} \log(x+1) = \frac{2}{(x+1)^3}.$$

(2) $f(x) = x^{-1} \log x$.

$$\frac{d^3}{dx^3} \frac{\log x}{x} = \frac{11 - 6 \log x}{x^4}.$$

(3) $f(x) = \frac{x^m}{1-x}, m \in \mathbb{Z}_{\geq 0}$.

$$\frac{d^3}{dx^3} \frac{x^m}{1-x} = \frac{(m-2)(m-1)mx^{m-3}}{1-x} + \frac{3(m-1)mx^{m-2}}{(1-x)^2} + \frac{6mx^{m-1}}{(1-x)^3} + \frac{6x^m}{(1-x)^4}.$$

(4) $f(x) = x^m e^x, m \in \mathbb{Z}_{\geq 0}$.

$$\frac{d^3}{dx^3}(x^m e^x) = e^x x^{m-3}(m^3 + 3m^2(x-1) + m(3x^2 - 3x + 2) + x^3).$$

(5) $f(x) = e^{ax} \sin(bx), a, b \in \mathbb{R}$.

$$\frac{d^3}{dx^3}(e^{ax} \sin(bx)) = e^{ax}((3a^2b - b^3) \cos(bx) + a(a^2 - 3b^2) \sin(bx)).$$

(6) $f(x) = e^{-x^2}$.

$$\frac{d^3}{dx^3}e^{-x^2} = -4e^{-x^2}x(2x^2 - 3).$$

6.1.8 A8) $f'(x_0) > 0$ does not imply f is increasing in a neighborhood of x_0 : consider

$$f(x) = \begin{cases} x + 2x^2 \sin \frac{1}{x}, & x \neq 0; \\ 0, & x = 0. \end{cases}$$

Prove that $f'(0) > 0$ but for any $\varepsilon > 0$, f is not monotonic on $(-\varepsilon, \varepsilon)$. Proof:

$$f'(0) = \lim_{h \rightarrow 0} \frac{1 + 2h \sin \frac{1}{h}}{h} = 1 > 0.$$

However, for any $n \in \mathbb{N}$, let $x_n = \frac{1}{(2n+1/2)\pi}$, $y_n = \frac{1}{(2n-1/2)\pi}$, then

$$f(x_n) = x_n + 2x_n^2, f(y_n) = y_n - 2y_n^2.$$

Note that $0 < x_n < y_n$, but

$$f(x_n) - f(y_n) = 2x_n^2 + 2y_n^2 - \pi x_n y_n > 0,$$

i.e. $f(x_n) > f(0), f(y_n)$, therefore f is not monotonic on any $(-\varepsilon, \varepsilon)$.

6.1.9 A9) $A \in \mathbf{M}_n(\mathbb{R})$, calculate

$$\left. \frac{d}{dx} \right|_{x=0} \det(\mathbf{I}_n + xA).$$

Solution: Let $\Phi(x) = I_n + xA$, then $\Phi(0) = I_n$. Denote $\Phi(t) = (\varphi_1(t), \dots, \varphi_n(t))$. Note that $\det$ is a multi-linear function for n rows, hence by Euler's formula:

$$\frac{d}{dt} \det \Phi(t) = \det(\varphi'_1(t), \varphi_2(t), \dots, \varphi_n(t)) + \dots + \det(\varphi_1(t), \varphi_2(t), \dots, \varphi'_n(t)).$$

When $t = 0$, the formula becomes

$$\left. \frac{d}{dt} \right|_{t=0} \det \Phi(t) = \varphi'_{1,1} + \dots + \varphi'_{n,n} = \text{tr } \Phi'(0) = \text{tr } A.$$

6.1.10 A10) Prove that the derivation of odd functions are even, and that of even functions are odd.

Proof: If f is an odd function then

$$f'(-x) = \lim_{h \rightarrow 0} \frac{f(-x+h) - f(-x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) - f(x-h)}{h} = f'(x),$$

so f' is even. If f is an even function then

$$f'(-x) = \lim_{h \rightarrow 0} \frac{f(-x+h) - f(-x)}{h} = - \lim_{h \rightarrow 0} \frac{f(x) - f(x-h)}{h} = -f'(x),$$

so f' is odd.

6.1.11 A11) Prove that

$$f(x) = \begin{cases} 1/q, & x = \frac{p}{q} \in \mathbb{Q}, q \geq 1, \gcd(p, q) = 1; \\ 0, & x \in \mathbb{Q}^C. \end{cases}$$

is nowhere differentiable on $\mathbb{R}$. Proof: For any $x \in \mathbb{Q}$, $f(x) \neq 0$, but for any $\varepsilon > 0$, there exists $y \in (x - \varepsilon, x + \varepsilon) \cap \mathbb{Q}^C$, such that $f(y) = 0$. Therefore f is not continuous at x , and clearly not differentiable. Consider any $x \in \mathbb{Q}^C$, there is a sequence of irrational numbers $\{y_n\}_{n \geq 1}$ that converges to x , then

$$\lim_{n \rightarrow \infty} \frac{f(x) - f(y_n)}{x - y_n} = 0.$$

Choose any sequence of rational numbers $\{r_n = p_n/q_n\}_{n \geq 1}$ that converges to x , then

$$\lim_{n \rightarrow \infty} \frac{f(x) - f(r_n)}{x - r_n} = \lim_{n \rightarrow \infty} \frac{1}{xq_n - p_n} = \infty.$$

Therefore f is nowhere differentiable on $\mathbb{R}$.

6.2 PSB

6.2.1 B1) Define the hyperbolic functions:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \cosh x = \frac{e^x + e^{-x}}{2}, \tanh x = \frac{\sinh x}{\cosh x}.$$

1. Prove that

$$\begin{aligned} (1) \cosh^2 x - \sinh^2 x &= 1 \text{ Proof: } \cosh^2 x - \sinh^2 x = \frac{e^{2x} + e^{-2x} + 2}{4} - \frac{e^{2x} + e^{-2x} - 2}{4} = 1. \\ (2) \sinh(x + y) &= \sinh x \cosh y + \cosh x \sinh y. \text{ Proof: } \sinh x \cosh y + \cosh x \sinh y = \frac{e^{x+y} - e^{y-x} + e^{x-y} - e^{-x-y}}{4} + \frac{e^{x+y} - e^{x-y} + e^{y-x} - e^{-x-y}}{4} = \frac{e^{x+y} - e^{-x-y}}{2} = \sinh(x+y) \\ (3) \cosh(x + y) &= \cosh x \cosh y + \sinh x \sinh y. \text{ Proof: Same as (2).} \\ (4) \tanh(x + y) &= \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}. \end{aligned}$$

$$\text{Proof: } \tanh(x + y) = \frac{\sinh x \cosh y + \cosh x \sinh y}{\cosh x \cosh y + \sinh x \sinh y} = \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}.$$

1. Calculate $\sinh'(x)$, $\cosh'(x)$ and $\tanh'(x)$.

$$\text{Solution: } \sinh'(x) = \cosh x, \cosh'(x) = \sinh x, \tanh'(x) = \frac{1}{\cosh^2 x}.$$

1. Prove that $\sinh : \mathbb{R} \rightarrow \mathbb{R}$ has an inverse $\operatorname{arcsinh} : \mathbb{R} \rightarrow \mathbb{R}$ and calculate $\operatorname{arcsinh}'(x)$.

Proof: $\sinh'(x) = \cosh x > 0$, so $\sinh$ is monotonically increasing over $\mathbb{R}$. Also $\lim_{x \rightarrow \infty} \sinh x = \infty$, $\lim_{x \rightarrow -\infty} \sinh x = -\infty$, therefore $\sinh : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection and hence has an inverse. $\operatorname{arcsinh}'(x) = \frac{1}{\sqrt{1+x^2}}$.

6.2.2 B2) $a, b \in \mathbb{R}, a > 0$. Consider $f : [-1, 1] \rightarrow \mathbb{R}$, where

$$f(x) = \begin{cases} x^a \sin(x^{-b}), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Prove that

1. $f \in C([-1, 1])$ iff $a > 0$;

Proof: $f \in C([-1, 1])$ iff $\lim_{x \rightarrow 0} x^a \sin(x^{-b}) = 0$. If $a > 0$ then $|x^a \sin(x^{-b})| \leq |x|^a \rightarrow 0$. If $a < 0$ then let $x = ((2n + 1/2)\pi)^{-1/b}$, when $n \rightarrow \infty$, $x \rightarrow 0$ but $|x^a \sin(x^{-b})| \rightarrow \infty$. If $a = 0$, then let $x = ((2n + 1/2)\pi)^{-1/b}$, $|x^a \sin(x^{-b})| = 1$. Therefore $f \in C([-1, 1])$ iff $a > 0$.

1. f is differentiable at 0 iff $a > 1$;

Proof: f is differentiable at 0 iff $\lim_{x \rightarrow 0} x^{-a} \sin(x^{-b})$ exists. By 1 we know that $a > 1$. ($a = 1$ is invalid since $x = (2n\pi)^{-1/b}$ and $x = ((2n + 1/2)\pi)^{-1/b}$ converge to different values.)

1. f' is bounded on $[-1, 1]$ iff $a \geq 1 + b$;

Proof: $f'(x) = ax^{a-1} \sin(x^{-b}) + x^a \cos(x^{-b})(-b)x^{-b-1}$ is bounded iff x^{a-1} and x^{a-b-1} are bounded, i.e. $a \geq 1 + b$.

1. $f \in C^1([-1, 1])$ iff $a > 1 + b$;

Proof: $f \in C^1([-1, 1])$ iff $f'(0) = 0 = \lim_{x \rightarrow 0} f'(x)$. By 1 we know it is equivalent to $a > 1 + b$.

1. f' is differentiable at 0 iff $a > 2 + b$;
2. f'' is bounded on $[-1, 1]$ iff $a \geq 2 + 2b$;
3. $f \in C^2([-1, 1])$ iff $a > 2 + 2b$.

Proof: 5,6,7 are exactly the same as 2,3,4.

6.3 PSC

If f satisfy $\lim_{x \rightarrow x_0} f(x) = 0$ near x_0 , we call f an infinitesimal when $x \rightarrow x_0$. Likewise when $\lim_{x \rightarrow x_0} f(x) = +\infty$ or $\lim_{x \rightarrow x_0} f(x) = -\infty$, we call f an infinite quantity when $x \rightarrow x_0$. Suppose f, g are both infinitesimal when $x \rightarrow x_0$, and $g(x)$ does not vanish near x_0 . We introduce the notations

- if $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0$, we say f is an infinitesimal of higher order than g , and denote $f(x) = o(g(x)), x \rightarrow x_0$;
- If $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \ell \neq 0$, we say f and g are of the same order;
- If $\ell = 1$, denote $f \sim g, x \rightarrow x_0$;
- If $\limsup_{x \rightarrow x_0} \left| \frac{f(x)}{g(x)} \right| < +\infty$, denote $f(x) = O(g(x)), x \rightarrow x_0$.

6.3.1 C1 Suppose $a(x) = o(1)$ when $x \rightarrow x_0$, prove that:

(1) $o(a) + o(a) = o(a)$ Proof: If $f, g = o(a)$, then

$$\lim_{x \rightarrow x_0} \frac{f(x) + g(x)}{a(x)} = \lim_{x \rightarrow x_0} \frac{f(x)}{a(x)} + \lim_{x \rightarrow x_0} \frac{g(x)}{a(x)} = 0,$$

hence $f + g = o(a)$. (2) $co(a) = o(ca), c \in \mathbb{R}$ Proof: If $f = o(a)$, then

$$\lim_{x \rightarrow x_0} \frac{cf(x)}{a(x)} = c \lim_{x \rightarrow x_0} \frac{f(x)}{a(x)} = 0,$$

hence $cf = o(a) = o(ca)$. (3) $o(a)^k = o(a^k)$ Proof: If $f = o(a)$ then

$$\lim_{x \rightarrow x_0} \frac{f(x)^k}{a(x)^k} = \left(\lim_{x \rightarrow x_0} \frac{f(x)}{a(x)} \right)^k = 0,$$

hence $f^k = o(a^k)$. (4) $1/(1+a) = 1-a+o(a)$ Proof:

$$\lim_{x \rightarrow x_0} \frac{1/(1+a) - 1 + a}{a(x)} = \lim_{x \rightarrow x_0} \frac{a(x)}{1+a(x)} = 0,$$

hence $1/(1+a) = 1-a+o(a)$.

6.3.2 C2 Suppose f, g are infinitesimals when $x \rightarrow x_0$, then

1. Prove that $f \sim g \iff f(x) - g(x) = o(g(x)), x \rightarrow x_0$.

Proof: $f \sim g \iff \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 1 \iff \lim_{x \rightarrow x_0} \frac{f(x) - g(x)}{g(x)} = 0$, i.e. $f(x) - g(x) = o(g(x))$.

1. If $f \sim cg^k$, we call cg^k the leading term of f . Find the leading terms of the following functions, compared to $x - x_0$ or x :

(1) $1/\sin \pi x, x \rightarrow 1. \frac{1}{\sin \pi x} = -\frac{1}{\pi(x-1)} + o(1).$ (2) $\sqrt{1+x} - \sqrt{1-x}, x \rightarrow 0. \sqrt{1+x} - \sqrt{1-x} = x + o(x).$ (3) $\sin(\sqrt{1+\sqrt{1+\sqrt{x}}} - \sqrt{2}), x \rightarrow 0^+. = \frac{\sqrt{2}x^{1/2}}{8} + o(x^{1/2}).$ (4) $\sqrt{1+\tan x} - \sqrt{1-\sin x}, x \rightarrow 0. = x + o(x).$ (5) $\sqrt{x+\sqrt{x+\sqrt{x}}}, x \rightarrow 0^+. = x^{1/8} + o(x^{1/8}).$ (6) $\sqrt{x+\sqrt{x+\sqrt{x}}}, x \rightarrow \infty. = \sqrt{x} + o(\sqrt{x}).$

1. Suppose $f \sim cx^k, x \rightarrow 0$, i.e. $f(x) = cx^k + o(x^k)$. If $f(x) - c^k$ has a leading term $c'x^{k'}$, we denote $f(x) = cx^k + c'x^{k'} + o(x^{k'})$. Expand the following terms to $o(x^2)$:

(1) $\sqrt{1+x} - 1. \sqrt{1+x} - 1 = \frac{1}{2}x - \frac{1}{8}x^2 + o(x^2).$ (2) $(1+x)^{1/m} - 1, m \in \mathbb{Z}_{\geq 1}. (1+x)^{1/m} - 1 = \frac{1}{m}x - \frac{m-1}{2m^2}x^2 + o(x^2).$

6.4 PST: Takagi Function

Define $\psi : [0, 1] \rightarrow \mathbb{R}$ as

$$\psi(x) = \begin{cases} x, & 0 \leq x < \frac{1}{2}; \\ 1-x, & \frac{1}{2} \leq x \leq 1. \end{cases}$$

For $x \in \mathbb{R}$, let $\psi(x) = \psi(\{x\})$, then $\psi \in C(\mathbb{R})$. Define the Takagi function $T : \mathbb{R} \rightarrow \mathbb{R}$ as follows:

$$T(x) = \sum_{k=0}^{\infty} \frac{1}{2^k} \psi(2^k x),$$

and the partial sum $T_n(x) = \sum_{k=0}^n \frac{1}{2^k} \psi(2^k x)$.

6.4.1 T1) Prove that $T(x)$ is a well-defined bounded continuous function on $\mathbb{R}$.

Proof: Note that $\psi(x) \in [0, 1/2]$ so the series $\sum_{k=0}^{\infty} 2^{-k} \psi(2^k x)$ converges absolutely, and hence $T(x)$ is well-defined and bounded by $T(x) \in [0, 1]$.

6.4.2 T2) For $x \in [0, 1]$, suppose $x = \sum_{n=1}^{\infty} a_n 2^{-n}$ is the binary form of x . Let $v_n = \sum_{k=1}^n a_k$, and $\sigma_n(y) = a_n + (1 - 2a_n)y$, where $y \in \{0, 1\}$. Prove that

$$\psi(2^m x) = \sum_{n=1}^{\infty} \frac{\sigma_{m+1}(a_{m+n})}{2^n}.$$

Proof:

$$\psi(2^m x) = \psi\left(\sum_{n=1}^{\infty} a_n 2^{m-n}\right) = \psi\left(\sum_{n=m+1}^{\infty} a_n 2^{m-n}\right) = \begin{cases} \sum_{n=1}^{\infty} a_{m+n} 2^{-n}, & a_{m+1} = 0; \\ 1 - \sum_{n=1}^{\infty} a_{m+n} 2^{-n}, & a_{m+1} = 1. \end{cases}$$

Therefore

$$\psi(2^m x) = \sigma_n\left(\sum_{n=1}^{\infty} a_{m+n} 2^{-n}\right) = \sum_{n=1}^{\infty} \sigma_{m+1}(a_{m+n}) 2^{-n}.$$

6.4.3 T3) $x = \sum_{n=1}^{\infty} a_n 2^{-n} \in [0, 1]$, prove that

$$T(x) = \sum_{n=1}^{\infty} \frac{(1 - a_n)v_n + a_n(n - v_n)}{2^n}.$$

Proof: By T2),

$$T(x) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \sigma_{m+1}(a_{m+n}) 2^{-m-n} = \sum_{n=1}^{\infty} \sum_{m=0}^{n-1} \sigma_{m+1}(a_n) 2^{-n} = \sum_{n=1}^{\infty} \frac{(1-a_n)v_n + a_n(n-v_n)}{2^n}.$$

6.4.3.1 T4) Suppose $x_0 = k_0 2^{-m_0} \in [0, 1]$, where $k_0 \in \mathbb{Z}_{\geq 1}$ is odd, $m_0 \in \mathbb{Z}_{\geq 0}$. Let $h_n = 2^{-n}$, where $n \in \mathbb{Z}_{\geq m_0}$. Prove that the sequence $\left\{ \frac{T(x+h_n) - T(x)}{h_n} \right\}_{n \geq m_0}$ does not converge.

Proof: By T3),

$$\frac{T(x+h_n) - T(x)}{h_n} = \frac{1}{h_n} \left(\frac{n-v_n}{2^n} - \frac{v_n}{2^n} \right) = n - 2 \sum_{k=1}^n a_k = n - 2 - 2S_2(k_0) \rightarrow \infty.$$

6.4.3.2 T5) $f : I \rightarrow \mathbb{R}$, where I is an open interval. If f is differentiable at a , prove that

$$\lim_{(h,h') \rightarrow (0,0), h,h' > 0} \frac{f(a+h) - f(a-h')}{h+h'} = f'(a).$$

i.e. it converges for any sequence $(h_n, h'_n) \rightarrow (0, 0)$, $h_n, h'_n > 0$. Proof: Consider any sequence $(h_n, h'_n) \rightarrow (0, 0)$, then

$$\frac{f(a+h) - f(a-h')}{h+h'} = \frac{f(a+h) - f(a)}{h} \cdot \frac{h}{h+h'} + \frac{f(a) - f(a-h')}{h'} \cdot \frac{h'}{h+h'} \rightarrow f'(a).$$

6.4.3.3 T6) Same as T5), if $f \in C^1(I)$, $a \in I$, prove that

$$\lim_{(h,h') \rightarrow (0,0), h+h' \neq 0} \frac{f(a+h) - f(a-h')}{h+h'} = f'(a).$$

Proof: For any $h+h' \neq 0$, there exists $\xi \in [a, a+h]$ and $\eta \in [a-h', a]$ such that $f(a+h) = f(a) + hf'(\xi)$ and $f(a-h') = f(a) - h'f'(\eta)$, then

$$\left| \frac{f(a+h) - f(a-h')}{h+h'} - f'(a) \right| \leq \frac{h}{h+h'} |f'(\xi) - f'(a)| + \frac{h'}{h+h'} |f'(\eta) - f'(a)| \rightarrow 0.$$

Hence

$$\lim_{(h,h') \rightarrow (0,0), h+h' \neq 0} \frac{f(a+h) - f(a-h')}{h+h'} = f'(a).$$

6.4.3.4 T7) Suppose $x \in [0, 1]$, such that for any $n \in \mathbb{N}$, $2^n x \notin \mathbb{Z}$. For every $n \in \mathbb{N}$, define $\{h_n\}_{n \geq 1}$ and $\{h'_n\}_{n \geq 1}$ as follows:

$$\lfloor 2^n x \rfloor = 2^n(x - h'_n), \quad \lfloor 2^n x \rfloor + 1 = 2^n(x + h_n).$$

Prove that for an arbitrary n , $h_n + h'_n = 2^{-n}$ and for every integer $1 \leq \ell \leq n-1$, the interval $(2^\ell(x - h'_n), 2^\ell(x + h_n))$ does not include integers or half-integers. Proof: $1 = 2^n(x + h_n) - 2^n(x - h'_n) = 2^n(h_n + h'_n)$, hence $h_n + h'_n = 2^{-n}$. For any integer $1 \leq \ell \leq n-1$, $2^\ell(x - h'_n) = \lfloor 2^n x \rfloor \cdot 2^{\ell-n}$ and $2^\ell(x + h_n) = (\lfloor 2^n x \rfloor + 1) 2^{\ell-n}$. Since $n - \ell \geq 1$, the interval does not include integers or half-integers.

6.4.3.5 T8) Follow the notations of T7), prove that the sequence $\left\{ \frac{T(x+h_n) - T(x-h'_n)}{h_n + h'_n} \right\}_{n \geq 1}$ diverges.

Proof: Let $t = \lfloor 2^n x \rfloor$, then

$$a_n = \frac{T(x+h_n) - T(x-h'_n)}{h_n + h'_n} = \sum_{k=0}^{n-1} 2^{n-k} \left(\psi\left(\frac{t+1}{2^{n-k}}\right) - \psi\left(\frac{t}{2^{n-k}}\right) \right).$$

Since the interval $(2^{k-n}(t+1), 2^{k-n}t)$ does not contain any integers or half-integers, $2^{n-k}(\psi(2^{k-n}(t+1)) - \psi(2^{k-n}t)) \in \{-1, 1\}$, so $a_n \in \mathbb{Z}$ and n, a_n have the same parity. Therefore the sequence $\{a_n\}_{n \geq 1}$ diverges.

6.4.3.6 T9) Prove that $T(x)$ is continuous but nowhere differentiable on $\mathbb{R}$.

Proof: For any $x \in [0, 1]$, if $x = k_0 \cdot 2^{-m_0}$ as in T4), by T4) the sequence $\left\{ \frac{T(x+h_n) - T(x)}{h_n} \right\}$ diverges, hence T is not differentiable at x . Otherwise for any $n \in \mathbb{N}$, $2^n x \notin \mathbb{Z}$. Define $\{h_n\}_{n \geq 1}$ and $\{h'_n\}_{n \geq 1}$ as in T7), then by T8), the sequence $\left\{ \frac{T(x+h_n) - T(x-h'_n)}{h_n + h'_n} \right\}_{n \geq 1}$ diverges. Combined with T5) we know that T is not differentiable at x . Therefore T is nowhere differentiable on $\mathbb{R}$, since T is periodic. For any x, y in $\mathbb{R}$,

$$|T(x) - T(y)| \leq \sum_{k=0}^N 2^{-k} |T(2^k x) - T(2^k y)| + \sum_{k=N+1}^{\infty} 2^{-k} \leq 2 \max_{0 \leq k \leq N} |T(2^k x) - T(2^k y)| + 2^{-N}.$$

Hence for any $N > 0$, when $\varepsilon \rightarrow 0$, $|T(x) - T(x + \varepsilon)| \leq 2^{1-N} \rightarrow 0$, so T is (uniformly) continuous on $\mathbb{R}$.

6.4.3.7 T10) Prove that

$$T(x) = \begin{cases} 2x + \frac{T(4x)}{4}, & 0 \leq x < \frac{1}{4}; \\ \frac{1}{2} + \frac{T(4x-1)}{4}, & \frac{1}{4} \leq x < \frac{1}{2}; \\ \frac{1}{2} + \frac{T(4x-2)}{4}, & \frac{1}{2} \leq x < \frac{3}{4}; \\ 2 - 2x + \frac{T(4x-3)}{4}, & \frac{3}{4} \leq x \leq 1. \end{cases}$$

Proof: If $0 \leq x < 1/4$, then

$$T(x) = \psi(x) + \psi(2x)/2 + \sum_{k=2}^{\infty} \psi(2^k x) 2^{-k} = 2x + \frac{T(4x)}{4}.$$

The other cases are exactly the same.

6.4.3.8 T11) Let $\Gamma = \{(x, T(x)) : 0 \leq x \leq 1\} \subset \mathbb{R}^2$. Define $\Phi_i : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$\Phi_0 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1/4 & 0 \\ 1/2 & 1/4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \Phi_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1/4 & 0 \\ 0 & 1/4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1/4 \\ 1/2 \end{pmatrix}, \quad (6.6)$$

$$\Phi_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1/4 & 0 \\ 0 & 1/4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}, \quad \Phi_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1/4 & 0 \\ 0 & 1/4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 3/4 \\ 1/2 \end{pmatrix}. \quad (6.7)$$

Prove that Φ_i maps Γ to $\{(x, T(x)) : x \in [\frac{i}{4}, \frac{i+1}{4}]\}$. Proof: Consider $(x, T(x)) \in \Gamma$, then by T10),

$$\Phi_0 \begin{pmatrix} x \\ T(x) \end{pmatrix} = \begin{pmatrix} x/4 \\ x/2 + T(x)/4 \end{pmatrix} = \begin{pmatrix} x/4 \\ T(x/4) \end{pmatrix}.$$

Hence $\Phi_0(\Gamma) = \{(x, T(x)) : x \in [0, 1/4]\}$. The cases $i = 1, 2, 3$ are similar.

6.4.3.9 T12) Let $S_0 = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 \leq y \leq 1\}$. For every $n \geq 0$, define

$S_{n+1} = \bigcup_{k=0}^3 \Phi_k(S_n)$. Prove that S_n is a sequence of monotonically decreasing compact sets and $\Gamma = \bigcap_{n \geq 0} S_n$.

Proof: Let $S_n(x) = \{y \in [0, 1] : (x, y) \in S_n\}$. We prove by induction that $S_n \subset S_{n-1}$ and $S_n(x)$ is a closed interval containing $T(x)$ for any $x \in [0, 1]$. The base $n = 0$ is trivial. Suppose $S_n \subset S_{n-1}$ and $S_n(x)$ is a closed interval containing $T(x)$, then consider S_{n+1} . Note that $\Phi_k(S_n)$ are disjoint, since for any $(x, y) \in \Phi_k(S_n)$, $x \in [k/4, (k+1)/4]$. Hence for any $x \in [0, 1/4]$, $S_{n+1}(x) = \{y : (x, y) = \Phi_0(4x, z), z \in S_n(4x)\} = \{2x + z/4 : z \in S_n(4x)\}$ is a closed

interval containing $T(x) = 2x + T(4x)/4$. By the induction hypothesis $S_n(x) = \{2x + z/4 : z \in S_{n-1}(4x)\}$ and $S_n(4x) \subset S_{n-1}(4x)$ so $S_{n+1}(x) \subset S_n(x)$. The case $x \in [1/4, 1]$ is similar. Therefore $S_{n+1} \subset S_n$ and S_{n+1} is compact, so by induction $S_n \subset S_{n-1}$ for all $n > 0$ and S_n is compact. Clearly $\Gamma \subset \bigcap_{n \geq 0} S_n$, so it suffices to show that $|S_n(x)| \rightarrow 0$ for all $x \in [0, 1]$. From the proof above we get $\sup_{x \in [0, 1]} |S_n(x)| \leq \sup_{x \in [0, 1]} |S_{n-1}(x)|/4$, hence $|S_n(x)| \rightarrow 0$, therefore

$$\Gamma = \bigcap_{n \geq 0} S_n.$$

6.4.3.10 T13) Prove that $\sup_{x \in \mathbb{R}} T(x) \geq \frac{2}{3}$.

Proof: For any $(x, y) \in \Gamma$, by T11) we know that $(x/4 + 1/4, y/4 + 1/2) \in \Gamma$, hence if $a = \sup_{x \in \mathbb{R}} T(x)$ then $a \geq a/4 + 1/2$, i.e. $a \geq 2/3$.

6.4.3.11 T14) Find a $c \in [0, 1]$ such that $T(c) = \frac{2}{3}$.

Solution: Consider $c = 1/3$, then by T10), $T(c) = T(c)/4 + 1/2$, hence $T(c) = \frac{2}{3}$.

6.4.3.12 T15) For $x \in [0, 1]$, suppose $x = \sum_{n=1}^{\infty} b_n 4^{-n}$, where $b_n \in \{0, 1, 2, 3\}$. Prove that

$$\left\{ x \in [0, 1] : T(x) = \frac{2}{3} \right\} = \left\{ x \in [0, 1] : x = \sum_{n=1}^{\infty} b_n 4^{-n}, b_n \in \{1, 2\} \right\}.$$

Proof: If $x = \sum_{n=1}^{\infty} b_n 4^{-n}$, where $b_n \in \{1, 2\}$, then by T10),

$$T(x) = \frac{1}{2} + \frac{1}{4} T\left(\sum_{n=1}^{\infty} b_{n+1} 4^{-n}\right) = \cdots = \frac{1}{2} \left(1 + \frac{1}{4} + \frac{1}{4^2} + \cdots\right) = \frac{2}{3}.$$

Otherwise take the least n such that $b_n \in \{0, 3\}$, denote $y = \sum_{k=1}^{\infty} b_{n+k-1} 4^{-n}$, then

$$T(x) = \frac{1}{2} \left(1 + \frac{1}{4} + \cdots + \frac{1}{4^{n-2}}\right) + \frac{\min\{2y, 2-2y\}}{4^{n-1}} + \frac{1}{4^n} T(4y - b_n) < \frac{2}{3},$$

since $T(4y - b_n) \leq 2/3$ and $\min\{2y, 2-2y\} < 1/2$. Therefore

$$\left\{ x \in [0, 1] : T(x) = \frac{2}{3} \right\} = \left\{ x \in [0, 1] : x = \sum_{n=1}^{\infty} b_n 4^{-n}, b_n \in \{1, 2\} \right\}.$$

6.4.3.13 T16) As in T11), study the self-similarity of Φ_1, Φ_2 on

$\{(x, T(x)) : x \in [0, 1], T(x) = \frac{2}{3}\}$, which is a cantor set of Hausdorff dimension $\frac{1}{2}$.

Solution: Same as T11), denote $\Gamma' = \{(x, T(x)) : x \in [0, 1], T(x) = \frac{2}{3}\}$, then

$$\Phi_1(\Gamma') = \left\{ (x, T(x)) : x \in \left[0, \frac{1}{2}\right], T(x) = \frac{2}{3} \right\}, \quad \Phi_2(\Gamma') = \left\{ (x, T(x)) : x \in \left[\frac{1}{2}, 1\right], T(x) = \frac{2}{3} \right\}.$$

Chapter 7 Émile Borel Lemma

7.1 PSA

f is a function on the interval I .

7.1.1 A1) Suppose f is twice-differentiable at x , prove that

$$f''(x) = \lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

Proof: For any $h > 0$, consider the function $g(t) = f(t) - f(t-h)$, then there exists $\xi \in [0, h]$ such that $g(x+h) = g(x) + hg'(\xi)$, and there exists $\eta \in [\xi-h, \xi] \subset [-h, h]$ such that $f'(\xi) - f'(\xi-h) = hf''(\eta)$

$$\frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = \frac{f'(\xi) - f'(\xi-h)}{h} = f''(\eta),$$

therefore

$$f''(x) = \lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

7.1.2 A2) Suppose $x_0 \in I$, and

$$f(x) = a_0 + a_1(x-x_0) + \cdots + a_n(x-x_0)^n + o(|x-x_0|^n) \quad (7.1)$$

$$= b_0 + b_1(x-x_0) + \cdots + b_n(x-x_0)^n + o(|x-x_0|^n). \quad (7.2)$$

when $x \rightarrow x_0$, then for any $i = 0, 1, \dots, n$, $a_i = b_i$. Proof: Otherwise let $c_i = a_i - b_i$ and take the least k such that $c_k \neq 0$, then

$$c_k(x-x_0)^k + \cdots + c_n(x-x_0)^n + o(|x-x_0|^n) = 0 \implies c_k = -c_{k+1}(x-x_0) - \cdots - c_n(x-x_0)^{n-k} + o(|x-x_0|^{n-k}),$$

which leads to contradiction when $x \rightarrow x_0$.

7.1.3 A3) Suppose f is n -times differentiable at 0. Prove that if f is an even (odd) function, then the Taylor expansion of f at 0 has only even (odd) terms.

Proof: Use the fact that if f is even (odd) then f' is odd (even).

7.1.4 A4) If f is differentiable on (a, b) and $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow b^-} f(x)$ prove that exists $x_0 \in (a, b)$ such that $f'(x_0) = 0$.

Proof: Otherwise if $f'(x) \neq 0$ for all $x \in (a, b)$, by Darboux's theorem $f'(x)$ have the same sign over (a, b) , hence f is monotonic and non-constant on (a, b) , contradicting $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow b^-} f(x)$.

7.1.5 A5) Suppose $f \in C([a, b])$ and is differentiable on (a, b) . Prove that f is strictly increasing on $[a, b]$ iff for any $x \in (a, b)$, $f'(x) \geq 0$ and on any sub-interval $(c, d) \subset (a, b)$, $f'(x)$ does not vanish.

Proof: $\implies$ For any $x \in (a, b)$, $(f(x+h) - f(x))/h \geq 0$ so

$$f'(x) = \lim_{h \rightarrow \infty} \frac{f(x+h) - f(x)}{h} \geq 0.$$

If $f'(x)$ vanish on some sub-interval (c, d) then $f(c) = f(d)$, a contradiction. $\Leftarrow$ For any $a \leq x < y \leq b$, there exists $\xi \in (a, b)$ such that $f(y) - f(x) = (y - x)f'(\xi)$, hence $f(y) \geq f(x)$ and f is increasing. If $f(x) = f(y)$ for some $x < y$ then $f(t)$ is constant on $[x, y]$ and hence f' vanish on (x, y) , a contradiction.

7.2 PSB

Use L'Hôpital theorem to calculate limits:

7.2.1 B1) $a > 0$, then

$$\lim_{x \rightarrow \infty} \frac{\log x}{x^a} = \lim_{x \rightarrow \infty} \frac{x^{-1}}{ax^{a-1}} = 0.$$

7.2.2 B2) $a > 0, b > 1$ then

$$\lim_{x \rightarrow \infty} \frac{x^a}{b^x} = \lim_{x \rightarrow \infty} \frac{ax^{a-1}}{b^x \ln b} = \cdots = \lim_{x \rightarrow \infty} \frac{a(a-1) \cdots \{a\}}{b^x (\ln b)^{[a]} x^{1-\{a\}}} = 0.$$

7.2.3 B3)

$$\lim_{x \rightarrow 0} \frac{e^{ax} - e^{bx}}{\sin ax - \sin bx} = \lim_{x \rightarrow 0} \frac{ae^{ax} - be^{bx}}{a \cos ax - b \cos bx} = 1.$$

7.2.4 B4)

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{x - \sin x} = \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{1 + \cos x}{\cos^2 x} = 2.$$

7.2.5 B5)

$$\lim_{x \rightarrow 0} \frac{1 - \cos x^2}{x^2 \sin x^2} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x} = \lim_{x \rightarrow 0} \frac{\sin x}{\sin x + x \cos x} = \lim_{x \rightarrow 0} \frac{\cos x}{2 \cos x - x \sin x} = \frac{1}{2}.$$

7.2.6 B6)

$$\lim_{x \rightarrow 1} \frac{\sqrt{2x - x^4} - \sqrt[3]{x}}{1 - x^{4/3}} = \lim_{x \rightarrow 1} \frac{(2x - x^4)^{-1/2}(1 - 2x^3) - x^{-2/3}/3}{-\frac{4}{3}x^{1/3}} = 1.$$

7.2.7 B7)

$$\lim_{x \rightarrow 1^-} (\log x)(\log(1 - x)) = \lim_{x \rightarrow 1^-} \frac{\log(1 - x)}{1/\log x} = \lim_{x \rightarrow 1^-} \frac{x \log^2 x}{1 - x} = 0.$$

7.2.8 B8)

$$\lim_{x \rightarrow 0^+} \frac{\log \sin ax}{\log \sin bx} = \lim_{x \rightarrow 0^+} \frac{\sin bx}{\sin ax} \cdot \frac{a \cos ax}{b \cos bx} = 1.$$

7.2.9 B9)

$$\lim_{x \rightarrow 0^+} x^x = \exp \lim_{x \rightarrow 0^+} \frac{\log x}{x^{-1}} = \exp \lim_{x \rightarrow 0^+} -x = 1.$$

7.2.10 B10)

$$\lim_{x \rightarrow 1} x^{1/(1-x)} = \exp \lim_{x \rightarrow 1} \frac{\log x}{1-x} = e^{-1}.$$

7.2.11 B11)

$$\lim_{x \rightarrow 1} \left(\frac{1}{\log x} - \frac{1}{x-1} \right) = \lim_{x \rightarrow 1} \frac{x-1-\log x}{(x-1)\log x} = \lim_{x \rightarrow 1} \frac{1-x^{-1}}{1-x^{-1}+\log x} \quad (7.3)$$

$$= \lim_{x \rightarrow 1} \frac{x-1}{x-1+x\log x} = \frac{1}{2}. \quad (7.4)$$

7.2.12 B12)

$$\lim_{x \rightarrow 0^+} (\sin x)^x = \exp \lim_{x \rightarrow 0^+} \frac{\log \sin x}{x^{-1}} = \exp \lim_{x \rightarrow 0^+} -\frac{x^2}{\tan x} = 1.$$

7.2.13 B13)

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/(1-\cos x)} = \exp \lim_{x \rightarrow 0} \frac{\log \sin x - \log x}{1-\cos x} = \exp \lim_{x \rightarrow 0} \frac{\cot x - x^{-1}}{\sin x} \quad (7.5)$$

$$= \exp \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x \sin^2 x} = \exp \lim_{x \rightarrow 0} \frac{-x \sin x}{\sin^2 x + x \sin 2x} \quad (7.6)$$

$$= e^{-1/3}. \quad (7.7)$$

7.2.14 B14)

$$\lim_{x \rightarrow a} \frac{a^x - x^a}{x - a} = \lim_{x \rightarrow a} \frac{a^x \log a - a x^{a-1}}{1} = a^a (\log a - 1).$$

7.2.15 B15)

$$\lim_{x \rightarrow \infty} \frac{(1+1/x)^x - e}{1/x} = \lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x} = \lim_{x \rightarrow 0} (1+x)^{1/x} \cdot \frac{x/(x+1) - \log(1+x)}{x^2} \quad (7.8)$$

$$= e \lim_{x \rightarrow 0} \frac{(x+1)^{-2} - (x+1)^{-1}}{2x} = \frac{e}{2}. \quad (7.9)$$

7.2.16 B16)

$$\lim_{x \rightarrow \infty} \frac{x^{\log x}}{(\log x)^x} = \exp \lim_{x \rightarrow \infty} (\log x)^2 - x \log \log x = 0.$$

7.2.17 B17)

$$\lim_{x \rightarrow \infty} (x+a)^{1+1/x} - x^{1+1/(x+a)} = \lim_{x \rightarrow \infty} \frac{(x+a)^{1+1/x} x^{-1} - x^{1/(x+a)}}{x^{-1}} \quad (7.10)$$

$$= a. \quad (7.11)$$

7.2.18 B18)

$$\lim_{x \rightarrow \infty} \sqrt[3]{x^3 + x^2 + x + 1} - \sqrt{x^2 + x + 1} \cdot \frac{\log(e^x + x)}{x} = -\frac{1}{6}. \quad (7.12)$$

(Using WolframAlpha)

7.3 PSC

Calculate the maximum and minimum values of the following functions: 1. $f(x) = x^4 - 2x^2 + 5$, $x \in [-2, 2]$. $f(x) = (x^2 - 1)^2 + 4 \in [4, 13]$. 2. $f(x) = \frac{2x}{1+x^2}$, $x \in \mathbb{R}$ $1 - f(x) = (1+x^2)^{-1}(x-1)^2 \geq 0$, $f(x) + 1 = (1+x^2)^{-1}(x+1)^2 \geq 0$, therefore $f(x) \in [-1, 1]$. 3. $f(x) = \arctan x - \frac{1}{2} \log(1+x^2)$, $x \in \mathbb{R}$. $f'(x) = \frac{1-x}{x^2+1}$, hence $\sup_{x \in \mathbb{R}} f(x) = f(1) = \frac{\pi}{4} - \frac{\log 2}{2}$, and f has no minimum. 4. $f(x) = x \log x$, $x \in (0, \infty)$. $f'(x) = \log x + 1$, hence $\inf_{x \in (0, \infty)} f(x) = f(e^{-1}) = -e^{-1}$, and f has no maximum. 5. $f(x) = \sqrt{x} \log x$, $x \in (0, \infty)$. $f'(x) = x^{-1/2} \left(1 + \frac{\log x}{2}\right)$, hence $\inf_{x \in (0, \infty)} f(x) = f(e^2) = -2e^{-1}$. 6. $f(x) = 2 \tan x - \tan^2 x$, $x \in [0, \pi/2)$. $f(x) = 1 - (1 - \tan x)^2 \in (-\infty, 1]$.

7.4 PSD

f is differentiable on (a, b) . Suppose $x_0 \in (a, b)$ and $f'(x_0) = 0$.

7.4.1 D1) Prove that $f(x_0)$ is a local maximum if there exists $(x_0 - \delta, x_0 + \delta) \subset (a, b)$ such that

$$f'(x) = \begin{cases} > 0, & \forall x \in (x_0 - \delta, x_0), \\ < 0, & \forall x \in (x_0, x_0 + \delta). \end{cases}$$

Proof: Trivial by Lagrange mean-value theorem.

7.4.2 D2) Prove that if $f''(x_0)$ exists and $f''(x_0) < 0$ then $f(x_0)$ is a local maximum.

Proof: $f''(x_0) < 0$ and $f'(x_0) = 0$ implies for some $\delta > 0$, $f'(x) < 0$ for $x \in (x_0, x_0 + \delta)$ and $f'(x) > 0$ for $x \in (x_0 - \delta, x_0)$. Hence by D1), $f(x_0)$ is a local maximum.

7.4.3 D3) Suppose f is n -times differentiable at x_0 , $f'(x_0) = \dots = f^{(n-1)}(x_0) = 0$ and $f^{(n)}(x_0) \neq 0$. Determine the conditions that $f(x_0)$ is a local maximum.

Solution: n is even and $f^{(n)}(x_0) < 0$.

7.5 PSE: Roots of Polynomials

7.5.1 E1) Prove that if all the roots of the polynomial $P_n(x) \in \mathbb{R}[x]$ are real numbers, then so are the polynomials $P'_n(x), P''_n(x), \dots, P_n^{n-1}(x)$, where $n = \deg P_n$.

Proof: We only need to prove that P'_n has $n - 1$ real roots. By Rolle's mean-value theorem, between any two roots of P_n there is a root of P'_n hence P'_n has $n - 1$ real roots.

7.5.2 E2) Prove that the Legendre polynomial $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$ has n different roots in the interval $(-1, 1)$.

Proof: We know that the polynomials $\sqrt{(2n+1)/2} P_n(x)$ form a set of orthogonal base on the space $L^2([-1, 1])$, hence it must have n different roots in the interval $(-1, 1)$.

7.5.3 E3) Prove that the Laguerre polynomial $L_n(x) = \frac{e^x}{n!} \frac{d^n}{dx^n} (e^{-x} x^n)$ has n different real roots.

Proof: We know that the Laguerre polynomials are orthogonal on the space $L^2([0, \infty))$ with weight e^{-x} , hence it must have n distinct roots. Or note that $f(x) = x^n e^{-x}$ has a root with multiplicity n at 0 and it vanishes at ∞ , hence use Rolle's theorem and induction we can show that $f^{(k)}(x)$ has a root with multiplicity $n - k$ at 0 and k roots between 0 and ∞ .

7.5.4 E4) Prove that the Hermite polynomial $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2})$ has n different real roots.

Proof: We know that the polynomials $H_n(x)/\sqrt{2^n n!} \sqrt{n}$ form a set of orthogonal base on the Hilbert space $L^2(\mu)$ where $\mu(dx) = e^{-x^2} dx$, hence it must have n distinct real roots.

7.6 PSF: Émile Borel's Lemma

7.6.1 Part 1:

7.6.1.1 F1) Define $\phi : \mathbb{R} \rightarrow \mathbb{R}$:

$$\phi(x) = \begin{cases} e^{-1/x^2}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Prove that $\phi \in C^\infty(\mathbb{R})$. Proof: We prove by induction that for any $n \in \mathbb{Z}_{\geq 0}$, there is a polynomial $P_n \in \mathbb{R}[x]$ such that

$$\phi^{(n)}(x) = \begin{cases} P_n(1/x) \cdot e^{-1/x^2}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

(Which implies $\phi^{(n)}$ is continuous.) The case $n = 0$ is trivial. Suppose it holds for n , then for any $x > 0$,

$$\phi^{(n+1)}(x) = e^{-1/x^2} \left(P_n(1/x) \frac{2}{x^3} - P'_n(1/x) \frac{1}{x^2} \right),$$

for any $x < 0$, $\phi^{(n+1)}(x) = 0$, and for $x = 0$,

$$\phi_+^{(n+1)}(0) = \lim_{x \rightarrow 0^+} e^{-1/x^2} P_n(1/x) \frac{1}{x} = 0.$$

Hence the claim holds for $n + 1$ too. Therefore $\phi \in C^\infty(\mathbb{R})$.

7.6.1.2 F2) Define $\chi : \mathbb{R} \rightarrow \mathbb{R}$:

$$\chi(x) = \frac{\phi(2 - |x|)}{\phi(2 - |x|) + \phi(|x| - 1)}.$$

Prove that $\chi(x) \in C^\infty(\mathbb{R})$ and $\chi|_{[-1,1]} \equiv 1$, $\chi|_{(-\infty,-2] \cup [2,\infty)} \equiv 0$, $0 \leq \chi(x) \leq 1$ and χ is an even function. **Proof:** $2 - |x|$ and $|x| - 1$ cannot be both negative, hence the denominator is always positive, so $\chi \in C^\infty(\mathbb{R})$. The fact that $\chi|_{[-1,1]} \equiv 1$, $\chi|_{(-\infty,-2] \cup [2,\infty)} \equiv 0$, $\chi(x) \in [0, 1]$ and χ is even is trivial.

7.6.1.3 F3) Prove that for any $0 < a < b$, there exists a smooth function $\rho(x) \in C^\infty(\mathbb{R})$ such that $\rho|_{[-a,a]} \equiv 1$, $\rho|_{(-\infty,-b] \cup [b,\infty)} \equiv 0$, and $0 \leq \rho(x) \leq 1$.

Proof: Same as F2), define

$$\rho(x) = \frac{\phi(b - |x|)}{\phi(b - |x|) + \phi(|x| - a)}.$$

7.6.1.4 F4) Prove that there exists an even function $\psi \in C^\infty(\mathbb{R}^n)$ such that $\psi|_{\{x: |x| \leq 1\}} \equiv 1$, $\psi|_{\{x: |x| \geq 2\}} \equiv 0$, and $0 \leq \psi(x) \leq 1$.

Proof: (A special case of Urysohn's lemma) Define $f : \mathbb{R}^n \rightarrow \mathbb{R}$ as $f(\mathbf{x}) = \phi(1 - |\mathbf{x}|^2)$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}$ as $g(\mathbf{x}) = \phi(|x|^2/4 - 1)$, then f vanishes on $B(0, 1)^C$ and g vanishes on $\bar{B}(0, 2)$. Therefore

$$\psi(\mathbf{x}) = \frac{f(\mathbf{x})}{f(\mathbf{x}) + g(\mathbf{x})}$$

satisfy the requirements.

7.6.2 Part 2: Interchanging $\sum$ and $\frac{d}{dx}$

$I = [a, b]$ is a closed interval, $\{f_k\}_{k \geq 0}$ is a sequence of functions in $C^1(I)$. Assume $\sum_{k=0}^{\infty} f_k$ converges point-wise on I , and let $f(x) = \sum_{k=0}^{\infty} f_k(x)$.

7.6.2.1 F5) Assume the series $\sum_{k=0}^{\infty} f'_k(x)$ converges absolutely on I , i.e. $\sum_{k=0}^{\infty} \|f'_k\|_\infty$ converges. Prove that f is differentiable and $f'(x) = \sum_{k=0}^{\infty} f'_k(x)$.

Proof: Note that

$$\frac{f(x+h) - f(x)}{h} = \sum_{k=0}^{\infty} \frac{f_k(x+h) - f_k(x)}{h} = \sum_{k=0}^{\infty} f'_k(x + \xi_k).$$

Hence

$$\left| \frac{f(x+h) - f(x)}{h} - \sum_{k=0}^{\infty} f'_k(x) \right| \leq \sum_{n=0}^N |f'_k(x + \xi_n) - f'_k(x)| + 2 \sum_{n=N+1}^{\infty} \|f'_k\|$$

Note that f'_k is uniformly continuous, so

$$\lim_{h \rightarrow 0} \sum_{n=0}^N |f'_k(x + \xi_n) - f'_k(x)| = 0, \quad \lim_{N \rightarrow \infty} 2 \sum_{n=N+1}^{\infty} \|f'_k\| = 0.$$

Hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \sum_{k=0}^{\infty} f'_k(x).$$

7.6.2.2 F6) Assume $\sum_{k=0}^{\infty} f'_k(x)$ converges uniformly on I , then f is differentiable and

$$f'(x) = \sum_{k=0}^{\infty} f'_k(x).$$

Proof: Let $g(x) = \sum_{k=0}^{\infty} f'_k(x)$, since the series converges uniformly, $g(x)$ is continuous on I . By Lebesgue's Dominated Convergence Theorem,

$$\int_{x_0}^x g(t) dt = \sum_{k=0}^{\infty} f_k(t) \Big|_{x_0}^x = f(x) - f(x_0).$$

Hence $f'(x) = g(x) = \sum_{k=0}^{\infty} f'_k(x)$.

7.6.2.3 F7) Calculate the derivative of e^x using F6).

Solution: On any closed interval $[-M, M]$,

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

converges uniformly. Hence

$$(e^x)' = \sum_{n=0}^{\infty} \left(\frac{x^n}{n!} \right)' = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x.$$

7.6.3 Part 3: Borel's Lemma

Given an arbitrary sequence $\{a_k\}_{k \geq 0}$.

7.6.3.1 F8) For any $t_k > 0$, let $f_k(x) = \frac{a_k}{k!} x^k \chi(t_k x)$, determine the derivatives of any order of f_k at $x = 0$.

Solution: Note that when $x = 0$, $\chi^{(m)}(t_k x) = 0$ for any $m \geq 1$ and $\chi(t_k x) = 1$. Hence

$$f_k^{(n)}(0) = \frac{a_k}{k!} \sum_{j=0}^n \binom{n}{j} (x^k)^{(j)} \chi^{(n-j)}(t_k x) \Big|_{x=0} = \frac{a_k}{k!} (x^k)^{(n)} \Big|_{x=0} = a_k \delta_{n,k}.$$

7.6.3.2 F9) Prove that when $k \geq 2n$,

$$f_k^{(n)}(x) = a_k \sum_{\ell=0}^n \binom{n}{\ell} \frac{t_k^{n-\ell}}{(k-\ell)!} x^{k-\ell} \chi^{(n-\ell)}(t_k x).$$

Proof: Leibniz's Formula.

7.6.3.3 F10) (Borel's lemma) Prove that for any sequence $\{a_k\}_{k \geq 0}$, there exists a smooth function f on $\mathbb{R}$, such that for any $k \geq 0$, $f^{(k)}(0) = a_k$.

Proof: Let $f_k(x) = \frac{a_k}{k!} x^k \chi(t_k x)$ where t_k is yet to be determined, and

$$f(x) = \sum_{k=0}^{\infty} f_k(x) = \sum_{k=0}^{\infty} \frac{a_k x^k}{k!} \chi(t_k x).$$

For any $n \geq 0$, we want to show that $\sum_{k=0}^{\infty} f_k^{(n)}(x)$ converges uniformly on $\mathbb{R}$. Suppose $M_n = \sup_{x \in \mathbb{R}, m \leq n} |\chi^{(m)}(x)|$, and

$$C_k = \sup_{n < k/2} \sum_{\ell=0}^n \frac{\binom{n}{\ell}}{(k-\ell)!},$$

then for any $x \in \mathbb{R}$,

$$|f_k^{(n)}(x)| \leq |a_k| C_k M_k t_k^{-k/2}.$$

Hence if we choose t_k such that

$$|a_k|C_k M_k t_k^{-k/2} < 2^{-k},$$

then the series

$$\sum_{k=0}^{\infty} f_k^{(n)}(x)$$

converges uniformly on $\mathbb{R}$. By F6) we know that $f(x) = \sum_{k=0}^{\infty} f_k(x)$ is smooth, and by F8) we obtain $f^{(n)}(0) = a_n$ for any $n \geq 0$,

7.6.4 Part 4: Peano's Proof

7.6.4.1 F11) $\{c_k\}$ and $\{b_k\}$ are two sequences, and $b_k > 0$. Prove that

$$\left(\frac{c_k x^k}{1 + b_k x^2} \right)^{(n)}(0) = \begin{cases} n!(-1)^j c_{n-2j} b_{n-2j}^j, & \text{if } k = n - 2j, j \in \mathbb{Z}_{\geq 0}; \\ 0, & \text{otherwise.} \end{cases}$$

Proof: For $x \rightarrow 0$,

$$\frac{c_k x^k}{1 + b_k x^2} = c_k \sum_{n=0}^{\infty} (-1)^n x^{2n+k} b_k^n.$$

Which converges absolutely on the interval $[-b_k^{-1/2}/2, b_k^{-1/2}/2]$, and so are its n -times derivations, hence by F5)

$$\left(\frac{c_k x^k}{1 + b_k x^2} \right)^{(n)}(0) = c_k \sum_{j=0}^{\infty} (-1)^j \frac{(2j+k)!}{(2j+k-n)!} x^{2j+k-n} b_k^j \Big|_{x=0} = \begin{cases} n!(-1)^j c_k b_k^j, & k = n - 2j, \\ 0, & \text{otherwise.} \end{cases}$$

7.6.4.2 F12) Prove that there is a constant C such that for any $k \geq n + 2$, and any x ,

$$\left| \left(\frac{c_k x^k}{1 + b_k x^2} \right)^{(n)}(x) \right| \leq C(n+1)! \frac{|c_k| k!}{b_k} |x|^{k-n-2}.$$

Proof: Use du Bois-Reymond, we can let $C = 1$.

7.6.4.3 F13) Prove that for a given $\{c_k\}$, we can choose $\{b_k\}$ such that b_k depends only on the value of c_k , and the function

$$f(x) = \sum_{k=0}^{\infty} \frac{c_k x^k}{1 + b_k x^2}$$

is infinitely differentiable. Proof: Let $b_k = (k!)^2 c_k$, then by F12),

$$\left| \sum_{k \geq n+2} \left(\frac{c_k x^k}{1 + b_k x^2} \right)^{(n)} \right| \leq (n+1)! \sum_{k \geq n+2} \frac{|x|^{k-n-2}}{k!}$$

hence the series

$$\sum_{k=0}^{\infty} \left(\frac{c_k x^k}{1 + b_k x^2} \right)^{(n)}$$

converges uniformly for any $n \geq 1$. By F6) the function $f(x)$ is infinitely differentiable, and

$$f^{(n)}(x) = \sum_{k=0}^{\infty} \left(\frac{c_k x^k}{1 + b_k x^2} \right)^{(n)}.$$

7.6.4.4 F14) Prove that $f(0) = c_0$, $f'(0) = c_1$ and when $n \geq 2$,

$$\frac{f^{(n)}(0)}{n!} = c_n + \sum_{j=1}^{\lfloor n/2 \rfloor} (-1)^j c_{n-2j} b_{n-2j}^j.$$

Proof: Combine F11) and F13).

7.6.4.5 F15) Prove that by carefully choosing $\{c_k\}$ and $\{b_k\}$, we can prove Borel's lemma.

Proof: Let $b_k = (k!)^2 c_k$ and define c_k inductively such that

$$c_n = \frac{a_n}{n!} + \sum_{j=1}^{\lfloor n/2 \rfloor} (-1)^j c_{n-2j} b_{n-2j}^j$$

Then let $f(x) = \sum_{k=0}^{\infty} \frac{c_k x^k}{1+b_k x^2}$

7.7 PSG: Midterm Test Part B

Consider $f: \mathbb{R} \rightarrow \mathbb{R}$.

- Let $\mathcal{B}$ be all bounded function on $\mathbb{R}$.
- Let $\mathcal{L}$ be all Lipschitz functions on $\mathbb{R}$.

Suppose $a, \lambda \in \mathbb{R}$, $f \in \mathcal{B} \cap \mathcal{L}$, the goal is to find a function $F \in \mathcal{L}$ to solve:

$$F(x) - \lambda F(x+a) = f(x), \quad x \in \mathbb{R}. \quad (\star)$$

7.7.1 Part 1: Basic Properties of Lipschitz Functions**7.7.1.1 B1) Prove that if $f, g \in \mathcal{B} \cap \mathcal{L}$, then $fg \in \mathcal{L}$.**

Proof: Suppose $|f(x) - f(y)|, |g(x) - g(y)| \leq A|x - y|$, and $|f(x)|, |g(x)| \leq C$, then for any $x, y \in \mathbb{R}$,

$$|f(x)g(x) - f(y)g(y)| \leq 2MA|x - y|.$$

Hence $fg \in \mathcal{L}$.

7.7.1.2 B2) Prove that if f is differentiable and $f \in \mathcal{L}$ then $f' \in \mathcal{B}$.

Proof: If $|f(x) - f(y)| \leq C|x - y|$ then for any $x \in \mathbb{R}$,

$$|f'(x)| = \lim_{h \rightarrow 0} \left| \frac{f(x+h) - f(x)}{h} \right| \leq C.$$

Hence $f' \in \mathcal{B}$.

7.7.1.3 B3) Prove that if f is differentiable and $f' \in \mathcal{B}$ then $f \in \mathcal{L}$.

Proof: For any $x, y \in \mathbb{R}$, there exists $\xi \in (x, y)$ such that

$$|f(x) - f(y)| = |x - y| \cdot |f'(\xi)| \leq \sup_{t \in \mathbb{R}} |f'(t)| \cdot |x - y|.$$

Hence $f \in \mathcal{L}$.

7.7.1.4 B4) If $f \in \mathcal{B}$ and there exists $B > 0$ such that for any $x, y \in \mathbb{R}$, $|x - y| \leq 1$ implies $|f(x) - f(y)| \leq B|x - y|$. Prove that $f \in \mathcal{L}$.

Proof: Suppose $M = \sup_{x \in \mathbb{R}} |f(x)|$, then for any $x, y \in \mathbb{R}$,

$$|f(x) - f(y)| \leq \max\{B, 2M\}|x - y|.$$

Hence $f \in \mathcal{L}$.

7.7.2 Part 2: Solution of $(\star)$ when $|\lambda| < 1$.

Suppose $f \in \mathcal{B} \cap \mathcal{L}$ and $|\lambda| < 1$.

7.7.2.1 B5) Suppose F satisfy $(\star)$. Prove that for any $x \in \mathbb{R}$ and $n \in \mathbb{Z}_{\geq 1}$,

$$F(x) = \lambda^n F(x + na) + \sum_{k=0}^{n-1} \lambda^k f(x + ka), \quad (7.13)$$

$$F(x) = \lambda^{-n} F(x - na) - \sum_{k=1}^n \lambda^{-k} f(x - ka). \quad (7.14)$$

Proof: Use induction and apply $(\star)$. (Let $n \rightarrow \infty$ and we can obtain F formally.)

7.7.2.2 B6) Prove that for any $x \in \mathbb{R}$, $\sum_{k \geq 0} \lambda^k f(x + ka)$ converges.

Proof: Since f is bounded,

$$\left| \sum_{k=n}^{n+p} \lambda^k f(x + ka) \right| \leq \frac{M\lambda^n}{1-\lambda}.$$

Hence the series converges.

7.7.2.3 B7-8) Let $F(x) = \sum_{k \geq 0} \lambda^k f(x + ka)$. Prove that $F \in \mathcal{L}$ and solve $(\star)$.

Proof: For any $x, y \in \mathbb{R}$,

$$|F(x) - F(y)| \leq \sum_{k=0}^{\infty} \lambda^k |f(x + ka) - f(y + ka)| \leq \sum_{k=0}^{\infty} \lambda^k C|x - y| = \frac{C}{1-\lambda}|x - y|.$$

Hence $F \in \mathcal{L}$. For any $x \in \mathbb{R}$,

$$F(x) - \lambda F(x + a) = \sum_{k \geq 0} \lambda^k f(x + ka) - \sum_{k \geq 1} \lambda^k f(x + ka) = f(x).$$

Therefore F solves $(\star)$. If F' also solves $(\star)$, let $G = F - F'$, then G is bounded and

$$G(x) = \lambda G(x + a), \quad x \in \mathbb{R}.$$

Therefore for any $x \in \mathbb{R}$,

$$|G(x)| = \lambda^n |G(x + na)| \leq M\lambda^n \rightarrow 0.$$

Hence $G \equiv 0$ and $F \equiv F'$, so the solution to $(\star)$ is F .

7.7.2.4 B9) Solve $(\star)$ when $f(x) \equiv 1$ and $f(x) = \cos x$.

Solution: When $f(x) \equiv 1$,

$$F(x) = \sum_{k=0}^{\infty} \lambda^k f(x + ka) = \frac{1}{1-\lambda}.$$

When $f(x) = \cos x$,

$$F(x) = \sum_{k=0}^{\infty} \lambda^k \cos(x + ka) = \sum_{k=0}^{\infty} \lambda^k \frac{e^{i(x+ka)} + e^{-i(x+ka)}}{2} = \frac{1}{2} \left(\frac{e^{ix}}{1 - \lambda e^{ia}} + \frac{e^{-ix}}{1 - \lambda e^{-ia}} \right) \quad (7.15)$$

$$= \frac{\cos x - \lambda \cos(x - a)}{1 - 2\lambda \cos a + \lambda^2}. \quad (7.16)$$

7.7.3 Part 3: Solution of $(\star)$ when $|\lambda| > 1$.

7.7.3.1 B10) Solve $(\star)$ as in Part 2.

Solution: By B5), the solution should be

$$F(x) = - \sum_{k=1}^{\infty} \lambda^{-k} f(x - ka).$$

$f \in \mathcal{B}$ implies the series converges. Same as B8) we can show that the solution to $(\star)$ is unique, and like B7) we can show that $F \in \mathcal{L}$ and F satisfy $(\star)$.

7.7.3.2 B11) Solve $(\star)$ for $f(x) \equiv 1$ and $f(x) = \cos x$.

Solution: When $f(x) \equiv 1$,

$$F(x) = - \sum_{k=1}^{\infty} \lambda^{-k} f(x - ka) = \frac{1}{1 - \lambda}.$$

When $f(x) = \cos x$, same as B9) we have

$$F(x) = - \sum_{k=1}^{\infty} \lambda^{-k} f(x - ka) = \frac{\cos x - \lambda \cos(x - a)}{1 - 2\lambda \cos a + \lambda^2}.$$

7.7.4 Part 4: The Case when $|\lambda| = 1$.

7.7.4.1 B12) Suppose $\lambda = 1$. Prove that there exists $F \in \mathcal{L}$ not identically zero, such that for any x , $F(x) - F(x + a) = 0$.

Proof: Let $F(x) = |\{x/a\} - 1/2|$, then $F(x) = F(x + a)$, and $F \in \mathcal{L} \cap \mathcal{B}$.

7.7.4.2 B13) Let $f(x) = \cos x$ in $(\star)$. Prove that if $\cos a \neq 1$, then there exists $F \in \mathcal{L}$ that solves $(\star)$. Determine whether the solution is unique.

Proof: The equation $(\star)$ becomes $F(x) = F(x + a) + \cos x$. Let

$$F(x) = \{x/a\} - \sum_{k=0}^{\lfloor x/a \rfloor - 1} \cos(k + \{x/a\})a,$$

(if $x < 0$ the sum is viewed as from $\lfloor x/a \rfloor - 1$ to 0) then clearly $F(x) = F(x + a) + \cos x$, and F is bounded since $\cos a \neq 1$. For any $x, y \in \mathbb{R}$, if $|x - y| < a/2$, then suppose $na \leq x < y < (n + 1)a$,

$$|F(x) - F(y)| \leq \left| \left\{ \frac{x}{a} \right\} - \left\{ \frac{y}{a} \right\} \right| + 2 \left| \sin \frac{\{x/a\} - \{y/a\}}{2} a \right| \cdot \left| \sum_{k=0}^{n-1} \sin(k + (\{x/a\} + \{y/a\})/2)a \right| \quad (7.17)$$

$$\leq \frac{|x - y|}{a} + \frac{|x - y|}{|\sin a|}. \quad (7.18)$$

Hence $F \in \mathcal{L}$ by B4), so F solves $(\star)$. The solution is clearly not unique since we can add any factor of the F in B12) to the solution.

7.7.4.3 B14) Following B13), if $a = 2\pi$, then $(\star)$ has no solution in $\mathcal{L}$.

Proof: If $a = 2\pi$ and F is a solution to $(\star)$, then for any $x, y \in \mathbb{R}$,

$$|F(x + 2\pi n) - F(y + 2\pi n)| = n|\cos x - \cos y| \rightarrow \infty.$$

Hence $F \notin \mathcal{L}$.

7.7.4.4 B15) Suppose $\lambda = -1$, Prove that there exists $F \in \mathcal{L}$ not identically zero, such that for any x , $F(x) + F(x + a) = 0$.

Proof: Let $F(x) = |2\{x/2a\} - 1| - 1/2$, then $F \in \mathcal{L}$ and $F(x) + F(x + a) = 0$.

7.7.4.5 B16) Suppose $\lambda = -1$, $a = 1$, $f \in \mathcal{L}$ is monotonically decreasing and $\lim_{x \rightarrow \infty} f(x) = 0$, f is differentiable and f' is increasing. Prove that there exists $F \in \mathcal{L}$ such that

$$F(x) + F(x + 1) = f(x), \quad x \in \mathbb{R}.$$

Further show that if we require $F \in \mathcal{L}$ and $\lim_{x \rightarrow \infty} F(x) = 0$, then the solution is unique. Proof: Since f is monotonically decreasing, for any $x \in \mathbb{R}$, the series

$$F(x) = \sum_{n=0}^{\infty} (-1)^n f(x + n)$$

converges. For any $x, y \in \mathbb{R}$, $|x - y| < 1$, there exists $\xi_n \in (x + n, y + n)$ such that $f(y + n) - f(x + n) = (y - x)f'(\xi_n)$, hence (by B3) f' is bounded)

$$|F(x) - F(y)| = |y - x| \cdot \left| \sum_{n=0}^{\infty} (-1)^n f'(\xi_n) \right| \leq \sup_{t \in \mathbb{R}} |f'(t)| \cdot |y - x|.$$

so $F \in \mathcal{L}$. Clearly $F(x) + F(x + 1) = f(x)$, so F solves $(\star)$, and $0 < F(x) < f(x)$ so $\lim_{x \rightarrow \infty} F(x) = 0$. If $F' \in \mathcal{L}$ also satisfy $(\star)$ and $\lim_{x \rightarrow \infty} F(x) = 0$, let $G = F - F'$, then $G(x) + G(x + 1) = 0$ and $\lim_{x \rightarrow \infty} G(x) = 0$. Hence $G(x) = \lim_{n \rightarrow \infty} (-1)^n G(x + n) = 0$ for any $x \in \mathbb{R}$, so $G \equiv 0$. Therefore F is the unique solution.

Chapter 8 Sturm-Liouville Theory

8.1 PSA: Convex functions

8.1.1 A1)

(1) $f(x) = |x|$, $I = \mathbb{R}$ is convex, since

$$|\lambda x + (1 - \lambda)y| \leq \lambda|x| + (1 - \lambda)|y|.$$

(2) $f(x) = x^p$, $p \in \mathbb{R}$, $I = \mathbb{R}_{>0}$ $f''(x) = p(p-1)x^{p-2}$ so f is concave if $p \in [0, 1]$ and convex if $p \in (-\infty, 0] \cup [1, \infty)$.

(3) $f(x) = \sin x$, $I = [0, \pi]$ is concave since $f''(x) = -\sin x \leq 0$ when $x \in [0, \pi]$. (4) $f(x) = x \log x$, $I = \mathbb{R}_{\geq 0}$ is (strictly) convex since $f''(x) = 1/x > 0$. (5) $f(x) = \mathbf{1}_{\{0,1\}}$, $I = [0, 1]$ is convex since

$$f(\lambda x + (1 - \lambda)y) = 0 \leq \lambda f(x) + (1 - \lambda)f(y).$$

8.1.2 A2) Prove the following properties:

1. If f, g are convex on I , then $f + g$ is convex on I .

Proof: By definition,

$$(f + g)(\lambda x + (1 - \lambda)y) \leq \lambda(f + g)(x) + (1 - \lambda)(f + g)(y),$$

so $f + g$ is convex.

1. If f, g are monotonically increasing, non-negative, convex functions on I , then fg is convex.

Proof: Note that

$$f(\lambda x + (1 - \lambda)y)g(\lambda x + (1 - \lambda)y) \leq (\lambda f(x) + (1 - \lambda)f(y)) \cdot (\lambda g(x) + (1 - \lambda)g(y))$$

and

$$\lambda f(x)g(x) + (1 - \lambda)f(y)g(y) - (\lambda f(x) + (1 - \lambda)f(y))(\lambda g(x) + (1 - \lambda)g(y)) \quad (8.1)$$

$$= \lambda(1 - \lambda)(f(x) - f(y))(g(x) - g(y)) \geq 0. \quad (8.2)$$

hence

$$(fg)(\lambda x + (1 - \lambda)y) \leq \lambda(fg)(x) + (1 - \lambda)(fg)(y).$$

1. If f is convex on I , g is a monotonically increasing convex function on $J \supset f(I)$, then $g \circ f$ is convex.

Proof: Note that

$$g(f(\lambda x + (1 - \lambda)y)) \leq g(\lambda f(x) + (1 - \lambda)f(y)) \leq \lambda g(f(x)) + (1 - \lambda)g(f(y)).$$

hence $g \circ f$ is convex.

1. If f, g are convex on I , then $h(x) = \max\{f(x), g(x)\}$ is convex.

Proof: For any x, y, λ and $t = \lambda x + (1 - \lambda)y$, suppose $h(t) = f(t)$, then

$$h(t) \leq \lambda f(x) + (1 - \lambda)f(y) \leq \lambda h(x) + (1 - \lambda)h(y)$$

hence h is convex.

8.1.3 A3) Suppose $f \in C((a, b))$. If for any $x, y \in (a, b)$, $f\left(\frac{x+y}{2}\right) \leq \frac{f(x)+f(y)}{2}$, prove that f is convex.

Proof: For any $x, y \in (a, b)$ and $\lambda \in [0, 1]$, we need to prove that

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Note that it holds for any dyadic number λ , since the cases $\lambda = 0, 1, 1/2$ is trivial, and for $\lambda = (2m+1)/2^t$, let $u = m/2^{t-1}, v = (m+1)/2^{t-1}$, then

$$f(\lambda x + (1-\lambda)y) \leq \frac{f(ux + (1-u)y) + f(vx + (1-v)y)}{2} \quad (8.3)$$

$$\leq \lambda f(x) + (1-\lambda)f(y). \quad (8.4)$$

Now since $f \in C((a, b))$, for any $\lambda \in (0, 1)$ there is a sequence of dyadic numbers λ_n such that $\lim_{n \rightarrow \infty} \lambda_n = \lambda$, hence

$$f(\lambda x + (1-\lambda)y) = \lim_{n \rightarrow \infty} f(\lambda_n x + (1-\lambda_n)y) \leq \lim_{n \rightarrow \infty} \lambda_n f(x) + (1-\lambda_n)f(y) \quad (8.5)$$

$$= \lambda f(x) + (1-\lambda)f(y). \quad (8.6)$$

8.1.4 A4) f is a convex function on $[a, b]$. Prove that if there exists $c \in (a, b)$ such that $f(c) \geq \max\{f(a), f(b)\}$ then f is constant.

Proof: For any $t \in (a, b)$, let $\lambda = (t-a)/(b-a)$ then

$$f(t) = f(\lambda a + (1-\lambda)b) \leq \lambda f(a) + (1-\lambda)f(b) \leq \max\{f(a), f(b)\}.$$

By $f(c) \geq \max\{f(a), f(b)\}$ we know that $f(a) = f(b)$. If for some $t \in (a, b)$, $f(t) \neq f(a)$, suppose $c \in (a, t)$, then

$$f(c) = \lambda f(a) + (1-\lambda)f(t) < f(a)$$

a contradiction. Hence $f(t) = f(a)$ for all $t \in [a, b]$.

8.1.5 A5) f is convex on $\mathbb{R}$. Prove that if f has an upper-bound, then f is constant.

Proof: Otherwise suppose that $f(a) < f(b)$, where $a < b$. (If $f(a) > f(b)$ let $g(x) = f(-x)$). Let $x_0 = a, x_1 = b, x_n = a + n(b-a)$, then

$$f(x_{n+1}) - f(x_n) \geq f(x_n) - f(x_{n-1}) \geq f(b) - f(a),$$

hence $f(x_n) \geq f(a) + n(f(b) - f(a)) \rightarrow \infty$, leading to contradiction.

8.1.6 A6) f is strictly convex on I . Suppose $f(x_0)$ is a local minimum of f , prove that x_0 is the unique global minimum point of f .

Proof: Suppose there is another $x_1 \neq x_0$ such that $f(x_1) \leq f(x_0)$, then let $x_n = x_0 + n(x_1 - x_0)$. Since f is strictly convex, $f(x_n) < \max\{f(x_1), f(x_0)\} = f(x_0)$, contradicting the fact that $f(x_0)$ is a local minimum.

8.1.7 A7) I is an open interval. Prove that f is convex on I , iff for any $x_0 \in I$, there exists $a \in \mathbb{R}$, such that for any $x \in I$, $f(x) \geq a(x - x_0) + f(x_0)$.

Proof: Suppose f is convex on I , then the any $x_0 \in I$, the function $g(x) = \frac{f(x)-f(x_0)}{x-x_0}$ is monotonically increasing. Hence we can let $a = \sup_{x < x_0} g(x) < \infty$. If for any $x_0 \in I$, and $x \in I$, $f(x) \geq g(x_0)(x - x_0) + f(x_0)$, then for any $x, y \in I$ and $\lambda \in (0, 1)$, let $t = \lambda x + (1-\lambda)y$,

$$\lambda f(x) + (1-\lambda)f(y) \geq \lambda(f(t) + (x-t)g(t)) + (1-\lambda)(f(t) + (1-\lambda)(y-t)g(t)) \quad (8.7)$$

$$= f(t) = f(\lambda x + (1-\lambda)y). \quad (8.8)$$

Hence f is convex.

8.2 PSB

8.2.1 B1) Prove the following inequalities:

(1)

$$x - \frac{x^2}{2} < \log(1+x) < x, \quad x > 0.$$

Proof: If $f(x) = \log(x+1) - x$, then $f'(x) = \frac{1}{x+1} - 1 < 0$ hence $f(x) < f(0) = 0$. Let $g(x) = \log(1+x) - x + x^2/2$, then $g'(x) = \frac{1}{x+1} + (x+1) - 2 \geq 0$, hence $g(x) > g(0) = 0$. (2)

$$(x^\alpha + y^\alpha)^{1/\alpha} > (x^\beta + y^\beta)^{1/\beta}, \quad x, y > 0, \beta > \alpha > 0.$$

Proof: Assume that $x^\alpha + y^\alpha = 1$, then $0 < x, y < 1$, so

$$x^\beta + y^\beta < x^\alpha + y^\alpha < 1 \implies (x^\beta + y^\beta)^{1/\beta} < (x^\alpha + y^\alpha)^{1/\alpha}.$$

(3)

$$x - \frac{x^3}{6} < \sin x < x, \quad x > 0.$$

Proof: Let $f(x) = \sin x - x$, then $f'(x) = \cos x - 1 \leq 0$, so $f(x) < f(0) = 0$. Let $g(x) = \sin x - x + x^3/6$, then $g'(x) = \cos x - 1 + x^2/2$, $g''(x) = x - \sin x > 0$, so $g'(x) > g'(0) = 0$ and $g(x) > g(0) = 0$. (4)

$$\left(\frac{1+x}{2}\right)^p + \left(\frac{1-x}{2}\right)^p \leq \frac{1}{2}(1+x^p), \quad p \in [2, \infty), x \in [0, 1].$$

Proof: ???

8.2.2 B2) Find all $a > 0$ such that $a^x \geq x^a$ for any $x > 0$.

Solution: $f(x) = x^{1/x}$ then $f'(x) = x^{1/x} \frac{1-\log x}{x^2}$ hence f has a unique minimum at e .

8.2.3 B3) Prove that for any $x_i, t_i, i = 1, 2, \dots, n, \sum_{i=1}^n t_i = 1$,

$$\left(\sum_{i=1}^n t_i x_i\right)^{\sum_{i=1}^n t_i x_i} \leq \prod_{i=1}^n x_i^{t_i x_i}.$$

Proof: Let $f(x) = x \log x$, then $f''(x) = 1/x > 0$, so f is convex. By Jensen's inequality,

$$\sum_{i=1}^n t_i f(x_i) \geq f\left(\sum_{i=1}^n t_i x_i\right)$$

hence

$$\left(\sum_{i=1}^n t_i x_i\right)^{\sum_{i=1}^n t_i x_i} \leq \prod_{i=1}^n x_i^{t_i x_i}.$$

and equality holds iff $x_i = x_1$.

8.2.4 B4) Prove that for any $a, b > 0, 1/p + 1/q = 1$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}, \quad \text{if } p > 1; \quad ab \geq \frac{a^p}{p} + \frac{b^q}{q}, \quad \text{if } p < 1.$$

Proof: The function $-\log x$ is convex, so when $p > 1, q > 1$, then

$$\log\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p} \log a^p + \frac{1}{q} \log b^q$$

so $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$. When $p < 1$, then $pq < 0$, so likewise $ab \geq \frac{a^p}{p} + \frac{b^q}{q}$.

8.2.5 B5) Prove that if $x_i, y_i \geq 0, i = 1, 2, \dots, n, 1/p + 1/q = 1$, then

$$\sum_{i=1}^n x_i y_i \leq \left(\sum_{i=1}^n x_i^p \right)^{1/p} \left(\sum_{i=1}^n y_i^q \right)^{1/q}, \text{ if } p > 1;$$

and the inequality reverses when $p < 1$. Proof: Assume that $\sum_{i=1}^n x_i^p = \sum_{i=1}^n y_i^q = 1$, then by B4), if $p > 1$,

$$\sum_{i=1}^n x_i y_i \leq \sum_{i=1}^n \frac{x_i^p}{p} + \frac{y_i^q}{q} = \frac{1}{p} + \frac{1}{q} = 1.$$

The case $p < 1$ is similar.

8.3 PSC

8.3.1 C1) Suppose $f \in C([0, 1])$, g is differentiable on $[0, 1]$ and $g(0) = 0$. If there is a constant $\lambda \neq 0$, such that for any $x \in [0, 1]$, $|g(x)f(x) + \lambda g'(x)| \leq |g(x)|$, prove that $g(x) \equiv 0$.

Proof: Otherwise assume that $\forall \varepsilon > 0 \exists t \in (0, \varepsilon)$, such that $g(t) \neq 0$. Let $C = (1 + \sup_{x \in [0, 1]} |f(x)|)/\lambda$, then $|g'(x)| \leq C|g(x)|, \forall x \in [0, 1]$. For any $t \in (0, 1)$, there exists $\xi \in [0, t]$ such that $g(t) = g(0) + tg'(\xi)$, hence

$$\frac{|g(t)|}{t} = |g'(\xi)| \leq C \sup_{\xi \in [0, t]} |g(\xi)|.$$

For any $t > 0$ suppose $|g(s)| = \sup_{\xi \in [0, t]} |g(\xi)| > 0$, then $|g(s)|/s \leq C|g(s)|$ hence $t \geq s \geq \frac{1}{C}$, a contradiction.

8.3.2 C2) f is twice differentiable on $(-1, 1)$, $f(0) = f'(0) = 0$. If for any $x \in (-1, 1)$, $|f''(x)| \leq |f(x)| + |f'(x)|$, prove that $f(x) \equiv 0$.

Proof: We prove that $f''(x) \equiv 0$. Otherwise suppose $\forall \varepsilon > 0, \exists x \in [0, \varepsilon], f''(x) \neq 0$. Note that

$$|f''(x)| \leq |f(x)| + |f'(x)| \leq \left(\frac{x^2}{2} + |x| \right) \sup_{y \in [0, x]} |f''(y)|.$$

Since $f''(0) = 0$, take $x \in [0, 1/2]$ such that $f''(x) \neq 0$, and suppose $|f''(t)| = \sup_{y \in [0, x]} |f''(y)|$, then $|f''(t)| \leq (t^2/2 + t)|f''(t)|$, a contradiction.

8.3.3 C3) f is n -times differentiable on $\mathbb{R}$, $f(0) = f'(0) = \dots = f^{(n-1)}(0) = 0$. If there exists $C \in \mathbb{R}_{>0}$ and $\ell \in \mathbb{Z}_{\geq 0}$ such that for any $x \in \mathbb{R}$, $|f^{(n)}(x)| \leq C|f^{(\ell)}(x)|$. Prove that $f(x) \equiv 0$.

Proof: We can assume that $\ell = 0$. Since $f^{(k)}(x) = 0, \forall 0 \leq k < n$, we have

$$|f^{(n)}(x)| \leq C|f(x)| \leq C \frac{x^n}{n!} \sup_{y \in [0, x]} |f^{(n)}(y)|.$$

Hence for any $x \in [0, \varepsilon], \varepsilon = (n!/C)^{1/n}$, $f^{(n)}(x) = 0$, so $f^{(k)}(x) = 0$ for all $x \in [0, \varepsilon], 0 \leq k < n$. Likewise we get $f(x) \equiv 0$.

8.3.4 C4) $n \in \mathbb{Z}_{>0}$, prove that the polynomial $P(x) = \sum_{k=0}^{n+1} \binom{n+1}{k} (-1)^k (x-k)^n \equiv 0$.

Proof: We know the identity

$$\sum_{k=0}^n \binom{n}{k} (-1)^k k^l = 0, \forall 0 \leq l \leq n-1.$$

Since $\Delta^n x^l \equiv 0$ Likewise by considering $f(t) = (x-t)^n$ we have $P(x) \equiv 0$. (Or we can use C3)

8.3.5 C5 $f \in C^\infty(\mathbb{R})$. Assume there exists $C > 0$ such that for any $n \in \mathbb{Z}_{\geq 0}$ and $x \in \mathbb{R}$,
 $|f^{(n)}(x)| \leq C$.

i. Prove that given an arbitrary $x_0 \in \mathbb{R}$,

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k, \forall x \in \mathbb{R}.$$

Proof: The Lagrange remainder

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$$

tends to zero as $n \rightarrow \infty$, hence the Taylor series

$$f(x) = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

ii. $E \subset \mathbb{R}$ is an infinite bounded set. Prove that if $f(E) = \{0\}$, then $f \equiv 0$. Proof: Suppose $E \subset [-M, M]$, then by Bolzano-Weierstrass theorem, there exists a sequence $\{z_n\}_{n \geq 1} \subset E$ such that $z = \lim_{n \rightarrow \infty} z_n$ exists. Since $f \in C(\mathbb{R})$, $f(z) = \lim_{n \rightarrow \infty} f(z_n) = 0$, so

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z)}{k!} (x - z)^k.$$

If f does not vanish on $\mathbb{R}$, then take the least $m > 0$ such that $f^{(m)}(z) \neq 0$. When $z_n \rightarrow z$,

$$0 = \frac{f^{(m)}(z)}{m!} + \sum_{k=m+1}^{\infty} \frac{f^{(k)}(z)}{k!} (x - z)^{k-m}$$

which leads to contradiction. Hence f vanishes on $\mathbb{R}$.

8.3.6 C6 Assume $f \in C^2((0, 1))$, $\lim_{x \rightarrow 1^-} f(x) = 0$. If there exists $C > 0$, such that for any $x \in (0, 1)$, $(1 - x)^2 |f''(x)| \leq C$. Prove that $\lim_{x \rightarrow 1^-} (1 - x) f'(x) = 0$.

Proof: For any $0 < x < y < 1$, there exists $\xi \in (x, y)$ such that

$$f(y) = f(x) + (y - x)f'(x) + \frac{(y - x)^2}{2} f''(\xi).$$

For any $\lambda > 0$, let $y = (\lambda + x)/(\lambda + 1) \in (x, 1)$, then

$$|(y - x)f'(x)| \leq |f(y)| + |f(x)| + \frac{\lambda^2}{2} (1 - y)^2 |f''(\xi)| \leq |f(y)| + |f(x)| + \frac{C\lambda^2}{2}.$$

Therefore

$$|(1 - x)f'(x)| \leq (|f(y)| + |f(x)|) \frac{\lambda + 1}{\lambda} + \frac{1}{2} \lambda(\lambda + 1)C$$

Hence for any $\lambda > 0$,

$$\lim_{x \rightarrow 1^-} |(1 - x)f'(x)| \leq \frac{1}{2} \lambda(\lambda + 1)C \rightarrow 0,$$

so $\lim_{x \rightarrow 1^-} (1 - x)f'(x) = 0$.

8.4 PSD

Calculate $\sup_{x \in I} f(x)$ and $\inf_{x \in I} f(x)$:

8.4.1 D1 $f(x) = \frac{(\log x)^2}{x}$, $I = \mathbb{R}_{>0}$

Solution: Let $y = \log x \in \mathbb{R}$, then $f(x) = y^2 e^{-y}$.

$$\frac{d}{dy} y^2 e^{-y} = y e^{-y} (2 - y).$$

Hence $\sup_{x \in I} f(x) = f(e^2) = 4e^{-2}$, $\inf_{x \in I} f(x) = \min\{f(0), f(\infty)\} = 0$.

8.4.2 D2) $f(x) = |x(x^2 - 1)|, I = \mathbb{R}$

Solution: $\sup = \infty, \inf = 0$.

8.4.3 D3)

$$f(x) = \frac{x(x^2 + 1)}{x^4 - x^2 + 1}, I = \mathbb{R}.$$

Solution: Note that

$$2(x^4 - x^2 + 1) - x(x^2 + 1) = (x^2 - 1)^2 + (x - 1)^2(x^2 + x + 1) \geq 0.$$

Therefore $f(x) \leq 2$ where equality holds at $x = 1$. Since $f(x) = f(-x)$, $\sup = 2, \inf = -2$.

8.4.4 D4)

$$f(x) = x^{1/3}(1 - x)^{2/3}, I = (0, 1).$$

Solution: By AM-GM, $f(x) \leq \frac{2^{2/3}}{3}$ where equality holds at $x = 1/3$. Hence $\sup = \frac{2^{2/3}}{3}, \inf = 0$.

8.4.5 D5)

$$f(x) = \left(1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!}\right) e^{-x}, I = \mathbb{R}.$$

Solution: $f'(x) = -e^{-x} \frac{x^n}{n!}$, so if n is even, $\sup = \infty, \inf = 0$, and if n is odd, $\sup = 1, \inf = -\infty$.

8.4.6 D6) $f(x) = \sin^{2m} x \cos^{2n} x, I = \mathbb{R}$.

Solution: Let $t = \sin^2 x \in [0, 1]$, then $f(x) = t^m(1 - t)^n \in [0, n^n m^n / (n + m)^{n+m}]$.

8.5 PSE

Compare the two functions (or real numbers).

8.5.1 E1) $f(x) = e^x, g(x) = 1 + xe^x, x > 0$.

Solution: The case $x \geq 1$ is trivial. If $x \in (0, 1)$, then

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \leq \sum_{n=0}^{\infty} x^n = \frac{1}{1 - x},$$

hence $f(x) \leq g(x)$. Therefore $f(x) \leq g(x)$ for all $x > 0$.

8.5.2 E2) $f(x) = xe^{x/2}, g(x) = e^x - 1, x > 0$.

Solution: ($x/2 \leq \sinh(x/2)$) Consider $h(x) = e^{x/2} - e^{-x/2} - x$, then $h(0) = 0$ and

$$h'(x) = \frac{1}{2}(e^{x/2} + e^{-x/2} - 2) \geq 0.$$

Hence $h(x) \geq 0$, i.e. $g(x) \geq f(x)$ for all $x > 0$.

8.5.3 E3) $f(x) = \left(\frac{x+1}{2}\right)^{(x+1)}, g(x) = x^x, x > 0$.

Solution: Consider $h(x) = x \log x - (x+1) \log \frac{x+1}{2}$, then $h(1) = 0$ and

$$h'(x) = \log \frac{2x}{x+1} \geq 0 \iff x \geq 1.$$

Hence $f(x) \leq g(x)$ for all $x > 0$.

8.5.4 E4) $2^{\sqrt{2}}$ and e .

Solution: Note that

$$\log 2 = \sum_{n=1}^{\infty} \frac{1}{2^n n} \leq \frac{2}{3} + \sum_{n=4}^{\infty} \frac{1}{2^n \cdot 4} = \frac{2}{3} + \frac{1}{32} < \frac{2}{3} + \frac{1}{30} = 0.7 < \frac{1}{\sqrt{2}},$$

hence $2^{\sqrt{2}} < e$.

8.5.5 E5) $f(x) = \log(1 + \sqrt{1+x^2}), g(x) = 1/x + \log x, x > 0$.

Solution: Consider $h(x) = \log x + 1/x - \log(1 + \sqrt{1+x^2})$, then

$$h'(x) = \frac{1}{x} - \frac{1}{x^2} - \frac{x}{(1 + \sqrt{1+x^2})\sqrt{1+x^2}} \leq 0.$$

($\iff (x-1)(\sqrt{1+x^2} + 1 + x^2) - x^3 \leq 0 \iff (x-1)\sqrt{1+x^2} \leq x^2$) Therefore $h(x) \geq \lim_{x \rightarrow \infty} h(x) = 0$.

8.5.6 E6) $\log 8$ and 2 .

Solution: Note that

$$\log 2 = \log \frac{1}{1 - \frac{1}{2}} = \sum_{n=1}^{\infty} \frac{1}{2^n n} \geq \sum_{n=1}^3 \frac{1}{2^n n} = \frac{2}{3},$$

hence $\log 8 > 2$.

8.6 PSF

If f satisfy $f(x) = (x - x_0)^r g(x)$ in a neighborhood of x_0 , where $r \in \mathbb{Z}_{\geq 0}$, g is continuous at x_0 and $g(x_0) \neq 0$, then we call x_0 an r -fold root of f .

8.6.1 F1) Suppose x_0 is an r -fold root of f where $r > 0$. Prove that if $g(x) = f(x)/(x - x_0)^r$ is continuous, then x_0 is an $(r - 1)$ -fold root of f' .

Proof: Suppose $f(x) = (x - x_0)^r g(x)$ in the neighborhood $O(x_0)$, then $f'(x) = (x - x_0)^r g'(x) + r(x - x_0)^{r-1} g(x)$ in $O(x_0)$. Therefore let $h(x) = (x - x_0)g'(x) + g(x)$, then $f'(x) = (x - x_0)^{r-1} h(x)$ and $h(x_0) = g(x_0) \neq 0$, so x_0 is an $(r - 1)$ -fold root of f' .

8.6.2 F2) Suppose f is n -times differentiable on $\mathbb{R}$. Prove that if $f(x) = 0$ has $n + 1$ distinct real roots, then $f^{(n)}(x) = 0$ has at least one root.

Proof: Use induction and Rolle's mean-value theorem to prove that $f^{(n-k)}(x)$ has at least $k + 1$ different real roots.

8.6.3 F3) f is differentiable on $\mathbb{R}$. Suppose $f(x) = 0$ has r roots (counting multiplicity), then $f'(x) = 0$ has at least $r - 1$ roots (counting multiplicity).

Proof: Combine F1) and F2).

8.6.4 F4) Suppose f is n -times differentiable on $\mathbb{R}$. Prove that if $f(x) = 0$ has exactly $n + 1$ roots counting multiplicity, then $f^{(n)}(x) = 0$ has at least one root.

Proof: Use F3) and induction.

8.7 PSG

Let $a \in \mathbb{R}$, $f : (a, \infty) \rightarrow \mathbb{R}$ twice differentiable on (a, ∞) , and

$$M_0 := \sup_{x \in (a, \infty)} |f(x)|, \quad M_1 := \sup_{x \in (a, \infty)} |f'(x)|, \quad M_2 := \sup_{x \in (a, \infty)} |f''(x)|,$$

are real numbers.

8.7.1 G1) Prove that $M_1^2 \leq 4M_0M_2$.

Proof: Let $h = \sqrt{M_0/M_2}$, then for any $x \in (a, \infty)$, there exists $\xi \in (x, x + 2h)$ such that

$$f(x + 2h) = f(x) + 2hf'(x) + 2h^2f''(\xi) \implies f'(x) = hf''(\xi) + \frac{f(x + 2h) - f(x)}{2h}.$$

Therefore $f'(x) \leq M_0/h + M_2h = 2\sqrt{M_0M_2}$, hence $M_1^2 \leq 4M_0M_2$.

8.7.2 G2) Let $a = -1$, consider the function

$$f(x) = \begin{cases} 2x^2 - 1, & x \in (-1, 0), \\ \frac{x^2 - 1}{x^2 + 1}, & x \in [0, \infty), \end{cases}$$

verify that f is twice differentiable, and $M_0 = 1, M_1 = 4, M_2 = 4$. Proof: Note that $\lim_{x \rightarrow 0^-} f(x) = -1 = f(0)$ so f is continuous, and

$$f'(x) = \begin{cases} 4x, & x \in (-1, 0), \\ \frac{4x}{(x^2 + 1)^2}, & x \in [0, \infty). \end{cases}$$

f' is also continuous, so

$$f''(x) = \begin{cases} 4, & x \in (-1, 0), \\ 4 \frac{1 - 3x^2}{(x^2 + 1)^2}, & x \in [0, \infty). \end{cases}$$

Therefore $f \in C^2((-1, \infty))$ and $M_0 = 1, M_1 = 4, M_2 = 4$.

8.7.3 G3) Suppose $f : (a, \infty) \rightarrow \mathbb{R}^n$ is twice differentiable, Let M_0, M_1, M_2 be the least upper bounds of $|f|, |f'|, |f''|$. Prove that $M_1^2 \leq 4M_0M_2$.

Proof: Use G1) and Cauchy-Schwarz inequality.

8.8 Problem S: Sturm-Liouville Theory

Assume the following uniqueness theorem holds:

Theorem: Suppose $a(t) \in C^1(\mathbb{R})$, $t_0 \in \mathbb{R}$. If $x(t), y(t) \in C^2(\mathbb{R})$ both satisfy the equation

$$x''(t) + a(t)x(t) = 0, \quad y''(t) + a(t)y(t) = 0,$$

and $(x(t_0), x'(t_0)) = (y(t_0), y'(t_0))$, then $x(t) \equiv y(t)$.

(Can be proved using Exercise C3?) For any $f : \mathbb{R} \rightarrow \mathbb{R}$, $t \geq 0$, denote

$$Z_t(f) = |\{x \in [0, t] : f(x) = 0\}|.$$

8.8.1 Part 1

Let $a(t), b(t) \in C^1(\mathbb{R})$ and for any $t \in \mathbb{R}$, $a(t) \leq b(t)$. Suppose $x(t), y(t) \in C^2(\mathbb{R})$ satisfy the following equation

$$x''(t) + a(t)x(t) = 0, \quad y''(t) + b(t)y(t) = 0.$$

Further assume that $x(t), y(t)$ are not identically zero.

8.8.1.1 S1) Assume $x(t_1) = 0$, if there exists $t > t_1$, such that $x(t) = 0$. Prove that there exists $t_2 > t_1$ such that $x(t_2) = 0$ and x has no roots in (t_1, t_2) . We call t_1, t_2 neighboring roots.

Proof: Consider the set $S = \{t > t_1 : x(t) = 0\}$, and let $t_2 = \inf S$. Note that $|x''(t)| \leq |a(t)| \cdot |x(t)|$, so by C3) $x'(t_1) \neq 0$. Assume $x'(t_1) > 0$, since $x \in C^2(\mathbb{R})$, there exists $\varepsilon > 0$ such that $x'(t) > 0$ for all $t \in (t_1, t_1 + \varepsilon)$, hence $x(t) > 0$ for all $t \in (t_1, t_1 + \varepsilon)$. Therefore $t_2 > t_1$, so by $x \in C(\mathbb{R})$, $x(t_2) = 0$ and x has no roots in (t_1, t_2) .

8.8.1.2 S2) If $t_2 > t_1$ are two neighboring roots of x , prove that y has a root in $(t_1, t_2]$.

Proof: Otherwise assume that x, y are positive on (t_1, t_2) , and $y(t_2) \neq 0$. Consider the function $h(t) = x'y - xy'$, then $h'(t) = (b - a)xy \geq 0$, so $h(t_2) \geq h(t_1) = x'(t_1)y(t_1) \geq 0$, but $h(t_2) = x'(t_2)y(t_2) < 0$, a contradiction.

8.8.1.3 S3) Prove that for any $t \geq 0$, $Z_t(y) \geq Z_t(x) - 1$.

Proof: Use S2).

8.8.1.4 S4) Suppose $t_2 > t_1$ and $x(t_1) = x'(t_2) = 0$, prove that

- If $y(t_1) = 0$, then there exists $t_3 \in [t_1, t_2]$, such that $y'(t_3) = 0$.

Proof: We can assume that $t_2 = \inf \{t > t_1 : x'(t) = 0\}$ ($t_2 > t_1$ since $x'(t_1) \neq 0$). If there is no such t_3 , we can further assume that $x'(t), y'(t), x(t), y(t) > 0$ for all $t \in (t_1, t_2)$. Again consider $h(t) = x'y - xy'$, then $h(t_1) = 0$, $h(t_2) = -x(t_2)y'(t_2) < 0$, but $h'(t) = (b - a)xy \geq 0$, leading to contradiction.

- If $y'(t_2) = 0$, then there exists $t_4 \in [t_1, t_2]$ such that $y(t_4) = 0$.

(The two theorems are similar.)

8.8.2 Part 2

In this section, $p(t) \in C^1(\mathbb{R})$ is a positive function. $x(t), y(t) \in C^2(\mathbb{R})$ are not identically zero and satisfy

$$x''(t) + p(t)x(t) = 0, \quad y''(t) + p(t)y(t) = 0.$$

8.8.2.1 S5) Prove that for any $t \geq 0$, $|Z_t(x) - Z_t(y)| \leq 1$.

Proof: Use S3).

8.8.2.2 S6) Prove that

- If t_1, t_2 are neighboring roots of x , then there exists a unique $t_3 \in [t_1, t_2]$ such that $x'(t_3) = 0$.

Proof: The existence of t_3 is given by Rolle's mean-value theorem. If there exists $t_3 < t_4 \in [t_1, t_2]$ such that $x'(t_3) = x'(t_4) = 0$, then $t_3, t_4 \neq t_1, t_2$ and there exists $t_5 \in [t_3, t_4]$ such that $x''(t_5) = 0$. Hence $x(t_5) = 0$, which contradicts the fact that t_1, t_2 are neighboring roots. Therefore t_3 is unique.

- If t'_1, t'_2 are neighboring roots of x' , then there exists a unique $t'_3 \in [t'_1, t'_2]$ such that $x(t'_3) = 0$.

Proof: Exactly the same.

8.8.2.3 S7) Prove that

- t_0 is a local maximum of $|x(t)|$ iff $x'(t_0) = 0$.

Proof: Trivial?

- t'_0 is a local maximum of $|x'(t)|$ iff $x(t'_0) = 0$.

8.8.3 Part 3

In this section, $p(t) \in C^1(\mathbb{R})$ is monotonically decreasing and $\lim_{t \rightarrow \infty} p(t) > 0$. Denote

$$p(\infty) := \lim_{t \rightarrow \infty} p(t).$$

$x(t) \in C^2(\mathbb{R})$ is not identically zero and

$$x''(t) + p(t)x(t) = 0.$$

8.8.3.1 *S8) Calculate

$$\lim_{t \rightarrow \infty} \frac{Z_t(x)}{t}.$$

Solution: By S5) we can ignore initial conditions. First consider the ODE $y''(t) + p(\infty)y(t) = 0$, where one solution is $y = \sin(t\sqrt{p(\infty)})$, so $\lim_{t \rightarrow \infty} Z_t(y)/t = \sqrt{p(\infty)}/\pi$. Since $p(t) \geq p(\infty)$, by S3) we know $\lim_{t \rightarrow \infty} Z_t(x)/t \geq \lim_{t \rightarrow \infty} Z_t(y)/t = \sqrt{p(\infty)}/\pi$. For any $\varepsilon > 0$, there exists $M > 0$ such that for any $t > M$, $p(t) < p(\infty) + \varepsilon$. By S3), $\lim_{t \rightarrow \infty} Z_t(x)/t \leq \sqrt{p(\infty) + \varepsilon}/\pi$. Therefore

$$\lim_{t \rightarrow \infty} \frac{Z_t(x)}{t} = \frac{\sqrt{p(\infty)}}{\pi}.$$

8.8.3.2 S9) Suppose $0 \leq t_1 < t_2 < t_3 < \dots$ are all the roots of $x(t)$ on $[0, \infty)$, $0 \leq t'_1 < t'_2 < \dots$ are all the roots of $x'(t)$ on $[0, \infty)$. Prove that the sequence $\{|x'(t_k)|\}_{k \geq 1}$ is monotonically decreasing and the sequence $\{|x(t'_k)|\}_{k \geq 1}$ is monotonically increasing, and

$$\lim_{k \rightarrow \infty} |x'(t_k)| = \sqrt{p(\infty)} \lim_{k \rightarrow \infty} |x(t'_k)|.$$

Proof: Consider the (energy) function $E(t) = p(t)x^2(t) + x'(t)^2$, then $E'(t) = p'x^2 \leq 0$ so E is monotonically decreasing. For $k \geq 1$, $E(t_k) = x'(t_k)^2$ is decreasing, so $\{|x'(t_k)|\}_{k \geq 1}$ is decreasing. Likewise, consider $F(t) = x(t)^2 + x'(t)^2/p(t)$, then $F'(t) = -p'(x'/p)^2 \geq 0$, so $F(t'_k) = x(t'_k)^2$ is increasing, and

$$\lim_{k \rightarrow \infty} |x'(t_k)| = \sqrt{\lim_{k \rightarrow \infty} E(t_k)} = \sqrt{p(\infty)} \lim_{k \rightarrow \infty} |x(t'_k)|.$$

8.8.3.3 *S10) Suppose $0 \leq \tilde{t}_1 < \tilde{t}_2 < \dots$ are all the roots of $x(t)x'(t)$ on $[0, \infty)$. Prove that the sequence $\{\tilde{t}_{k+1} - \tilde{t}_k\}_{k \geq 1}$ is monotonically increasing and calculate its limit.

Proof: By S6), the roots of x and x' appear alternating in $\{\tilde{t}_k\}$. Since t is a root of x iff t is a root of x'' , we only need to prove that if t_1, t_2 are neighboring roots of x , and $t_3 \in [t_1, t_2]$ satisfy $x'(t_3) = 0$, then $t_3 - t_1 \leq t_2 - t_3$. Same as before we can prove that, for $p(t), q(t), x(t), y(t)$ such that $p(0) = q(0), p(t) \leq q(t), x'(0) = y'(0) = 0, x(0) = y(0)$ and

$$x''(t) + p(t)x(t) = 0, y''(t) + q(t)y(t) = 0,$$

then the first roots a, b of x, y satisfy $a \leq b$. Since the sequence is increasing, by S8) we know that

$$\lim_{k \rightarrow \infty} \tilde{t}_{k+1} - \tilde{t}_k = \frac{1}{2} \lim_{t \rightarrow \infty} Z_t(x)/t = \sqrt{p(\infty)}/2\pi.$$

Chapter 9 Stone-Weierstrass Theorem

9.1 PSA

Assume $I = [a, b] \subset \mathbb{R}$, V is a normed linear space.

9.1.1 A1) $\sigma_1, \sigma_2 \in \mathcal{S}$ are two partitions. Prove that for any $\varepsilon > 0$, there exists a partition σ such that $\sigma \prec \sigma_1, \sigma \prec \sigma_2$ and $|\sigma| < \varepsilon$.

Proof: Take $n > 1/\varepsilon$, and let

$$\sigma = \sigma_1 \cup \sigma_2 \cup \left\{ \frac{k}{n}a + \frac{n-k}{n}b : 0 \leq k \leq n \right\}.$$

9.1.2 A2) Consider the space of simple functions $\mathcal{E}(I)$ with range V . Prove that it is a linear space on $\mathbb{R}$, and the integration operator $\int_a^b : \mathcal{E}(I) \rightarrow V$ is well-defined and is linear. Use this to define Riemann integrable functions with range V .

Proof: For any simple function $f = \sum_{i=1}^n c_i \mathbf{1}_{A_i}$ (where A_i are disjoint), let

$$\int_a^b f = \sum_{i=1}^n c_i \mu(A_i)$$

For any function $f : I \rightarrow V$, partition $\mathcal{C} = \{x_0, x_1, \dots, x_n\}$ and $\xi_i \in [x_{i-1}, x_i]$, define

$$\mathcal{R}(f; \mathcal{C}, \xi) = \sum_{k=0}^n f(\xi_k)(x_k - x_{k-1}).$$

Then f is Riemann integrable iff $\lim_{|\mathcal{C}| \rightarrow 0} \mathcal{R}(f; \mathcal{C}, \xi)$ exists.

9.1.3 A3) Suppose $f : I \rightarrow \mathbb{R}^n$ and f_i be the components of f , then $f \in \mathcal{R}(I)$ iff for every i , $f_i \in \mathcal{R}(I)$.

Proof: Note that

$$\max\{|x_k|\} \leq |(x_1, \dots, x_n)|_{\mathbb{R}^n} \leq |x_1| + \dots + |x_n|.$$

Hence the limit $|\underline{S}(f; \sigma) - \overline{S}(f; \sigma)| = 0$ iff the components of f are all Riemann integrable.

9.1.4 A4) Assume $a < c < b$, then for any $f \in \mathcal{R}(I)$, $f|_{[a,c]}$ and $f|_{[c,b]}$ are both Riemann integrable, and

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Proof: They are both obviously Riemann integrable, and for any partition σ , let $\sigma' = \sigma \cup \{c\} = \sigma_1 \cup \sigma_2$, where σ_1 and σ_2 are partitions of $[a, c]$ and $[c, b]$, then

$$\underline{S}(f; \sigma) \leq \underline{S}(f; \sigma') = \underline{S}(f|_{[a,c]}; \sigma_1) + \underline{S}(f|_{[c,b]}; \sigma_2),$$

and the other side is the same. Hence

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

9.1.5 A5) Prove that for any two partition σ and σ' , $\underline{S}(f; \sigma) \leq \overline{S}(f; \sigma')$. Use this to prove that if $f \in \mathcal{R}(I)$, then $\lim_{|\sigma| \rightarrow 0} |\underline{S}(f; \sigma) - \overline{S}(f; \sigma)| = 0$.

Proof: Let $\sigma'' = \sigma \cup \sigma'$, then

$$\underline{S}(f; \sigma) \leq \underline{S}(f; \sigma'') \leq \overline{S}(f; \sigma'') \leq \overline{S}(f; \sigma').$$

If $f \in \mathcal{R}(I)$, then $\sup_{\sigma} \underline{S}(f; \sigma) = \inf_{\sigma} \overline{S}(f; \sigma)$ hence

$$\lim_{|\sigma| \rightarrow 0} |\underline{S}(f; \sigma) - \overline{S}(f; \sigma)| = 0.$$

9.1.6 A6) $f \in \mathcal{R}(I)$. Prove that if we change the value of f at a finite number of points to g , then g is Riemann integrable and $\int_I g = \int_I f$.

Proof: We can assume that f and g differ only at the point $c \in (a, b)$. Let $M = \sup_{x \in I} |f(x)|$. For any $\varepsilon > 0$, and any partition σ , let $\sigma' = \sigma \cup \{c - \varepsilon, c + \varepsilon\}$, then $|\underline{S}(f; \sigma') - \underline{S}(f; \sigma)| \leq 4\varepsilon M \rightarrow 0$.

9.1.7 A7) $f \in C([a, b])$. Assume for any $x \in I$, $f(x) \geq 0$ and there exists $x_0 \in I$ such that $f(x_0) > 0$. Prove that $\int_a^b f > 0$.

Proof: Since f is continuous and $f(x_0) > 0$, there is an $\varepsilon > 0$ such that for all $y \in (x_0 - \varepsilon, x_0 + \varepsilon)$, $f(y) > 0$. Hence for any partition $\sigma = \{x_0, x_1, \dots, x_n\}$ such that $|\sigma| < \varepsilon/2$, there is a $k \in \{1, \dots, n\}$ such that $(x_{k-1}, x_k) \subset (x_0 - \varepsilon, x_0 + \varepsilon)$. Hence $\mathcal{R}(f; \sigma, \xi) > 0$ whenever $|\sigma| < \varepsilon/2$, so $\int_a^b f(x) dx > 0$.

9.1.8 A8) Suppose $f, g \in C^1(I)$, then

$$\int f' \cdot g = f \cdot g - \int f \cdot g'.$$

Proof:

$$d(f \cdot g) = df \cdot g + f \cdot dg.$$

9.1.9 A9) Suppose $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable, f is a continuous function, then

$$\int (f \circ \Phi) \Phi' = \int f.$$

Proof:

$$(f(\Phi(x)))' = f'(\Phi(x))\Phi'(x).$$

9.2 PSB: Calculating Integrals

9.2.1 (1)

$$\int \frac{x^5}{1+x} dx = \int x^4 - x^3 + x^2 - x + 1 - \frac{1}{1+x} dx \quad (9.1)$$

$$= \frac{x^5}{5} - \frac{x^4}{4} + \frac{x^3}{3} - \frac{x^2}{2} + x - \log(x+1) + C. \quad (9.2)$$

9.2.2 (2)

$$\int \sqrt{x} \sqrt{x} \sqrt{x} \, dx = \int x^{7/8} \, dx = \frac{8}{15} x^{15/8}.$$

9.2.3 (3)

$$\int \left(\frac{1+x}{1-x} + \frac{1-x}{1+x} \right) \, dx = \int \left(\frac{2}{1-x} + \frac{2}{1+x} - 2 \right) \, dx \quad (9.3)$$

$$= -2x + 2 \log \frac{1+x}{1-x} + C. \quad (9.4)$$

9.2.4 (4)

$$\int \frac{e^{3x} + 1}{1 + e^x} \, dx = \int 1 - e^x + e^{2x} \, dx = x - e^x + \frac{e^{2x}}{2} + C.$$

9.2.5 (5)

$$\int \sqrt{1 - \sin(2x)} \, dx = \int \sqrt{2} \sin \left(x - \frac{\pi}{4} \right) \, dx = -\sqrt{2} \cos \left(x - \frac{\pi}{4} \right) + C.$$

9.2.6 (6)

$$\int \frac{\cos(2x)}{\cos x - \sin x} \, dx = \int \cos x + \sin x \, dx = \sin x - \cos x + C.$$

9.2.7 (7)

$$\int \tan^2 x \, dx = -x + \tan x + C.$$

9.2.8 (8)

$$\int |x| \, dx = \frac{x|x|}{2} + C.$$

9.2.9 (9)

$$\int e^{-|x|} \, dx = -\operatorname{sgn}(x) e^{-|x|} + C.$$

9.2.10 (10)

$$\int \frac{x^2}{(1-x)^{2018}} \, dx = \frac{1}{2017(1-x)^{2017}} - \frac{1}{1013(1-x)^{2016}} + \frac{1}{2015(1-x)^{2015}}.$$

9.2.11 (11)

$$\int |x-1| \, dx = \frac{(x-1)|x-1|}{2} + C.$$

9.2.12 (12)

$$\int \frac{1}{\sqrt{b^2+x^2}} \, dx = \frac{1}{b} \log \frac{1 + \tan \frac{\arctan \frac{x}{b}}{2}}{1 - \tan \frac{\arctan \frac{x}{b}}{2}} + C.$$

9.2.13 (13)

Let $x = t^2$, then

$$\int \frac{dx}{\sqrt{x}(1+x)} = 2 \arctan \sqrt{x} + C.$$

9.2.14 (14)

$$\int \frac{x^4}{(1-x^5)^4} \, dx = \frac{1}{5} \int \frac{dx^5}{(1-x^5)^4} = \frac{1}{15(1-x^5)^3} + C.$$

9.2.15 (15)

$$\int \left(\frac{1}{\sqrt{3-x^2}} + \frac{1}{1-3x^2} \right) dx = \arcsin \frac{x}{\sqrt{3}} + \frac{1}{2\sqrt{3}} \log \frac{1+\sqrt{3}x}{1-\sqrt{3}x} + C.$$

9.2.16 (16)

$$\int \frac{2x-3}{x^2-3x+8} \, dx = \log(x^2-3x+8) + C.$$

9.2.17 (17)

$$\int \frac{dx}{\sin^2(2x + \frac{\pi}{4})} = \frac{\tan(2x - \pi/4)}{2} + C.$$

9.2.18 (18)

$$\int \frac{dx}{1+\cos x} = \tan \frac{x}{2} + C.$$

9.2.19 (19)

$$\int \frac{1}{x^2} \sin \frac{1}{x} \, dx = \cos \frac{1}{x} + C.$$

9.2.20 (20)

$$\int \cos^5 x \, dx = \frac{\sin^5 x}{5} - \frac{2 \sin^3 x}{3} + \sin x + C.$$

9.2.21 (21)

$$\int \cos(ax) \sin(bx) \, dx = \frac{\cos(a-b)x}{2(a-b)} - \frac{\cos(a+b)x}{2(a+b)} + C.$$

9.2.22 (22)

$$\int \frac{dx}{a \cos x + b \sin x} = \frac{2}{\sqrt{a^2 + b^2}} \tanh^{-1} \frac{a \tan(x/2) - b}{\sqrt{a^2 + b^2}} + C.$$

9.2.23 (23)

$$\int \frac{\sin(2x)}{a^2 \cos^2 x + b^2 \sin^2 x} \, dx = \frac{\log((b^2 - a^2) \sin^2 x + a^2)}{b^2 - a^2} + C.$$

9.2.24 (24)

$$\int \frac{dx}{2 - \sin^2 x} = \frac{1}{\sqrt{2}} \arctan \left(\frac{\tan x}{\sqrt{2}} \right) + C.$$

9.2.25 (25)

$$\int \frac{dx}{x \ln x \ln \ln x} = \ln \ln \ln x + C.$$

9.2.26 (26)

$$\int \frac{\log x}{x \sqrt{1 + \log x}} \, dx = \frac{2}{3} (1 + \log x)^{3/2} - 2 \sqrt{1 + \log x} + C.$$

9.2.27 (27)

$$\int \frac{\cos x + \sin x}{(\sin x - \cos x)^{1/3}} \, dx = \frac{3}{2} (\sin x - \cos x)^{2/3} + C.$$

9.2.28 (28)

$$\int e^{\sqrt{x}} \, dx = 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C.$$

9.2.29 (29)

$$\int \frac{x^{n/2}}{1 + x^{n+2}} \, dx = \frac{2}{n+2} \arctan x^{n/2+1} + C.$$

9.2.30 (30)

$$\int \frac{\sqrt{x}}{1-x^{1/3}} dx = 6 \arctan x^{1/6} - \frac{6}{5}x^{5/6} - \frac{6}{7}x^{7/6} - 2x^{1/2} - 6x^{1/6} + C.$$

9.2.31 (31)

$$\int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2 \sqrt{a^2 + x^2}} + C.$$

9.2.32 (32)

$$\int \frac{dx}{\cos^4 x} = \frac{\sin x}{2 \cos^3 x} + \frac{\sin(3x)}{6 \cos^3 x} + C.$$

9.2.33 (33)

$$\int \arcsin^2 x dx = x \arcsin^2 x + 2\sqrt{1-x^2} \arcsin x - 2x + C.$$

9.2.34 (34)

$$\int x \arcsin x dx = \frac{x\sqrt{1-x^2}}{4} - \frac{1}{4} \arcsin x(1-2x^2) + C.$$

9.2.35 (35)

$$\int x \arctan x = \frac{1}{2}(x^2 + 1) \arctan x - \frac{1}{2}x + C.$$

9.2.36 (36)

$$\int \frac{\arctan x}{x^2} = \log x - \frac{\arctan x}{x} - \frac{1}{2} \log(1+x^2) + C.$$

9.2.37 (37)

$$\int x^2 \sin x = -x^2 \cos x + 2x \sin x + 2 \cos x + C.$$

9.2.38 (38)

$$\int \frac{x}{\cos^2 x} = x \tan x + \log \cos x + C.$$

9.2.39 (39)

$$\int \log(x + \sqrt{1+x^2}) = x \log(x + \sqrt{1+x^2}) - \sqrt{1+x^2} + C.$$

9.2.40 (40)

$$\int \sin \log x = \frac{x}{2}(\sin \log x - \cos \log x) + C.$$

9.2.41 (41)

$$\int \sqrt{x^2 + a^2} = \frac{1}{2}x\sqrt{a^2 + x^2} + \frac{a^2}{4} \log \frac{x + \sqrt{a^2 + x^2}}{\sqrt{a^2 + x^2} - x^2} + C.$$

9.2.42 (42)

$$\int \frac{x^2}{\sqrt{x^2 - a^2}} = \frac{1}{2}x\sqrt{x^2 - a^2} + \frac{a^2}{4} \log \frac{x + \sqrt{x^2 - a^2}}{x - \sqrt{x^2 - a^2}} + C.$$

9.2.43 (43)

$$\int \frac{x \log(x + \sqrt{1 + x^2})}{\sqrt{1 + x^2}} = \sqrt{x^2 + 1} \log(x + \sqrt{1 + x^2}) - x + C.$$

Let $u = \sqrt{x^2 + 1} + x$ then $du/dx = 1 + x/\sqrt{1 + x^2}$, so it becomes

$$\int \frac{(u^2 - 1) \log u}{2u^2} dx = -\frac{u}{2} + \frac{1}{2u} + \frac{1}{2}u \log u + \frac{\log u}{2u} + C.$$

9.2.44 (44)

$$\int \frac{1}{\sqrt{x^2 + a^2}} = \log \frac{\sin t + \cos t}{\sin t - \cos t} + C = \tanh^{-1} \frac{x}{\sqrt{x^2 + a^2}} + C.$$

where $t = \frac{1}{2} \arctan(x/a)$.

9.2.45 (45)

$$\int \frac{xe^x}{(1+x)^2} = \frac{e^x}{1+x} + C.$$

9.2.46 (46)

$$\int \arctan(1 + \sqrt{x}) = x \arctan(1 + \sqrt{x}) - \sqrt{x} + \log(2 + 2\sqrt{x} + x) + C.$$

9.2.47 (47)

$$\int \left(1 - \frac{2}{x}\right)^2 e^x = e^x - \frac{4e^x}{x} + C.$$

since $\int e^x/x^2 dx = -e^x/x + \int e^x/x dx$.

9.2.48 (48)

$$\int \sqrt{2 + \tan^2 x} = \theta + \log \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} + C.$$

where $\theta = \arcsin(\sin x / \sqrt{2})$.

9.2.49 (49)

$$\int \frac{1}{1+x^3} = -\frac{1}{6} \log(x^2 - x + 1) + \frac{1}{3} \log(x + 1) + \frac{1}{\sqrt{3}} \arctan \frac{2x-1}{\sqrt{3}} + C.$$

9.2.50 (50)

$$\int \frac{x^7}{x^4 + 2} = \frac{x^4}{4} - \frac{1}{2} \log(2 + x^4).$$

9.2.51 (51)

$$\int \frac{2x^2 + 1}{(x+3)(x-1)(x-4)} = -\frac{1}{4} \log(1-x) + \frac{11}{7} \log(4-x) + \frac{19}{28} \log(x+3) + C.$$

9.2.52 (52)

$$\int \frac{1+x^2}{1+x^4} = \frac{1}{\sqrt{2}} (\arctan(\sqrt{2}x+1) - \arctan(1-\sqrt{2}x)) + C.$$

Note that

$$\frac{1+x^2}{1+x^4} = \frac{1}{2(x^2 + \sqrt{2}x + 1)} - \frac{1}{2(-x^2 + \sqrt{2}x + 1)}.$$

9.2.53 (53)

Let $x = y^6 - 1$ then

$$\int \frac{x}{\sqrt{x+1} + (x+1)^{1/3}} = \int (y^3 - 1)(1 - y + y^2)6y^3 dy \quad (9.5)$$

$$= \frac{2x\sqrt{x+1}}{3} - \frac{3x(x+1)^{1/3}}{4} + \frac{6x(x+1)^{1/6}}{7} - x + \frac{6}{5}(x+1)^{5/6} \quad (9.6)$$

$$- \frac{3}{2}(x+1)^{2/3} + \frac{2\sqrt{x+1}}{3} - \frac{3(x+1)^{1/3}}{4} + \frac{6(x+1)^{1/6}}{7} + C. \quad (9.7)$$

9.2.54 (54)

Let $x = y^2$, then

$$\int \frac{1}{\sqrt{x+x^2}} = \int \frac{dy}{\sqrt{y^2+1}} = \tanh^{-1} \left(\sqrt{\frac{x}{x+1}} \right) + C.$$

9.2.55 (55)

The Poisson kernel

$$\int \frac{1-r^2}{1-2r \cos x + r^2} = 2 \arctan \left(\frac{1+r}{1-r} \tan \frac{x}{2} \right) + C.$$

9.2.56 (56)

Let $x = \tan \theta$ then

$$\int \frac{1}{x\sqrt{1+x^2}} = \int \frac{d\theta}{\sin \theta} = \log \tan \frac{\arctan x}{2} + C.$$

9.2.57 (57)

Let $t = \tan x/2$ then

$$\int \frac{1}{5-3\cos x} = \frac{1}{2} \arctan \left(2 \tan \frac{x}{2} \right) + C.$$

9.2.58 (58)

Let $t = \tan x$, then

$$\int \frac{1}{2+\sin^2 x} = \frac{1}{\sqrt{6}} \arctan \left(\sqrt{\frac{3}{2}} \tan x \right) + C.$$

9.2.59 (59)

$$\int \frac{\sin^3 x}{\cos^4 x} = \frac{1}{3 \cos^3 x} - \frac{1}{\cos x} + C.$$

9.2.60 (60)

Let $t = \cos x$ then

$$\int \frac{1}{\sin x \cos^4 x} = - \int t^{-4} + t^{-2} + \frac{1}{1-t^2} = \frac{1}{3 \cos^3 x} + \frac{1}{\cos x} + \frac{1}{2} \log \frac{1+\cos x}{1-\cos x} + C.$$

9.3 Problem W: Stone-Weierstraß Theorem**9.3.1 Part 1: Approximating $|x|$**

9.3.1.1 W1) (Dini) Suppose $K \subset \mathbb{R}^n$ is compact, $f_n : K \rightarrow \mathbb{R}$ is a sequence of continuous functions, which converges point-wise to $f : K \rightarrow \mathbb{R}$. If f is continuous and $f_n \leq f_{n+1}$, then f_n converges uniformly to f .

Proof: For any $\varepsilon > 0$, and any $x \in K$, there is an integer $n_x > 0$ such that $|f_{n_x}(x) - f(x)| < \varepsilon/4$. There exists $\delta > 0$, such that $\forall y \in B(x, \delta) \cap K$, $|f(x) - f(y)| < \varepsilon/4$ and $|f_{n_x}(x) - f_{n_x}(y)| < \varepsilon/4$, then $|f_{n_x}(y) - f(y)| < \varepsilon/4$. Note that $K \subset \bigcup_{x \in K} B(x, \delta_x)$ hence we can choose a finite set of $x_1, x_2, \dots, x_N$ such that $K \subset \bigcup_{i=1}^N B(x_i, \delta_{x_i})$. Let $M = \max\{n_{x_i} : i = 1, 2, \dots, N\}$ then for any $m \geq M$ and $x \in K$, $|f_m(x) - f(x)| < \varepsilon$. Hence f_n converges uniformly to f .

9.3.1.2 W2) Consider the interval $[-1, 1]$. Define inductively a sequence of polynomials:

$$P_0(x) = 0, P_{n+1}(x) = P_n(x) + \frac{1}{2}(x^2 - P_n^2(x)).$$

Prove that for any n, x , $0 \leq P_n(x) \leq P_{n+1}(x) \leq |x|$. Proof: Assume $x > 0$, we prove by induction. If $t = P_n(x) \in [0, x]$, then

$$P_{n+1}(x) = \frac{1}{2}x^2 - \frac{1}{2}(t-1)^2 + \frac{1}{2} \leq \frac{1}{2}(x^2 - (1-x)^2 + 1) = x,$$

and $P_{n+1}(x) \geq P_n(x) = t$, hence $P_{n+1}(x) \in [0, x]$.

9.3.1.3 W3) Prove that $|x|$ can be uniformly approximated by polynomials on the interval $[-1, 1]$, i.e. for any $\varepsilon > 0$, there exists a polynomial $P_\varepsilon(x)$ such that

$$\sup_{x \in [-1, 1]} ||x| - P_\varepsilon(x)| < \varepsilon.$$

Proof: By W2), the sequence of polynomials $\{P_n\}$ converge point-wise to $|x|$, hence by W1) P_n converge uniformly to $|x|$.

9.3.2 Part 3: Bernstein Polynomial

Assume $I = [0, 1]$, and n is an integer.

9.3.2.1 W4) For any $0 \leq k \leq n$, define $p_{n,k}(x) = \binom{n}{k} x^k (1-x)^{n-k}$. Prove that

$$\sum_{0 \leq k \leq n} p_{n,k}(x) \left(x - \frac{k}{n}\right)^2 = \frac{x(1-x)}{n}.$$

Proof: Note that

$$\begin{aligned} \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} x^2 &= x^2, \\ \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} \frac{k}{n} &= \sum_{k=1}^n \binom{n-1}{k-1} x^k (1-x)^{n-k} = x, \\ \sum_{k=0}^n p_{n,k}(x) k(k-1) &= n(n-1) \sum_{k=2}^n \binom{n-2}{k-2} x^k (1-x)^{n-k} = n(n-1)x^2. \end{aligned} \tag{9.8}$$

Therefore

$$\sum_{k=0}^n p_{n,k}(x) \left(x - \frac{k}{n}\right)^2 = \frac{x(1-x)}{n}.$$

9.3.2.2 W5) For any $f \in C([0, 1])$, define

$$B_{f,n} = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

For $x \in [0, 1]$, prove that

$$|f(x) - B_{f,n}(x)| \leq \sum_{k=0}^n \left|f(x) - f\left(\frac{k}{n}\right)\right| p_{n,k}(x).$$

Proof: Note that

$$\sum_{k=0}^n f(x) \binom{n}{k} x^k (1-x)^{n-k} = f(x).$$

9.3.2.3 W6) For any $f \in C([0, 1])$, prove that for any $\varepsilon > 0$, there exists n such that

$$\|f - B_{f,n}\|_\infty < \varepsilon.$$

Proof: Let

$$I = \sum_{|m-nx| < n^{3/4}} \left(f(x) - f\left(\frac{m}{n}\right)\right) p_{n,m}(x), \tag{9.9}$$

$$II = \sum_{|m-nx| > n^{3/4}} \left(f(x) - f\left(\frac{m}{n}\right)\right) p_{n,m}(x). \tag{9.10}$$

Then $|f - B_{f,n}| \leq |I| + |II|$. For any $\varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall x \in [0, 1]$, $n \geq N \implies |I| < \varepsilon$, since

$$|I| \leq \sup_{|x - m/n| < n^{-1/4}} |f(x) - f(m/n)| \rightarrow 0.$$

Suppose $M = \sup_{x \in [0, 1]} |f(x)|$, then

$$|II| \leq 2M \sum_{|m - nx| > n^{3/4}} p_{n,m}(x) \leq 2M \sqrt{n} \sum_{m=0}^n (x - m/n)^2 p_{n,m}(x) = \frac{2Mx(1-x)}{\sqrt{n}}.$$

Hence $\|f - B_{f,n}\|_{\infty} \rightarrow 0$.

9.3.3 Part 3: Stone-Weierstrass Theorem

9.3.3.1 W7-14):

Let X be a compact Hausdorff space, $\mathcal{A} \subset C(X, \mathbb{R})$ satisfy the following properties: (a) $\forall c \in \mathbb{R}$, $c \cdot 1_X \in \mathcal{A}$, (b) $\forall f, g \in \mathcal{A}$, $f + g, f - g, fg \in \mathcal{A}$, (c) $\mathcal{A}$ can separate any pair of points in X . Then $\bar{\mathcal{A}} = C(X, \mathbb{R})$.

9.3.3.2 Lemma 1

There is a list of polynomials $\{P_n(x)\}$ that converges uniformly to $|x|$ on $[-1, 1]$.

9.3.3.3 Lemma 2

If $\mathcal{A}$ is a subspace of $C(X, \mathbb{R})$, such that (a) $\mathcal{A}$ is a lattice, (b) $1_X \in \mathcal{A}$, and (c) $\mathcal{A}$ can separate any pair of points, then $\bar{\mathcal{A}} = C(X, \mathbb{R})$.

9.3.3.4 Proof of main theorem

Assume WLOG $\mathcal{A}$ is closed, then by Lemma 1, $\forall f \in \mathcal{A}$, $P_n(f) \in \mathcal{A}$, hence $|f| \in \mathcal{A}$. (Since X is compact, $|f|$ is bounded.) Note that

$$\max\{f, g\} = \frac{1}{2}(|f + g| + |f - g|), \min\{f, g\} = \frac{1}{2}(|f + g| - |f - g|).$$

Hence $\mathcal{A}$ is a lattice, by Lemma 2 $\bar{\mathcal{A}} = C(X, \mathbb{R})$.

9.3.3.5 Proof of Lemma 1

Proof 1: Let

$$Q_n(x) = \int_0^x (1 - t^2)^n dt / \int_0^1 (1 - t^2)^n dt. \quad (9.11)$$

$$P_n(x) = \int_0^x Q_n(t) dt. \quad (9.12)$$

Note that

$$\int_{\varepsilon}^1 (1 - t^2)^n dt \leq (1 - \varepsilon^2)^n (1 - \varepsilon) \rightarrow 0$$

Hence (combined with Wallis's formula), $P_n(x)$ converges uniformly to $|x|$ on $[a, b]$. Proof 2: WLOG change the interval to $[-1/2, 1/2]$. The series

$$(1 - t)^{1/2} = 1 + \sum_{n=1}^{\infty} (-t)^n \binom{\frac{1}{2}}{n} = 1 - \sum_{n=1}^{\infty} c_n t^n.$$

converges when $|t| < 1$. Hence $\forall \varepsilon > 0$, there exists $Q \in \mathbb{R}[x]$ such that $\sup_{|t| \leq 1/2} |Q(t) - (1 - t)^{1/2}| < \varepsilon/2$. Let $t = 1 - x^2$, then $|Q(1 - x^2) - |x|| < \varepsilon/2$, so $P(x) = Q(1 - x^2) - Q(1)$ converges to $|x|$ uniformly on $[-1/2, 1/2]$.

9.3.3.6 Proof of Lemme 2

Step 1: Take any $f \in C(X, \mathbb{R})$, and any $x, y \in X$, we can find $g_{xy} \in \mathcal{A}$, such that $g_{xy}(x) = f(x), g_{xy}(y) = f(y)$. Since there exists $u \in \mathcal{A}$ such that $u(x) \neq u(y)$,

$$\begin{pmatrix} u(x), 1 \\ u(y), 1 \end{pmatrix} \begin{pmatrix} \lambda \\ \mu \end{pmatrix} = \begin{pmatrix} f(x) \\ f(y) \end{pmatrix}$$

has a solution. (If $x = y$ it is trivial.) Step 2: For all $\varepsilon > 0$, $x, y \in X$, there is an open neighborhood $O_{x,y}$ of y , such that $\forall z \in O_{x,y}, f(z) - g_{xy}(z) \leq \varepsilon$. Note that $\bigcup_{y \in X} O_{x,y} = X$, so by X is compact, there is a list $y_1, \dots, y_N$ such that $\bigcup_{k \leq N} O_{x,y_k} = X$. Let $h_x = \max \{g_{xy_k} : k \leq N\}$, then $h_x(y) - f(y) \geq -\varepsilon$, and $f(x) = h_x(x)$. Step 3: For all $x \in X$, there is an open neighborhood G_x of x , such that $\forall z \in G_x, h_x(z) - f(z) \leq \varepsilon$. Note that $\bigcup_{x \in X} G_x = X$, so by X is compact, there is a list $x_1, \dots, x_M$ such that $\bigcup_{k \leq M} G_{x_k} = X$. Let $F = \min \{h_{x_k} : k \leq M\}$, then $|F(x) - f(x)| \leq \varepsilon, \forall x \in X$. Therefore $\bar{\mathcal{A}} = C(X, \mathbb{R})$.

For complex numbers, there is an additional requirement: for any $f \in \mathcal{A}, \bar{f} \in \mathcal{A}$.

9.3.3.7 W15-16):

It is easy to see that polynomials and trigonometric polynomials both satisfy the requirements of the theorem.

Chapter 10 Irrationality of Pi

10.1 PSA

10.1.1 A1) Construct continuous functions $f_n, f \in C([0, 1])$, such that for every $x \in [0, 1]$, when $n \rightarrow \infty, f_n(x) \rightarrow f(x)$, but

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx \neq \int_0^1 f(x) dx.$$

Solution: Let $f_n(x) = nxe^{-nx^2}$, $f(x) = \lim_{n \rightarrow \infty} f_n(x) = 0$. Then

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \lim_{n \rightarrow \infty} \frac{n}{2} \int_0^1 e^{-nx^2} d(x^2) = \lim_{n \rightarrow \infty} \frac{n}{2} \left(\frac{1}{n} - \frac{1}{ne^n} \right) = \frac{1}{2}$$

Hence $\lim_{n \rightarrow \infty} \int_0^1 f_n = 1/2 \neq 0 = \int_0^1 f$.

10.1.2 A2) $\alpha \in \mathbb{R}_{\geq 0}$. Prove that $\int_{100}^{\infty} \frac{dx}{x \log^{\alpha}(x)}$ converges iff $\alpha > 1$.

Proof: Substitute $y = \log x$, then

$$\int_{100}^{\infty} \frac{dx}{x \log^{\alpha}(x)} = \int_{\log 100}^{\infty} \frac{dy}{y^{\alpha}}$$

which converges iff $\alpha > 1$.

10.1.3 A3) f, F are defined on I , and for every bounded closed interval $J \subset I$, f, F are both Riemann integrable on J . Assume for all $x \in I$, $|f(x)| \leq F(x)$. Then if the improper integral of F on I converges, so does f .

Proof: This is because

$$\int_I f(x) dx \text{ converges} \iff \forall \varepsilon > 0 \exists N \forall u, v \in I, N < u < v, \left| \int_u^v f(x) dx \right| < \varepsilon.$$

10.1.4 A4) Prove the integrals below converge:

(1) $\int_0^{\infty} e^{-x^2} dx$ (2) $\int_0^1 \frac{dx}{\sqrt{1-x^3}}$ (3) $\int_1^{\infty} \frac{(\log x)^2}{1+x(\log x)^5} dx$ (1):

$$\int_0^{\infty} e^{-x^2} dx \leq 1 + \int_1^{\infty} e^{-x} dx \leq 1 + \frac{1}{e}.$$

(2):

$$\int_0^1 \frac{dx}{\sqrt{1-x^3}} \leq \int_0^1 \frac{dx}{\sqrt{1-x}} = 2.$$

(3):

$$\int_1^{\infty} \frac{(\log x)^2}{1+x(\log x)^5} dx \leq 5000 + \int_{100}^{\infty} \frac{1}{x(\log x)^3} dx, \text{ which converges by A2.}$$

10.1.5 A5) Prove the series below converge:

(1) $\sum_{n=1}^{\infty} e^{-n}(n^2 + \log n)$ (2) $\sum_{n=1}^{\infty} \frac{\log n}{1+n(\log n)^3}$ (1):

$$\sum_{n=1}^{\infty} e^{-n}(n^2 + \log n) \leq \sum_{n=1}^{\infty} \frac{2n^2}{e^n} \leq 2 \int_0^{\infty} x^2 e^{-x} dx = 4.$$

(2):

$$\sum_{n=1}^{\infty} \frac{\log n}{1 + n(\log n)^3} \leq \sum_{n=2}^{\infty} \frac{1}{n(\log n)^2} \leq \frac{1}{2(\log 2)^2} + \int_2^{\infty} \frac{1}{x(\log x)^2} dx \leq 3.$$

10.1.6 A6) Calculate

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^\alpha}{n^{\alpha+1}}, \alpha > -1.$$

Solution:

$$\sum_{k=1}^n k^\alpha \leq \int_1^{n+1} x^\alpha dx = \frac{1}{\alpha+1} ((n+1)^{\alpha+1} - 1). \quad (10.1)$$

$$\sum_{k=1}^n k^\alpha \geq 1 + \int_1^n x^\alpha dx = 1 + \frac{1}{\alpha+1} n^{\alpha+1}. \quad (10.2)$$

Therefore

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^\alpha}{n^{\alpha+1}} = \frac{1}{\alpha+1}.$$

10.1.7 A7) Calculate $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$, to show that $\pi = 3.14 \dots$

Solution:

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \int_0^1 (x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}) dx = \frac{22}{7} - \pi. \quad (10.3)$$

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx \leq \int_0^1 \frac{x^3(1-x)^4}{2} dx = \frac{1}{560} < 0.02, \frac{22}{7} > 3.1428. \quad (10.4)$$

10.1.8 A8) Assume $a, b, n \in \mathbb{Z}$, let

$$f_{a,b;n} = \frac{x^n(a-bx)^n}{n!}.$$

- Prove that for $k = 0, 1, \dots, 2n$, $f_{a,b;n}^{(k)}(x) \in \mathbb{Z}$ when $x = 0, \frac{a}{b}$.

See B10)

- If $\pi = \frac{a}{b} \in \mathbb{Q}$, then for every $n \in \mathbb{N}$,

$$\int_0^\pi f_{a,b;n}(x) \sin x dx$$

is an integer. Proof: By Darboux's formula of integration of parts

$$\int_0^\pi f_{a,b;n}(x) \sin x dx = \sum_{k=0}^{2n} f_{a,b;n}^{(k)}(x) \sin \left(x - \frac{(k+1)\pi}{2} \right) \Big|_0^\pi \in \mathbb{Z}.$$

- Prove that $\pi \notin \mathbb{Q}$.

Proof: Let $n = 2a^4 + 10$, then $\forall 0 \leq x \leq a/b$,

$$f_{a,b;n} \leq \frac{a^{2n}}{n!} < \frac{1}{2} \frac{(a^4)^{n/2}}{n \cdot (n-1) \cdots \left(\frac{n}{2}\right)} < \frac{1}{2}.$$

Hence

$$0 < \int_0^\pi f_{a,b;n}(x) \sin x dx < \frac{1}{2} \int_0^\pi \sin x dx = 1,$$

leading to contradiction.

10.1.9 A9) Let $I_n = \int_0^{\pi/2} \sin^n x \, dx$, prove that $I_n \sim \sqrt{\frac{\pi}{2n}}$.

Proof: Since $I_n = \frac{n-1}{n} I_{n-2}$,

$$I_n = \begin{cases} \frac{(n-1)!!}{n!!}, & n \text{ is even,} \\ \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2}, & n \text{ is odd.} \end{cases}$$

Combined with $I_{2n+1} < I_{2n} < I_{2n-1}$, we get

$$\left[\frac{(2n)!!}{(2n-1)!!} \right]^2 \frac{1}{2n+1} < \frac{\pi}{2} < \left[\frac{(2n)!!}{(2n-1)!!} \right]^2 \frac{1}{2n},$$

where

$$0 < - \left[\frac{(2n)!!}{(2n-1)!!} \right]^2 \frac{1}{2n+1} + \left[\frac{(2n)!!}{(2n-1)!!} \right]^2 \frac{1}{2n} = \left[\frac{(2n)!!}{(2n-1)!!} \right]^2 \frac{1}{2n(2n+1)} < \frac{\pi}{4n}.$$

Therefore

$$\lim_{n \rightarrow \infty} \left[\frac{(2n)!!}{(2n-1)!!} \right]^2 \frac{1}{2n} = \frac{\pi}{2}.$$

Hence $I_n \sim \sqrt{\frac{\pi}{2n}}$.

10.1.10 A10) Assume $f : [0, 1] \rightarrow [0, 1]$ is monotonously increasing, $g = f^{-1} : [0, 1] \rightarrow [0, 1]$ is its inverse, and f, g are both continuously differentiable, then

$$\int_0^1 f(x) \, dx + \int_0^1 g(x) \, dx = 1.$$

Proof: We show that

$$\int_0^x f(t) \, dt + \int_0^{f(x)} g(t) \, dt = xf(x), \forall 0 \leq x \leq 1. (1)$$

$x = 0$ is trivial, hence it suffices to show that the derivatives of the two sides match.

$$\frac{d}{dx} \int_0^x f(t) \, dt = f(x), \frac{d}{dx} \int_0^{f(x)} g(t) \, dt = f'(x) \cdot g(f(x)) = xf'(x).$$

Hence (1) holds.

10.1.11 A11) Prove that

$$\lim_{\varepsilon \rightarrow 0} \sum_{k=0}^{\infty} \frac{(-1)^k (1-\varepsilon)^{2k+1}}{2k+1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}.$$

Proof: By Dirichlet's test, $\sum_{k=0}^{\infty} (-1)^k (1-\varepsilon)^{2k+1} / (2k+1)$ converges uniformly. Hence for any $\delta > 0$, there exists an $N \in \mathbb{Z}$ such that

$$\left| \sum_{k=N}^{\infty} \frac{(-1)^k x^{2k}}{2k+1} \right| < \delta, \forall x \in [0, 1].$$

Then $\forall \varepsilon < \frac{\delta}{N}$,

$$\left| \sum_{k=0}^{\infty} \frac{(-1)^k (1-\varepsilon)^{2k+1}}{2k+1} - \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} \right| \tag{10.5}$$

$$\leq \sum_{k=0}^{N-1} \frac{|(1-\varepsilon)^{2k+1} - 1|}{2k+1} + \left| \sum_{k=N}^{\infty} \frac{(-1)^k (1-\varepsilon)^{2k+1}}{2k+1} \right| + \left| \sum_{k=N}^{\infty} \frac{(-1)^k}{2k+1} \right| < 3\delta. \tag{10.6}$$

Hence

$$\lim_{\varepsilon \rightarrow 0} \sum_{k=0}^{\infty} \frac{(-1)^k (1-\varepsilon)^{2k+1}}{2k+1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}.$$

10.1.12 A12) For any continuous function $f : [a, b] \times [c, d] \rightarrow \mathbb{R}$, $(x, y) \mapsto f(x, y)$, where $a, b, c, d \in \mathbb{R}$, show that f is uniformly continuous on $[a, b] \times [c, d]$.

Proof: $K = [a, b] \times [c, d]$ is a compact set. Consider an arbitrary $\varepsilon > 0$. For any $x \in K$, there is an open ball $B(x, 2r_x)$ with center x such that $\forall y \in B(x, 2r_x)$, $|f(x) - f(y)| < \varepsilon/2$. Let $O_x = B(x, r_x)$. Note that $\bigcup_{x \in K} O_x = K$ and K is compact, hence we can find $x_1, \dots, x_n$ such that $\bigcup_{k=1}^n O_{x_k} = K$. Let $\delta = \min\{r_{x_k} : k \leq n\}$, then $\forall |u - v| < \delta$, suppose $u \in O_{x_1}$, then

$$|v - x_1| \leq |v - u| + |u - x_1| < 2r_{x_1} \implies v \in B(x_1, 2r_{x_1}).$$

Hence

$$|f(u) - f(v)| \leq |f(u) - f(x_1)| + |f(v) - f(x_1)| < \varepsilon.$$

Therefore f is uniformly continuous on K .

10.2 PSB: On $\zeta(2)$

10.2.1 Part 1: The sequence $\{\sum_{k=1}^n 1/k^p\}$

Define the sequence $S_n(p) = \sum_{k=1}^n 1/k^p$ where $p \in \mathbb{Z}_{\geq 1}$.

10.2.1.1 B1) Prove that for any $k \in \mathbb{Z}_{\geq 1}$, we have

$$\frac{1}{(k+1)^p} \leq \int_k^{k+1} \frac{1}{x^p} dx \leq \frac{1}{k^p}.$$

Proof: $\frac{1}{(k+1)^p} \leq \frac{1}{x^p} \leq \frac{1}{k^p}, \forall k \leq x \leq k+1$.

10.2.1.2 B2) Prove that for any $n \in \mathbb{Z}_{\geq 2}$, we have

$$S_n(p) - 1 \leq \int_1^n \frac{1}{x^p} dx \leq S_{n-1}(p).$$

Proof:

$$S_n(p) - 1 = \sum_{k=1}^{n-1} \frac{1}{(k+1)^p} \leq \sum_{k=1}^{n-1} \int_k^{k+1} \frac{1}{x^p} dx = \int_1^n \frac{1}{x^p} dx.$$

Likewise we have $\int_1^n \frac{1}{x^p} dx \leq S_{n-1}(p)$.

10.2.1.3 B3) Let $p \in \mathbb{Z}_{\geq 1}$. Prove that $x \mapsto \frac{1}{x^p}$ is integrable on $[1, \infty)$ iff $p \geq 2$.

Proof: For $p \geq 2$,

$$\lim_{n \rightarrow \infty} \int_1^n \frac{1}{x^p} dx = \lim_{n \rightarrow \infty} \frac{1}{1-p} x^{1-p} \Big|_1^n = \frac{1}{1-p}.$$

If $p = 1$, $\lim_{n \rightarrow \infty} \int_1^n \frac{1}{x} dx = \lim_{n \rightarrow \infty} \log x \Big|_1^n = \infty$.

10.2.1.4 B4) Prove that $\{S_n(p)\}_{n \geq 1}$ converges iff $p \geq 2$. For $p \geq 2$ let

$$\zeta(p) = \lim_{n \rightarrow \infty} S_n(p) = \sum_{k=1}^{\infty} \frac{1}{k^p}.$$

Proof: If $p = 1$, $S_n(p) \geq \int_1^{n+1} \frac{1}{x} dx \rightarrow \infty$. For $p \geq 2$, $S_n(p) \leq S_{n+1}(p)$, and $S_n(p) \leq 1 + \int_1^n \frac{1}{x^p} dx \leq 1 + \int_1^{\infty} \frac{1}{x^p} dx$. Hence $\lim_{n \rightarrow \infty} S_n(p)$ exists.

10.2.2 Part 2: Calculate $\zeta(2)$

(We can also use Bernolli numbers and the Taylor expansion of $\tan x$).

Let $h(t) = \frac{t^2}{2\pi} - t$, $\varphi : [0, \pi] \rightarrow \mathbb{R}$:

$$\varphi(x) = \begin{cases} -1, & x = 0; \\ \frac{h(x)}{2 \sin(\frac{x}{2})}, & 0 < x \leq \pi. \end{cases}$$

10.2.2.1 B5) Prove that $\varphi \in C^1([0, \pi])$.

Proof:

$$\lim_{x \rightarrow 0} \frac{h(x)}{2 \sin(\frac{x}{2})} = \lim_{x \rightarrow 0} \frac{-x + o(x)}{2 \sin(\frac{x}{2})} = -1 = \varphi(0).$$

Hence $\varphi \in C^1([0, \pi])$.

10.2.2.2 B6) For all $k \geq 1$, calculate

$$\int_0^\pi h(x) \cos(kx) \, dx.$$

Solution:

$$\int_0^\pi \left(\frac{x^2}{2\pi} - x \right) \cos(kx) \, dx = \frac{1}{k} \int_0^\pi \left(\frac{x^2}{2\pi} - x \right) \, d \sin(kx) \quad (10.7)$$

$$= -\frac{1}{k} \int_0^\pi \sin(kx) \left(\frac{x}{\pi} - 1 \right) \, dx \quad (10.8)$$

$$= \frac{1}{k^2} \int_0^\pi \left(\frac{x}{\pi} - 1 \right) \, d \cos(kx) \quad (10.9)$$

$$= \frac{1}{k^2} - \frac{1}{\pi k^2} \int_0^\pi \cos(kx) \, dx = \frac{1}{k^2}. \quad (10.10)$$

10.2.2.3 B7) Prove that there is a constant λ , such that for any $x \in (0, \pi)$,

$$\sum_{k=1}^n \cos(kx) = \frac{\sin(n + \frac{1}{2})x}{2 \sin(\frac{x}{2})} - \lambda.$$

Proof: Note that $2 \cos(kx) \sin(\frac{x}{2}) = \sin(k + 1/2)x - \sin(k - 1/2)x$, hence

$$\sum_{k=1}^n \cos(kx) \cdot 2 \sin \frac{x}{2} = \sin \left(n + \frac{1}{2} \right) x - \sin \frac{x}{2}, \lambda = \frac{1}{2}.$$

10.2.2.4 B8) Prove that for any $\psi \in C^1([0, \pi])$,

$$\lim_{n \rightarrow \infty} \int_0^\pi \psi(x) \sin(n + 1/2)x \, dx = 0.$$

Proof: Since $\sin(n + 1/2)x = c_1 \sin nx + c_2 \cos nx$, where c_1, c_2 are constant, it suffices to show that

$$\lim_{n \rightarrow \infty} \int_0^\pi \psi(x) \sin(2nx) \, dx = \lim_{n \rightarrow \infty} \int_0^\pi \psi(x) \cos(2nx) \, dx = 0.$$

Note that

$$\int_0^\pi \psi(x) \sin(2nx) \, dx = \sum_{k=1}^n \int_{(k-1)\pi/n}^{k\pi/n} \psi(x) \sin(2nx) \, dx \quad (10.11)$$

$$= \sum_{k=1}^n \frac{1}{2n} \int_0^{2\pi} \psi\left(\frac{x}{2n} + \frac{(k-1)\pi}{n}\right) \sin x \, dx \quad (10.12)$$

$$\left(t = \frac{(k-1)\pi}{n}\right) \leq \sum_{k=1}^n \frac{\pi}{n} \sup_{x \leq \pi} \left| \psi\left(\frac{x+\pi}{2n} + t\right) - \psi\left(\frac{x}{2n} + t\right) \right| \quad (10.13)$$

$$\leq \pi \sup_{0 \leq x \leq \pi - \pi/2n} \left| \psi\left(x + \frac{\pi}{2n}\right) - \psi(x) \right| \rightarrow 0. \quad (10.14)$$

since ψ is uniformly continuous on $[0, \pi]$.

10.2.2.5 B9) Prove that $\zeta(2) = \frac{\pi^2}{6}$.

Proof:

$$\zeta(2) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k^2} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_0^\pi h(x) \cos(kx) \, dx \quad (10.15)$$

$$= \lim_{n \rightarrow \infty} \int_0^\pi \psi(x) \sin(n + 1/2)x - \frac{1}{2} \left(\frac{x^2}{2\pi} - x \right) \, dx \quad (10.16)$$

$$(B8) = \frac{1}{2} \int_0^\pi \left(x - \frac{x^2}{2\pi} \right) \, dx = \frac{\pi^2}{6}. \quad (10.17)$$

10.2.3 Part 3: $\zeta(2)$ is irrational

Otherwise assume $\pi^2 = \frac{a}{b}$ where $a, b \in \mathbb{Z}$.

10.2.3.1 B10) Define a sequence of polynomials $f_n(x) = \frac{x^n(1-x)^n}{n!}$, where $n \in \mathbb{Z}_{\geq 1}$. Prove that for any $k \in \mathbb{Z}$, $f_n^{(k)}(0), f_n^{(k)}(1) \in \mathbb{Z}$.

Proof: If $k \leq n-1$, then $f_n^{(k)}(0) = f_n^{(k)}(1) = 0$. If $k \geq n$, then

$$\text{if } x^n(1-x)^n = \sum_{k=n}^{2n} c_k x^k, \text{ then } f_n^{(k)}(x) = \sum_{m=n}^{2n} c_k \binom{m}{k} x^{m-k} \in \mathbb{Z}[x].$$

Hence $f_n^{(k)}(0), f_n^{(k)}(1) \in \mathbb{Z}$.

10.2.3.2 B11) Define the sequence

$$F_n(x) = b^n(\pi^{2n} f_n(x) - \pi^{2n-2} f_n^{(2)}(x) + \cdots + (-1)^n f_n^{(2n)}(x)).$$

Prove that $F_n(0), F_n(1) \in \mathbb{Z}$. Proof: For $0 \leq k \leq n$, $b^n \pi^{2n-2k}, f_n^{(2k)}(x) \in \mathbb{Z}$, when $x \in \{0, 1\}$.

10.2.3.3 B12) For $n \geq 1$, define $\{g_n\}_{n \geq 1}, \{A_n\}_{n \geq 1}$ as below:

$$g_n(x) = F'_n(x) \sin(\pi x) - \pi F_n(x) \cos(\pi x), \quad A_n = \pi \int_0^1 a^n f_n(x) \sin(\pi x) \, dx.$$

Prove that $A_n \in \mathbb{Z}$ and $g'_n = \pi^2 a^n f_n(x) \sin(\pi x)$. Proof: Note that

$$g'_n(x) = b^n \pi^{2n} \sum_{k=0}^n \left(f_n^{(2k)}(x) \sin(\pi x) - \pi f_n^{(2k+1)}(x) \cos(\pi x) \right)' (-\pi^2)^k \quad (10.18)$$

$$= b^n \pi^{2n+2} f_n(x) \sin(\pi x). \quad (10.19)$$

And

$$A_n = \frac{1}{\pi} \int_0^1 dg_n(x) = \frac{1}{\pi} (g_n(1) - g_n(0)) \quad (10.20)$$

$$= F_n(0) + F_n(1) \in \mathbb{Z}. \quad (10.21)$$

10.2.3.4 B13) Prove that there exists $n \in \mathbb{Z}$ such that for all $x \in [0, 1]$, $a^n f_n(x) < 1/2$.

Proof:

$$f_n(x) = \frac{1}{n!} (x(1-x))^n \leq \frac{1}{n! 4^n} \rightarrow 0.$$

10.2.3.5 B14) Prove that there exists $n \in \mathbb{Z}$ such that $A_n \in (0, 1)$, leading to contradiction.

Proof: $f_n, \sin(\pi x) \geq 0$, when $x \in [0, 1]$, hence $A_n > 0$. Take n such that $a^n f_n < 1/2$ then $A_n < \frac{\pi}{2} \int_0^1 \sin(\pi x) dx = 1$. Therefore $A_n \in (0, 1)$, contradicting with $A_n \in \mathbb{Z}$.

10.3 PSC: Calculation of Integrals

$$a \neq 0, b \neq 0$$

10.3.1 (1) $\int_0^\pi \sin^3 x dx$

$$\int_0^\pi \sin^3(x) dx = -2 \int_0^{\pi/2} \sin^2(x) d \cos(x) = 2 \int_0^1 (1-x^2) dx = \frac{4}{3}.$$

10.3.2 (2) $\int_{-\pi}^\pi x^2 \cos x dx$

$$\int_{-\pi}^\pi x^2 \cos(x) dx = (x^2 - 2) \sin(x) + 2x \cos(x) \Big|_{-\pi}^\pi = -4\pi.$$

10.3.3 (3) $\int_0^1 \frac{x}{1+\sqrt{1+x}} dx$

$$\int_0^1 \frac{x}{1+\sqrt{1+x}} dx = \int_0^1 \sqrt{1+x} - 1 dx = \frac{2}{3} (1+x)^{3/2} - x \Big|_0^1 = \frac{4\sqrt{2}-5}{3}.$$

10.3.4 (4) $\int_0^{\sqrt{3}} x \arctan x dx$

$$\int_0^{\sqrt{3}} x \arctan x dx = \frac{1}{2} \int_0^{\sqrt{3}} \arctan x dx^2 \quad (10.22)$$

$$= \frac{1}{2} x^2 \arctan x \Big|_0^{\sqrt{3}} - \frac{1}{2} \int_0^{\sqrt{3}} \frac{x^2}{1+x^2} dx \quad (10.23)$$

$$= \frac{3}{2} \frac{\pi}{3} - \frac{\sqrt{3}}{2} + \frac{1}{2} \arctan \sqrt{3} = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}. \quad (10.24)$$

10.3.5 (5) $\int_{-1}^0 (2x+1)\sqrt{1-x-x^2} \, dx$

$$\int_{-1}^0 (2x+1)\sqrt{1-x-x^2} \, dx = \int_{-1}^1 \frac{y}{4}\sqrt{5-y^2} \, dy = 0$$

10.3.6 (6) $\int_{\frac{1}{e}}^e |\log x| \, dx$

$$\int_{\frac{1}{e}}^e |\log x| \, dx = \int_1^e \log x \, dx + \int_1^{1/e} \log x \, dx \quad (10.25)$$

$$= (x \log x - x) \Big|_1^e + (x \log x - x) \Big|_1^{1/e} = 2 - \frac{2}{e} \quad (10.26)$$

10.3.7 (7) $\int_0^a x^2 \sqrt{a^2 - x^2} \, dx$

$$\int_0^a x^2 \sqrt{a^2 - x^2} \, dx = a^4 \int_0^{\pi/2} \sin^2 t \cos^2 t \, dt = \frac{a^4 \pi}{16}$$

10.3.8 (8) $\int_0^{\log 2} \sqrt{e^x - 1} \, dx$

$$\int_0^{\log 2} \sqrt{e^x - 1} \, dx = \int_1^2 \frac{\sqrt{y-1}}{y} \, dy = \int_0^1 \frac{\sqrt{x}}{1+x} \, dx \quad (10.27)$$

$$= \int_0^{\pi/4} 2 \tan^2 \theta \, d\theta = 2 - \frac{\pi}{2}. \quad (10.28)$$

10.3.9 (9) $\int_1^2 x^{100} \log x \, dx$

$$\int_1^2 x^{100} \log x \, dx = \int_1^2 \log x \, d\frac{x^{101}}{101} = \frac{2^{101} \log 2}{101} - \int_1^2 \frac{x^{100}}{101} \, dx \quad (10.29)$$

$$= \frac{2^{101} \log 2}{101} - \frac{2^{101} - 1}{101^2}. \quad (10.30)$$

10.3.10 (10) $\int_0^a \log(x + \sqrt{x^2 + a^2}) \, dx$

$$\int_0^a \log(x + \sqrt{x^2 + a^2}) \, dx =$$

$$\int_0^a \log(x + \sqrt{x^2 + a^2}) \, dx = a \int_0^1 \log a + \log(t + \sqrt{t^2 + 1}) \, dt \quad (10.31)$$

$$= a \log a + a \int_0^1 \log(t + \sqrt{t^2 + 1}) \, dt \quad (10.32)$$

$$= a \log a + (\log(1 + \sqrt{2}) + \sqrt{2} - 1)a. \quad (10.33)$$

$$\int_0^1 \log(x + \sqrt{x^2 + 1}) \, dx = x \log(x + \sqrt{x^2 + 1}) \Big|_0^1 - \int_0^1 \frac{x}{\sqrt{1+x^2}} \, dx \quad (10.34)$$

$$(x = \tan \theta) = \log(1 + \sqrt{2}) + \int_0^{\pi/4} \frac{1}{\cos^2 \theta} \, d \cos \theta \quad (10.35)$$

$$= \log(1 + \sqrt{2}) + \sqrt{2} - 1. \quad (10.36)$$

$$\mathbf{10.3.11} \quad (11) \int_0^{\pi/2} \frac{\cos x \sin x}{a^2 \sin^2 x + b^2 \cos^2 x} dx$$

$$\int_0^{\pi/2} \frac{\cos x \sin x}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \int_0^{\pi/2} \frac{\sin 2x}{a^2 + b^2 + (b^2 - a^2) \cos 2x} dx \quad (10.37)$$

$$= \frac{1}{2} \int_{-1}^1 \frac{1}{(a^2 + b^2) + (b^2 - a^2)t} dt \quad (10.38)$$

$$= \frac{1}{2(a^2 - b^2)} \log \left(\frac{a^2}{b^2} \right). \quad (10.39)$$

$$\mathbf{10.3.12} \quad (12) \int_0^{\pi/2} \frac{\cos^2 x}{\cos x + \sin x} dx$$

$$\int_0^{\pi/2} \frac{\cos^2 x}{\cos x + \sin x} dx = \int_0^{\pi/2} \frac{\sin^2 x}{\cos x + \sin x} dx \quad (10.40)$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{1}{\cos x + \sin x} dx \quad (10.41)$$

$$= \int_{\pi/4}^{\pi/2} \frac{1}{\sqrt{2} \sin x} dx \quad (10.42)$$

$$= -\frac{\log \tan \left(\frac{\pi}{8} \right)}{\sqrt{2}}. \quad (10.43)$$

$$\mathbf{10.3.13} \quad (13) \int_0^{\pi/2} \frac{\sin^2 x}{\cos x + \sin x} dx$$

See (12)

$$\mathbf{10.3.14} \quad (14) \int_0^{\pi/4} \log(1 + \tan x) dx$$

$$\int_0^{\pi/4} \log(1 + \tan x) dx = \int_0^{\pi/4} \log \frac{\sin x + \cos x}{\cos x} dx \quad (10.44)$$

$$= \int_0^{\pi/4} \log \frac{\sqrt{2} \sin(x + \pi/4)}{\cos x} dx \quad (10.45)$$

$$= \frac{\pi}{8} \log 2. \quad (10.46)$$

$$\mathbf{10.3.15} \quad (15) \int_0^4 \frac{|x-1|}{|x-2|+|x-3|} dx$$

$$\int_0^1 \frac{|x-1|}{|x-2|+|x-3|} dx = \int_0^1 \frac{1-x}{5-2x} dx = \frac{1}{2} - \frac{3}{4} \log \frac{5}{3}, \quad (10.47)$$

$$\int_1^2 \frac{|x-1|}{|x-2|+|x-3|} dx = \int_0^1 \frac{x}{3-2x} dx = -\frac{1}{2} + \frac{3}{4} \log 3, \quad (10.48)$$

$$\int_2^3 \frac{|x-1|}{|x-2|+|x-3|} dx = \int_0^1 (x+1) dx = \frac{3}{2}, \quad (10.49)$$

$$\int_3^4 \frac{|x-1|}{|x-2|+|x-3|} dx = \int_0^1 \frac{x+2}{2x+1} dx = \frac{1}{2} + \frac{3}{4} \log 3, \quad (10.50)$$

$$\Rightarrow \int_0^4 \frac{|x-1|}{|x-2|+|x-3|} dx = 2 + \frac{3}{4} \log \frac{27}{5} \quad (10.51)$$

$$10.3.16 \quad (16) \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$$

$$\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \int_0^\pi x d \arctan \cos x \quad (10.52)$$

$$= -x \arctan \cos x \Big|_0^\pi + \int_0^\pi \arctan \cos x dx \quad (10.53)$$

$$= \frac{\pi^2}{4} + 0 = \frac{\pi^2}{4}. \quad (10.54)$$

$$10.3.17 \quad (17) \int_0^{\pi/4} \frac{\sin x}{\sin x + \cos x} dx$$

$$\int_0^{\pi/4} \frac{\sin x}{\sin x + \cos x} dx = \frac{1}{2} \int_{\pi/4}^{\pi/2} \frac{\sin t - \cos t}{\sin t} dt \quad (10.55)$$

$$= \frac{\pi}{8} - \frac{\log 2}{4}. \quad (10.56)$$

$$10.3.18 \quad (18) \int_0^{\pi/2} \frac{\sin 2019x}{\sin x} dx$$

$$\int_0^{\pi/2} \frac{\sin(2n+1)x}{\sin x} dx = \int_0^{\pi/2} 1 + \sum_{k=1}^n \cos(2kx) dx = \frac{\pi}{2}. \quad (10.57)$$

$$10.3.19 \quad (19) \int_2^4 \frac{\log \sqrt{9-x}}{\log \sqrt{9-x} + \log \sqrt{x+3}} dx$$

$$\int_2^4 \frac{\log \sqrt{9-x}}{\log \sqrt{9-x} + \log \sqrt{x+3}} dx = \int_{-1}^1 \frac{\log \sqrt{6+t}}{\log \sqrt{6+t} + \log \sqrt{6-t}} dt = 1. \quad (10.58)$$

$$10.3.20 \quad (20) \int_0^1 \frac{1}{\sqrt{1+x^2} + \sqrt{1-x^2}} dx$$

$$\int_0^1 \frac{dx}{\sqrt{1+x^2} + \sqrt{1-x^2}} = \int_0^1 \frac{1}{2} (\sqrt{1+x^2} - \sqrt{1-x^2}) dx \quad (10.59)$$

$$= -\frac{\pi}{8} + \frac{\sqrt{2}}{4} + \frac{1}{8} \log \frac{\sqrt{2}+1}{\sqrt{2}-1}. \quad (10.60)$$

$$10.3.21 \quad (21) \int_0^1 \sqrt{x + \sqrt{x+1}} dx$$

$$\int_0^1 \sqrt{x + \sqrt{x+1}} dx = \int_1^{1+\sqrt{2}} \sqrt{y} d \frac{2y+1-\sqrt{4y+5}}{2} \quad (10.61)$$

$$= \int_1^{1+\sqrt{2}} \sqrt{y} - \frac{\sqrt{y}}{\sqrt{4y+5}} dy \quad (10.62)$$

$$\left(y = \frac{z^2-5}{4} \right) = \frac{2}{3} y^{3/2} \Big|_1^{1+\sqrt{2}} - \int_3^{1+2\sqrt{2}} \frac{\sqrt{z^2-5}}{4} dz \quad (10.63)$$

$$= \frac{2}{3} ((1+\sqrt{2})^{3/2} - 1) - \frac{3\sqrt{2}-1}{8} + \frac{5}{32} \log \frac{3+\sqrt{2}}{5}. \quad (10.64)$$

10.3.22 (22) $\int_{-1}^1 \frac{\sin \sin \sin x}{x^{800} + 1} dx$

$$\int_{-1}^1 \frac{\sin \sin \sin x}{x^{800} + 1} dx = 0. (\text{by symmetry}) \quad (10.65)$$

Chapter 11 Density of Sum of Squares

11.1 PSA: Riemann Integral

11.1.1 A1) $f \in C([a, b])$, $g \in \mathcal{R}([a, b])$, where g is positive. Prove that there exists $\xi \in (a, b)$, such that

$$\int_a^b fg = f(\xi) \int_a^b g.$$

Proof: Since g is positive on $[a, b]$,

$$\inf_{x \in [a, b]} f(x) \int_a^b g \leq \int_a^b fg \leq \sup_{x \in [a, b]} f(x) \int_a^b g.$$

By $f \in C([a, b])$, there exists such an $\xi \in (a, b)$.

11.1.2 A2) Prove without using Lebesgue theorem: if f is monotonously increasing on $[a, b]$, then $f \in \mathcal{R}([a, b])$.

Proof: For any $\varepsilon > 0$ let $n = \lceil 1/\varepsilon \rceil + 1$, and

$$\mathcal{C} = \left\{ x_k = a + (b - a) \frac{k}{n} : k = 0, 1, \dots, n \right\}.$$

Then

$$g(x) = \max_{x_k \leq x} \{f(x_k)\} \leq f, h(x) = \min_{x_k \geq x} \{f(x_k)\} \geq f.$$

and both are monotonous simple functions. Therefore

$$\overline{\int_a^b f} - \underline{\int_a^b f} \leq \overline{S}(f; \mathcal{C}) - \underline{S}(f; \mathcal{C}) = \frac{1}{n} (f(b) - f(a)) \rightarrow 0.$$

Hence f is Riemann integrable.

11.1.3 A3) Prove that $1_{\mathbb{Q}}$ is not Riemann integrable on $[0, 1]$.

Proof: Let $\varepsilon = \frac{1}{2}$. For any $\mathcal{C} = \{0 = x_0 < \dots < x_n = 1\}$, $\omega(x_{k-1}, x_k) = 1$, hence

$$\sum_{k=1}^n \omega(x_{k-1}, x_k) (x_k - x_{k-1}) = 1 > \varepsilon.$$

Therefore $1_{\mathbb{Q}}$ is not Riemann integrable.

11.1.4 A4) Prove that if $f \in \mathcal{R}([a, b])$, then $|f|^p \in \mathcal{R}([a, b])$, where $p \geq 0$.

Proof: Since $x \mapsto |x|^p$ is continuous, $|f|^p$ is continuous as x whenever f is continuous at x . Hence

$$f \in \mathcal{R}([a, b]) \implies |f|^p \in \mathcal{R}([a, b]).$$

11.1.5 A5) Prove Hölder's Inequality: if $f, g \in \mathcal{R}([a, b])$, $p, q > 0$, $1/p + 1/q = 1$, then

$$\left| \int_a^b fg \right| \leq \left(\int_a^b |f|^p \right)^{1/p} \left(\int_a^b |g|^q \right)^{1/q}.$$

Proof: By A4) the functions are integrable. We can assume that

$$\int_a^b |f|^p = \int_a^b |g|^q = 1.$$

Then by Young's inequality,

$$\left| \int_a^b fg \right| \leq \int_a^b |f| \cdot |g| \leq \int_a^b \frac{1}{p} |f|^p + \frac{1}{q} |g|^q = \frac{1}{p} + \frac{1}{q} = 1.$$

11.1.6 A6) Prove Minkowski's inequality: if $f, g \in \mathcal{R}([a, b])$, $p \geq 1$, then

$$\left(\int_a^b |f + g|^p \right)^{1/p} \leq \left(\int_a^b |f|^p \right)^{1/p} + \left(\int_a^b |g|^p \right)^{1/p}.$$

Proof: Note that if $1/p + 1/q = 1$, then

$$\int_a^b |f + g|^p = \int_a^b |f| \cdot |f + g|^{1-p} + \int_a^b |g| \cdot |f + g|^{1-p} \quad (11.1)$$

$$\leq \left(\left(\int_a^b |f|^p \right)^{1/p} + \left(\int_a^b |g|^p \right)^{1/p} \right) \left(\int_a^b |f + g|^{(1-p)q} \right)^{1/q} \quad (11.2)$$

$$(11.3)$$

Hence

$$\left(\int_a^b |f + g|^p \right)^{1/p} \leq \left(\int_a^b |f|^p \right)^{1/p} + \left(\int_a^b |g|^p \right)^{1/p}.$$

The equality holds, when $|f|/|f + g|^{1-p}$, $|g|/|f + g|^{1-p}$ are both constant, which is equivalent to $|f|/|g|$ is constant.

11.2 PSB: Convex Functions

11.2.1 B1) Assume $f \in \mathcal{R}([a, b])$ and f is convex, prove that

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx \leq \frac{f(a) + f(b)}{2}.$$

Proof: Note that $f\left(\frac{a+b}{2}\right) \leq \frac{f(x) + f(a+b-x)}{2} \leq \frac{f(a) + f(b)}{2}$, and

$$\int_a^b f(x) dx = \int_a^b \frac{f(x) + f(a+b-x)}{2} dx.$$

Hence

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx \leq \frac{f(a) + f(b)}{2}.$$

11.2.2 B2) Assume f is twice differentiable on $[a, b]$ and for any x , $f''(x) > 0$, $f(x) \leq 0$. Prove that for any x ,

$$f(x) \geq \frac{2}{b-a} \int_a^b f(y) dy.$$

Proof: For any $x \leq y \leq b$,

$$f(y) \leq \frac{b-y}{b-x} f(x) + \frac{y-x}{b-x} f(b) \leq \frac{b-y}{b-x} f(x),$$

hence

$$\int_x^b f(y) \, dy \leq f(x) \int_x^b \frac{b-y}{b-x} \, dy = \frac{b-x}{2} f(x).$$

Likewise,

$$\int_a^x f(y) \, dy \leq f(x) \int_a^x \frac{y-a}{x-a} \, dy = \frac{x-a}{2} f(x).$$

Therefore

$$f(x) \geq \frac{2}{b-a} \int_a^b f(y) \, dy.$$

11.2.3 B3) Assume f is twice differentiable on $\mathbb{R}$ and $f''(x) \geq 0$, $\varphi \in C([a, b])$. Prove that

$$\frac{1}{b-a} \int_a^b (f \circ \varphi)(t) \, dt \geq f\left(\frac{1}{b-a} \int_a^b \varphi(t) \, dt\right).$$

Proof: We prove the proposition for any convex function f and φ on the set X . Let

$$\langle g \rangle = \frac{1}{\mu(X)} \int_X g \, d\mu.$$

Then since f is convex, there is a constant K such that $f(y) - f(\langle \varphi \rangle) \geq K(y - \langle \varphi \rangle)$. Hence

$$\langle f(\varphi) \rangle = \frac{1}{\mu(X)} \int_X f(\varphi(t)) \, d\mu \tag{11.4}$$

$$\geq \frac{1}{\mu(X)} \int_X f(\langle \varphi \rangle) \, d\mu + \frac{1}{\mu(X)} \int_X K(\varphi(t) - \langle \varphi \rangle) \, d\mu \tag{11.5}$$

$$= f(\langle \varphi \rangle). \tag{11.6}$$

11.2.4 B4) Assume $f \in C([a, b])$ and for any x , $f(x) > 0$. Prove that

$$\log\left(\frac{1}{b-a} \int_a^b f\right) \geq \frac{1}{b-a} \int_a^b \log f.$$

Proof: Since $-\log x$ is convex, we can use B3).

11.2.5 B5) Prove that if f is convex on $\mathbb{R}$, $\varphi \in C([0, 1])$, then

$$f\left(\int_0^1 \varphi\right) \leq \int_0^1 f \circ \varphi.$$

Proof: A special case of what we proved in B3).

11.3 PSC: Integrals and Derivatives

11.3.1 C1) Assume $f \in C^1([0, 2])$, $|f'| \leq 1$, $f(0) = f(2) = 1$. Prove that

$$1 \leq \int_0^2 f \leq 3.$$

Proof: Note that for $0 \leq x \leq 1$,

$$|f(x) - 1| = x|f'(\xi)| \leq x.$$

and for $1 \leq x \leq 2$,

$$|f(x) - 1| = (2-x)|f'(\xi)| \leq 2-x.$$

Hence

$$\int_0^2 |f(x) - 1| dx \leq \int_0^1 x dx + \int_1^2 (2 - x) dx = 1.$$

11.3.2 C2) Assume that $f \in C^2([0, 1])$. Prove that $\exists \xi \in [0, 1]$, such that

$$\int_0^1 f(x) dx = f\left(\frac{1}{2}\right) + \frac{1}{24}f''(\xi).$$

Proof: Let $g(x) = f(x) + f(1 - x)$, then

$$\int_0^1 f(x) dx - f\left(\frac{1}{2}\right) = \int_0^{1/2} g(x) - 2f\left(\frac{1}{2}\right) dx \quad (11.7)$$

$$(\text{integration by parts}) = - \int_0^{1/2} x g'(x) dx = -\frac{1}{2} \int_0^{1/2} g'(x) dx^2 \quad (11.8)$$

$$(\text{integration by parts}) = \frac{1}{2} \int_0^{1/2} x^2 g''(x) dx. \quad (11.9)$$

Note that $g'' \in C([0, 1])$ hence by A1), $\exists \eta \in (0, \frac{1}{2})$,

$$\int_0^1 f(x) dx - f\left(\frac{1}{2}\right) = g''(\eta) \frac{1}{2} \int_0^{1/2} x^2 dx = \frac{1}{48} g''(\eta).$$

Since $f'' \in C([0, 1])$, there exists $\xi \in (\eta, 1 - \eta)$, such that

$$f''(\xi) = \frac{f''(\eta) + f''(1 - \eta)}{2} = \frac{g''(\eta)}{2}.$$

Therefore

$$\int_0^1 f(x) dx = f\left(\frac{1}{2}\right) + \frac{1}{24}f''(\xi).$$

11.3.3 C3) Assume $f \in C^1([0, 1])$. Prove that

$$\max_{x \in [a, b]} |f(x)| \leq \frac{1}{b - a} \left| \int_a^b f(x) dx \right| + \int_a^b |f'(x)| dx.$$

Proof: For any $t \in [a, b]$,

$$(b - a)|f(t)| \leq \left| \int_a^b f(x) dx \right| + \left| \int_a^b f(x) - f(t) dx \right|$$

where

$$\left| \int_a^b f(x) - f(t) dx \right| = \left| \int_a^b \left(\int_t^x f'(u) du \right) dx \right| \quad (11.10)$$

$$\leq \int_a^b \int_t^x |f'(u)| du dx \quad (11.11)$$

$$(11.12)$$

$$\leq (b - a) \int_a^b |f'(u)| du. \quad (11.13)$$

11.3.4 C4) Suppose $f \in C([0, 1])$ and for any $g \in C([0, 1])$, $g(0) = g(1) = 0$, we have

$$\int_0^1 f(x)g(x) dx = 0.$$

Prove that $f(x) \equiv 0$. Proof: Otherwise assume $f(t) > 0$ for some $t \in (0, 1)$, then there exists an $\varepsilon > 0$ such that $(t - \varepsilon, t + \varepsilon) \subset [0, 1]$ and $\forall x \in (t - \varepsilon, t + \varepsilon), f(x) > f(t)/2$. Let

$$g(x) = \begin{cases} 0, & x \notin (t - \varepsilon, t + \varepsilon), \\ 1 - \frac{|x-t|}{\varepsilon}, & x \in (t - \varepsilon, t + \varepsilon). \end{cases}$$

Then

$$\int_0^1 f(x)g(x) dx > \int_{t-\varepsilon}^{t+\varepsilon} \frac{f(t)}{2} g(x) dx > 0,$$

leading to contradiction. Hence $f(x) \equiv 0$.

11.3.5 C5) Suppose $f \in C([0, 1])$ and for any $n \in \mathbb{Z}_{\geq 0}$,

$$\int_0^1 f(x)x^n dx = 0.$$

Prove that $f(x) \equiv 0$. Proof: Otherwise, $\int_0^1 f^2 > 0$. By Stone-Weierstrass theorem, for any $\varepsilon > 0$, there is a polynomial P such that $\sup_{x \in [0,1]} |f(x) - P(x)| < \varepsilon$. Hence

$$0 = \int_0^1 f(x)P(x) dx = \int_0^1 f^2 - \int_0^1 f(x)(f(x) - P(x)) dx \geq \int_0^1 f^2 - \sup_{x \in [0,1]} |f(x)|\varepsilon > 0$$

when $\varepsilon \rightarrow 0$, leading to contradiction.

11.3.6 C6) (Gronwall's Inequality) Suppose $\varphi \in C([0, T])$ and for any $t \in [0, T]$,

$$|\varphi(t)| \leq M + k \int_0^t |\varphi(s)| ds, \text{ where } M, k \text{ are positive real numbers. Prove that } \\ \forall t \in [0, T], |\varphi(t)| \leq Me^{kt}.$$

Proof: Let

$$f : \left[0, \frac{T}{k}\right] \rightarrow \mathbb{R}, t \mapsto \frac{e^{-t}|\varphi(t/k)|}{M},$$

then for any $t \in [0, T/k]$,

$$f(t) \leq e^{-t} + e^{-t} \int_0^t f(s)e^s ds.$$

Let $f(t) = \sup_{s \in [0, T/k]} \{f(s)\}$ then

$$f(t) \leq e^{-t} + e^{-t} \int_0^t f(t)e^s dx = e^{-t} + f(t)(1 - e^{-t}).$$

Hence $f(s) \leq f(t) \leq 1, \implies |\varphi(t)| \leq Me^{kt}$.

11.3.7 C7) Assume $a, b > 0, f \in C([-a, b])$. If for any $x \in (-a, b), f(x) > 0$ and

$$\int_{-a}^b xf(x) dx = 0. \text{ Prove that}$$

$$\int_{-a}^b x^2 f(x) dx \leq ab \int_{-a}^b f(x) dx.$$

Proof: Note that

$$\int_{-a}^b (x+a)(x-b)f(x) dx \leq 0.$$

Combined with $\int_{-a}^b xf(x) dx = 0$ we get

$$\int_{-a}^b x^2 f(x) dx \leq ab \int_{-a}^b f(x) dx.$$

11.3.8 C8) Assume $f \in C([-1, 1])$. Prove that

$$\lim_{\lambda \rightarrow 0^+} \int_{-1}^1 \frac{\lambda}{\lambda^2 + x^2} f(x) dx = \pi f(0).$$

Proof: Let $M = \sup_{|x| \leq 1} |f(x)|$ and

$$g(x) = \frac{\lambda}{\lambda^2 + x^2},$$

then (g is sort of a good kernel)

$$\int_{-1}^1 g(x) dx = 2 \arctan \frac{1}{\lambda}.$$

Hence

$$\left| \int_{-1}^1 f(x)g(x) dx - \pi f(0) \right| \quad (11.14)$$

$$\leq \left| \pi - 2 \arctan \frac{1}{\lambda} \right| f(0) + \int_{-\varepsilon}^{\varepsilon} |f(x) - f(0)|g(x) dx + \int_{\varepsilon \leq x \leq 1} M g(x) dx \quad (11.15)$$

$$\leq \left| \pi - 2 \arctan \frac{1}{\lambda} \right| f(0) + \sup_{|x| \leq \varepsilon} |f(x) - f(0)| \pi + 2M \left| \arctan \frac{1}{\lambda} - \arctan \frac{\varepsilon}{\lambda} \right| \quad (11.16)$$

$$\rightarrow 0 \quad (11.17)$$

since

$$\arctan \frac{1}{\lambda} - \arctan \frac{\varepsilon}{\lambda} = \arctan \frac{\lambda(1 - \varepsilon)}{\lambda^2 + \varepsilon} \rightarrow 0, \text{ when } \lambda \rightarrow 0^+.$$

and $\sup_{|x| \leq \varepsilon} |f(x) - f(0)| \rightarrow 0$ when $\varepsilon \rightarrow 0$.

11.3.9 C9) Assume f is differentiable on $[1, \infty)$ and both $\int_1^\infty f(x) dx$ and $\int_1^\infty f'(x) dx$ converges. Prove that

$$\lim_{x \rightarrow \infty} f(x) = 0.$$

Proof: For any $\varepsilon > 0$, there exists $N > 1$, such that $\forall u, v > N$,

$$\left| \int_u^v f'(x) dx \right| < \varepsilon, \text{ i.e. } |f(u) - f(v)| < \varepsilon$$

Hence for any $u > N$, if $|f(u)| > \varepsilon$,

$$\left| \int_u^M f(x) dx \right| \geq (M - u)(|f(u) - \varepsilon|) \rightarrow \infty, \text{ as } M \rightarrow \infty,$$

which contradicts the fact that $\int_1^\infty f(x) dx$ converges. Therefore $|f(u)| < \varepsilon$ for any $u > N$, which implies $\lim_{x \rightarrow \infty} f(x) = 0$.

11.3.10 C10) Prove that

$$\int_0^\infty \frac{\sin^2 x}{x^2} dx = \int_0^\infty \frac{\sin x}{x} dx, \int_0^\infty \frac{\cos x}{1+x} dx = \int_0^\infty \frac{\sin x}{(1+x)^2} dx.$$

Proof:

$$\int_0^\infty \frac{\sin^2 x}{x^2} dx = - \int_0^\infty \sin^2 x d\frac{1}{x} = \int_0^\infty \frac{\sin 2x}{x} dx = \int_0^\infty \frac{\sin x}{x} dx. \quad (11.18)$$

$$\int_0^\infty \frac{\cos x}{1+x} dx = \int_0^\infty \frac{1}{1+x} d\sin x = \int_0^\infty \frac{\sin x}{(1+x)^2} dx. \quad (11.19)$$

11.4 PSD: Calculation of improper integrals

11.4.1 D1)

$$\int_0^1 \log x \, dx = (x \log x - x) \Big|_0^1 = -1.$$

11.4.2 D2)

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2} \, dx = \arctan x \Big|_{-\infty}^{\infty} = \pi.$$

11.4.3 D3)

Calculating residues, we get

$$\int_{-\infty}^{\infty} \frac{dx}{x^4+1} = 2\pi i \cdot (\operatorname{Res}(f; e^{i\pi/4}) + \operatorname{Res}(f; e^{3i\pi/4})) = \frac{\pi}{\sqrt{2}}.$$

Hence

$$\int_0^{\infty} \frac{dx}{x^4+1} = \frac{\pi}{2\sqrt{2}}.$$

11.4.4 D4)

Same as D3)

$$\int_{-\infty}^{\infty} \frac{1+x^2}{1+x^4} \, dx = \sqrt{2}\pi.$$

Hence

$$\int_0^{\infty} \frac{1+x^2}{1+x^4} \, dx = \frac{\pi}{\sqrt{2}}.$$

11.4.5 D5)

$$\int_{-\infty}^0 x e^x \, dx = \int_{-\infty}^0 x \, de^x = - \int_{-\infty}^0 e^x \, dx = -1.$$

11.4.6 D6)

$$\int_0^{\infty} e^{-\sqrt{x}} \, dx = 2 \int_0^{\infty} y e^{-y} \, dy = 2 \int_0^{\infty} e^{-y} \, dy = 2.$$

11.4.7 D7)

$$\int_0^{\infty} \frac{dx}{(a^2+x^2)^{3/2}} = \frac{1}{a^2} \int_0^{\infty} \frac{dx}{(1+x^2)^{3/2}} = \frac{1}{2a^2} B\left(\frac{1}{2}, 1\right) = \frac{1}{a^2}.$$

(We can also substitute $x = a \tan \theta$).

11.4.8 D8)

$$\int_2^{\infty} \frac{dx}{x^2+x-2} = \frac{1}{3} \log \frac{x-1}{x+2} \Big|_2^{\infty} = \frac{\log 3}{3}.$$

11.4.9 D9)

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2 + x + 1)^2} = \frac{8}{3\sqrt{3}} \int_{-\infty}^{\infty} \frac{du}{(1 + u^2)^2} \quad (11.20)$$

$$(u = \tan \theta) = \frac{8}{3\sqrt{3}} \int_{-\pi/2}^{\pi/2} \cos^2 \theta \, d\theta = \frac{4\sqrt{3}\pi}{9}. \quad (11.21)$$

11.4.10 D10)

$$\int_{-1}^1 \frac{dx}{\sqrt{1-x^2}} = \int_{-\pi/2}^{\pi/2} 1 \, d\theta = \pi.$$

11.4.11 D11)

$$\int_{-1}^1 \frac{\arcsin x}{\sqrt{1-x^2}} \, dx = \int_{-\pi/2}^{\pi/2} \theta \, d\theta = 0.$$

11.4.12 D12)

Let γ be the unit circle, then

$$\int_{-1}^1 \frac{dx}{(2-x)^2 \sqrt{1-x^2}} = \int_{-\pi/2}^{\pi/2} \frac{d\theta}{(2-\sin \theta)^2} = \frac{1}{2} \int_{-\pi}^{\pi} \frac{d\theta}{(2-\sin \theta)^2} \quad (11.22)$$

$$= \frac{1}{2} \int_{\gamma} -\frac{4}{i} \frac{z \, dz}{(z^2 - 4iz - 1)^2} \quad (11.23)$$

$$= -4\pi \operatorname{Res} \left(\frac{z}{(z^2 - 4iz - 1)^2}; (2 - \sqrt{3})i \right) \quad (11.24)$$

$$= \frac{2\pi}{3\sqrt{3}}. \quad (11.25)$$

11.4.13 D13)

$$\int_0^1 \frac{\arcsin \sqrt{x}}{x(1-x)} \, dx > \int_{1/4}^1 \frac{\pi}{6} \frac{1}{1-x} \, dx \text{ which diverges.}$$

11.4.13.1 D14)

$$\int_0^1 (1-x)^n x^{1/2-1} \, dx = B\left(n+1, \frac{1}{2}\right) = \frac{\Gamma(n+1)\Gamma(\frac{1}{2})}{\Gamma(n+\frac{3}{2})} = \frac{n!2^{n+1}}{(2n+1)!}.$$

11.4.13.2 D15)

$$\int_0^1 \frac{x^n}{\sqrt{1-x^2}} \, dx = \int_0^{\pi/2} \sin^n x \, dx = \begin{cases} \frac{(n-1)!!}{n!!}, n \text{ is odd,} \\ \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2}, n \text{ is even.} \end{cases}$$

11.4.13.3 D16)

Using integration by parts, and substitute $x = e^{-y}$,

$$\int_0^1 x^m (\log x)^n dx = (-1)^n \int_0^\infty e^{-(m+1)y} y^n dy \quad (11.26)$$

$$= (-1)^n \frac{n!}{(m+1)^n} \int_0^\infty e^{-(m+1)y} dy = \frac{(-1)^n n!}{(m+1)^{n+1}}. \quad (11.27)$$

11.4.13.4 D17)

$$\int_2^\infty \frac{dx}{x(\log x)^p} = \int_{\log 2}^\infty \frac{dy}{y^p} = \frac{(\log 2)^{1-p}}{p-1}.$$

11.4.13.5 D18)

Substitute $x = ay$, then

$$\int_0^\infty \frac{\log x}{x^2 + a^2} dx = \frac{\pi \log a}{2a} + \frac{1}{a} \int_0^\infty \frac{\log y}{1 + y^2} dy = \frac{\pi \log a}{2a}. \quad (11.28)$$

since by substituting $y = 1/z$,

$$\int_0^\infty \frac{\log y}{1 + y^2} dy = - \int_0^\infty \frac{\log z}{1 + z^2} dz = 0.$$

11.4.13.6 D19)

$$\int_0^\infty x^n e^{-x} dx = \Gamma(n) = (n-1)!.$$

11.4.13.7 D20)

$$\int_{-\infty}^\infty \frac{dx}{(ax^2 + 2bx + c)^n} = \frac{1}{d^n} \sqrt{\frac{d}{a}} \int_{-\infty}^\infty \frac{du}{(1 + u^2)^n} = \frac{1}{d^n} \sqrt{\frac{d}{a}} \pi \frac{(2n-3)!!}{(2n-2)!!}.$$

where $d = \frac{ac-b^2}{a}$

11.4.13.8 D21)

$$\int_0^\infty x^{2n-1} e^{-x^2} dx = \frac{1}{2} \int_0^\infty y^{n-1} e^{-y} dy = \frac{(n-1)!}{2}.$$

11.4.13.9 D22)

The Poisson kernel

$$\frac{1-r^2}{1-2r \cos x + r^2} = \frac{1-r^2}{(1-re^{ix})(1-re^{-ix})} \quad (11.29)$$

$$= (1-r^2) \sum_{n=0}^\infty r^n e^{inx} \sum_{m=0}^\infty r^m e^{-imx} \quad (11.30)$$

$$= \sum_{k=-\infty}^\infty r^{|k|} e^{ikx}. \quad (11.31)$$

Hence

$$\int_{-\pi}^\pi \frac{1-r^2}{1-2r \cos x + r^2} dx = 2\pi.$$

11.4.13.10 D23)

$$\int_0^\infty e^{-ax} \cos bx \, dx = \frac{1}{b} \int_0^\infty e^{-ax} \, d \sin bx = \frac{a}{b} \int_0^\infty e^{-ax} \sin bx \, dx \quad (11.32)$$

$$= -\frac{a}{b^2} \int_0^\infty e^{-ax} \, d \cos bx = \frac{a}{b^2} - \frac{a^2}{b^2} \int_0^\infty e^{-ax} \cos bx \, dx \quad (11.33)$$

$$= \frac{a}{a^2 + b^2}. \quad (11.34)$$

11.4.13.11 D24)

Same as (23),

$$\int_0^\infty e^{-ax} \sin bx \, dx = \frac{b}{a^2 + b^2}.$$

11.4.13.12 D25)

$$\int_1^\infty \frac{dx}{x(x+1) \cdots (x+n)} = \lim_{N \rightarrow \infty} \int_0^N \sum_{k=0}^n \frac{(-1)^k}{x+k} \binom{n}{k} dx \quad (11.35)$$

$$= \lim_{N \rightarrow \infty} \sum_{k=0}^n (-1)^k \binom{n}{k} \log \left(\frac{N+k}{(k+1)} \right) \quad (11.36)$$

$$= - \sum_{k=0}^n (-1)^k \binom{n}{k} \log(k+1) + \lim_{N \rightarrow \infty} \sum_{k=0}^n (-1)^k \binom{n}{k} \log \left(1 + \frac{k}{N} \right) \quad (11.37)$$

$$= - \sum_{k=0}^n (-1)^k \binom{n}{k} \log(k+1). \quad (11.38)$$

11.4.13.13 D26)

$$\int_0^\pi \log \sin x \, dx = 2 \int_0^{\pi/2} \log \sin x \, dx = 2 \int_0^{\pi/2} \log \cos x \, dx \quad (11.39)$$

$$= \int_0^{\pi/2} \log \sin 2x - \log 2 \, dx = \frac{1}{2} \int_0^\pi \log \sin x \, dx - \frac{\pi}{2} \log 2 \quad (11.40)$$

$$= -\pi \log 2. \quad (11.41)$$

11.4.13.14 D27)

$$\int_0^\infty e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}.$$

Note that

$$\max\{0, 1 - x^2\} < e^{-x^2} < \frac{1}{1 + x^2}.$$

Hence

$$\frac{(2n)!!}{(2n+1)!!} < \int_0^\infty e^{-nx^2} \, dx < \frac{(2n-3)!!}{(2n-2)!!} \cdot \frac{\pi}{2}.$$

Therefore

$$\sqrt{n} \frac{(2n)!!}{(2n+1)!!} < \int_0^\infty e^{-x^2} \, dx < \sqrt{n} \frac{(2n-3)!!}{(2n-2)!!} \cdot \frac{\pi}{2}.$$

By Wallis's formula,

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

11.5 PSE: Density of sum of squares

Let $I = (0, \infty)$.

11.5.1 Part 1

11.5.1.1 E1) Prove that $e^{-u}/\sqrt{u}$ is integrable on I , and let $K = \int_0^\infty e^{-u}/\sqrt{u} du$.

Proof:

$$\int_1^\infty e^{-u}/\sqrt{u} du < \int_1^\infty e^{-u} du = \frac{1}{e}. \quad (11.42)$$

$$\int_0^1 e^{-u}/\sqrt{u} du < \int_0^1 u^{-1/2} du = \frac{1}{2}. \quad (11.43)$$

11.5.1.2 E2) Prove that for any $x \in I$,

$$F(x) = \int_0^\infty \frac{e^{-u}}{\sqrt{u}(u+x)} du$$

is well-defined. Proof:

$$F(x) < \int_0^\infty \frac{e^{-u}}{x\sqrt{u}} du \text{ converges.}$$

11.5.1.3 E3) Prove that $F \in C^1(I)$ and calculate $F'(x)$.

Solution: Let $f(x, u) = \frac{e^{-u}}{\sqrt{u}(u+x)}$, then f is uniformly continuous on any closed subinterval of I , and so is

$$\frac{d}{dx} f(x, u) = -\frac{e^{-u}}{\sqrt{u}(u+x)^2}.$$

Also,

$$\int_0^\infty \frac{d}{dx} f(x, u) du$$

converges uniformly. Hence F is continuously differentiable and

$$F'(x) = -\int_0^\infty \frac{e^{-u}}{\sqrt{u}(u+x)^2} du.$$

11.5.1.4 E4) Prove that for any $x \in I$,

$$xF'(x) - \left(x - \frac{1}{2}\right) F(x) = -K.$$

Proof: We show that

$$x \int_0^\infty \frac{e^{-u}}{\sqrt{u}(u+x)^2} du + \left(x - \frac{1}{2}\right) \int_0^\infty \frac{e^{-u}}{\sqrt{u}(u+x)} du = \int_0^\infty \frac{e^{-u}}{\sqrt{u}} du.$$

Note that, by substituting $u \rightarrow ux$,

$$x \int_0^\infty \frac{e^{-u}}{\sqrt{u}(u+x)^2} du = \frac{1}{\sqrt{x}} \int_0^\infty \frac{e^{-ux}}{\sqrt{u}(1+u)^2} du, \quad (11.44)$$

$$\left(x - \frac{1}{2}\right) \int_0^\infty \frac{e^{-u}}{\sqrt{u}(u+x)} du = \left(\sqrt{x} - \frac{1}{2\sqrt{x}}\right) \int_0^\infty \frac{e^{-ux}}{\sqrt{u}(1+u)} du, \quad (11.45)$$

$$\int_0^\infty \frac{e^{-u}}{\sqrt{u}} du = \sqrt{x} \int_0^\infty \frac{e^{-ux}}{\sqrt{u}} du. \quad (11.46)$$

Hence it is equivalent to

$$x \int_0^\infty \frac{e^{-ux}}{\sqrt{u}} du = \int_0^\infty \frac{e^{-ux}}{\sqrt{u}(1+u)^2} du + \left(x - \frac{1}{2}\right) \int_0^\infty \frac{e^{-ux}}{\sqrt{u}(1+u)} du.$$

Note that $de^{-ux}\sqrt{u} = -e^{-ux}\left(x\sqrt{u} - \frac{1}{2\sqrt{u}}\right) du$, hence

$$x \int_0^\infty \frac{e^{-ux}}{\sqrt{u}} du - \left(x + \frac{1}{2}\right) \int_0^\infty \frac{e^{-ux}}{\sqrt{u}(1+u)} du \quad (11.47)$$

$$= \int_0^\infty e^{-ux} \left(x\sqrt{u} - \frac{1}{2\sqrt{u}}\right) \frac{du}{1+u} \quad (11.48)$$

$$= - \int_0^\infty \frac{de^{-ux}\sqrt{u}}{1+u} = - \int_0^\infty e^{-ux} \sqrt{u} \frac{du}{(1+u)^2} \quad (11.49)$$

$$= \int_0^\infty \frac{e^{-ux}}{\sqrt{u}} \frac{du}{(1+u)^2} - \int_0^\infty \frac{e^{-ux}}{\sqrt{u}} \frac{du}{1+u}. \quad (11.50)$$

$$(\sqrt{u} = \frac{1}{\sqrt{u}}((1+u) - 1))$$

11.5.1.5 E5) Define $G : I \rightarrow \mathbb{R}, x \mapsto \sqrt{x}e^{-x}F(x)$. Prove that $\exists C \in \mathbb{R}$ such that

$$G(x) = C - K \int_0^x \frac{e^{-t}}{\sqrt{t}} dt.$$

Proof: By B4)

$$G'(x) = \sqrt{x}e^{-x}F'(x) + \left(\frac{1}{2\sqrt{x}} - \sqrt{x}\right)e^{-x}F(x) = -K\frac{e^{-x}}{\sqrt{x}}.$$

Hence let $C = G(0)$, then

$$G(x) = C + \int_0^x G'(t) dt = C - K \int_0^x \frac{e^{-t}}{\sqrt{t}} dt.$$

11.5.1.6 E6) Calculate the value of K .

Solution: Note that when $x \rightarrow \infty$, $F(x) \rightarrow 0$ hence $G(x) \rightarrow 0$. Therefore

$$0 = \lim_{x \rightarrow \infty} G(x) = G(0) - K \int_0^\infty \frac{e^{-t}}{t} dt = G(0) - K^2.$$

Where

$$G(0) = \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{e^x} \int_0^\infty \frac{e^{-u}}{\sqrt{u}(x+u)} du = \lim_{x \rightarrow 0} \int_0^\infty \frac{e^{-ux}}{\sqrt{u}(1+u)} du \quad (11.51)$$

$$= \int_0^\infty \frac{1}{\sqrt{u}(1+u)} du = \int_0^\infty \frac{2dt}{1+t^2} = \pi. \quad (11.52)$$

Hence $K = \sqrt{\pi}$.

11.5.2 Part 2

Define

$$f(x) = \sum_{n=1}^\infty \frac{e^{-nx}}{\sqrt{n}}, g(x) = \sum_{n=0}^\infty \sqrt{n}e^{-nx}.$$

11.5.2.1 E7) Prove that f, g are well-defined on I and are both continuous on I .

Proof: Let $C = \sup_{x \geq 0} x^3 e^{-x}$, then

$$\sum_{n=1}^{\infty} \frac{e^{-nx}}{\sqrt{n}} < \sum_{n=0}^{\infty} \sqrt{n} e^{-nx} \leq \sum_{n=1}^{\infty} \frac{C}{(nx)^2 \sqrt{x}} \text{ converges.}$$

On any closed sub-interval of I , the two series both converge uniformly, and e^{-nx} is continuous, hence f, g are both continuous on I .

11.5.2.2 E8) Prove that $\forall x \in I$,

$$\int_1^{\infty} \frac{e^{-ux}}{\sqrt{u}} du \leq f(x) \leq \int_0^{\infty} \frac{e^{-ux}}{\sqrt{u}} du.$$

Proof: The function $e^{-ux}/\sqrt{u}$ is monotonously decreasing by u , hence

$$\int_1^N \frac{e^{-ux}}{\sqrt{u}} du \leq \sum_{n=1}^{N-1} \frac{e^{-nx}}{\sqrt{n}} \leq f(x). \quad (11.53)$$

$$\sum_{n=1}^N \frac{e^{-nx}}{\sqrt{n}} \leq \int_0^N \frac{e^{-ux}}{\sqrt{u}} du \leq \int_0^{\infty} \frac{e^{-ux}}{\sqrt{u}} du. \quad (11.54)$$

Therefore

$$\int_1^{\infty} \frac{e^{-ux}}{\sqrt{u}} du \leq f(x) \leq \int_0^{\infty} \frac{e^{-ux}}{\sqrt{u}} du.$$

11.5.2.3 E9) Prove that $\exists C_0$ such that

$$\lim_{x \rightarrow 0^+} \sqrt{x} f(x) = C_0.$$

Proof: By E8)

$$\sqrt{x} f(x) \leq \int_0^{\infty} \frac{e^{-ux}}{\sqrt{ux}} du x = \int_0^{\infty} \frac{e^{-t}}{\sqrt{t}} dt = \sqrt{\pi}. \quad (11.55)$$

$$\sqrt{x} f(x) \geq \int_1^{\infty} \frac{e^{-ux}}{\sqrt{ux}} du x = \int_x^{\infty} \frac{e^{-t}}{\sqrt{t}} dt \rightarrow \sqrt{\pi}. \quad (11.56)$$

Hence

$$\lim_{x \rightarrow 0^+} \sqrt{x} f(x) = \sqrt{\pi}.$$

11.5.2.4 E10) Define the sequence $\{a_n\}_{n \geq 1}$ as follows:

$$a_n = \left(\sum_{k=1}^n \frac{1}{\sqrt{k}} \right) - 2\sqrt{n}.$$

Prove that $\{a_n\}$ converges. Proof: By Euler-Maclaurin formula, for $f(x) = 1/\sqrt{x}$,

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} = \frac{f(1) + f(n)}{2} + \int_1^n \frac{1}{\sqrt{x}} dx + \int_1^n \tilde{B}_1(x) f'(x) dx \quad (11.57)$$

$$= 2\sqrt{n} - \frac{3}{2} + \frac{1}{2\sqrt{n}} + \int_1^n \tilde{B}_1(x) f'(x) dx \quad (11.58)$$

$$(11.59)$$

Hence

$$\lim_{n \rightarrow \infty} a_n = - \int_1^{\infty} \frac{\tilde{B}_1(x)}{2x^{3/2}} dx - \frac{3}{2}.$$

11.5.2.5 E11) Prove that for any $x \in I$, the function

$$h(x) = \sum_{n \geq 1} \left(\sum_{k=1}^n \frac{1}{\sqrt{k}} \right) e^{-nx}$$

is well-defined. Proof: By E10), $|a_n|$ is bounded, hence

$$h(x) = \sum_{n \geq 1} 2\sqrt{n}e^{-nx} + a_n e^{-nx} = 2g(x) + \sum_{n \geq 1} a_n e^{-nx} \leq 2g(x) + \sup_n |a_n| \cdot \frac{1}{e^x - 1}.$$

11.5.2.6 E12) Express $h(x)$ using $f(x)$ and find a constant C_1 such that

$$\lim_{x \rightarrow 0^+} x^{\frac{3}{2}} h(x) = C_1.$$

Proof: Since $e^{-nx}/k > 0$, we can interchange the sums

$$h(x) = \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} \sum_{n=k}^{\infty} e^{-nx} = \sum_{k=1}^{\infty} \frac{e^{-kx}}{\sqrt{k}} \frac{1}{1 - e^{-x}} = \frac{1}{1 - e^{-x}} f(x).$$

Therefore

$$\lim_{x \rightarrow 0^+} x^{3/2} h(x) = \lim_{x \rightarrow 0^+} \sqrt{x} f(x) = \sqrt{\pi}.$$

11.5.2.7 E13) Prove that

$$\lim_{x \rightarrow 0^+} x^{\frac{3}{2}} g(x) = \frac{\sqrt{\pi}}{2}.$$

Proof:

$$\lim_{x \rightarrow 0^+} x^{3/2} |h(x) - 2g(x)| \leq \lim_{x \rightarrow 0^+} \sup_n |a_n| \cdot \frac{x^{3/2}}{e^x - 1} = 0.$$

Hence

$$\lim_{x \rightarrow 0^+} x^{3/2} g(x) = \frac{1}{2} \lim_{x \rightarrow 0^+} x^{3/2} h(x) = \frac{\sqrt{\pi}}{2}.$$

11.5.3 Part 3

Given $A \subset \mathbb{Z}_{\geq 0}$, we can define a sequence $\{a_n\}_{n \geq 0}$:

$$a_n = \begin{cases} 1, & \text{if } n \in A; \\ 0, & \text{if } n \notin A. \end{cases}$$

Define the set $I_A \subset \mathbb{R}_{\geq 0}$ as follows:

$$I_A = \left\{ x \geq 0 : \text{the series } \sum_{n \geq 0} a_n e^{-nx} \text{ converges} \right\}.$$

Define the function $f_A : I_A \rightarrow \mathbb{R}$ as follows:

$$f_A(x) = \sum_{n \geq 0} a_n e^{-nx}.$$

Let $\Phi(A) = \lim_{x \rightarrow 0} x f_A(x)$ (if the limit exists) and let

$$\mathcal{S} = \{A \subset \mathbb{Z}_{\geq 0} : \lim_{x \rightarrow 0^+} x f_A(x) \text{ exists}\}.$$

For example, let

$$A_1 = \{n^2 : n \in \mathbb{Z}_{\geq 1}\}, A_2 = \{p^2 + q^2 : p, q \in \mathbb{Z}_{\geq 1}\}.$$

11.5.3.1 E14) Determine the set I_A .

Solution: If A is finite, then $I_A = \mathbb{R}_{\geq 0}$. Otherwise $I_A = \mathbb{R}_{> 0} = I$.

11.5.3.2 E15) Given $A \subset \mathbb{Z}_{\geq 0}$, for any $n \geq 0$, define the set $A_{\leq n}$:

$$A_{\leq n} = \{\ell \in A : \ell \leq n\}.$$

Prove that for any $x > 0$, the series

$$\sum_{n=0}^{\infty} |A_{\leq n}| \cdot e^{-nx}$$

converges, and satisfy

$$\sum_{n=0}^{\infty} |A_{\leq n}| \cdot e^{-nx} = \frac{f_A(x)}{1 - e^{-x}}.$$

Proof: $|A_{\leq n}| \leq n + 1$, hence

$$\sum_{n=0}^{\infty} |A_{\leq n}| \cdot e^{-nx} \text{ converges.}$$

Therefore

$$\sum_{n=0}^{\infty} |A_{\leq n}| \cdot e^{-nx} = \sum_{n=0}^{\infty} \sum_{k=0}^n a_k \cdot e^{-nx} = \sum_{k=0}^{\infty} a_k \cdot \frac{e^{-kx}}{1 - e^{-x}} = \frac{f_A(x)}{1 - e^{-x}}.$$

11.5.3.3 E16) Prove that for any $x > 0$

$$\frac{f_{A_1}(x)}{1 - e^{-x}} = \sum_{n=0}^{\infty} \lfloor \sqrt{n} \rfloor e^{-nx}.$$

Proof: By E15),

$$|A_{1 \leq n}| = \sum_{k=0}^n [\sqrt{k} \in \mathbb{Z}_{\geq 1}] = \lfloor \sqrt{n} \rfloor.$$

11.5.3.4 E17) Prove that

$$\lim_{x \rightarrow 0^+} \sqrt{x} f_{A_1}(x)$$

exists and calculate the value of $\Phi(A_1)$. Proof:

$$\lim_{x \rightarrow 0^+} \sqrt{x} f_{A_1}(x) = \lim_{x \rightarrow 0^+} \sqrt{x} (1 - e^{-x}) \left(g(x) - \sum_{n=0}^{\infty} \lfloor \sqrt{n} \rfloor e^{-nx} \right).$$

Since $1 - e^{-x} \sim x$, $g(x) \sim \frac{\sqrt{\pi}}{2} x^{-3/2}$, and

$$\left| \sum_{n=0}^{\infty} \lfloor \sqrt{n} \rfloor e^{-nx} \right| \leq \frac{1}{1 - e^{-x}}.$$

Hence

$$\lim_{x \rightarrow 0^+} \sqrt{x} f_{A_1}(x) = \frac{\sqrt{\pi}}{2}.$$

and

$$\Phi(A_1) = \lim_{x \rightarrow 0^+} x f_{A_1}(x) = 0.$$

11.5.3.5 E18) Let $v(n) = \#\{(p, q) \in \mathbb{Z}_{\geq 1}^2 : p^2 + q^2 = n\}$. Prove that for any $x > 0$, the series

$$\sum_{n \geq 1} v(n) e^{-nx}$$

converges and

$$\sum_{n \geq 1} v(n) e^{-nx} = (f_{A_1}(x))^2.$$

Proof: Since $v(n) \leq n$, $\sum_{n \geq 1} v(n) e^{-nx}$ converges.

$$\sum_{n \geq 1} v(n) e^{-nx} = \sum_{n \geq 1} \sum_{k=0}^n a_k a_{n-k} e^{-nx} = \sum_{n \geq 1} \sum_{k=0}^n a_k e^{-kx} \cdot a_{n-k} e^{-(n-k)x} = (f_{A_1}(x))^2.$$

11.5.3.6 E19) Prove that for any $x > 0$

$$f_{A_2}(x) \leq (f_{A_1}(x))^2$$

and give an upper-bound of $\Phi(A_2)$ (assuming it exists). Proof:

$$f_{A_2}(x) = \sum_{n \geq 1} [v(n) \geq 1] \cdot e^{-nx} \leq \sum_{n \geq 1} v(n) e^{-nx} = (f_{A_1}(x))^2.$$

Hence

$$\Phi(A_2) = \lim_{x \rightarrow 0^+} x f_{A_2}(x) \leq \lim_{x \rightarrow 0^+} (\sqrt{x} f_{A_1}(x))^2 = \frac{\pi}{4}.$$

11.5.4 Part 4

Assume $\{a_n\}_{n \geq 0}$ is a sequence of non-negative numbers, such that for any $x > 0$ the series

$$S(x) = \sum_{n \geq 0} a_n e^{-nx}$$

converges. Moreover, assume that the limit below exists:

$$\lim_{x \rightarrow 0^+} x S(x) = \lim_{x \rightarrow 0^+} x \sum_{n \geq 0} a_n e^{-nx} = \ell \in [0, +\infty).$$

Let $F = \{f : [0, 1] \rightarrow \mathbb{R}\}$, $E_0 = C([0, 1])$. Let E be the space of piecewise continuous functions, and define the norm on E :

$$\|\psi\|_\infty = \sup_{x \in [0, 1]} |\psi(x)|.$$

11.5.4.1 E20) Define $L : E \rightarrow F$ as follows:

$$(L(\psi))(x) = \sum_{n=0}^{\infty} a_n e^{-nx} \psi(e^{-nx}), \psi \in E.$$

Prove that L is well-defined and is linear. Moreover, if for any $x \in [0, 1]$, $\psi_1(x) \leq \psi_2(x)$, then for any $x \in [0, 1]$,

$$(L(\psi_1))(x) \leq (L(\psi_2))(x).$$

Proof: Since $\psi \in E$, ψ is bounded, hence L is well-defined and is clearly linear. The inequality holds since a_n are non-negative.

11.5.4.2 E21) Define the subspace of E

$$E_1 = \{\psi \in E : \lim_{x \rightarrow 0^+} x(L(\psi))(x) \text{ exists}\}.$$

Define the linear map $\Delta : E_1 \rightarrow \mathbb{R}$ as follows:

$$\Delta(\psi) = \lim_{x \rightarrow 0^+} x(L(\psi))(x), \psi \in E_1.$$

Prove that E_1 is a subspace of E and there is a constant $M > 0$ such that for any $\psi \in E_1$,

$$|\Delta(\psi)| \leq M \|\psi\|_\infty.$$

Proof: Since L is linear, so is Δ , thus E_1 is clearly a subspace of E .

$$|\Delta(\psi)| = \left| \lim_{x \rightarrow 0^+} x \sum_{n=0}^{\infty} a_n e^{-nx} \psi(e^{-nx}) \right| \leq \|\psi\|_\infty \cdot \left| \lim_{x \rightarrow 0^+} x S(x) \right| = \ell \|\psi\|_\infty.$$

11.5.4.3 E22) For the polynomial $P_n(x) = x^n$, prove that $P_n \in E_1$ and calculate $\Delta(P_n)$.

Proof:

$$\Delta(P_n) = \lim_{x \rightarrow 0^+} x \sum_{k=0}^{\infty} a_k e^{-kx} e^{-nkx} = \frac{1}{n+1} \ell.$$

11.5.4.4 E23) Prove that $E_0 \subset E_1$ and for every $\psi \in E_0$ calculate $\Delta(\psi)$.

Proof: Since Δ is linear, by E22) we know that for any polynomial P ,

$$\Delta(P) = \int_0^1 P(x) dx.$$

By Stone-Weierstraß theorem, any continuous function on $[0, 1]$ can be uniformly approximated with polynomials, hence (same as E24)

$$\Delta(\psi) = \int_0^1 \psi(x) dx, \forall \psi \in E_0.$$

11.5.4.5 E24) For $a \in (0, 1)$, $\varepsilon \in (0, \min(a, 1 - a))$, define the functions

$$g_-(x) = \begin{cases} 1, & x \in [0, a - \varepsilon]; \\ \frac{a-x}{\varepsilon}, & x \in (a - \varepsilon, a); \\ 0, & x \in [a, 1] \end{cases}, g_+(x) = \begin{cases} 1, & x \in [0, a]; \\ \frac{a+\varepsilon-x}{\varepsilon}, & x \in (a, a + \varepsilon); \\ 0, & x \in [a + \varepsilon, 1] \end{cases}.$$

Prove that $g_\pm \in E_0$ and calculate $\Delta(g_\pm)$. Further prove that $\mathbf{1}_{[0,a]} \in E_1$ and calculate $\Delta(\mathbf{1}_{[0,a]})$. Proof: $g_\pm \in E_0$ is trivial, and $\Delta(g_\pm) = \ell \int_0^1 g_\pm = \ell(a \pm \varepsilon/2)$. Since $g_- \leq \mathbf{1}_{[0,a]} \leq g_+$,

$$x(L(g_-))(x) \leq x(L(\mathbf{1}_{[0,a]}))(x) \leq x(L(g_+))(x)$$

For any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $0 < x < \delta$,

$$|x(L(g_-))(x) - \Delta(g_-)|, |x(L(g_+))(x) - \Delta(g_+)| < \frac{\ell\varepsilon}{2}.$$

Hence for any $0 < x < \delta$,

$$x(L(\mathbf{1}_{[0,a]}))(x) \leq x(L(g_+))(x) \leq \Delta(g_+) + \frac{\ell\varepsilon}{2} = a + \ell\varepsilon. \quad (11.60)$$

$$x(L(\mathbf{1}_{[0,a]}))(x) \geq x(L(g_-))(x) \geq \Delta(g_-) - \frac{\ell\varepsilon}{2} = a - \ell\varepsilon. \quad (11.61)$$

Therefore

$$\Delta(\mathbf{1}_{[0,a]}) = \lim_{x \rightarrow 0^+} x(L(\mathbf{1}_{[0,a]}))(x) = a.$$

11.5.4.6 E25) Prove that $E_1 = E$ and for $\psi \in E$ determine the formula of $\Delta(\psi)$.

Proof: Use the same method as E24) applied to Darboux's sum. Hence

$$E_1 = E, \text{ and } \Delta(\psi) = \ell \int_0^1 \psi(x) dx.$$

11.5.4.7 E26) Define the function

$$\psi(x) = \begin{cases} 0, & x \in [0, e^{-1}); \\ \frac{1}{x}, & x \in [e^{-1}, 1]. \end{cases}$$

Prove the following equation by calculating $L(\psi) \left(\frac{1}{N} \right)$:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=0}^N a_k = \ell.$$

Proof:

$$(L(\psi)) \left(\frac{1}{N} \right) = \sum_{n=0}^{\infty} a_n e^{-n/N} \psi(e^{-n/N}) = \sum_{n=0}^N a_n.$$

Hence by E25),

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^N a_n = \Delta(\psi) = \ell \int_0^1 \psi(x) dx = \ell.$$

11.5.4.8 E27) Consider $A \in \mathcal{S}$, and calculate

$$\lim_{n \rightarrow \infty} \frac{|A_{\leq n}|}{n}.$$

which is called the asymptomatic density of A on $\mathbb{Z}_{\geq 0}$. Solution:

$$\lim_{n \rightarrow \infty} \frac{|A_{\leq n}|}{n} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^N a_n = \lim_{x \rightarrow 0^+} x \sum_{n=0}^{\infty} a_n e^{-nx} = \Phi(A).$$

11.5.4.9 E28) Calculate

$$\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n v(k)}{n},$$

and give an upper-bound of the asymptomatic density of A_2 . Solution:

$$\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n v(k)}{n} = \lim_{x \rightarrow 0^+} x \sum_{n=0}^{\infty} v(n) e^{-nx} = \lim_{x \rightarrow 0^+} x (f_{A_1}(x))^2 = \frac{\pi}{4}.$$

From E19) $\Phi(A_2) \leq \frac{\pi}{4}$.

Quote: God does not care about our mathematical difficulties. He integrates empirically. —Albert Einstein

Chapter 12 Oscillatory Integral

12.1 PSA: Stieltjes Integral

Let μ be a monotonic function on $I = [a, b]$.

12.1.1 A1) For any pair of partitions $\sigma, \sigma' \in \mathcal{S}(I)$,

$$\underline{S}_\mu(f; \sigma) \leq \overline{S}_\mu(f; \sigma').$$

Proof: Suppose $\mathcal{C} = \sigma \cup \sigma'$, then

$$\underline{S}_\mu(f; \sigma) \leq \underline{S}_\mu(f; \mathcal{C}) \leq \overline{S}_\mu(f; \mathcal{C}) \leq \overline{S}_\mu(f; \sigma').$$

12.1.2 A2) For any $\rho \in C([a, b])$, $\rho \geq 0$, $\mu(x) = \int_a^x \rho(t) dt$. Prove that for any $f \in \mathcal{R}([a, b])$, $f \in \mathcal{R}([a, b]; \mu)$ and

$$\int_a^b f d\mu = \int_a^b f(x) \rho(x) dx.$$

Proof: Consider any $\mathcal{C} = \{x_0, x_1, \dots, x_n\}$, then if we denote $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$, $u_i = \inf_{x \in [x_{i-1}, x_i]} \rho(x)$, $v_i = \inf_{x \in [x_{i-1}, x_i]} f(x) \rho(x)$, $M = \sup_{x \in [a, b]} f(x)$, $v_i - m_i u_i \leq f(t) \rho(t) - f(t) u_i \leq M \omega_\rho(x_{i-1}, x_i)$. Hence for any $\varepsilon > 0$ there exists a $\delta > 0$, for any $\max\{x_i - x_{i-1}\} < \delta$, $\sup_{x, y \in [x_{i-1}, x_i]} |\rho(x) - \rho(y)| < \varepsilon$. Then

$$\underline{S}(f\rho; \mathcal{C}) = \sum_{k=1}^n v_k(x_k - x_{k-1}) \leq \sum_{k=1}^n u_k m_k(x_k - x_{k-1}) + M\varepsilon(b-a) \quad (12.1)$$

$$\leq M\varepsilon(b-a) + \sum_{k=1}^n m_k \int_{x_{k-1}}^{x_k} \rho(t) dt = M\varepsilon(b-a) + \underline{S}_\mu(f; \mathcal{C}). \quad (12.2)$$

The other side is similar, hence $\sup\{\underline{S}_\mu(f; \mathcal{C})\} = \inf\{\overline{S}_\mu(f; \mathcal{C})\}$ so $f \in \mathcal{R}([a, b]; \mu)$ and

$$\int_a^b f d\mu = \int_a^b f(x) \rho(x) dx.$$

12.1.3 A3) Prove that $\mathcal{R}(I; \mu)$ is a linear space on $\mathbb{R}$ and the integration operator

$$\int_a^b \cdot d\mu : \mathcal{R}(I; \mu) \rightarrow \mathbb{R}, f \mapsto \int_a^b f d\mu.$$

is linear. Proof: Since $\underline{S}_\mu(\cdot; \mathcal{C})$ and $\overline{S}_\mu(\cdot; \mathcal{C})$ is linear for any $\mathcal{C}$, $\mathcal{R}(I; \mu)$ is clearly a linear space on $\mathbb{R}$, and $\int_a^b \cdot d\mu$ is a linear operator.

12.1.4 A4) Suppose $f_1, f_2 \in \mathcal{R}(I; \mu)$. If the any $x \in I$, $f_1(x) \leq f_2(x)$, then

$$\int_a^b f_1 d\mu \leq \int_a^b f_2 d\mu.$$

Proof: By A3), we can assume $f_1 = 0$. Then for any $\mathcal{C}$, $\underline{S}_\mu(f; \mathcal{C}) \geq 0$ since $f \geq 0$, hence $\int_a^b f d\mu = \sup\{\underline{S}_\mu(f; \mathcal{C})\} \geq 0$.

12.1.5 A5) If $f \in \mathcal{R}([a, b]; \mu)$, then for any $c \in [a, b]$, $f|_{[a, c]}$ and $f|_{[c, b]}$ are both Stieltjes integrable and

$$\int_a^b f \, d\mu = \int_a^c f \, d\mu + \int_c^b f \, d\mu.$$

Proof: For any partition σ , let $\sigma' = \sigma \cup \{c\}$, then σ' can be split into two partitions of the intervals $[a, c]$ and $[c, b]$: $\sigma' = \sigma_1 \cup \sigma_2$, such that $\underline{S}_\mu(f; \sigma') = \underline{S}_\mu(f; \sigma_1) + \underline{S}_\mu(f; \sigma_2)$ and $\bar{S}_\mu(f; \sigma') = \bar{S}_\mu(f; \sigma_1) + \bar{S}_\mu(f; \sigma_2)$. Hence

$$\inf \underline{S}_\mu(f; \sigma_1) + \inf \underline{S}_\mu(f; \sigma_2) \leq \inf \underline{S}_\mu(f; \sigma') \leq \sup \bar{S}_\mu(f; \sigma') \leq \sup \bar{S}_\mu(f; \sigma_1) + \sup \bar{S}_\mu(f; \sigma_2).$$

Therefore

$$\int_a^b f \, d\mu = \int_a^c f \, d\mu + \int_c^b f \, d\mu.$$

12.1.6 A6) If $f, g \in \mathcal{R}([a, b]; \mu)$, then $f \cdot g \in \mathcal{R}([a, b]; \mu)$.

Proof: Same as in the case of the Riemann integral.

12.1.7 A7) Define Stieltjes integral on the interval $[0, \infty)$: Suppose $f \in C([0, \infty))$ is continuous and bounded, define

$$\int_0^\infty f \, d\mu = \lim_{M \rightarrow \infty} \int_0^M f \, d\mu.$$

Suppose $\{\alpha_n\}_{n \geq 1}$ is a sequence of positive real numbers and $\sum_{n=1}^\infty \alpha_n$ converges, define the monotonic function $\mu = \sum_{n=1}^\infty \alpha_n \mathbf{1}_{\geq n}$, then

$$\int_1^\infty f \, d\mu = \sum_{n=1}^\infty \alpha_n f(n).$$

Proof: Note that

$$\mu(x+0) - \mu(x-0) = \begin{cases} 0, & x \notin \mathbb{Z}, \\ \alpha_x, & x \in \mathbb{Z}. \end{cases}$$

Hence

$$\int_0^N f \, d\mu = \sum_{n=1}^{N-1} f(n) \alpha_n.$$

By definition,

$$\int_0^\infty f \, d\mu = \sum_{n=1}^\infty \alpha_n f(n).$$

12.1.8 A8) $f, g \in \mathcal{R}([a, b]; \mu)$ are real-valued Riemann integrable functions. Suppose for any $x \in [a, b]$, $g(x) \geq 0$. Let

$$m = \inf_{x \in I} f(x), \quad M = \sup_{x \in I} f(x).$$

Then there exists $\ell \in [m, M]$ such that

$$\int_a^b fg \, d\mu = \ell \int_a^b g \, d\mu.$$

Proof: Note that $mg \leq fg \leq Mg$, and use A4).

12.1.9 A9) Construct a Stieltjes integral to show that Abel summation method is a special case of integration by parts.

Proof: The Abel summation formula states that

$$\sum_{i=1}^n T_i(S_i - S_{i-1}) = T_n S_n - T_1 S_0 - \sum_{i=1}^{n-1} S_i(T_{i+1} - T_i).$$

Consider the monotonically increasing function $\mu : [0, n] \rightarrow \mathbb{R}, x \mapsto T_{\lceil x \rceil}, \mu(0) = T_1$, and f be a polynomial such that $f(k) = S_k$ for $k = 0, 1, \dots, n$. Then

$$\int_0^n f' \mu \, dx = \sum_{k=1}^n \int_{k-1}^k f' \mu = \sum_{k=1}^n \int_{k-1}^k f'(x) T_k \, dx = \sum_{k=1}^n T_k (S_k - S_{k-1}).$$

While

$$\int_0^n f \, d\mu = \sum_{k=1}^{n-1} f(k)(\mu(k+0) - \mu(k)) = \sum_{k=1}^{n-1} S_k (T_{k+1} - T_k).$$

and

$$f\mu \Big|_0^n = T_n S_n - T_1 S_0.$$

Hence the formula is a special case of integration by parts.

12.2 PSB: Convergence of Improper Integrals

b can be ∞ .

12.2.1 B1) Assume $f : [a, b) \rightarrow \mathbb{R}$, and for any $b^- < b$, f is integrable on $[a, b^-]$. Prove that the integral $\int_a^b f(x) \, dx$ exists iff: for any $\varepsilon > 0$, $\exists b(\varepsilon) \in (a, b)$ such that for any $b', b'' > b(\varepsilon)$, $\left| \int_{b'}^{b''} f(x) \, dx \right| < \varepsilon$.

Proof: Let

$$F(t) = \int_a^t f(x) \, dx, \forall t \in [a, b).$$

Then $\int_a^b f(x) \, dx$ exists iff $\lim_{t \rightarrow b^-} F(t)$ exists, which is equivalent to

$$\forall \varepsilon > 0, \exists b(\varepsilon) \in (a, b), \forall b', b'' > b(\varepsilon), \left| \int_{b'}^{b''} f(x) \, dx \right| = |F(b'') - F(b')| < \varepsilon.$$

12.2.2 B2) If $|f(x)| \leq F(x), x \in [a, b)$ and $\int_a^b F(x) \, dx$ converges, then $\int_a^b f(x) \, dx$ converges.

Proof: Use B1) and

$$\left| \int_u^v f(x) \, dx \right| \leq \int_u^v F(x) \, dx.$$

12.2.3 B3) Prove the Dirichlet test for convergence: if $f, g : [a, \infty) \rightarrow \mathbb{R}$ satisfy

- f is continuous and there exists $A > 0$, such that for any $M \geq a$,

$$\left| \int_a^M f(x) \, dx \right| \leq A.$$

- g is monotonic and $\lim_{x \rightarrow \infty} g(x) = 0$.

Then $\int_a^\infty f(x)g(x) \, dx$ converges.

12.2.4 Lemma: The Second Integral Mean Value Theorem

If f is integrable and g is monotonic and non-negative(or non-positive) on $[a, b]$, then there exists $c \in (a, b)$ such that

$$\int_a^b f(x)g(x) \, dx = g(a) \int_a^c f(x) \, dx + g(b) \int_c^b f(x) \, dx.$$

Proof: Assume that g is non-negative and monotonically decreasing. It is easy to see that there exists $\xi \in (a, b)$ such that

$$\int_a^b f(x)g(x) \, dx = g(a) \int_a^\xi f(x) \, dx.$$

Apply the above formula to $f(x)$ and $g(x) - g(b)$ and we get

$$\int_a^b f(x)g(x) \, dx = g(a) \int_a^\xi f(x) \, dx + g(b) \int_\xi^b f(x) \, dx.$$

Proof of B3): Since $|\int_u^v f(x) \, dx| \leq 2A$, by lemma

$$\left| \int_u^v f(x)g(x) \, dx \right| \leq 2A(|g(u)| + |g(v)|).$$

By B1), the integral converges.

12.2.5 B4) Prove the Abel test of convergence:

If $f, g : [a, \infty) \rightarrow \mathbb{R}$ satisfy:

- $\int_a^\infty f(x) \, dx$ exists.
- g is monotonic and g is bounded.

Then $\int_a^\infty f(x)g(x) \, dx$ converges. Proof: Suppose g is monotonically increasing, then

$$\left| \int_u^v f(x)(g(x) - g(a)) \, dx \right| \leq 2M \left(\left| \int_u^\xi f(x) \, dx \right| + \left| \int_\xi^v f(x) \, dx \right| \right) \rightarrow 0$$

since $\int_a^\infty f(x) \, dx$ converges. Therefore both $\int_a^\infty f(x)(g(x) - g(a)) \, dx$ and $\int_a^\infty f(x)g(a) \, dx$ converges, hence $\int_a^\infty f(x)g(x) \, dx$ converges.

12.2.6 B5) Determine whether the following integrals converges:

(1)

$$\int_0^\infty \frac{\log(1+x)}{x^p} \, dx$$

(absolutely) convergent when $1 < p < 2$, diverges when $p \leq 1$ or $p \geq 2$. (2)

$$\int_1^\infty \frac{\sin x}{x^p} \, dx$$

Absolutely convergent when $p > 1$, conditionally convergent when $0 < p \leq 1$, diverges when $p \leq 0$. (3)

$$\int_1^\infty \sin x^2 \, dx = \frac{1}{2} \int_1^\infty \frac{\sin y}{y^{1/2}} \, dy$$

is conditionally convergent. (4)

$$\int_0^\infty \frac{\sin^2 x}{x} \, dx$$

diverges (5) $p, q > 0$,

$$\int_0^{2\pi} \sin^{-p} x \cos^{-q} x \, dx$$

Absolutely convergent when $p, q < 1$, diverges when $p \geq 1$ or $q \geq 1$. (6)

$$\int_0^\infty x^p \sin(x^q) \, dx$$

If $q = 0$ the integral diverges. Assume $q \neq 0$ below.

$$\int_0^\infty x^p \sin(x^q) dx = \frac{1}{q} \int_0^\infty y^{(p+1)/q-1} \sin y dy.$$

Let $\alpha = \frac{p+1}{q} - 1$, then the integral

- diverges if $\alpha \leq -2$ or $\alpha \geq 0$,
- converges absolutely if $-2 < \alpha < -1$.
- converges conditionally if $-1 \leq \alpha < 0$.

(7) $q \geq 0$,

$$\int_0^\infty \frac{x^p \sin x}{1+x^q} dx$$

If $p \leq -2$, then the integral diverges near 0, since $x^p \sin x \sim x^{p+1}$. The integral converges (absolutely) near 0 otherwise. Assume $p > -2$ below. If $p - q < -1$ then the integral converges absolutely when it tends to infinity, since $\frac{x^p}{1+x^q} \sim x^{p-q}$. If $-1 \leq p - q < 0$ then the integral converges conditionally, since the integral of $(x^{p-q})'$ converges. (8)

$$\int_0^{\pi/2} \frac{\log \sin x}{\sqrt{x}} dx = 2 \int_0^{\pi/2} \log \sin x d\sqrt{x} \quad (12.3)$$

$$= 2\sqrt{x} \log \sin x \Big|_0^{\pi/2} - 2 \int_0^{\pi/2} \sqrt{x} \cot x dx \quad (12.4)$$

$$= -2 \int_0^{\pi/2} \frac{\sqrt{x}}{\sin x} \cos x dx \quad (12.5)$$

converges, since $\int_0^1 \frac{1}{\sqrt{x}} dx$ converges. (9)

$$\int_e^\infty \frac{\log \log x}{\log x} \sin x dx = \int_1^\infty \frac{\log y}{y} e^y \sin e^y dy. \quad (12.6)$$

It is easy to see the integral does not converge absolutely. Meanwhile

$$f'(x) = \left(\frac{\log \log x}{\log x} \right)' = \frac{1 - \log \log x}{(\log x)^2 x},$$

and

$$\int_e^\infty \frac{\log \log x - 1}{(\log x)^2 x} dx = \int_1^\infty \frac{\log y - 1}{y^2} dy = \int_0^\infty \frac{t-1}{e^t} dt.$$

converges. By Lagrange mean value theorem,

$$\int_{2\pi}^\infty \frac{\log \log x}{\log x} \sin x dx = \sum_{n=1}^\infty \int_{2n\pi}^{(2n+1)\pi} (f(x) - f(x+\pi)) \sin x dx \quad (12.7)$$

$$\leq \sum_{n=1}^\infty -2\pi f'(2n\pi) \leq 2\pi \int_e^\infty -f'(x) dx \quad (12.8)$$

converges.

12.3 PSC: Oscillatory Integral

$F(t), G(t) : [1, \infty) \rightarrow \mathbb{R}$, $\lim_{t \rightarrow \infty} G(t) = 0$. Assume that for any $t \geq 1$, $G(t) \neq 0$. If

$$\lim_{t \rightarrow \infty} \frac{F(t)}{G(t)} = 1.$$

Then we say F, G have the same order, and $F \sim G$.

12.3.1 Part 1

12.3.1.1 C1) $d > 0$ is a given real number. Assume $g \in C^1([0, d])$. Prove that there is a constant C , such that

$$\left| \int_0^d e^{-tx} g(x) dx \right| \leq \frac{C}{t}.$$

Proof: Let $C = \sup_{x \in [0, d]} |g(x)|$, then

$$\left| \int_0^d e^{-tx} g(x) dx \right| \leq C \int_0^d e^{-tx} dx = \frac{C}{t} (1 - e^{-td}) \leq \frac{C}{t}.$$

12.3.1.2 C2) Assume $d > 0, g \in C([0, d])$ and $g(0) \neq 0$. Prove that

$$\int_0^d e^{-tx} g(x) dx \sim \frac{g(0)}{t}.$$

Proof: Let $M = \sup_{x \in [0, d]} |g(x)|$, then

$$\left| \int_0^d e^{-tx} t \frac{g(x)}{g(0)} dx - 1 \right| = \left| \int_0^{td} e^{-u} \frac{g(u/t)}{g(0)} du - \int_0^\infty e^{-u} du \right| \quad (12.9)$$

$$\leq \int_{td}^\infty e^{-u} du + \int_0^N e^{-u} \left| \frac{g(u/t)}{g(0)} - 1 \right| du + \int_N^{td} e^{-u} \left| \frac{g(u/t)}{g(0)} - 1 \right| du \quad (12.10)$$

$$\leq e^{-td} + \sup_{0 \leq x \leq N/t} \left| \frac{g(x)}{g(0)} - 1 \right| + \left(\frac{M}{|g(0)|} + 1 \right) \int_N^{td} e^{-u} du \rightarrow 0. \quad (12.11)$$

(let $t \rightarrow \infty$ then let $N \rightarrow \infty$).

12.3.1.3 C3) $d > 0, g \in C([0, d]), g(0) \neq 0$. Prove that

$$\int_0^d e^{-tx^2} g(x) dx \sim \frac{\sqrt{\pi} \cdot g(0)}{2\sqrt{t}}.$$

Proof: Same as C2), let $M = \sup_{x \in [0, d]} |g(x)/g(0)|$, then

$$\left| \int_0^d e^{-tx^2} \sqrt{t} \frac{g(x)}{g(0)} dx - \frac{\sqrt{\pi}}{2} \right| = \left| \int_0^{d\sqrt{t}} e^{-u^2} \frac{g(u/\sqrt{t})}{g(0)} du - \int_0^\infty e^{-u^2} du \right| \quad (12.12)$$

$$\leq \int_{d\sqrt{t}}^\infty e^{-u^2} du + \int_0^N e^{-u^2} \left| \frac{g(u/\sqrt{t})}{g(0)} - 1 \right| dx + \int_N^{d\sqrt{t}} e^{-u^2} (M + 1) du. \quad (12.13)$$

which tends to 0, same as C2).

For $t \geq 1, f, \varphi \in C([a, b])$, define the function

$$F(t) = \int_a^b e^{-t\varphi(x)} f(x) dx.$$

Our goal is to study $F(t)$ when $t \rightarrow \infty$.

12.3.1.4 C4) Assume $\varphi \in C^1([a, b])$, and for any $x \in [a, b]$, $\varphi'(x) \neq 0$. Further assume that $\varphi'(x) > 0$. Let $d = \varphi(b) - \varphi(a)$. Prove that

$$\Psi : [a, b] \rightarrow [0, d], \quad x \mapsto \varphi(x) - \varphi(a),$$

is a continuously differentiable bijection on $[a, b]$. Proof: φ is monotonic by $\varphi'(x) > 0$, hence Ψ is a bijection and $\Psi' = \varphi'$.

12.3.1.5 C5) Assume $\varphi \in C^1([a, b])$, and for any $x \in [a, b]$, $\varphi'(x) \neq 0$. Prove that if $f(a) \neq 0$, then when $t \rightarrow \infty$,

$$F(t) \sim \frac{f(a)}{\varphi'(a)} \frac{e^{-t\varphi(a)}}{t}.$$

Proof: Let $g(x) = f(x)/\Psi'(x)$, and $h = (t\Psi)^{-1}$ then

$$\left| F(t) \frac{t}{e^{-t\varphi(a)}} - \frac{f(a)}{\varphi'(a)} \right| = \left| \int_a^b e^{-t\Psi(x)} t f(x) dx - \frac{f(a)}{\Psi'(a)} \right| = \left| \int_a^b e^{-t\Psi(x)} g(x) dt\Psi(x) - g(a) \right| \quad (12.14)$$

$$= \left| \int_0^{t\Psi(b)} e^{-u} g(h(u)) du - g(h(0)) \int_0^\infty e^{-u} du \right| \quad (12.15)$$

$$= |g(h(0))| \int_{t\Psi(b)}^\infty e^{-u} du + \int_0^{t\Psi(b)} e^{-u} |g(h(u)) - g(h(0))| du \quad (12.16)$$

$$+ \int_{N\Psi(b)}^{t\Psi(b)} e^{-u} |g(h(u)) - g(h(0))| du \quad (12.17)$$

$$\leq |g(a)| e^{-t\Psi(b)} + \sup_{x \in [a, \Psi^{-1}(N\Psi(b)/t)]} |g(x) - g(a)| + \int_{N\Psi(b)}^{t\Psi(b)} e^{-u} 2M du. \quad (12.18)$$

which tends to 0 since g is continuous. ($M = \sup_{x \in [a, b]} |g(x)|$).

12.3.1.6 C6) Assume that $\varphi \in C^2([a, b])$, $\varphi'(a) = 0$, $\varphi''(x) > 0$ and for any $x \in (a, b]$, $\varphi'(x) > 0$. Let $d = \sqrt{\varphi(b) - \varphi(a)}$. Prove that

$$\Psi : [a, b] \rightarrow [0, d], \quad x \mapsto \sqrt{\varphi(x) - \varphi(a)}.$$

is a continuously differentiable bijection on $[a, b]$, and calculate $\Psi'(a)$. Proof: Trivial. $\Psi' = \frac{\varphi'}{2\Psi}$, hence

$$\Psi'(a) = \lim_{x \rightarrow a^+} \frac{\varphi'(x)}{2\sqrt{\varphi(x) - \varphi(a)}} = \lim_{x \rightarrow a^+} \frac{\varphi''(x)}{\varphi'(x)/\sqrt{\varphi(x) - \varphi(a)}} = \sqrt{\frac{\varphi''(a)}{2}}.$$

12.3.1.7 C7) Assume $\varphi \in C^2([a, b])$, $\varphi'(a) = 0$, $\varphi''(a) > 0$. Prove that if $f(a) \neq 0$, when $t \rightarrow \infty$,

$$F(t) \sim \frac{\sqrt{\pi} f(a)}{\sqrt{2\varphi''(a)}} \frac{e^{-t\varphi(a)}}{\sqrt{t}}.$$

Proof: Let $g = f/\Psi'$, $h = (\sqrt{t}\Psi)^{-1}$, then

$$F(t) \frac{\sqrt{t}}{e^{-t\varphi(a)}} = \int_a^b e^{-t\Psi^2(x)} f(x) \sqrt{t} dx = \int_a^b e^{-t\Psi^2(x)} g(x) d\sqrt{t}\Psi(x) = \int_0^{\sqrt{t}\Psi(b)} e^{-u^2} g(h(u)) du.$$

Hence

$$\left| F(t) \frac{\sqrt{t}}{e^{-t\varphi(a)}} - \frac{\sqrt{\pi}}{2} g(a) \right| = \left| \int_0^{\sqrt{t}\Psi(b)} e^{-u^2} g(h(u)) du - \int_0^\infty e^{-u^2} g(h(0)) du \right| \quad (12.19)$$

$$\leq g(a) \int_{\sqrt{t}\Psi(b)}^\infty e^{-u^2} du + \int_0^{N\Psi(b)} e^{-u^2} |g(h(u)) - g(h(0))| du \quad (12.20)$$

$$+ \int_{N\Psi(b)}^{\sqrt{t}\Psi(b)} e^{-u^2} 2M du \quad (12.21)$$

$$\leq g(a) e^{-\sqrt{t}\Psi(b)} + \sqrt{\pi} \sup_{x \in [a, \Psi^{-1}(N\Psi(b)/\sqrt{t})]} |g(x) - g(a)| + 2Me^{-N\Psi(b)}. \quad (12.22)$$

which tends to 0 as $t \rightarrow \infty$ and $N \rightarrow \infty$, since g is continuous. (A much simpler solution can be given using the Laplace method)

12.3.1.8 C8) Given $f \in C((0, \infty))$, $\varphi \in C^2((0, \infty))$. Assume that

- exists a unique $a \in (0, \infty)$ such that $\varphi'(a) = 0$;
- $\varphi''(a) > 0$, $f(a) \neq 0$;
- $\int_0^\infty e^{-\varphi(x)} |f(x)| dx$ converges.

Prove that when $t \rightarrow \infty$, the function $G(t) = \int_0^\infty e^{-t\varphi(x)} f(x) dx$ satisfy

$$G(t) \sim \frac{\sqrt{2\pi} f(a)}{\sqrt{\varphi''(a)}} \frac{e^{-t\varphi(a)}}{\sqrt{t}}.$$

Proof: (Simple application of the Laplace method) Apply C7) to the intervals $[a/2, a]$ and $[a, 2a]$, then

$$\int_{a/2}^{2a} e^{-t\varphi(x)} f(x) dx \sim \frac{\sqrt{2\pi} f(a)}{\sqrt{\varphi''(a)}} \cdot \frac{e^{-t\varphi(a)}}{\sqrt{t}}.$$

While the integral on the intervals $(0, a/2)$, $(2a, \infty)$ converges rapidly. Hence

$$G(t) \sim \frac{\sqrt{2\pi} f(a)}{\sqrt{\varphi''(a)}} \frac{e^{-t\varphi(a)}}{\sqrt{t}}.$$

12.3.1.9 C9) $\Gamma(n) = (n-1)!$.

Proof:

$$\Gamma(n+1) = \int_0^\infty t^n e^{-t} dt = - \int_0^\infty t^n de^{-t} = n \int_0^\infty t^{n-1} e^{-t} dt = n\Gamma(n).$$

12.3.1.10 C10) Prove Stirling's approximation

$$n! \sim \sqrt{2\pi} \frac{n^{n+1/2}}{e^n}.$$

Proof: Note that, by substituting $t = ns$,

$$n! = \Gamma(n+1) = \int_0^\infty e^{-t} t^n dt = n^{n+1} \int_0^\infty e^{-n(s-\log s)} ds.$$

Hence

$$\frac{\Gamma(t+1)}{t^{t+1}} \sim \sqrt{2\pi} \frac{e^{-t}}{\sqrt{t}}.$$

12.3.2 Part 2

For $\lambda \geq 1$, $f, \varphi \in C^\infty([a, b])$, define the function

$$I(\lambda) = \int_a^b e^{i\lambda\varphi(x)} f(x) dx.$$

Our goal is to study $I(\lambda)$ when $\lambda \rightarrow \infty$.

12.3.2.1 C11) Assume that for any $x \in [a, b]$, $\varphi'(x) \neq 0$. Define the maps

$$L : C^\infty([a, b]) \rightarrow C^\infty([a, b]), h \mapsto \frac{1}{i\lambda\varphi'(x)} h'(x), \quad (12.23)$$

$$M : C^\infty([a, b]) \rightarrow C^\infty([a, b]), h \mapsto - \left(\frac{h}{i\varphi'} \right)'(x). \quad (12.24)$$

Assume that $f, g \in C^\infty([a, b])$. Prove that if there exists $c > 0$ such that for any $x \in [a, a+c] \cup [b-c, b]$, $h(x) = 0$, then M/λ is the adjoint of L :

$$\int_a^b h \cdot Lg = \frac{1}{\lambda} \int_a^b g \cdot Mh.$$

Proof: By integration of parts,

$$\int_a^b h \cdot Lg = \int_a^b \frac{h}{i\lambda\varphi'} dg = - \int_a^b g d \left(\frac{h}{i\lambda\varphi'} \right) = \frac{1}{\lambda} \int_a^b g \cdot Mh.$$

12.3.2.2 C12) Assume that for any $x \in [a, b]$, $\varphi'(x) \neq 0$ and f vanishes near a and b . prove that for any $n \in \mathbb{Z}_{\geq 1}$, there is a constant c_n independent of λ such that

$$|I(\lambda)| \leq \frac{c_n}{\lambda^n}.$$

Proof: Let $g = e^{i\lambda\varphi}$, then $Lg = g$. $f \in C^\infty([a, b])$ vanishes near a, b hence $M^n f$ vanishes near a, b for any $n \in \mathbb{Z}_{\geq 0}$. Therefore

$$|I(\lambda)| = \left| \int_a^b fg \right| = \frac{1}{\lambda} \left| \int_a^b g \cdot Mf \right| = \dots = \frac{1}{\lambda^n} \left| \int_a^b g \cdot M^n f \right|.$$

so $c_n = \left| \int_a^b g \cdot M^n f \right|$ is valid.

12.3.2.3 C13) If there exists $\delta > 0$, such that for any $x \in [a, b]$, $|\varphi'(x)| \geq \delta$ and $\varphi'(x)$ is monotonic on $[a, b]$. Prove that there is a constant C_1 independent of λ, φ, a, b such that

$$\left| \int_a^b e^{i\lambda\varphi(x)} dx \right| \leq \frac{C_1}{\lambda\delta}.$$

Proof: Let $C_1 = 4$ then (since φ' maintains the same sign)

$$\left| \int_a^b e^{i\lambda\varphi(x)} dx \right| = \left| \int_a^b \frac{de^{i\lambda\varphi}}{\lambda\varphi'} \right| \leq \left| \frac{e^{i\lambda\varphi}}{\lambda\varphi'} \right|_a^b + \frac{1}{\lambda} \left| \int_a^b e^{i\lambda\varphi} \frac{\varphi''}{(\varphi')^2} dx \right| \quad (12.25)$$

$$\leq \frac{2}{\lambda\delta} + \frac{1}{\lambda} \int_a^b \left| \frac{\varphi''}{(\varphi')^2} \right| \quad (12.26)$$

$$= \frac{2}{\lambda\delta} + \frac{1}{\lambda} \int_a^b d \frac{1}{\varphi'} \leq \frac{4}{\lambda\delta}. \quad (12.27)$$

12.3.2.4 C14) Suppose for any $x \in [a, b]$, $|\varphi''(x)| \geq 1$. Prove that there is a unique $c \in [a, b]$ such that

$$|\varphi'(c)| = \inf_{x \in [a, b]} |\varphi'(x)|.$$

Further prove that for any $x \in [a, b]$,

$$|\varphi'(x)| \geq |x - c|.$$

Proof: Since $\varphi \in C^\infty([a, b])$ and $|\varphi''| \geq 1$, φ'' maintains the same sign. Assume that $\forall x \in [a, b]$, $\varphi''(x) \geq 1$, then φ' is monotonically increasing. Therefore, if $\varphi' \neq 0$, then $c \in \{a, b\}$, otherwise, c is the unique null point of φ' . Either $\varphi'(c) = 0$ or $c = a$, when φ' maintains the same sign, so we always have $|\varphi'(x)| \geq |\varphi'(x) - \varphi'(c)|$, and

$$\forall x \in [a, b], \exists \xi \in [x, c], |\varphi'(x) - \varphi'(c)| \geq |x - c| \cdot \varphi'(\xi) \geq |x - c|.$$

Therefore $|\varphi'(x)| \geq |x - c|$.

12.3.2.5 !C15) Assume that for any $x \in [a, b]$, $|\varphi''(x)| \geq 1$. Prove that there is a constant C_2 independent of λ, φ, a, b , such that

$$\left| \int_a^b e^{i\lambda\varphi(x)} dx \right| \leq \frac{C_2}{\sqrt{\lambda}}.$$

Proof: Since $\varphi \in C^\infty([a, b])$, we can assume $\varphi''(x) \geq 1$. For an arbitrary $\delta > 0$, divide the interval $[a, b]$ into two parts: $E_1 = \{x : |\varphi'(x)| \leq \delta\}$ and $E_2 = \{x : |\varphi'(x)| > \delta\}$. Note that $\forall x, y \in E_1$, $|\varphi'(x) - \varphi'(y)| \leq 2\delta$, but $|\varphi'(x) - \varphi'(y)| \geq \left| \int_x^y \varphi''(t) dt \right| \geq |x - y|$. Therefore E_1 is an interval of length at most 2δ , so

$$\left| \int_{E_1} e^{i\lambda\varphi(x)} dx \right| \leq 2\delta.$$

Now consider the integral on E_2 , which is the union of one or two intervals. By C13),

$$\left| \int_{E_2} e^{i\lambda\varphi(x)} dx \right| \leq 2 \cdot \frac{4}{\lambda\delta}.$$

Therefore

$$\left| \int_a^b e^{i\lambda\varphi(x)} dx \right| \leq 2\delta + \frac{8}{\lambda\delta} = \frac{8}{\sqrt{\lambda}}.$$

(if we let $\delta = \sqrt{4/\lambda}$.)

12.3.2.6 C16) Assume that for any $x \in [a, b]$, $|\varphi''(x)| \geq 1$. Prove that there is a constant C_2 independent of $\lambda, \varphi, f, a, b$ such that

$$\left| \int_a^b e^{i\lambda\varphi(x)} f(x) dx \right| \leq \frac{C_2}{\sqrt{\lambda}} \left(|f(a)| + \int_a^b |f'(x)| dx \right).$$

Proof: By C15),

$$\left| \int_a^b e^{i\lambda\varphi(x)} f(x) dx \right| \leq \left| \int_a^b e^{i\lambda\varphi(x)} f(a) dx \right| + \left| \int_a^b e^{i\lambda\varphi(x)} \int_a^x f'(t) dt dx \right| \quad (12.28)$$

$$\leq |f(a)| \frac{C_2}{\sqrt{\lambda}} + \left| \int_a^b f'(t) \int_t^b e^{i\lambda\varphi(x)} dx dt \right| \quad (12.29)$$

$$\leq \frac{C_2}{\sqrt{\lambda}} \left(|f(a)| + \int_a^b |f'(x)| dx \right). \quad (12.30)$$

Chapter 13 Midterm Exam

13.1 Problem A

13.1.1 A1)

Prove using $\varepsilon - N$:

$$\lim_{n \rightarrow \infty} \frac{n}{n^2 + 1} = 0.$$

Proof: For any $\varepsilon > 0$, there exists $N = \lfloor 1/\varepsilon \rfloor + 10$ such that for any $n > N$,

$$0 < \frac{n}{n^2 + 1} < \frac{1}{n} < \varepsilon.$$

Hence $\lim_{n \rightarrow \infty} \frac{n^2}{n^2 + 1} = 0$.

13.1.2 A2)

Prove using $\varepsilon - \delta$ that $f(x) = \begin{cases} \sin(1/x), & x \neq 0; \\ 0, & x = 0 \end{cases}$ is not continuous at x . Proof: Let $\varepsilon = 1$, for any $\delta > 0$, there

exists $x_1 = (2\pi n)^{-1}$ and $x_2 = (2\pi n + \pi/2)^{-1}$ where $n = \lfloor 1/\delta \rfloor + 10$ such that $0 < x_2 < x_1 < \delta$ and $|\sin x_1 - \sin x_2| = 1$, so f is not continuous at x .

13.1.3 A3)

Calculate

$$\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n k^2 \right)^{1/n}.$$

Solution: Clearly $\sum_{k=1}^n k^2 < \sum_{k=1}^n n^2 = n^3$ so $0 < \left(\sum_{k=1}^n k^2 \right)^{1/n} < (n^3)^{1/n} = n^{3/n}$. Since $\lim_{n \rightarrow \infty} n^{1/n} = 0$,

$$\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n k^2 \right)^{1/n} = 0.$$

13.1.4 A6)

Prove that the series $\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+100)^2}$ converges. Proof: Note that

$$\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+100)^2} < \sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2} = 1.$$

Hence the series converges.

13.1.5 A5)

Given a real sequence $\{a_n\}$, if $\sum_{n=1}^{\infty} a_n$ converges, prove that for any $x \in (-1, 1)$, $\sum_{n=1}^{\infty} a_n x^n$ converges. Proof: $\sum_{n=1}^{\infty} a_n$ converges implies $\lim_{n \rightarrow \infty} a_n = 0$. For any $n, m \geq 0$,

$$\left| \sum_{k=n}^{n+m} a_k x^k \right| \leq \sum_{k=n}^{\infty} |a_k| x^k \leq \frac{x^n}{1-x} \sup_{k \geq n} |a_k|.$$

Since $x^n \rightarrow 0$ and $\sup_{k \geq n} |a_k| \rightarrow 0$, the series is Cauchy hence it converges.

13.1.6 A6)

Suppose $f \in C((a, b))$, prove that for any $x_1, x_2 \in (a, b)$, there exists $x_0 \in (a, b)$ such that

$$f(x_0) = \frac{1}{2}(f(x_1) + f(x_2)).$$

Proof: Let $g(x) = f(x) - \frac{1}{2}(f(x_1) + f(x_2))$, then $g(x_1)g(x_2) \leq 0$ so g has a solution $x_0 \in (a, b)$.

13.1.7 A7)

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function with period 1, prove that f is bounded and can reach $\sup f(x)$. Proof: $\sup_{x \in \mathbb{R}} f(x) = \sup_{x \in [0, 1]} f(x) < \infty$ and there exists $x \in [0, 1]$ such that $f(x) = \sup_{y \in [0, 1]} f(y)$.

13.1.8 A8)

Prove that $f(x) = \sqrt{x+1}$ is uniformly continuous on $\mathbb{R}_{\geq 0}$. Proof: Note that

$$|f(x) - f(y)| = |\sqrt{x+1} - \sqrt{y+1}| = \frac{|x - y|}{\sqrt{x+1} + \sqrt{y+1}} \leq |x - y|$$

Hence f is uniformly continuous.

13.2 Problem B

Given a sequence $\{a_k\}$ of complex numbers, let

$$a_n^* = \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} a_k.$$

13.2.1 Part 1

Suppose $a_k = z^k$ for any $k \geq 0$, where $z \in \mathbb{C}$.

13.2.1.1 B1)

Prove that if $|z| < 1$, then $\sum_{k=0}^{\infty} a_k$ converges. Denote the limit by $A(z)$. Proof: Note that

$$\sum_{k=0}^n a_k = \frac{1 - z^{n+1}}{1 - z},$$

and $|z^{n+1}| = |z|^{n+1} \rightarrow 0$ so $\sum_{k=0}^{\infty} a_k$ converges and $A(z) = 1/(1 - z)$.

13.2.1.2 B2)

Prove that if $|z| < 1$, then $\sum_{k=0}^{\infty} a_k^*$ converges, denote it by $A^*(z)$. Proof: Since $a_k^* = \left(\frac{1+z}{2}\right)^k$, and $\left|\frac{1+z}{2}\right| \leq \frac{1+|z|}{2} < 1$, $\sum_{k=0}^{\infty} a_k^*$ converges and $A^*(z) = 1/\left(1 - \frac{1+z}{2}\right) = \frac{2}{1-z}$.

13.2.1.3 B3)

Prove that if $|z| \geq 1$, then $\sum_{k=0}^{\infty} a_k$ does not converge. Proof: $|a_k| = 1$ does not tend to 0, so $\sum_{k=0}^{\infty} a_k$ does not converge.

13.2.1.4 B4)

Find $z \in \mathbb{C}$ such that $|z| > 1$ and $\sum_{k=0}^{\infty} a_k^*$ converges. Solution: Let $z = i\sqrt{2}$, then $\left|\frac{1+z}{2}\right| = \frac{\sqrt{3}}{2} < 1$ so $\sum_{k=0}^{\infty} a_k^*$ converges.

13.2.1.5 B5)

Prove that if $|z| = 1$ and $z \neq \pm 1$, then $\sum_{k=0}^{\infty} a_k^*$ converges. Proof: Likewise $|\frac{1+z}{2}| \leq 1$ and equality holds iff $z = 1$, so $\sum_{k=0}^{\infty} a_k^*$ converges.

13.2.2 Part 2

Suppose $\{a_k\}_{k \geq 0}$ is a sequence of real numbers.

13.2.2.1 B6)

Prove that if $k \in \mathbb{Z}_{\geq 0}$ is fixed,

$$\lim_{n \rightarrow \infty} \binom{n}{k} / \frac{n^k}{k!} = 1, \quad \lim_{n \rightarrow \infty} \binom{n}{k} / 2^n = 0.$$

Proof: Note that

$$\binom{n}{k} / \frac{n^k}{k!} = \prod_{j=0}^{k-1} \frac{n-j}{n}$$

hence $\lim_{n \rightarrow \infty} \binom{n}{k} / \frac{n^k}{k!} = 1$. For $n > 2k$, $\binom{n}{k} < \binom{n}{k+1} < \cdots < \binom{n}{\lfloor n/2 \rfloor}$ so

$$\binom{n}{k} / 2^n < \frac{1}{\lfloor n/2 \rfloor - k} \rightarrow 0.$$

13.2.2.2 B7)

Given any non-negative integer $n > q$, we define

$$a_{n,q}^* = \frac{1}{2^n} \sum_{k=0}^q \binom{n}{k} a_k.$$

For any fixed q , calculate $\lim_{n \rightarrow \infty} a_{n,q}^*$. Solution: By B6) $\lim_{n \rightarrow \infty} \binom{n}{k} / 2^n = 0$ uniformly for any $0 \leq k \leq q$ so $\lim_{n \rightarrow \infty} a_{n,q}^* = 0$.

13.2.2.3 B8)

If $\lim_{n \rightarrow \infty} a_n = 0$, prove that $\lim_{n \rightarrow \infty} a_n^* = 0$. Proof: Note that

$$|a_n^*| \leq |a_{n,q}^*| + \frac{1}{2^n} \sum_{k=q+1}^n |a_k| \leq |a_{n,q}^*| + \sup_{k > q} |a_k|.$$

Since $\lim_{n \rightarrow \infty} a_{n,q}^* = 0$, and $\sup_{k > q} |a_k| \rightarrow 0$ as $q \rightarrow \infty$, $\lim_{n \rightarrow \infty} a_n^* = 0$.

13.2.2.4 B9)

If $\lim_{n \rightarrow \infty} a_n$ exists, prove that $\lim_{n \rightarrow \infty} a_n^*$ exists and is exactly $\lim_{n \rightarrow \infty} a_n$. Proof: Suppose $a = \lim_{n \rightarrow \infty} a_n$, let $b_n = a_n - a$ and $b_n^* = a_n^* - a$, then $b_n^* = \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} a_k$ and $\lim_{n \rightarrow \infty} b_n = 0$. Hence $\lim_{n \rightarrow \infty} b_n^* = 0$ so $\lim_{n \rightarrow \infty} a_n^* = \lim_{n \rightarrow \infty} a_n$.

13.2.2.5 B10)

If $\lim_{n \rightarrow \infty} a_n^*$ exists, does $\lim_{n \rightarrow \infty} a_n$ exist? Solution: Not necessarily. If $a_n = (-1)^n$ then $a_n^* = 0$ so $\lim_{n \rightarrow \infty} a_n^* = 0$ but $\lim_{n \rightarrow \infty} a_n$ does not exist.

13.2.2.6 B11)

For $n \geq 0$ define

$$S_n = \sum_{k=0}^n a_k, S_n^* = \sum_{k=0}^n a_k^*, U_n = 2^n S_n^*.$$

Prove that for any $n \geq 0$,

$$U_n = \sum_{k=0}^n \binom{n+1}{k+1} S_k.$$

Proof: Note that

$$U_n = 2^n \sum_{k=0}^n a_k^* = 2^n \sum_{k=0}^n 2^{-k} \sum_{j=0}^k \binom{k}{j} a_j = \sum_{j=0}^n a_j \sum_{k=0}^n 2^{n-k} \binom{k}{j}$$

while

$$\sum_{k=0}^n \binom{n+1}{k+1} S_k = \sum_{k=0}^n \binom{n+1}{k+1} \sum_{j=0}^k a_j = \sum_{j=0}^n a_j \sum_{k=j}^n \binom{n+1}{k+1}.$$

It suffices to show that for any j and $n \geq j$,

$$\sum_{k=j}^n 2^{n-k} \binom{k}{j} = \sum_{k=j}^n \binom{n+1}{k+1}.$$

Note that

$$\sum_{k=j}^n 2^{n-k} \binom{k}{j} - \sum_{k=j}^{n-1} 2^{n-1-k} \binom{k}{j} = \binom{n}{j} + \sum_{k=j}^{n-1} 2^{n-1-k} \binom{k}{j}$$

and

$$\sum_{k=j}^n \binom{n+1}{k+1} - \sum_{k=j}^{n-1} \binom{n}{k+1} = 1 + \sum_{k=j}^{n-1} \binom{n}{k} = \binom{n}{j} + \sum_{k=j}^{n-1} \binom{n}{k+1}.$$

Hence we can prove this by induction.

13.2.2.7 B12)

Prove that if $\sum_{k=0}^{\infty} a_k$ converges, then $\sum_{k=0}^{\infty} a_k^*$ converges. Proof: Note that $S_n^* = 2 \cdot \frac{1}{2^{n+1}} \sum_{k=0}^n \binom{n+1}{k+1} S_k$ so S_n converges implies S_n^* converges, and $\sum_{k=0}^{\infty} a_k^* = 2 \sum_{k=0}^{\infty} a_k$.

13.3 Problem C: Maximal Ideal of $C([a, b])$

Suppose $a < b$ are real numbers, we study the properties of the ring $C([a, b])$.

13.3.1 C1)

For any subset $A \subset [a, b]$, let $I(A) = \{f \in C([a, b]) : f(A) = \{0\}\}$. Prove that $I(A)$ is an ideal of $C([a, b])$. What is $I([a, b])$? Prove that if $A \subset B$ then $I(B) \subset I(A)$. Does there exist $A \subsetneq [a, b]$ such that $I(A) = \{0\}$? Proof: For any $f, g \in I(A)$ and any $x \in A$, $(f+g)(x) = 0$ and for $f \in I(A)$, $g \in C([a, b])$, $(fg)(x) = 0$ so $I(A)$ is an ideal. If $A \subset B$ and $f \in I(B)$ then $f(A) \subset f(B) = \{0\}$ so $f \in I(A)$. Let $A = (a, b]$, then $f(A) = 0$ implies $f(a) = \lim_{x \rightarrow a^+} f(x) = 0$ so $I(A) = \{0\}$.

13.3.2 C2)

Prove that $I \subset C([a, b])$ is an ideal, then $1 \notin I$. Further prove that for any $f \in I$, $f(x) = 0$ has a root in $[a, b]$. Proof: If $f \neq 0$ on $[a, b]$, then $f^{-1} \in C([a, b])$, so $1 = f \cdot f^{-1} \in I$. Hence for all $g \in C([a, b])$, $g = 1 \cdot g \in I$, implying $I = C([a, b])$, a contradiction.

13.3.3 C3)

For $f \in C([a, b])$, the set $V(f) = f^{-1}(\{0\})$ is closed. Prove that for $I \subset C([a, b])$, $V(I) = \{x \in [a, b] : \forall f \in I, f(x) = 0\}$ is closed. If $V(I) = [a, b]$, can I be determined? Proof: $V(I) = \bigcap_{f \in I} f^{-1}(\{0\})$ is the union of closed sets, hence $V(I)$ is closed. If $V(I) = [a, b]$, then for all $f \in I$ and $x \in [a, b]$, $f(x) = 0$ so $I = \{0\}$.

13.3.4 C4)

For any $x \in [a, b]$, let $A = \{x\}$ and $\mathfrak{m}_x = I(A) = I(\{x\})$. Prove that $\mathfrak{m}_x$ is a maximal ideal. Proof: $\mathfrak{m}_x = \{f \in C([a, b]) : f(x) = 0\}$. If there is a larger ideal $\mathfrak{m}_k \subset \mathfrak{m}$, then there exists $g \in \mathfrak{m} \setminus \mathfrak{m}_k$ so $g(x) \neq 0$. For any $h \in C([a, b])$, note that $h = \frac{h(x)}{g(x)}g + \left(h - \frac{h(x)}{g(x)}g\right)$, where $\frac{h(x)}{g(x)} \cdot g \in \mathfrak{m}$ and $h - \frac{h(x)}{g(x)}g$ vanish at x so is in $\mathfrak{m}$. Hence $h \in \mathfrak{m}$ and $\mathfrak{m} = C([a, b])$, a contradiction.

13.3.5 C5)

Prove that if $\mathfrak{m}$ is a maximal ideal of $C([a, b])$, then there exists $x \in [a, b]$ such that $\mathfrak{m} = \mathfrak{m}_x$. Proof: If for some $x \in [a, b]$, $f(x) = 0$ for all $f \in \mathfrak{m}$, then $\mathfrak{m} \subset \mathfrak{m}_x$. So we suppose $V(\mathfrak{m}) = \emptyset$. For every $x \in [a, b]$ take $f_x \in \mathfrak{m}$ such that $f_x(x) \neq 0$. Note $f_x(x) \neq 0$ implies there exists $\varepsilon_x > 0$ such that for any $y \in (x - \varepsilon_x, x + \varepsilon_x) \cap [a, b]$, $f_x(y) \neq 0$. Since $[a, b] \subset \bigcup_{x \in [a, b]} (x - \varepsilon_x, x + \varepsilon_x)$ and $[a, b]$ is compact, there exists $y_1, \dots, y_n \in [a, b]$ such that $[a, b] \subset \bigcup_{i=1}^n (y_i - \varepsilon_{y_i}, y_i + \varepsilon_{y_i})$. For ever $1 \leq i \leq n$, take the continuous function $g_i(x) = \begin{cases} 0, & |x - y_i| \geq \varepsilon_i, \\ \varepsilon_i - |x - y_i|, & \text{otherwise} \end{cases}$ such that $\text{supp} g_i = [y_i - \varepsilon_{y_i}, y_i + \varepsilon_{y_i}]$, then $h_i = g_i f_{y_i} \in \mathfrak{m}$ and $\text{supp} h_i = [y_i - \varepsilon_i, y_i + \varepsilon_i]$. Suppose $h_i \geq 0$, otherwise replace it with $-h_i$ (since $h_i \neq 0$ on $(y_i - \varepsilon_i, y_i + \varepsilon_i)$, it does not change signs). Then let $f = \sum_{i=1}^n h_i$, we have $f(x) \neq 0, \forall x \in [a, b]$ since $[a, b] \subset \bigcup_{i=1}^n (y_i - \varepsilon_i, y_i + \varepsilon_i)$.

13.3.6 C6)**

Suppose A is closed, prove that $V(I(A)) = A$. Proof: Clearly $A \subset V(I(A))$, consider any $x \notin A$, there exists $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \cap A = \emptyset$. Take $f(y) = \begin{cases} \varepsilon - |y - x|, & |y - x| < \varepsilon \\ 0, & |y - x| \geq \varepsilon \end{cases}$, then $\text{supp} f = [x - \varepsilon, x + \varepsilon]$, so $f \in I(A)$ but $f(x) \neq 0$ so $x \notin V(I(A))$. Therefore $A = V(I(A))$.

[!note] This result can be applied to compact metric spaces: Suppose (X, d) is a compact metric space, $C(X)$ are all complex valued continuous functions on X , then we have the following bijection:

$$X \longrightarrow \{\text{maximal ideal of } C(X)\}, x \mapsto \mathfrak{m}_x,$$

where $\mathfrak{m}_x = \{f \in C(X) : f(x) = 0\}$.

Chapter 14 Final Exam

14.1 PSA (60pts)

14.1.1 A1) (5pts) Compute the maximum value of $f(x) = x^3 - 12x + 1$ on the interval $[0, 1]$.

Solution: For any $x \in [0, 1]$,

$$f(x) = 1 + x(x^2 - 12) \leq 1,$$

and equality holds when $x = 0$. Hence the maximum value is 1.

14.1.2 A2) (5pts) Prove that the polynomial $f(x) = x^3 + ax^2 + bx + c$ cannot be convex.

Proof: $f''(x) = 3x + a$ cannot be positive on $\mathbb{R}$.

14.1.3 A3) (5pts)

Solution:

$$\sqrt{1+x} = 1 + \binom{1/2}{1}x + \binom{1/2}{2}x^2 + \binom{1/2}{3}x^3 + o(x^3) \quad (14.1)$$

$$= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + o(x^3). \quad (14.2)$$

14.1.4 A4) (5pts) Consider the function on $[-1, 1]$:

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \in [-1, 0) \cup (0, 1]; \\ 0, & x = 0. \end{cases}$$

Prove that $f \in C([-1, 1])$ but is not differentiable at 0. Proof: Since $\sin(1/x) \in [-1, 1]$,

$$\lim_{x \rightarrow 0^-} x \sin\left(\frac{1}{x}\right) = \lim_{x \rightarrow 0^+} x \sin\left(\frac{1}{x}\right) = 0.$$

so $f \in C([-1, 1])$. However,

$$f'_+(0) = \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \sin \frac{1}{h},$$

so f is not differentiable at 0.

14.1.5 A5) (5pts) Calculate the integral

$$\int_0^1 \frac{x^2 + 4}{x^2 + 3x + 2} dx.$$

Solution:

$$\int_0^1 \frac{x^2 + 4}{x^2 + 3x + 2} dx = \int_0^1 1 - \frac{8}{x+2} + \frac{5}{x+1} dx = 1 + 13 \log 2 - 8 \log 3.$$

14.1.6 A6) (5pts) Calculate the integral

$$\int_0^1 x^2 e^x dx.$$

Solution:

$$\int_0^1 x^2 e^x dx = \int_0^1 x^2 de^x = x^2 e^x \Big|_0^1 - \int_0^1 2xe^x dx \quad (14.3)$$

$$= e - 2 \int_0^1 x de^x = e - 2xe^x \Big|_0^1 + 2 \int_0^1 e^x dx \quad (14.4)$$

$$= e - 2e + 2e^x \Big|_0^1 = e - 2. \quad (14.5)$$

14.1.7 A7) (5pts) Calculate the integral

$$\int_0^\pi \sin^3(x) + \sin(2x) dx.$$

Solution:

$$\int_0^\pi \sin(2x) dx = -\frac{\cos 2x}{2} \Big|_0^\pi = 0.$$

while

$$\int_0^\pi \sin^3 x dx = -2 \int_0^{\pi/2} \sin^2 x d \cos x \quad (14.6)$$

$$= 2 \int_0^1 (1 - t^2) dt = \frac{4}{3}. \quad (14.7)$$

14.1.8 A8) (5pts) Calculate the integral

$$\int_{-\pi}^\pi \sin(x^3 + 2x) dx.$$

Solution: Note that $\sin(x^3 + 2x)$ is an odd function, so the integral is 0.

14.1.9 A9) (5pts) Calculate the limit

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^{2019}}{n^{2020}}.$$

Solution:

$$\int_0^1 x^{2019} dx \leq \sum_{k=1}^n \left(\frac{k}{n}\right)^{2019} \frac{1}{n} \leq \int_{1/n}^{(n+1)/n} x^{2019} dx$$

Hence

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^{2019}}{n^{2020}} = \frac{1}{2020}.$$

14.1.10 A10) (5pts) Determine whether the following improper integral converges:

$$\int_0^1 \frac{\log x}{x^{1/2}} dx.$$

Solution: Substitute $x = y^2$, then

$$\int_0^1 \frac{\log x}{x^{1/2}} dx = \int_0^1 4 \log y dy = -4.$$

14.1.11 A11) (5pts) Prove that for any $n \geq 1$,

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \geq \log n.$$

Proof: Note that

$$\frac{1}{n} \geq \int_n^{n+1} \frac{1}{x} dx.$$

Hence

$$\sum_{k=1}^n \frac{1}{k} \geq \int_1^{n+1} \frac{1}{x} dx = \log(n+1) \geq \log n.$$

14.1.12 A12) (5pts) Prove that the following improper integral converges:

$$\int_2^{\infty} \frac{\sin x}{\log x} dx.$$

Proof:

$$\int_{2\pi}^{\infty} \frac{\sin x}{\log x} dx = \sum_{n=1}^{\infty} \int_{2n\pi}^{(2n+1)\pi} \left(\frac{1}{\log x} - \frac{1}{\log(x+\pi)} \right) \sin x dx.$$

Note that

$$\left(\frac{1}{\log x} \right)' = -\frac{1}{x \log^2 x}.$$

Hence by Lagrange's mean value theorem,

$$\int_{2\pi}^{\infty} \frac{\sin x}{\log x} dx \leq \sum_{n=1}^{\infty} 2\pi^2 \frac{1}{2n\pi \log^2(2n\pi)}$$

converges.

Or use integration by parts

$$\int_2^{\infty} \frac{\sin x}{\log x} dx = C + \int_2^{\infty} \frac{\cos x}{x \log^2 x} dx.$$

14.2 PSB (21pts) The Arithmetic-Geometric Mean**14.2.1 B1) (2pts) Suppose $a, b > 0$. Prove that the following improper integral converges:**

$$I(a, b) = \int_0^{\infty} \frac{1}{\sqrt{x^2 + a^2} \sqrt{x^2 + b^2}} dx.$$

Proof:

$$I(a, b) \leq \frac{1}{ab} + \int_1^{\infty} \frac{1}{x^2} dx \leq 1 + \frac{1}{ab}.$$

14.2.2 B2) (2pts) Prove that

$$I(a, b) = \int_0^{\pi/2} \frac{1}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} d\theta.$$

Proof: Let $x = b \tan \theta$, then

$$I(a, b) = \int_0^{\pi/2} \frac{1}{\sqrt{a^2 + b^2 \tan^2 \theta} \cdot b \sec \theta} db \tan \theta = \int_0^{\pi/2} \frac{d\theta}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}}.$$

14.2.3 B3) (2pts) Prove that the map defined by $I(a, b)$

$$I : (0, \infty) \times (0, \infty) \rightarrow \mathbb{R}, (a, b) \mapsto I(a, b)$$

is a continuous function. Proof: For any $\theta \in \mathbb{R}$, $a^2 \cos^2 \theta + b^2 \sin^2 \theta \geq \min\{a^2, b^2\}$. Hence for $\varepsilon, \delta \rightarrow 0$,

$$\left| \frac{1}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} - \frac{1}{\sqrt{(a - \varepsilon)^2 \cos^2 \theta + (b - \delta)^2 \sin^2 \theta}} \right| \quad (14.8)$$

$$\leq \frac{1}{2 \min\{a^2/2, b^2/2\}^{3/2}} (|a^2 - (a - \varepsilon)^2| + |b^2 - (b - \delta)^2|) \rightarrow 0. \quad (14.9)$$

14.2.4 B4) (2pts) Prove that for any $\lambda > 0$,

$$I(a, b) = \int_0^\infty \frac{\lambda}{ab} \frac{1}{\sqrt{x^2 + \left(\frac{\lambda}{a}\right)^2} \sqrt{x^2 + \left(\frac{\lambda}{b}\right)^2}} dx.$$

Proof: Let $y = \lambda/x$, then

$$I(a, b) = \int_0^\infty \frac{1}{\sqrt{(\lambda/y)^2 + a^2} \sqrt{(\lambda/y)^2 + b^2}} d\frac{\lambda}{y} = \int_0^\infty \frac{\lambda}{ab} \frac{1}{\sqrt{y^2 + \left(\frac{\lambda}{a}\right)^2} \sqrt{y^2 + \left(\frac{\lambda}{b}\right)^2}} dy$$

14.2.5 B5) (2pts) Prove that

$$I(a, b) = 2 \int_{\sqrt{ab}}^\infty \frac{1}{\sqrt{x^2 + a^2} \sqrt{x^2 + b^2}} dx.$$

Proof: By substituting $y = ab/x$,

$$\int_0^{\sqrt{ab}} \frac{dx}{\sqrt{x^2 + a^2} \sqrt{x^2 + b^2}} = \int_{\sqrt{ab}}^\infty \frac{ab}{y^2} \frac{dy}{\sqrt{(ab/y)^2 + a^2} \sqrt{(ab/y)^2 + b^2}} \quad (14.10)$$

$$= \int_{\sqrt{ab}}^\infty \frac{dy}{\sqrt{y^2 + a^2} \sqrt{y^2 + b^2}}. \quad (14.11)$$

14.2.6 B6) (2pts) Prove that the map

$$\varphi : (\sqrt{ab}, \infty) \rightarrow (0, \infty), x \mapsto \varphi(x) = \frac{1}{2} \left(x - \frac{ab}{x} \right)$$

is a continuously differentiable bijection. Proof: $\varphi \in C^1$ is trivial. Since $\varphi'(x) = \frac{1}{2} \left(1 + \frac{ab}{x^2} \right) > 0$, φ is injective. $\lim_{x \rightarrow \sqrt{ab}} \varphi(x) = 0$, $\lim_{x \rightarrow \infty} \varphi(x) = \infty$, hence φ is bijective.

14.2.7 B7) (3pts) Use the substitution $y = \frac{1}{2} \left(x - \frac{ab}{x} \right)$ to prove that

$$I(a, b) = \int_0^\infty \frac{1}{\sqrt{y^2 + \left(\frac{a+b}{2}\right)^2} \sqrt{y^2 + (\sqrt{ab})^2}} dy.$$

Hence

$$I(a, b) = I\left(\frac{a+b}{2}, \sqrt{ab}\right).$$

Proof: Note that

$$I\left(\frac{a+b}{2}, \sqrt{ab}\right) = \int_0^\infty \frac{1}{\sqrt{y^2 + \left(\frac{a+b}{2}\right)^2} \sqrt{y^2 + ab}} dy = \frac{1}{2} \int_{\sqrt{ab}}^\infty \frac{dx/(1+ab/x^2)}{\sqrt{y^2 + \left(\frac{a+b}{2}\right)^2} \sqrt{y^2 + ab}} \quad (14.12)$$

$$= 2 \int_{\sqrt{ab}}^\infty \frac{dx}{\sqrt{x^2 + a^2} \sqrt{x^2 + b^2}} = I(a, b). \quad (14.13)$$

14.2.8 B8) (2pts) Inductively define the sequence: $a_1 = a, b_1 = b$; for $n \geq 1$, define

$a_{n+1} = \frac{a_n+b_n}{2}, b_{n+1} = \sqrt{a_n b_n}$. Prove that the limits $\lim_{n \rightarrow \infty} a_n$ and $\lim_{n \rightarrow \infty} b_n$ exist.

Proof: Clearly $\min\{a, b\} \leq a_n, b_n \leq \max\{a, b\}$, and $a_n \geq b_n$ for $n \geq 2$ so $a_{n+1} \leq a_n, b_{n+1} \geq b_n$. Hence $\lim_{n \rightarrow \infty} a_n$ and $\lim_{n \rightarrow \infty} b_n$ exist.

14.2.9 B9) (2pts) Prove that $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n$. Denote this value by $M(a, b)$ and call it the arithmetic-geometric mean of a and b .

Proof: Use $2a_{n+1} = a_n + b_n$, we have $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n$.

14.2.10 B10) (2pts) Prove that for any $a, b > 0$, their arithmetic-geometric mean $M(a, b)$ and the elliptic integral $I(a, b)$ have the following formula:

$$2M(a, b)I(a, b) = \pi.$$

Proof: Let $F(a, b) = 2M(a, b)I(a, b)$, then $F(a, b) = F\left(\frac{a+b}{2}, \sqrt{ab}\right)$ and

$$F(a, a) = 2a \int_0^\infty \frac{dx}{x^2 + a^2} = \pi.$$

Hence it suffices to show that F is continuous on $(0, \infty)^2$, then $F(a, b) = \lim_{n \rightarrow \infty} F(a_n, b_n) = F(M(a, b), M(a, b)) = \pi$. Consider any $(a, b), (a', b')$ and the deriving sequences $\{a_n\}, \{b_n\}, \{a'_n\}, \{b'_n\}$. Suppose $b < b'$, then $a_n \leq a'_n$ and $b_n \leq b'_n$. Let $d_n = a'_n - a_n$ and $\lambda_n = b'_n/b_n$, then $d_{n+1} \leq \max\{d_n, b_n(\lambda_n - 1)\}$ and $\lambda_{n+1} \leq \max\{\lambda_n, d_n/a_n + 1\}$. Hence $d_{n+1} \leq \max\{d_n, b_n d_{n-1}/a_{n-1}\} \leq \max\{d_n, d_{n-1}\}$ since $a_{n-1} \geq a_n \geq b_n$, and likewise $\lambda_{n+1} \leq \max\{\lambda_n, \lambda_{n-1}\}$. Therefore $M(a, b') - M(a, b) \leq |b - b'|$. Likewise we obtain $|M(a, b) - M(a', b')| \leq |a - a'| + |b - b'|$.

14.3 PSC (24pts) The Calculation of $\int_0^\infty \frac{1}{1+x^\alpha} dx$

Assume $\alpha > 1, \beta \in (0, 1), \alpha\beta = 1$.

14.3.1 C1) (2pts) Prove that the following improper integral converges:

$$I(\alpha) = \int_0^\infty \frac{1}{1+x^\alpha} dx.$$

Proof:

$$I(\alpha) \leq 1 + \int_1^\infty \frac{1}{x^\alpha} dx = 1 + \frac{1}{\alpha-1}.$$

14.3.2 C2) (2pts) Prove that the following two improper integrals converge:

$$J_1(\beta) = \int_0^1 \frac{x^{\beta-1}}{1+x} dx, \quad J_2(\beta) = \int_0^1 \frac{x^{-\beta}}{1+x} dx.$$

Proof:

$$J_1(\beta) \leq \int_0^1 x^{\beta-1} dx = \frac{1}{\beta}, \quad J_2(\beta) \leq \int_0^1 x^{-\beta} dx = \frac{1}{1-\beta}.$$

14.3.3 C3) (2pts) Prove that

$$\int_0^1 \frac{1}{1+x^\alpha} dx = \beta J_1(\beta).$$

Proof:

$$\beta J_1(\beta) = \int_0^1 \frac{1}{1+x} dx^\beta = \int_0^1 \frac{1}{1+y^\alpha} dy.$$

14.3.4 C4) Prove that

$$\int_1^\infty \frac{1}{1+x^\alpha} dx = \beta J_2(\beta).$$

Proof: By substituting $x = 1/y$,

$$\beta J_2(\beta) = \beta \int_1^\infty \frac{y^{\beta-1}}{1+y} dy = \int_1^\infty \frac{1}{1+y} dy^\beta = \int_1^\infty \frac{1}{1+t^\alpha} dt.$$

14.3.5 C5) For integers $n \geq 1$, define

$$h_n(x) = \sum_{k=0}^n (-1)^k x^k.$$

Prove that for any $x \in [0, 1]$,

$$\left| h_n(x) - \frac{1}{1+x} \right| \leq x^n.$$

Proof: Note that

$$h_n(x) = \frac{1 - (-x)^{n+1}}{1+x}$$

hence

$$\left| h_n(x) - \frac{1}{1+x} \right| = \frac{x^{n+1}}{1+x} \leq x^n.$$

14.3.6 C6) (2pts) Let

$$J_{1,n}(\beta) = \int_0^1 x^{\beta-1} h_n(x) dx, \quad J_{2,n}(\beta) = \int_0^1 x^{-\beta} h_n(x) dx$$

Prove that

$$\lim_{n \rightarrow \infty} J_{1,n}(\beta) = J_1(\beta), \quad \lim_{n \rightarrow \infty} J_{2,n}(\beta) = J_2(\beta).$$

Proof:

$$|J_{1,n}(\beta) - J_1(\beta)| \leq \int_0^1 x^{\beta-1} \left| h_n(x) - \frac{1}{1+x} \right| dx \leq \int_0^1 x^{n+\beta-1} dx = \frac{1}{n+\beta} \rightarrow 0.$$

The other equation can be proved in the same way.

14.3.7 C7) (2pts) Define the function g on $[0, \pi]$:

$$g(x) = \begin{cases} \frac{\cos(x/\alpha)-1}{\sin(x/2)}, & x \in (0, \pi]; \\ 0, & x = 0. \end{cases}$$

Prove that $g \in C^1([0, \pi])$. Proof: Since $\cos(x/\alpha) - 1 = O(x^2)$, $\sin(x/2) = O(x)$, $\lim_{x \rightarrow 0^+} g(x) = 0 = g(0)$, so g is continuous. The limit

$$\lim_{h \rightarrow 0^+} \frac{g(h)}{h} = \lim_{h \rightarrow 0^+} \frac{-h^2/2\alpha^2 + o(h^2)}{h^2/2 + o(h^2)} = -\frac{1}{\alpha^2}$$

exists, and

$$\lim_{x \rightarrow 0^+} g'(x) = \lim_{x \rightarrow 0^+} \frac{\sin(x/2)(-\sin(x/\alpha)/\alpha) + (1 - \cos(x/\alpha))\cos(x/2)/2}{\sin^2(x/2)} = -\frac{1}{\alpha^2}.$$

Hence $g \in C^1([0, \pi])$.

14.3.8 C8) (3pts) For any $n \geq 1$, let

$$a_n = \int_0^\pi g(x) \sin\left((2n+1)\frac{x}{2}\right) dx.$$

Prove that there is a constant C such that for any n ,

$$|a_n| \leq \frac{C}{2n+1}.$$

Proof: Consider

$$a_n = \frac{2}{2n+1} \int_0^{(2n+1)\pi/2} g\left(\frac{2y}{2n+1}\right) \sin y dy.$$

Note that

$$\left| \int_a^{a+2\pi} g\left(\frac{2y}{2n+1}\right) \sin y dy \right| = \left| \int_a^{a+\pi} \left(g\left(\frac{2y}{2n+1}\right) - g\left(\frac{2(y+\pi)}{2n+1}\right) \right) \sin y dy \right| \quad (14.14)$$

$$\leq \int_a^{a+\pi} g'(\xi) \frac{2\pi}{2n+1} dx \leq M \frac{2\pi^2}{2n+1}. \quad (14.15)$$

Hence $a_n = O(1/n)$.

14.3.9 C9) (2pts) For any $n \geq 1$, let

$$\varphi_n(x) = \cos x + \cos 2x + \cdots + \cos nx.$$

Define the integral

$$A_n = \int_0^\pi \varphi_n(x) \cos\left(\frac{x}{\alpha}\right) dx.$$

Prove that

$$A_n = \frac{\alpha}{2} \sin\left(\frac{\pi}{\alpha}\right) \sum_{k=1}^n (-1)^k \left(\frac{1}{1+\alpha k} + \frac{1}{1-\alpha k} \right).$$

Proof: Trivial, since

$$\int_0^\pi \cos kx \cos\left(\frac{x}{\alpha}\right) dx = \frac{\alpha}{2} \sin \frac{\pi}{\alpha} (-1)^k \left(\frac{1}{1+\alpha k} + \frac{1}{1-\alpha k} \right).$$

14.3.10 C10) (3pts) Prove that

$$\varphi_n(x) = -\frac{1}{2} + \frac{1}{2} \frac{\sin((2n+1)x/2)}{\sin(x/2)}.$$

And use this to prove that

$$\lim_{n \rightarrow \infty} A_n = -\frac{\alpha}{2} \sin\left(\frac{\pi}{\alpha}\right) + \frac{\pi}{2}.$$

Proof: $2 \sin(x/2) \varphi_n(x) = \sum_{k=1}^n 2 \sin(x/2) \cos kx = \sin((2n+1)x/2) - \sin(x/2)$. So

$$A_n = \frac{1}{2} \int_0^\pi \frac{\sin((2n+1)x/2)}{\sin(x/2)} - \cos(x/\alpha) dx + \frac{1}{2} \int_0^\pi g(x) \sin((2n+1)x/2) dx.$$

Clearly

$$\int_0^\pi \frac{\sin((2n+1)x/2)}{\sin(x/2)} dx = \pi + 2 \int_0^\pi \varphi_n(x) dx = \pi$$

hence

$$\lim_{n \rightarrow \infty} A_n = \frac{\pi}{2} - \frac{\alpha}{2} \sin\left(\frac{\pi}{\alpha}\right).$$

14.3.11 C11) (2pts) Prove that

$$I(\alpha) = \frac{\pi}{\alpha \sin\left(\frac{\pi}{\alpha}\right)}.$$

Proof: We proved that $I(\alpha) = \beta(J_1(\beta) + J_2(\beta))$, and $J_i(\beta) = \lim_{n \rightarrow \infty} J_{i,n}(\beta)$. Note that

$$\frac{\pi}{2} - \frac{\alpha}{2} \sin\left(\frac{\pi}{\alpha}\right) = \lim_{n \rightarrow \infty} A_n = \sum_{n=1}^{\infty} \frac{\alpha}{2} \sin\left(\frac{\pi}{\alpha}\right) (-1)^k \left(\frac{1}{1+\alpha k} + \frac{1}{1-\alpha k} \right)$$

so

$$\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{n+\beta} = \frac{\pi}{\sin \pi \beta}.$$

Clearly

$$J_{1,n}(\beta) + J_{2,n}(\beta) = \sum_{k=-n-1}^n \frac{(-1)^k}{k+\beta} \rightarrow \frac{\pi}{\sin \pi \beta}.$$

Therefore $I(\alpha) = \frac{\pi}{\alpha \sin(\pi/\alpha)}$.

14.4 Extra Problems:

14.5 PSD Stone-Weierstrass Theorem (10pts)

14.5.1 D1) (2pts) Suppose $f \in C^1([0, 1])$ and f is monotonically increasing, then there exists a sequence of real polynomials $\{P_n\}_{n \geq 1} \subset \mathbb{R}[X]$ such that

- for any n , P_n is monotonically increasing on $[0, 1]$.
- P_n converges uniformly to f .

Proof: By Stone-Weierstrass theorem, there exists a sequence of real polynomials $\{Q_n\}_{n \geq 1}$ such that $Q_n(x) \geq 0$ for $x \in [0, 1]$ and Q_n converges uniformly to $g = f'$ (e.g., taking the Bernstein polynomials). Let $P_n(x) = f(0) + \int_0^x Q_n(t) dt$, then $|P_n(x) - f(x)| \leq x \sup_{t \in [0, 1]} |g(t) - Q_n(t)|$ so $\|f - P_n\| \leq \|g - Q_n\| \rightarrow 0$, and P_n is monotonically increasing.

14.5.2 D2) (1pts) Suppose $f \in C([0, 1])$ and is monotonically increasing. For any $x \in (1, 2]$, let $f(x) = f(1)$. Define the function $f_n : [0, 1] \rightarrow \mathbb{R}$, such that

$$f_n(x) = n \int_x^{x+1/n} f(y) dy.$$

Prove that sequence $\{f_n\}_{n \geq 1}$ satisfy:

- for any n , $f_n \in C^1([0, 1])$;
- for any n , f_n is monotonically increasing on $[0, 1]$.
- f_n converges uniformly to f .

Proof: Clearly $f'_n(x) = n(f(x + 1/n) - f(x))$ is continuous on $[0, 1]$. For any $x > x'$, $f_n(x) - f_n(x') = n \int_0^{1/n} f(t + x) - f(t + x') dt \geq 0$ so f_n is monotonically increasing. For any $x \in [0, 1]$,

$$|f(x) - f_n(x)| \leq \int_0^1 |f(x + t/n) - f(x)| dx \leq \sup_{|u-v| \leq 1/n} |f(u) - f(v)|$$

hence f_n converges uniformly to f .

14.5.3 D3) (1pts) Suppose $f \in C([0, 1])$ and is monotonically increasing, then there exists a sequence of real polynomials $\{P_n\}_{n \geq 1} \subset \mathbb{R}[X]$ such that

- for any n , P_n is monotonically increasing on $[0, 1]$.
- P_n converges uniformly to f .

Proof: Take f_n such that $f_n \in C^1([0, 1])$, f_n is monotonically increasing, and $\|f - f_n\|_\infty < 2^{-n}$. Take polynomials $P_{n,k}$ such that $P_{n,k}$ is monotonically increasing and $\|P_{n,k} - f_n\|_\infty < 2^{-k}$, then $\|f - P_{n,n}\|_\infty < 2^{1-n}$ so $P_{n,n}$ converges uniformly to f where $P_{n,n}$ are monotonically increasing polynomials.

14.5.4 D4) (2pts) (Walsh) Suppose $f \in C([0, 1])$, and $x_1, x_2, \dots, x_m$ are m given points on the interval $[0, 1]$, then there exists a sequence of real polynomials $\{P_n\}_{n \geq 1} \subset \mathbb{R}[X]$ such that

- for any $n \geq 1$ and any $1 \leq i \leq m$, $P_n(x_i) = f(x_i)$;
- P_n converges uniformly to f .

Proof: Let Q be the polynomial of degree $m - 1$ such that $Q(x_i) = f(x_i)$. Now we find a sequence of polynomials P_n such that $P_n(x_i) = 0$ and P_n converges uniformly to $g = f - Q$. By Stone-Weierstrass theorem, we can take a sequence of polynomials Q_n that converges uniformly to g on $[0, 1]$. Consider

$$R_n(x) = Q_n(x) - \sum_{i=1}^m Q_n(x_i) L_i(x), \text{ where } L_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$$

then $R_n(x_i) = 0$ for any $1 \leq i \leq m$. Let $M = \sup\{|L_i(x)| : x \in [0, 1], 1 \leq i \leq m\}$, then for any $x \in [0, 1]$, $|R_n(x) - g(x)| \leq |Q_n(x) - g(x)| + M \sum_{i=1}^m |Q_n(x_i)| \leq (1 + Mm) \|Q_n - g\|_\infty$ since $g(x_i) = 0$. Therefore R_n converges uniformly to g and R_n has roots $x_1, \dots, x_m$, and we obtain a sequence $P_n = R_n + Q$ that converges uniformly to f and $P_n(x_i) = f(x_i)$.

14.5.5 D5) (1pts) Suppose $I = [a, b] \subset (0, 1)$, the polynomial $p(x) = 2x(1 - x)$, let

$Q_n = p \circ p \circ \dots \circ p$, prove that $\{Q_n\}$ converges uniformly to the constant function $1/2$ on I .

Proof: Note that $|p(x) - 1/2| = 2|x - 1/2|^2 \leq \lambda|x - 1/2|$ where $\lambda = 2 \max\{a, 1 - b\} < 1$. Hence $\|Q_n - 1/2\|_\infty \leq \lambda^n \rightarrow 0$ so Q_n converges uniformly to $1/2$ on I .

14.5.6 D6) (1pts) Suppose $I = [a, b] \subset (0, 1)$, for any $k \in \mathbb{Z}$, prove that there exists a sequence of polynomials $\{P_n\}_{n \geq 1} \subset \mathbb{Z}[X]$, such that $\{P_n\}_{n \geq 1}$ converges uniformly to the constant function 2^k on I .

Proof: The case $k \geq 0$ is trivial since $2^k \in \mathbb{Z}[X]$. If $k = -m \leq 0$, let $P_n = Q_n^m$, then $|P_n(x) - 2^{-m}| = |Q_n - 1/2| \cdot |Q_n^{m-1} + \dots + 2^{-(m-1)}|$. For n large enough, $|Q_n| < 1$, so $|P_n(x) - 2^{-m}| \leq m|Q_n - 2^{-1}|$ therefore P_n converges uniformly to 2^k .

14.5.7 D7) (2pts) (Chudnovsky) Suppose $f \in C(I)$, where $I = [a, b] \subset (0, 1)$. Prove that there exists a sequence of polynomials $\{P_n\}_{n \geq 1} \subset \mathbb{Z}[X]$, such that $\{P_n\}_{n \geq 1}$ converges uniformly to f on I .

[!!!]Proof: By D6) and binary representation we know the case $f = C \cdot \chi_{[a,b]}$ holds. Also if $P_n \rightarrow f$ uniformly then $xP_n \rightarrow xf$ uniformly. Hence all polynomials with real coefficients can be approximated. By Stone-Weierstrass theorem, all continuous functions can be uniformly approximated.

[!note] For further results on the approximation of continuous functions by polynomials with integer coefficients, see the article by Hervé Pépin and Nicolas Tosel: *_Approximation par des polynômes à coefficients dans $\mathbb{Z}_-$, RMS, 114ème année, 2003-2004.*

14.6 PSE

Find all functions $f \in C(\mathbb{R})$ such that for any $x, y \in \mathbb{R}$,

$$f(x)f(y) = \int_{x-y}^{x+y} f(t) dt.$$

Solution: $f \equiv 0$ is a trivial solution, now suppose otherwise. Let $F(x) = \int_0^x f(t) dt$, then $f(x)f(y) = F(x+y) - F(x-y)$ where F is differentiable, so f is also differentiable (take y such that $f(y) \neq 0$), and $f'(x)f(y) = f(x+y) - f(x-y)$. Take y such that $f(y) \neq 0$, then $f'(x) = \frac{f(x+y)-f(x-y)}{f(y)}$ is differentiable on $\mathbb{R}$. Hence $f''(x)f(y) = f'(x+y) - f'(x-y)$, and $f(x)f'(y) = f(x+y) + f(x-y)$ so $f(x)f''(y) = f'(x+y) - f'(x-y)$. Therefore $f''(x)f(y) = f''(y)f(x)$ for all $x, y \in \mathbb{R}$. Since $f(y) \neq 0$ for some y , there exists c such that $f''(x) = cf(x)$. If $c = 0$ then $f(x) = kx$ so $f(x) = 2x$. If $c \neq 0$ then $f(x) = Ae^{\sqrt{c}x} + Be^{-\sqrt{c}x}$ (let $\sqrt{-t} = i\sqrt{t}$), hence $f(x) = a \sin bx$ or $f(x) = a \sinh bx$ where $ab = 2$.

14.7 PSF

Given an arbitrary set of distinct real numbers $\{\alpha_1, \alpha_2, \dots, \alpha_{2020}\}$, and non-zero real numbers $a_1, a_2, \dots, a_{2020}$, consider the function defined on $(0, \infty)$:

$$f(x) = a_1x^{\alpha_1} + a_2x^{\alpha_2} + \dots + a_{2020}x^{\alpha_{2020}}.$$

Prove that $f(x)$ has at most 2019 roots on $(0, \infty)$. Proof: We prove by induction that for $n \geq 1$ and distinct non-zero real numbers $a_1, \dots, a_n$, the function $f(x) = \sum_{k=1}^n a_k x^{\alpha_k}$ has at most $n-1$ roots on $(0, \infty)$. The base $n = 1$ is trivial. Suppose $\alpha_1 < \dots < \alpha_n$, and f has n roots $r_1 < \dots < r_n$. Then they are also roots of $\tilde{f}(x) = a_1 + \sum_{k=2}^n a_k x^{\alpha_k - \alpha_1}$. By Rolle's theorem, the function $g = \tilde{f}' = \sum_{k=2}^n a_k (\alpha_k - \alpha_1) x^{\alpha_k - \alpha_1 - 1}$ has $n-1$ distinct roots, leading to contradiction.

Chapter 15 Winter Holiday

15.1 A. Young's Inequality

$f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is continuously differentiable. Assume $f(0) = 0$, $\lim_{x \rightarrow \infty} f(x) = \infty$, and for any $x \geq 0$, $f'(x) > 0$. Denote f^{-1} by $g(x)$.

15.1.1 A1) Prove that for any $a \geq 0$,

$$af(a) = \int_0^a f(x) dx + \int_0^{f(a)} g(y) dy.$$

Proof: $a = 0$ is trivial. Take derivatives on both sides, then

$$\frac{d}{da} \int_0^a f(x) dx = f(a), \quad \frac{d}{da} \int_0^{f(a)} g(y) dy = g(f(a))f'(a) = af'(a),$$

and $(af(a))' = f(a) + af'(a)$.

15.1.2 A2) Prove Young's inequality: for any $a, b \geq 0$,

$$ab \leq \int_0^a f(x) dx + \int_0^b g(y) dy.$$

Proof: Assume that $b \geq f(a)$, then

$$ab \leq \int_0^a f(x) dx + \int_0^{f(a)} g(y) dy + a(b - f(a)) \quad (15.1)$$

$$\leq \int_0^a f(x) dx + \int_0^b g(y) dy. \quad (15.2)$$

15.1.3 A3) Suppose $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is strictly monotonically increasing, and $f(0) = 0$, $\lim_{x \rightarrow \infty} f(x) = \infty$. Prove the inequality in A2)

Proof: It suffices to prove the equality in A1). For any $a, b \geq 0$, f can be uniformly approximated by functions P such that $P(0) = 0$, P is continuously differentiable and monotonically increasing. Then

$$aP(a) = \int_0^a P(x) dx + \int_0^{P(a)} P^{-1}(y) dy.$$

Note that $P(a) - f(a) \rightarrow 0$, $\int_0^a P(x) dx - \int_0^a f(x) dx \rightarrow 0$, $\int_0^b P^{-1}(y) dy - \int_0^b g(y) dy \rightarrow 0$ (since P is continuously differentiable), and $\int_b^{P(a)} P^{-1}(y) dy \rightarrow 0$ since P^{-1} is bounded by a . Hence the equality in A1) also holds for f .

15.1.4 A4) Assume $a, b \geq 0$, $p > 1$, $q > 1$, and $1/p + 1/q = 1$. Prove that

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof: Consider $f(x) = x^{p-1}$, $g(x) = x^{1/(p-1)} = x^{q-1}$, then by A2),

$$ab \leq \int_0^a f(x) dx + \int_0^b g(y) dy = \frac{a^p}{p} + \frac{b^q}{q}.$$

where equality holds iff $b = f(a) = a^{p-1}$, i.e. $a^p = b^q$.

15.2 B. Sobolev's Inequality

Consider any compact interval $[a, b]$.

15.2.1 B1) (Cauchy-Schwarz Inequality) Assume $f, g \in \mathcal{R}([a, b])$, prove that

$$\left| \int_a^b f(x)g(x) \, dx \right| \leq \left(\int_a^b |f(x)|^2 \, dx \right)^{1/2} \left(\int_a^b |g(x)|^2 \, dx \right)^{1/2}.$$

Proof: Consider the inner product

$$\langle f, g \rangle = \int_a^b f(x)g(x) \, dx.$$

Then

$$0 \leq \langle f + tg, f + tg \rangle = \langle f, f \rangle + 2t\langle f, g \rangle + t^2\langle g, g \rangle.$$

Hence

$$\Delta = 4(\langle f, g \rangle)^2 - \langle f, f \rangle \langle g, g \rangle \leq 0.$$

15.2.2 B2) Prove that for any $\varepsilon > 0$, there is a constant $C_\varepsilon > 0$ such that for any $f \in C^1([a, b])$, and any $x \in [a, b]$,

$$|f(x)^2 - f(a)^2| \leq C_\varepsilon \int_a^b f(x)^2 \, dx + \varepsilon \int_a^b f'(x)^2 \, dx.$$

Proof: Let $C_\varepsilon = 4/\varepsilon$, then

$$|f(x)^2 - f(a)^2| = \left| \int_a^x d f^2 \right| = \left| \int_a^x 2f f' \, dx \right| \quad (15.3)$$

$$\leq \varepsilon \int_a^b f'(x)^2 \, dx + \frac{4}{\varepsilon} \int_a^b f^2 \, dx. \quad (15.4)$$

15.2.3 B3) Prove that for any $\varepsilon > 0$, there is a constant $D_\varepsilon > 0$ such that for any $f \in C^1([a, b])$,

$$\sup_{x \in [a, b]} |f(x)|^2 \leq D_\varepsilon \int_a^b f(x)^2 \, dx + \varepsilon \int_a^b f'(x)^2 \, dx.$$

Proof: Let $D_\varepsilon = \frac{1}{b-a} + C_\varepsilon$, then

$$\sup_{x \in [a, b]} |f(x)|^2 \leq \inf_{x \in [a, b]} |f(x)|^2 + \sup_{x, y \in [a, b]} |f(x)^2 - f(y)^2| \quad (15.5)$$

$$\leq \frac{1}{b-a} \int_a^b f^2 + C_\varepsilon \int_a^b f^2 + \varepsilon \int_a^b f'(x)^2 \, dx. \quad (15.6)$$

15.3 C. Wirtinger's Inequality

Let $E = \{f \in C^1([0, 1]) : f(0) = f(1) = 0\}$.

15.3.1 C1) For any $f \in E$, define the improper integral

$$\mathbf{I}_1 = \int_0^1 \frac{f(x)f'(x)}{\tan(\pi x)} \, dx, \quad \mathbf{I}_2 = \int_0^1 \frac{f(x)^2}{\tan^2(\pi x)} (1 + \tan^2(\pi x)) \, dx.$$

Prove that they converge and determine the value I_1/I_2 . Proof:

$$\lim_{x \rightarrow 0^+} \frac{f(x)f'(x)}{\tan(\pi x)} = f'(0) \lim_{x \rightarrow 0^+} \frac{f'(x)}{\pi/\cos^2(\pi x)} = \frac{f'(0)^2}{\pi}.$$

Hence $ff'/\tan(\pi x) \in C([0, 1])$, so the integral converges.

$$I_1 = \int_0^1 \frac{df^2}{2\tan(\pi x)} = \frac{f^2}{2\tan \pi x} \Big|_0^1 + \int_0^1 f^2 \frac{\pi}{2} \sec^2(\pi x) dx \quad (15.7)$$

$$= \frac{\pi}{2} I_2. \quad (15.8)$$

15.3.2 C2C3) (Wirtinger's Inequality) For any $f \in E$,

$$\int_0^1 f^2 \leq \pi^{-2} \int_0^1 f'(x)^2 dx.$$

Proof:

$$I_2 = \int_0^1 f(x)^2 dx + \int_0^1 \frac{f^2}{\tan^2(\pi x)} dx = \frac{2}{\pi} I_1 \quad (15.9)$$

$$= \frac{2}{\pi} \int_0^1 \frac{f}{\tan(\pi x)} f' dx \leq \frac{2}{\pi} \sqrt{\int_0^1 \frac{f^2}{\tan^2(\pi x)} dx} \cdot \int_0^1 (f')^2 \quad (15.10)$$

$$(15.11)$$

$$\leq \int_0^1 \frac{f^2}{\tan^2(\pi x)} dx + \pi^{-2} \int_0^1 (f')^2. \quad (15.12)$$

Therefore

$$\int_0^1 f(x)^2 dx \leq \pi^{-2} \int_0^1 f'(x)^2 dx.$$

If equality holds, then $f' = \frac{\pi f}{\tan(\pi x)}$, hence $df/f = \frac{d\pi x}{\tan(\pi x)}$, so $f = C \sin(\pi x)$. Combined with $f \in E$, we obtain $f = C \sin(\pi x)$, $\forall C \in \mathbb{R}$.

15.3.3 C4) Assume $f \in \mathcal{R}([0, 2\pi])$, determine the value of $A \in \mathbb{R}$ which minimizes the integral

$$\int_0^{2\pi} |f(x) - A|^2 dx.$$

Solution:

$$\langle f - A, f - A \rangle = 2\pi A^2 + \langle f, f \rangle - 2A \int_0^{2\pi} f(x) dx.$$

Hence

$$A = \frac{1}{2\pi} \int_0^{2\pi} f(x) dx.$$

(Also, since e^{inx} forms a base of the linear space $\mathcal{R}([0, 1])$, A is the projection of f onto e^{i0x} , hence $A = \tilde{f}(0)$.)

15.3.4 C5) Another version of Wirtinger's inequality: for any $f \in C^1([0, 2\pi])$, if $f(0) = f(2\pi)$, and $\int_0^{2\pi} f(x) dx = 0$, then

$$\int_0^{2\pi} f(x)^2 dx \leq \int_0^{2\pi} f'(x)^2 dx.$$

Proof: Consider $g(x) = f(x) - f(x + \pi)$, then $g(0) = -g(\pi)$, hence $g(x) = 0$ for some $x \in [0, \pi]$. Assume that $g(0) = 0$, i.e. $f(0) = f(\pi)$. Apply C2) to $f|_{[0, \pi]} - f(0)$ and $f|_{[\pi, 2\pi]} - f(0)$, then by C4),

$$\int_0^{2\pi} f(x)^2 dx \leq \int_0^{2\pi} |f(x) - f(0)|^2 dx \leq \int_0^{2\pi} f'(x)^2 dx.$$

Proof using Fourier series: $\tilde{f}(0) = 0$, hence

$$\int_0^{2\pi} f^2 = 2\pi \sum_n |\tilde{f}(n)|^2 \leq 2\pi \sum_n n^2 |\tilde{f}(n)|^2 = \int_0^{2\pi} |f'|^2.$$

15.3.5 C6) (Isoperimetric inequality) Assume $\gamma : [0, 2\pi] \rightarrow \mathbb{R}^2$ is a continuously differentiable parameterization of a closed non-intersecting curve. Let $\gamma(t) = (x(t), y(t))$,

$$L = \int_0^{2\pi} \sqrt{x'(t)^2 + y'(t)^2} dt, \quad A = \int_0^{2\pi} x'(t)y(t) dt.$$

Prove that $L^2 \geq 4\pi A$, and equality holds iff γ forms a circle. Proof: We can choose the parameterization γ such that $|\gamma'(t)| = 1$ for any $t \in [0, 2\pi]$. Furthermore, we can let $\gamma(0) = \gamma(\pi) = 0$. Then $L = 2\pi$ and

$$\int_0^\pi x'y \leq \frac{1}{2} \int_0^\pi |x'|^2 + y^2 = \frac{1}{2} \int_0^\pi y^2 + 1 - |y'|^2 \leq \frac{\pi}{2}.$$

Where the last inequality comes from C2). If inequality holds, then by C2), $y = C \sin t$, $x = C \cos t$, hence γ forms a circle.

15.4 D. Gauss-Legendre Quadrature

For any $n \geq 1$, define the Legendre polynomial

$$P_n(x) = \frac{1}{2^n n!} \left(\frac{d}{dx} \right)^n ((x^2 - 1)^n).$$

Assume $P_0 = 1$.

15.4.1 D1) Prove that for any $\varphi(x) \in C([-1, 1])$, and any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ and $c_1, c_2, \dots, c_N \in \mathbb{R}$ such that

$$\|\varphi(x) - \sum_{k=1}^N c_k P_k(x)\|_\infty < \varepsilon.$$

Proof: $\deg P_n = n$, hence P_n are linearly independent. By Stone-Weierstrass theorem, there exists such N and $c_1, c_2, \dots, c_N$.

15.4.2 D2) Prove that for any $n \geq 1$, P_n satisfy the following differential equation:

$$(1 - x^2)f'' - 2xf' + n(n+1)f = 0.$$

Proof: Let $g(x) = (x^2 - 1)^n$, then

$$(1 - x^2)g'' + 2ng' + 2g = 0.???$$

We can prove by induction that $g^{(k)}$ satisfy

$$(1 - x^2)f'' + 2(n - k)f' + k(k+1)f = 0.$$

Let $k = n$ and we obtain the required equation.

15.4.3 D3) Prove that for any $n, m \geq 1$,

$$\int_{-1}^1 P_n(x)P_m(x) dx = \begin{cases} 0, & m \neq n; \\ \frac{2}{2n+1}, & m = n. \end{cases}$$

Proof: Note that for all $0 \leq m < n$, $\frac{d^m}{dx^m}((x^2 - 1)^n) = 0$ when $x \in \{-1, 1\}$. Hence by Darboux's integration by parts formula,

$$\langle P_n 2^n n!, P_m 2^m m! \rangle = \sum_{k=0}^m (-1)^k ((x^2 - 1)^n)^{(n-k-1)} ((x^2 - 1)^m)^{m+k} \Big|_{x=-1}^{x=1} = 0$$

Hence $\{P_n\}$ form an orthogonal base, and also

$$\langle P_n, P_n \rangle = \frac{(2n)!}{2^n n!} \int_{-1}^1 (1 - x^2)^n dx = \frac{2}{2n+1}. \text{ (Wallis formula)}$$

15.4.4 D4) Given $n \geq 1$, Prove that if $Q(x)$ is a polynomial with degree at most $n - 1$, then

$$\int_{-1}^1 Q(x)P_n(x) dx = 0.$$

Proof: $Q \in \text{Span}\langle P_0, P_1, \dots, P_{n-1} \rangle$ hence $\langle Q, P_n \rangle = 0$.

15.4.5 D5) Prove that for any $n \geq 1$, $P_n(x)$ has exactly n roots on the interval $(-1, 1)$. Denote them by $x_1^{(n)} < x_2^{(n)} < \dots < x_n^{(n)}$.

Proof: If P_n has less than n distinct roots on $(-1, 1)$, take all roots with odd multiplicity $r_1, \dots, r_k, k < n$. Let $Q(x) = \prod_{i=1}^k (x - r_i)$ then $P_n(x)$ and $Q(x)$ always have the same sign on $(-1, 1)$. Therefore $\langle P_n, Q \rangle \neq 0$, a contradiction.

15.4.6 D6) Prove that for any $n \geq 1$, there exists $\alpha_1^{(n)}, \alpha_2^{(n)}, \dots, \alpha_n^{(n)}$ such that for any polynomial Q with degree at most $2n - 1$,

$$\int_{-1}^1 Q(x) dx = \sum_{i=1}^n \alpha_i^{(n)} Q(x_i^{(n)}).$$

Proof: Suppose $Q(x) = P_n(x)T(x) + R(x)$, where $\deg R < \deg P_n = n$. Since $\deg Q \leq 2n - 1$, $\deg T \leq n - 1$, so $\langle P_n, T \rangle = 0$, i.e. $Q(x_i^{(n)}) = R(x_i^{(n)})$ and

$$\int_{-1}^1 Q(x) dx = \int_{-1}^1 R(x) dx.$$

By Lagrange interpolation, let

$$L_i(x) = \prod_{j \neq i} \frac{x - x_j^{(n)}}{x_i^{(n)} - x_j^{(n)}}$$

then $R(x) = \sum_{i=1}^n L_i(x)R(x_i^{(n)})$. Let $\alpha_i^{(n)} = \int_{-1}^1 L_i(x) dx$, then

$$\int_{-1}^1 R(x) dx = \sum_{i=1}^n \alpha_i^{(n)} R(x_i^{(n)}).$$

Hence

$$\int_{-1}^1 Q(x) dx = \sum_{i=1}^n \alpha_i^{(n)} Q(x_i^{(n)}).$$

15.4.7 D7) An approximation of integrals by Gauss: for any $\varphi \in C([-1, 1])$, let

$$G_n(\varphi) = \sum_{i=1}^n \alpha_i^{(n)} \varphi(x_i^{(n)}).$$

Prove that $\lim_{n \rightarrow \infty} G_n(\varphi) = \int_{-1}^1 \varphi(x) dx$. Proof: We show that $\alpha_i^{(n)}$ are all non-negative, hence take a sequence of polynomials $P_n(x)$ uniformly convergent to φ where $\deg P_n = n$, then

$$\left| G_n(\varphi) - \int_{-1}^1 \varphi(x) dx \right| \leq \int_{-1}^1 |P_n(x) - \varphi(x)| dx + \sum_{i=1}^n \alpha_i^{(n)} |\varphi(x_i^{(n)}) - P_n(x_i^{(n)})| \rightarrow 0.$$

since $\sum_{i=1}^n \alpha_i^{(n)} = \int_{-1}^1 1 dx = 2$. Note that $\alpha_i = \int_{-1}^1 L_i(x) dx$ where

$$L_i(x) = \prod_{j \neq i} \frac{x - x_j^{(n)}}{x_i^{(n)} - x_j^{(n)}}.$$

Since $\deg L = n - 1$,

$$\int_{-1}^1 L_i^2(x) dx = \sum_{j=1}^n \alpha_j^{(n)} L_j^2(x) = \alpha_i^{(n)},$$

Hence $\alpha_i^{(n)} \geq 0$.

15.5 E. Equidistribution (Weyl)

Given a sequence $\{x_k\}_{k \geq 1} \subset [0, 1]$, for any $0 \leq a < b \leq 1$, let

$$S_n([a, b]) = |\{x_k : k \leq n, x_k \in [a, b]\}|.$$

If for any $0 \leq a < b \leq 1$,

$$\lim_{n \rightarrow \infty} \frac{S_n([a, b])}{n} = b - a,$$

we say that $\{x_k\}_{k \geq 1}$ is equidistributed on $[0, 1]$.

15.5.1 E1) Prove that if the sequence $\{x_k\}_{k \geq 1}$ is equidistributed on $[0, 1]$, then $\{x_k\}_{k \geq 1}$ is dense on $[0, 1]$.

Proof: For any $x \in (0, 1)$ and $\min\{x, 1 - x\} > \varepsilon > 0$, since $\{x_k\}_{k \geq 1}$ is equidistributed on $[0, 1]$, $\lim_{n \rightarrow \infty} S_n([x - \varepsilon, x + \varepsilon])/n = 2\varepsilon > 0$, hence $\{x_k\}_{k \geq 1} \cap [x - \varepsilon, x + \varepsilon] \neq \emptyset$ so $\{x_k\}$ is dense.

15.5.2 E2) Construct a dense subset $\{x_k\}_{k \geq 1}$ of $[0, 1]$ such that it is not equidistributed.

Solution: List all rational numbers in $(0, \frac{1}{2})$ and $(\frac{1}{2}, 1)$ as $\{q_1, q_2, \dots\}$ and $\{r_1, r_2, \dots\}$. Let

$$x_n = \begin{cases} q_{3k+1}, & n = 4k + 1, \\ q_{3k+2}, & n = 4k + 2, \\ q_{3k+3}, & n = 4k + 3, \\ r_{k+1}, & n = 4k + 4. \end{cases}$$

Then $\{x_n\}$ includes all rational numbers in $(0, \frac{1}{2})$ and $(\frac{1}{2}, 1)$ so it is dense in $[0, 1]$, but

$$\lim_{n \rightarrow \infty} \frac{S_n([0, \frac{1}{2}])}{n} = \frac{3}{4}$$

so it is not equidistributed.

15.5.3 E3) For an arbitrary sequence $\{x_k\}_{k \geq 1} \subset [0, 1]$, let

$$D_n = \sup_{0 \leq a < b \leq 1} \left| \frac{S_n([a, b])}{n} - (b - a) \right|, \quad D_n^* = \sup_{0 < b < 1} \left| \frac{S_n([0, b])}{n} - b \right|.$$

Prove that $D_n^* \leq D_n \leq 2D_n^*$. Proof: $D_n \geq D_n^*$ is trivial. Note that

$$\frac{S_n([a, b])}{n} - (b - a) = \frac{S_n([0, b])}{n} - b - \left(\frac{S_n([0, a])}{n} - a \right).$$

Hence $D_n \leq 2D_n^*$.

15.5.4 E4) Prove that the sequence $\{x_k\}_{k \geq 1}$ is equidistributed on $[0, 1]$ iff $\lim_{n \rightarrow \infty} D_n^* = 0$.

Proof: By E3) $\lim_{n \rightarrow \infty} D_n^* = 0 \iff \lim_{n \rightarrow \infty} D_n = 0$. If $\lim_{n \rightarrow \infty} D_n = 0$, then for any $0 \leq a < b \leq 1$,

$$\lim_{n \rightarrow \infty} \left| \frac{S_n([a, b])}{n} - (b - a) \right| = 0.$$

Hence $\{x_k\}_{k \geq 1}$ is equidistributed. Suppose $\{x_k\}_{k \geq 1}$ is equidistributed, then for any $0 < b < 1$, $\lim_{n \rightarrow \infty} |S_n([0, b])/n - b| = 0$.

15.5.5 E5) Prove that the sequence $\{x_k\}_{k \geq 1}$ is equidistributed on $[0, 1]$ iff for any $f \in R([0, 1])$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(x_k) = \int_0^1 f(x) dx. \quad (1)$$

Proof: See E6).

15.5.6 E6) The sequence $\{x_k\}_{k \geq 1}$ is equidistributed on $[0, 1]$ iff for any $p \in \mathbb{Z}_{\geq 1}$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n e^{2\pi i p x_k} = 0. \quad (2)$$

Proof: i. Equidistribution $\implies$ (1): $\{x_k\}_{k \geq 1}$ is equidistributed on $[0, 1]$ implies that for any $\chi = \mathbf{1}_{[a, b]}$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \chi(x_k) = \int_0^1 \chi(x) dx.$$

(which implies (1) $\implies$ equidistribution) Hence (1) holds for any $\varphi = \sum_{k=1}^{n-1} c_k \mathbf{1}_{[x_k, x_{k+1}]}$. For any Riemann integrable function f and any $\varepsilon > 0$, take $\varphi_1 \leq f \leq \varphi_2$ where φ_1, φ_2 are step functions such that

$$\int_0^1 \varphi_2(x) dx - \varepsilon \leq \int_0^1 f(x) dx \leq \int_0^1 \varphi_1(x) dx + \varepsilon$$

There exists N such that for any $n \geq N$,

$$\left| \frac{1}{n} \sum_{k=1}^n \varphi_1(x_k) - \int_0^1 \varphi_1(x) dx \right|, \left| \frac{1}{n} \sum_{k=1}^n \varphi_2(x_k) - \int_0^1 \varphi_2(x) dx \right| < \varepsilon.$$

Hence

$$\int_0^1 f(x) dx \leq \int_0^1 \varphi_1(x) dx + \varepsilon < \frac{1}{n} \sum_{k=1}^n \varphi_1(x_k) + 2\varepsilon \leq \frac{1}{n} \sum_{k=1}^n f(x_k) + 2\varepsilon.$$

Likewise

$$\int_0^1 f(x) dx \geq \frac{1}{n} \sum_{k=1}^n f(x_k) - 2\varepsilon.$$

Therefore

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(x_k) = \int_0^1 f(x) dx$$

for any $f \in R([0, 1])$. ii. (1) $\implies$ (2): Since $e^{2\pi i p x} \in R([0, 1])$ and

$$\int_0^1 e^{2\pi i p x} dx = 0.$$

iii. (2) $\implies$ equidistribution: From (2) we know that (1) holds for trigonometric polynomials. By Stone-Weierstrass theorem, continuous functions can be uniformly approximated by trigonometric polynomials, hence (1) holds for continuous functions. Likewise, step functions can be uniformly approximated by continuous functions, hence we obtain $\{x_k\}$ is equidistributed.

15.5.7 Ex: (Van Der Corput) Suppose $\{\xi_n\}_{n \geq 1}$ is a sequence on $[0, 1]$. If for any $h \geq 1$, the sequence $\{\xi_{n+h} - \xi_n\}_{n \geq 1}$ is equidistributed, then $\{\xi_n\}_{n \geq 1}$ is equidistributed.

Proof: From E5) and E6) we know that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e^{2\pi i p (\xi_{n+h} - \xi_n)} = 0, \forall p \in \mathbb{Z} - \{0\}, h \geq 1.$$

We only need to prove the following lemma: Suppose $\{u_n\}_{n \geq 1} \subset \mathbb{C}$ is bounded, and for any $h \geq 1$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N u_{n+h} \bar{u}_n = 0,$$

then

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N u_n = 0.$$

Suppose $M = \sup_{n \in \mathbb{N}} |u_n|$. Note that for any $N > D > 0$,

$$\left| \frac{1}{N} \sum_{n=1}^N u_n - \frac{1}{D} \frac{1}{N} \sum_{h=1}^D \sum_{n=1}^N u_{n+h} \right| \leq \frac{1}{N} \sum_{k=1}^D \frac{D+1-k}{D} (|u_k| + |u_{N+k}|) \leq \frac{(D+1)M}{N}.$$

For a constant D , and any $\varepsilon > 0$, there exists N_0 such that for any $n > N_0$, and any $d_1 \neq d_2 \in [1, D]$, $\left| \sum_{n=1}^N u_{n+d_1} \bar{u}_{n+d_2} / N \right| < \varepsilon^2/2$, then

$$\left| \frac{1}{ND} \sum_{h=1}^D \sum_{n=1}^N u_{n+h} \right|^2 \leq \frac{1}{N} \sum_{n=1}^N \left| \frac{1}{D} \sum_{h=1}^D u_{n+h} \right|^2 = \frac{1}{N} \sum_{n=1}^N \frac{1}{D^2} \sum_{h,k} u_{n+h} \bar{u}_{n+k} \quad (15.13)$$

$$= \frac{1}{ND^2} \left(\sum_{h=1}^D \sum_{n=1}^N |u_{n+h}|^2 + \frac{D(D-1)}{2} \varepsilon^2 \right) \leq \frac{M^2}{D} + \frac{D-1}{2D} \varepsilon^2. \quad (15.14)$$

Hence for any $\varepsilon > 0$ and $D \geq 1$, there exists N_0 such that for any $n > N_0$,

$$\left| \frac{1}{N} \sum_{n=1}^N u_n \right| \leq \frac{(D+1)M}{N} + \sqrt{\frac{M^2}{D} + \frac{D-1}{2D} \varepsilon^2} \rightarrow 0.$$

Therefore

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N u_n = 0.$$

This implies that any polynomial $\sum_{i=0}^n c_i x^i$ with $c_n \in \mathbb{Q}^C$ is equidistributed.

15.5.8 Ex: (Fejér)

Suppose $g(t)$ ($t \geq 1$) satisfy: (a) $g \in C^1$; (b) g is monotonically increasing and $\lim_{t \rightarrow \infty} g(t) = +\infty$; (c) g' is monotonically decreasing and $\lim_{t \rightarrow \infty} g'(t) = 0$; (d) $\lim_{t \rightarrow \infty} t g'(t) = +\infty$. Prove that $\{\langle g(n) \rangle\}$ is equidistributed.

Proof: Consider

$$\left| \sum_{n=1}^{N-1} e^{2\pi i g(n)} - \int_1^N e^{2\pi i g(x)} dx \right| \leq \sum_{n=1}^{N-1} \int_n^{n+1} |\cos(2\pi g(n)) - \cos(2\pi g(x))| + |\sin(2\pi g(n)) - \sin(2\pi g(x))| dx \quad (15.15)$$

$$\leq \sum_{n=1}^{N-1} \int_n^{n+1} 4\pi |g'(\xi_x)| dx \leq 4\pi \sum_{n=1}^{N-1} \sup_{\xi \in [n, n+1]} |g'(\xi)| \quad (15.16)$$

and

$$\int_1^N e^{2\pi i g(x)} dx = \int_{g(1)}^{g(N)} \frac{e^{2\pi i t}}{g'(g^{-1}(t))} dt = \frac{1}{g'(N)} \int_{\gamma}^{g(N)} e^{2\pi i t} dt$$

(using mean value theorem for integrals). Hence

$$\lim_{N \rightarrow \infty} \frac{1}{N} \left| \sum_{n=1}^{N-1} e^{2\pi i g(n)} - \int_1^N e^{2\pi i g(x)} dx \right| \leq \lim_{N \rightarrow \infty} \frac{4\pi}{N} \sum_{n=1}^{N-1} \sup_{\xi \in [n, n+1]} |g'(\xi)| = 0 \text{ (Cesaro sum)} \quad (15.17)$$

and

$$\frac{1}{N} \left| \int_1^N e^{2\pi i g(x)} dx \right| \leq \frac{1}{2Ng'(N)} \rightarrow 0.$$

15.5.9 E7) Suppose $\theta > 0$, then the sequence $\{\langle n\theta \rangle\}_{n \geq 1}$ is equidistributed on $[0, 1]$ iff θ is irrational.

Proof: Note that

$$\frac{1}{N} \sum_{n=1}^N e^{2\pi i p n \theta} = \frac{e^{2\pi i p \theta}}{N} \frac{1 - e^{2\pi i p N \theta}}{1 - e^{2\pi i p \theta}} \rightarrow 0.$$

Hence by Weyl's Theorem $\{\langle n\theta \rangle\}_{n \geq 1}$ is equidistributed.

15.5.10 E8) Prove that the sequence $\{\xi_n = \langle \sqrt{n} \rangle\}_{n \geq 1}$ is equidistributed on $[0, 1]$.

Proof: See E8).

15.5.11 E9) For an arbitrary $a \neq 0$, $\sigma \in (0, 1)$, prove that the sequence $\{\xi_n = \langle a n^\sigma \rangle\}_{n \geq 1}$ is equidistributed on $[0, 1]$.

Proof: Use Fejér's theorem above. Note: Using Van Der Corput, we can prove the statement for any $\sigma \in \mathbb{R}_{>0} - \mathbb{Z}$, by considering $\Delta^k(a n^\sigma)$.

15.5.12 E10) Prove that for any $a \in \mathbb{R}$, the sequence $\{\xi_n = \langle a \log n \rangle\}_{n \geq 1}$ is not equidistributed on $[0, 1]$.

Proof: Let $f(x) = a \log x$, then $\lim_{x \rightarrow \infty} f'(x) = 0$. Consider

$$\left| \sum_{n=1}^{N-1} e^{2\pi i f(n)} - \int_1^N e^{2\pi i f(x)} dx \right| \leq 4\pi \sum_{n=1}^{N-1} \sup_{\xi \in [n, n+1]} |g'(\xi)| = o(N)$$

and

$$\frac{1}{N} \left| \int_1^N e^{2\pi i f(x)} dx \right| = \frac{1}{N} \left| \int_0^{\log N} e^{(2\pi i a + 1)t} dt \right| = \left| \frac{e^{2\pi i \log N} - N^{-1}}{2\pi i a + 1} \right| \rightarrow \frac{1}{|2\pi i a + 1|} \neq 0.$$

Hence $\{\xi_n\}$ is not equidistributed on $[0, 1]$.

15.6 F: Winding Number of Closed Curve

Let $E = \{f : \mathbb{R} \rightarrow \mathbb{C}^\times : f \in C^1, f(x+2\pi) = f(x)\}$, where $\mathbb{C}^\times = \mathbb{C} - \{0\}$. For $f \in E$, $f|_{[0,2\pi]}$ represents a closed curve on $\mathbb{C}$ that does not contain 0. For any $f \in E$, define

$$d(f) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f'(t)}{f(t)} dt.$$

15.6.1 F1) Prove that $d(f)$ is well-defined and calculate $d(f_n)$ where

$$f_n : \mathbb{R} \rightarrow \mathbb{C}^\times, x \mapsto f_n(x) = e^{inx}.$$

Proof: Since $f'_n(x) = ine^{inx}$, then

$$d(f_n) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f'_n(t)}{f_n(t)} dt = \frac{1}{2\pi i} \int_0^{2\pi} in dt = n.$$

15.6.2 F2) Represent $d(f)$ in polar coordinates.

Suppose $f(t) = \rho(t)e^{i\theta(t)}$ where $\rho(t), \theta(t) \in \mathbb{R}$, then $f'(t) = \rho'(t)e^{i\theta(t)} + i\rho(t)\theta'(t)e^{i\theta(t)}$, hence

$$d(f) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{\rho'(t)}{\rho(t)} dt + \frac{1}{2\pi i} \int_0^{2\pi} \theta'(t) dt = \frac{\log \rho(2\pi) - \log \rho(0)}{2\pi i} + \frac{\theta(2\pi) - \theta(0)}{2\pi} \quad (15.18)$$

$$= \frac{\theta(2\pi) - \theta(0)}{2\pi}. \quad (15.19)$$

Which is the number of times f circles around 0, counterclockwise.

15.6.3 F3) Use the function

$$\psi(t) = \exp \int_0^t \frac{f'(s)}{f(s)} ds$$

to show that $d(f) \in \mathbb{Z}$. Proof: Note that

$$\psi'(t) = \psi(t) \cdot \frac{f'(t)}{f(t)}.$$

Hence let $g(t) = \psi(t)/f(t)$, then $g'(t) = (\psi'(t)f(t) - \psi(t)f'(t))/f^2(t) = 0$, so $\psi(t) = Cf(t)$ for all $t \in [0, 2\pi]$. Therefore

$$1 = \psi(0) = Cf(0) = Cf(2\pi) = \psi(2\pi) = e^{2\pi id(f)},$$

i.e. $d(f) \in \mathbb{Z}$.

15.6.4 F4) Prove that for any $f \in E$, there exists $\varepsilon > 0$ such that for any $g \in E$, if

$$\|f - g\|_\infty < \varepsilon, \text{ then } d(g) = d(f).$$

Proof: Suppose $f = \rho_1 e^{i\theta_1}$ and $g = \rho_2 e^{i\theta_2}$. Note that for any $z = \rho e^{i\theta}$ and $w = re^{it}$, $|z - w| \geq \sup_{r>0} |z - re^{it}| = \rho|\sin(\theta - t)|$. Hence $\|f - g\|_\infty < \varepsilon$ implies $\|\sin(\theta_1 - \theta_2)\|_\infty < \delta = \varepsilon / \inf \rho_1$. Let $\theta(t) = \theta_1(t) - \theta_2(t)$, then when $\delta < 1$, $\theta(t)$ lies in $\bigcup (2k\pi - \alpha, 2k\pi + \alpha)$ for some $\alpha < \pi$. Since $\theta(t)$ is continuous, it must stay in the same interval, hence $|\theta(0) - \theta(2\pi)| < 2\pi$, i.e. $|d(f) - d(g)| < 1$ so $d(f) = d(g)$. Hence $\varepsilon = \inf \min \rho_1/2$ is sufficient.

15.6.5 F5) Try to define $d(f)$ for $f \in E$ using F4) and Weierstrass-Stone for trigonometric polynomials.

Solution: For $f_n(x) = e^{inx}$, let $d(f) = n$. For a trigonometric polynomial $P(x) = \sum_{k=-n}^n c_k f_k(x)$, let $Q(x) = \sum_{k=0}^{2n} c_{k-n} x^k$ and $d(P)$ be the number of roots of Q on the unit circle. For an arbitrary $f \in E$, take a sequence of trigonometric polynomials $\{P_n\}_{n \geq 1}$ uniformly convergent to f . The sequence $\{P_n\}_{n \geq 1}$ is Cauchy, so by F4) $d(P_n)$ is eventually constant, and let $d(f) = \lim_{n \rightarrow \infty} d(P_n)$. From F4) this definition is the same as the original one.

15.6.6 F6) Prove invariance under homotopy: Suppose $F(t, \tau) : \mathbb{R} \times [0, 1] \rightarrow \mathbb{C}^\times$ is continuous and for any $\tau \in [0, 1]$, $F(\cdot, \tau) \in E$, then

$$d(F(x, 0)) = d(F(x, 1)).$$

Proof: Let $S = \{\tau \in [0, 1] : d(F(x, 0)) = d(F(x, \tau))\}$, and $s = \sup S$. ($0 \in S$ so S is non-empty). Using F4), there is an $\varepsilon > 0$ such that for any $g \in E$, if $\|F(\cdot, s) - g\|_\infty < \varepsilon$, then $d(F(\cdot, s)) = d(g)$. Since $F(t, \tau)$ is continuous, there exists $\delta > 0$ such that $r \in (s - \delta, s + \delta)$ implies $\|F(\cdot, s) - F(\cdot, r)\|_\infty < \varepsilon$ so $d(F(\cdot, s)) = d(F(\cdot, r))$. Hence we can obtain $s \in S$ and $s = 1$, i.e.

$$d(F(x, 0)) = d(F(x, 1)).$$

15.6.7 F7) Suppose $P(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0$ is a complex polynomial, and $P(0) \neq 0$. Prove that there exists $\varepsilon_0 > 0$, such that for any $\varepsilon \in (0, \varepsilon_0)$, the function $f_\varepsilon(x) = P(\varepsilon e^{ix}) \in E$ and calculate $d(f_\varepsilon)$.

Proof: Note that $\|f_\varepsilon - c_0\|_\infty \rightarrow 0$ so by F4) such ε_0 exists and $d(f_\varepsilon) = 0$.

15.6.8 F8) Following F7), prove that there exists $R_0 > 0$ such that for any $R \in (R_0, \infty)$, $f_R(x) = P(Re^{ix}) \in E$ and calculate $d(f_R)$.

Proof: Let $g_R(x) = f_R(x)/R^n$, then $d(f_R) = d(g_R)$ and $\|g_R - e^{nix}\|_\infty \rightarrow 0$, hence $d(f_R) = d(g_R) = n$.

15.6.9 F9) Prove that every complex polynomial of degree n has at least one root.

Proof: Let $F(x, t) = P(Rte^{ix})$ where $F : \mathbb{R} \times [0, 1] \rightarrow \mathbb{C}^\times$ and R as in F7). If P has no roots, then $F(\cdot, t) \in E$ for any $t \in [0, 1]$, but for some $\varepsilon > 0$, $d(F(\cdot, \varepsilon)) = 0$ while $d(F(\cdot, 1)) = n$, leading to contradiction with F6).

15.7 G: Bolzano Curve

Define the sequence of functions on $[0, 1]$ $\{f_n\}_{n \geq 0}$ inductively, where $f_0(x) = x$, and for $n \geq 0$, $0 \leq k \leq 3^n$,

$$\begin{cases} f_{n+1}(k3^{-n}) &= f_n(k3^{-n}), \\ f_{n+1}(k3^{-n} + 3^{-n-1}) &= f_n(k3^{-n} + 2 \cdot 3^{-n-1}), \\ f_{n+1}(k3^{-n} + 2 \cdot 3^{-n-1}) &= f_n(k3^{-n} + 3^{-n-1}). \end{cases}$$

and f_{n+1} is linear on every interval $[k3^{-n-1}, (k+1)3^{-n-1}]$.

15.7.1 G1) Prove that for any $n \geq 0$, and any $0 \leq x, y \leq 1$,

$$|f_n(x) - f_n(y)| \leq 2^n |x - y|.$$

Proof: We can assume that $x = k \cdot 3^{-n}$ and $y = (k+1) \cdot 3^{-n}$, then $|x - y| = 3^{-n}$. Suppose $f_n(x) = f_{n-1}(v)$ and $f_n(y) = f_{n-1}(u)$, we obtain $|f_n(x) - f_n(y)| \leq 2^{n-1}|u - v| \leq 2^n \cdot 3^{-n} = 2^n|x - y|$.

15.7.2 G2) Prove that for any $n \geq 0$ and any $x \in [0, 1]$,

$$|f_{n+1}(x) - f_n(x)| \leq \frac{2^n}{3^{n+1}}.$$

Proof: The function $f_{n+1}(x) - f_n(x)$ is linear on every interval $[k3^{-n-1}, (k+1)3^{-n-1}]$, so we can assume that $x = k \cdot 3^{-n} + r \cdot 3^{-n-1}$ for some $k \leq 3^m$ and $r \in \{0, 1, 2\}$. By G1), in all three cases $|f_{n+1}(x) - f_n(x)| \leq 2^n \cdot 3^{-n-1}$.

15.7.3 G3) Prove that $\{f_n\}_{n \geq 1}$ converges uniformly to some $f \in C([0, 1])$.

Proof: For any N , any $x \in [0, 1]$ and $n > m > N$,

$$|f_n(x) - f_m(x)| \leq \frac{1}{3} \sum_{n=N}^{\infty} \left(\frac{2}{3}\right)^n = \frac{2^{N-1}}{3^N} \rightarrow 0.$$

Hence $\{f_n\}_{n \geq 1}$ converges uniformly to f , and $f \in C([0, 1])$ since f_n are all continuous.

15.7.4 G4) Prove that for any $n \geq 1$, and any $0 \leq k \leq 3^n$, $f(k3^{-n}) = f_n(k3^{-n})$.

Proof: Trivial since for any $m > n$, $f_m(k \cdot 3^{-n}) = f_{m-1}(k \cdot 3^{-n}) = f_n(k \cdot 3^{-n})$.

15.7.5 G5) Prove that for any $n \geq 1$ and any $0 \leq k \leq 3^n$, f is not differentiable at $k3^{-n}$.

Proof: As in G1), we can show by induction that for any $m > k$, $|f(k \cdot 3^{-n}) - f(k \cdot 3^{-n} + 3^{-m})| = 2^m 3^{-m}$ so $f'(k \cdot 3^{-n})$ does not exist. (Same for $f'(1)$).

15.7.6 G6) Prove that f is nowhere differentiable on $[0, 1]$.

Proof: If f is differentiable at x , then for any $\{h_n\}$ and $\{k_n\}$ such that $h_n, k_n > 0$ and $\lim_{n \rightarrow \infty} h_n = \lim_{n \rightarrow \infty} k_n = 0$,

$$\lim_{n \rightarrow \infty} \frac{f(x + h_n) - f(x - k_n)}{h_n + k_n} = f'(x).$$

Define h_n and k_n such that $x + h_n$ and $x - k_n$ are in the form $\{(k+1) \cdot 3^{-n}, k \cdot 3^{-n}\}$, then we infer f is not differentiable at x .

15.8 Problem W: e is Transcendental

Suppose $P(x)$ is a real polynomial of degree n . Let

$$I(t) = \int_0^t e^{t-x} P(x) dx.$$

15.8.1 W1) Prove that $I(t) = e^t \sum_{i=0}^n P^{(i)}(0) - \sum_{i=0}^n P^{(i)}(t)$.

Proof: Note that $(e^{t-x})^{(n)} = (-1)^n e^{t-x}$, then using integration by part we obtain

$$I(t) = (-1)^n \sum_{i=0}^n (-1)^i (e^{t-x})^{(n-i+1)} P^{(i)}(x) \Big|_0^t = e^t \sum_{i=0}^n P^{(i)}(0) - \sum_{i=0}^n P^{(i)}(t).$$

15.8.2 W2) Suppose there exist integers $a_0, a_1, \dots, a_n$, where $a_0 \neq 0$ such that

$$a_0 + a_1 e + a_2 e^2 + \dots + a_n e^n = 0.$$

For $p \in \mathbb{Z}_{\geq 0}$, let

$$P(x) = x^{p-1}(x-1)^p(x-2)^p \cdots (x-n)^p$$

and define

$$J = a_0 I(0) + a_1 I(1) + \cdots + a_n I(n).$$

Prove that $J \in \mathbb{Z}$ and $(p-1)! \mid J$. Proof: Denote $C = \sum_{i=0}^{\infty} P^{(i)}(0)$, then from W1),

$$J = \sum_{k=0}^n a_k I(k) = \sum_{k=0}^n a_k \left(e^k C - \sum_{i=0}^{\infty} P^{(i)}(k) \right) = - \sum_{k=0}^n \sum_{i=0}^{\infty} a_k P^{(i)}(k) \in \mathbb{Z},$$

Denote $F(k) = \sum_{i=0}^{\infty} P^{(i)}(k)/(p-1)!$, we show that $F(k) \in \mathbb{Z}$ for any $0 \leq k \leq n$. When $k = 0$, let $u(x) = x^{p-1}/(p-1)!$ and $v(x) = (x-1)^p \cdots (x-n)^p$, then

$$P^{(m)}(0) = \sum_{j=0}^m u^{(j)}(0) v^{(m-j)}(0) \binom{m}{j}$$

so

$$F(0) = \sum_{j=p-1}^{\infty} v^{(j-p+1)}(0) \binom{j}{p-1} \in \mathbb{Z}.$$

If p is a prime greater than n , then $F(0) = v(0) + p(\cdots) \in \mathbb{Z} - p\mathbb{Z}$. When $1 \leq k \leq n$, let $u(x) = (x-k)^p/(p-1)!$ and $v(x) = P(x)/(x-k)^p$, then

$$P^{(m)}(k) = \sum_{j=0}^m u^{(j)}(k) v^{(m-j)}(0) \binom{m}{j}$$

so

$$F(k) = \sum_{j=p}^{\infty} p v^{(j-p)}(0) \binom{j}{p} \in p\mathbb{Z}.$$

Therefore $(p-1)! \mid J$.

15.8.3 W3) Prove that if p is a prime greater than n then $J \neq 0$ hence $|J| \geq (p-1)!$.

Proof: From W2) we know that $F(0) \not\equiv 0 \pmod{p}$, and $p \mid F(k)$ for any $1 \leq k \leq n$. hence $J \not\equiv 0 \pmod{p}$ so $J \neq 0$.

15.8.4 W4) Prove that there exists $C > 0$ such that for any $p \in \mathbb{Z}_{\geq 1}$, $|J| \leq C^p$.

Proof: For any $0 \leq k \leq n$,

$$|I(k)| \leq \int_0^k |e^{k-x} P(x)| dx \leq e^n \cdot n^{(n+1)p},$$

hence

$$|J| \leq e^n \cdot n^{(n+1)p} \cdot \sum_{j=0}^n |a_n| < C^p.$$

where $C = n^{n+1} \cdot e \cdot (1 + \sum_{j=0}^n |a_n|)$.

15.8.5 W5) Prove that e is transcendental.

Proof: Otherwise by W3) and W4), there exists $C > 0$ such that for any prime $p > n$, $C^p \leq (p-1)!$ which contradicts with the infinity of primes.

15.8.6 Ex: π is also transcendental

Consider the identity $e^{i\pi} + 1 = 0$. Suppose πi is algebraic (with degree n), then

$$0 = \prod_{i=1}^n (1 + e^{\gamma_i}) = \sum_{\varepsilon_i \in \{0,1\}} e^{\sum_i \varepsilon_i \gamma_i} = a + \sum_{i=1}^m e^{\alpha_i},$$

where γ_i are the conjugates of πi , $a = 2^n - m \geq 1$ are the number of zero exponents in the first sum, and α_i are all the non-zero exponents. Note that

$$\phi(x) = \prod_{\varepsilon \in \{0,1\}} \left(x - \sum_{i=1}^n \varepsilon_i \gamma_i \right) \in \mathbb{Q}[x].$$

Let

$$\psi(x) = C \frac{\phi(x)}{x^a} = \sum_{i=0}^m b_i x^i \in \mathbb{Z}[x], b_m > 0, b_0 \neq 0,$$

whose roots are exactly α_i . Furthermore, $b_m \alpha_i$ are all algebraic integers. Apply the identity W1) to the polynomial

$$f(x) = \frac{b_m^{(m-1)p}}{(p-1)!} x^{p-1} \psi^p(x) = \frac{b_m^{mp}}{(p-1)!} x^{p-1} \prod_{i=1}^m (x - \alpha_i)^p.$$

Plug in $x = \alpha_i$ and sum over i , we obtain

$$-aF(0) - \sum_{i=1}^m F(\alpha_i) = \sum_{i=1}^m e^{\alpha_i} \int_0^{\alpha_i} f(t) e^{-t} dt. \quad (1)$$

Note that

$$F(0) = (-1)^{mp} b_m^{mp} \left(\prod_i \alpha_i \right)^p \in \mathbb{Z} - p\mathbb{Z}.$$

Also

$$\sum_{i=1}^m F(\alpha_i) = p b_m^{mp} \sum_i \alpha_i^{p-1} \prod_{j \neq i} (\alpha_i - \alpha_j)^p \in p\mathbb{Z}$$

for large p since it is symmetric in α_i and the denominator is cleared by b_m^{mp} . Therefore the LHS of (1) is a non-zero integer. We estimate the integral:

$$\left| e^{\alpha_i} \int_0^{\alpha_i} f(t) e^{-t} dt \right| \leq (|\alpha_i b_m^{m-1} \psi(\alpha_i)|)^p / (p-1)! \rightarrow 0$$

as $p \rightarrow \infty$, reaching contradiction.

15.8.6.1 Further Theorems:

1. Hilbert's 7th question: For any algebraic number $a \notin \{0, 1\}$, and irrational number b , a^b is transcendental.
2. Hermite-Lindemann: e^α is transcendental for any $\alpha \in \bar{\mathbb{Q}} - \{0\}$.

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